technische universität dortmund

Chair of Enterprise Computing
Department of Computer Science

Business Process Management

Introduction

WWU MÜNSTER

SAP

KIT
Karlsruher Institut für Technologie

Julius-Maximilians-
UNIVERSITÄT
WÜRZBURG

tu

Christian's CV

Your Expectations?

http://www.nichtlustig.de/

Code of Conduct

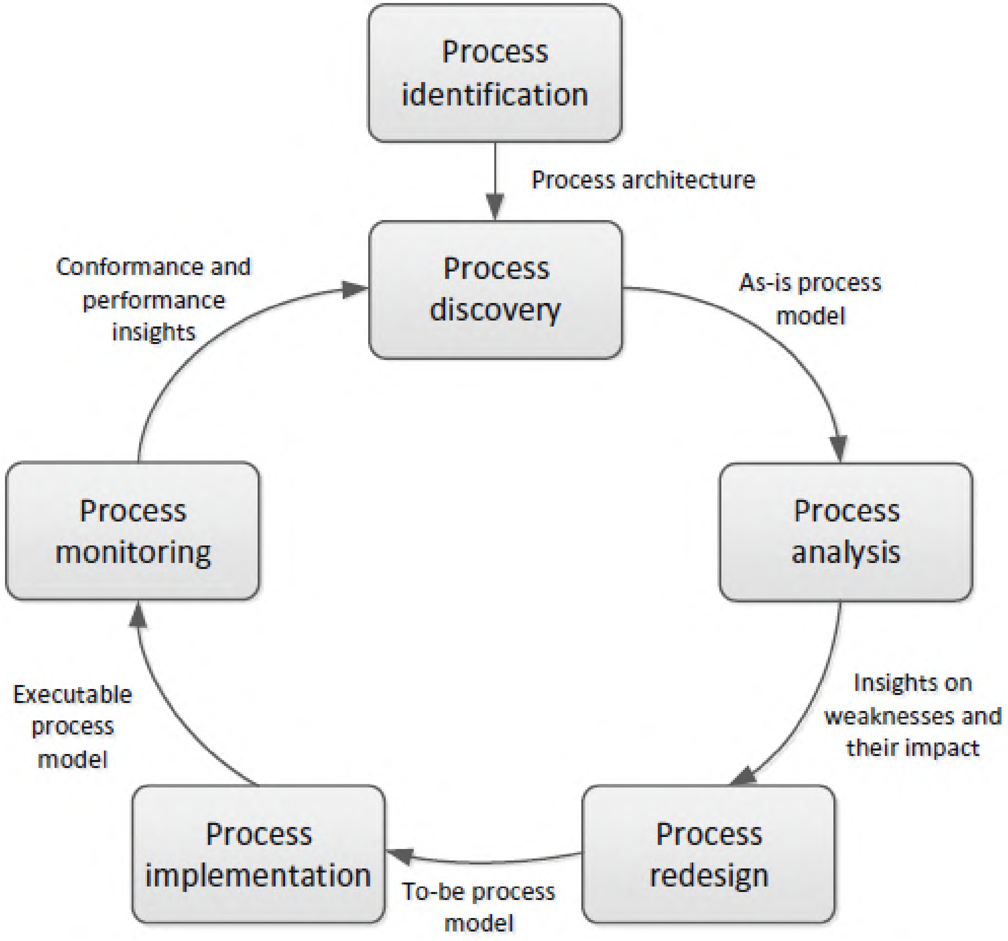
- Please ask **questions** if something is unclear or if there is a discussion
- Please use the platform **moodle**, to ask questions, download material and hand in assignments
- Please prepare the lectures and assignments, by looking at and reading the material **prior to coming to the lecture**
- Please **do not distribute** the material provided to you in this course

Let’s Get Organized!

Questions you should be able to answer after the lecture

- Why do we **need** business processes?
- How do I **structure and model** business processes?
- What techniques can be used to **discover** processes?
- How can I use **qualitative** and **quantitative** process **analysis**?
- How do I **redesign** processes?
- How do I make process models **executable**?
- What does a **BPMS** look like?
- What is **RPA**?
- How can I realize process **monitoring** and **analytics**?
- How can I install BPM as an **enterprise capability**?

The BPM Lifecycle



Dumas et al. (2018)

Roadmap

Introduction	10.04. Introduction	17.04. Process Identification			
Process Discovery & Modeling	24.04. Essentials of BPMN	01.05. Public Holiday	08.05. Advanced BPMN I	15.05. Advanced BPMN II	22.05. Process Discovery
Process Analysis & Redesign	29.05. Public Holiday	05.06. Qualitative Analysis	12.06. Quantitative Analysis	19.06. Public Holiday	26.06.2025 Process Redesign
Application of BPM	03.07. Process Technology	10.07. Process Monitoring and Mining	15.07. BPM as an Enterprise Capability		

Anwendungsfach Enterprise Computing (Bachelor)

Pflicht BIS
(Janiesch)

Pflicht BPM
(Janiesch)

Pflicht MNP
(Rehof)

Wahlpflicht EC 8 LP

Wahlpflicht WiWi 15 LP

Fachprojekt EC

Proseminar

Bachelorarbeit

Unsere Veranstaltungen im neuen Anwendungsfach Enterprise Computing (Bachelor)

Pflicht BIS (Janiesch)

Pflicht BPM (Janiesch)

Pflicht MNP (Rehof)

Wahlpflicht EC 8 LP
(z.B. 4 ECTS ERCIS Winter School)

Wahlpflicht WiWi 15 LP

Fachprojekt EC – Process Automation

Proseminar – EC

Bachelorarbeit

Mandatory Reading

- Dumas, La Rosa, Mendling & Reijers (2018): Fundamentals of Business Process Management. 2nd Edition. Springer.
  - https://doi.org/10.1007/978-3-662-56509-4

- Dumas, La Rosa, Mendling & Reijers (2021): Grundlagen des Geschäftsprozessmanagements. Springer.
  - https://doi.org/10.1007/978-3-662-58736-2

Suggested Further Reading

- **Further Business Process Management Textbooks**
  - Becker, Kugeler & Rosemann (2012): Prozessmanagement: Ein Leitfaden zur prozessorientierten Organisationsgestaltung. 7th Edition. Springer. (LINK)
  - vom Brocke & Rosemann (2015): Handbook on Business Process Management 1: Introduction, Methods, and Information Systems. 2nd Edition. Springer. (LINK)
  - vom Brocke & Rosemann (2015): Handbook on Business Process Management 2: Strategic Alignment, Governance, People and Culture. 2nd Edition. Springer. (LINK)

- **Process Technology**
  - Weske (2024): Business Process Management: Concepts, Languages, Architectures. Springer. (LINK)

Suggested Further Reading

- **Process Mining**
  - van der Aalst (2016): Process Mining: Data Science in Action. 2nd Edition. Springer. (LINK)
  - van der Aalst & Carmona (2022): Process Mining Handbook. Springer. (LINK)

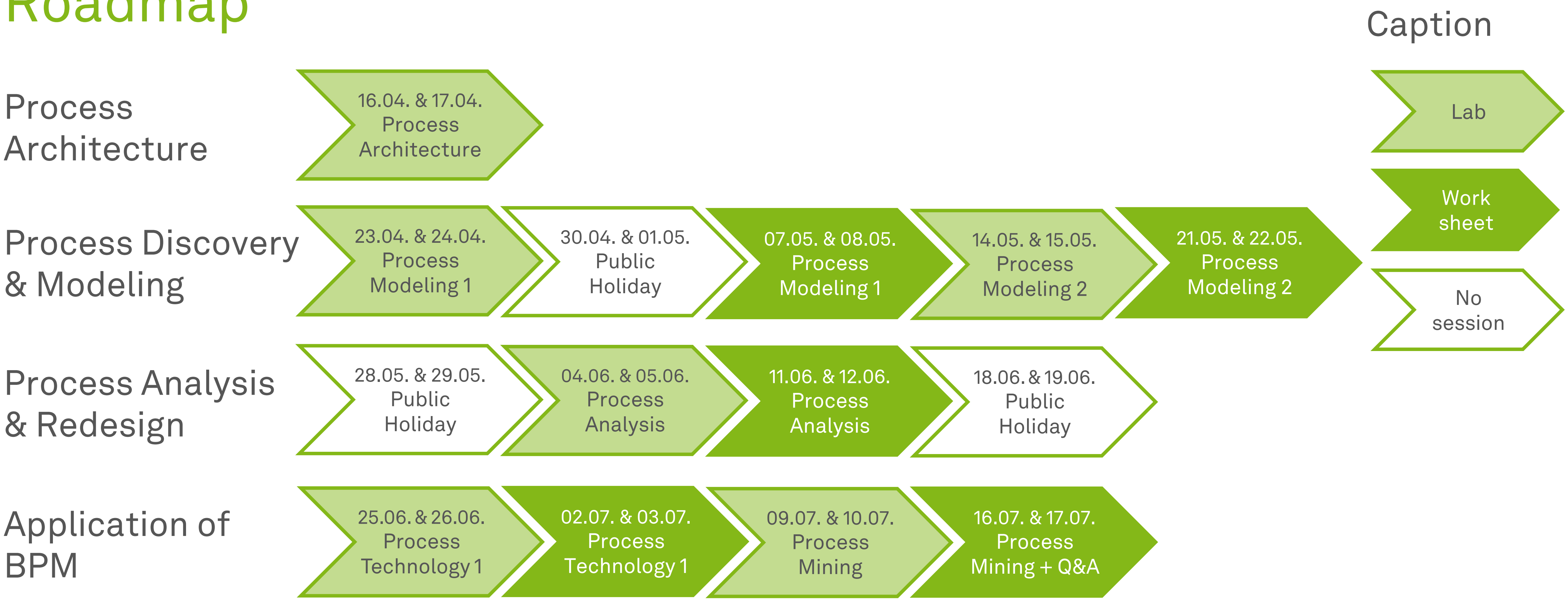
- **Business Process Management in Practice**
  - vom Brocke & Mendling (2018): Business Process Management Cases: Digital Innovation and Business Transformation in Practice. Springer. (LINK)
  - vom Brocke, Mendling & Rosemann (2021): Business Process Management Cases Vol. 2: Digital Transformation - Strategy, Processes and Execution. Springer. (LINK)
  - vom Brocke, Mendling & Rosemann (2025): Business Process Management Cases Vol. 3: Implementation in Practice. Springer. (LINK)

Lab Class

- Concurrent to the lecture
  - Tutors **Seyyid A. Ciftci**
  - Lab classes with exercises

- Attend one of two available lab classes
  - Starting from **16.04.2025** and **17.04.2025**
  - Every Wednesday – 14:15 - 15:45h
    - Location: **OH12, 2.063**
  - Every Thursday – 10:15 - 11:45h (if need be)
    - Location: **OH14, 3.04**

Roadmap


Caption
Lab
Work sheet
No session
Process Architecture
16.04. & 17.04. Process Architecture
Process Discovery & Modeling
23.04. & 24.04. Process Modeling 1
30.04. & 01.05. Public Holiday
07.05. & 08.05. Process Modeling 1
14.05. & 15.05. Process Modeling 2
21.05. & 22.05. Process Modeling 2
Process Analysis & Redesign
28.05. & 29.05. Public Holiday
04.06. & 05.06. Process Analysis
11.06. & 12.06. Process Analysis
18.06. & 19.06. Public Holiday
Application of BPM
25.06. & 26.06. Process Technology 1
02.07. & 03.07. Process Technology 1
09.07. & 10.07. Process Mining
16.07. & 17.07. Process Mining + Q&A

Assignments

- ...are available at the end of the preceding lab class
- ...must be handed in by **midnight on Tuesday** before the next class
- ...should be submitted in a team of **three**
- ...more next Wednesday and Thursday with Seyyid

- Example

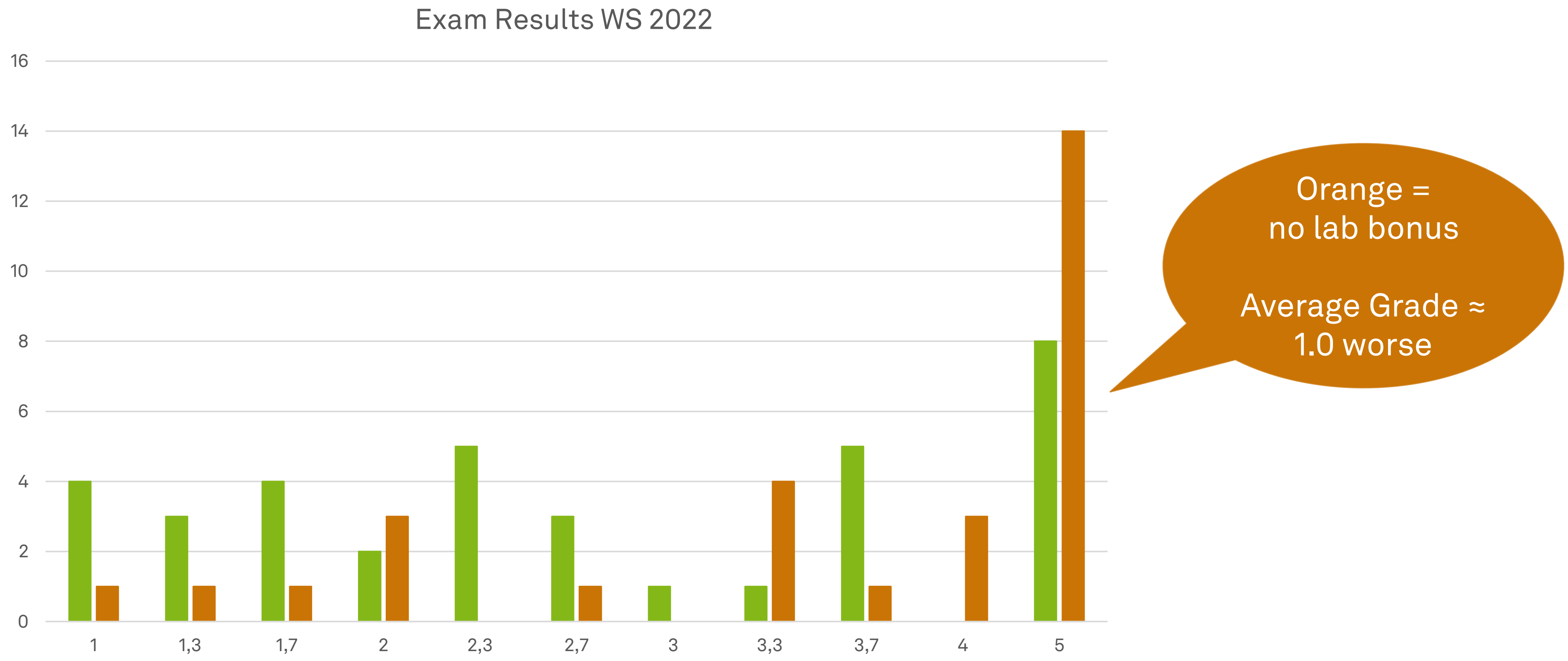
2	23.04.2025	2nd Lab Class and 1st Assignment
3	07.05.2025	3rd Lab Class and 1st Presentations
...	...	...

Assignment will be online after class

Hand in until 06.05.2025 at midnight

Discussion of 1st assignment

Reason 1 why you should do your assignments



Reason 2 why you should do your assignments

You need to **successfully*** complete **all** assignments to improve your grade of the **passed** exam by one step

* in time and complete + presented at least once

Introduction and Definitions

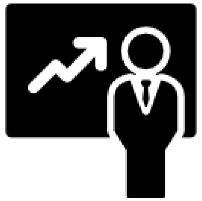
Learning Goals of Today

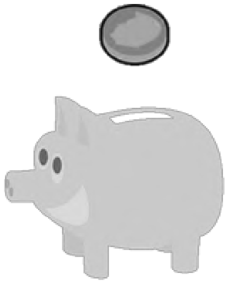
- Get to know the elements and structure of a business process
- Get an initial understanding of business process management and the business process lifecycle
- (Review (in brief) of the history of BPM)

What is a Process?


Vacation Planning

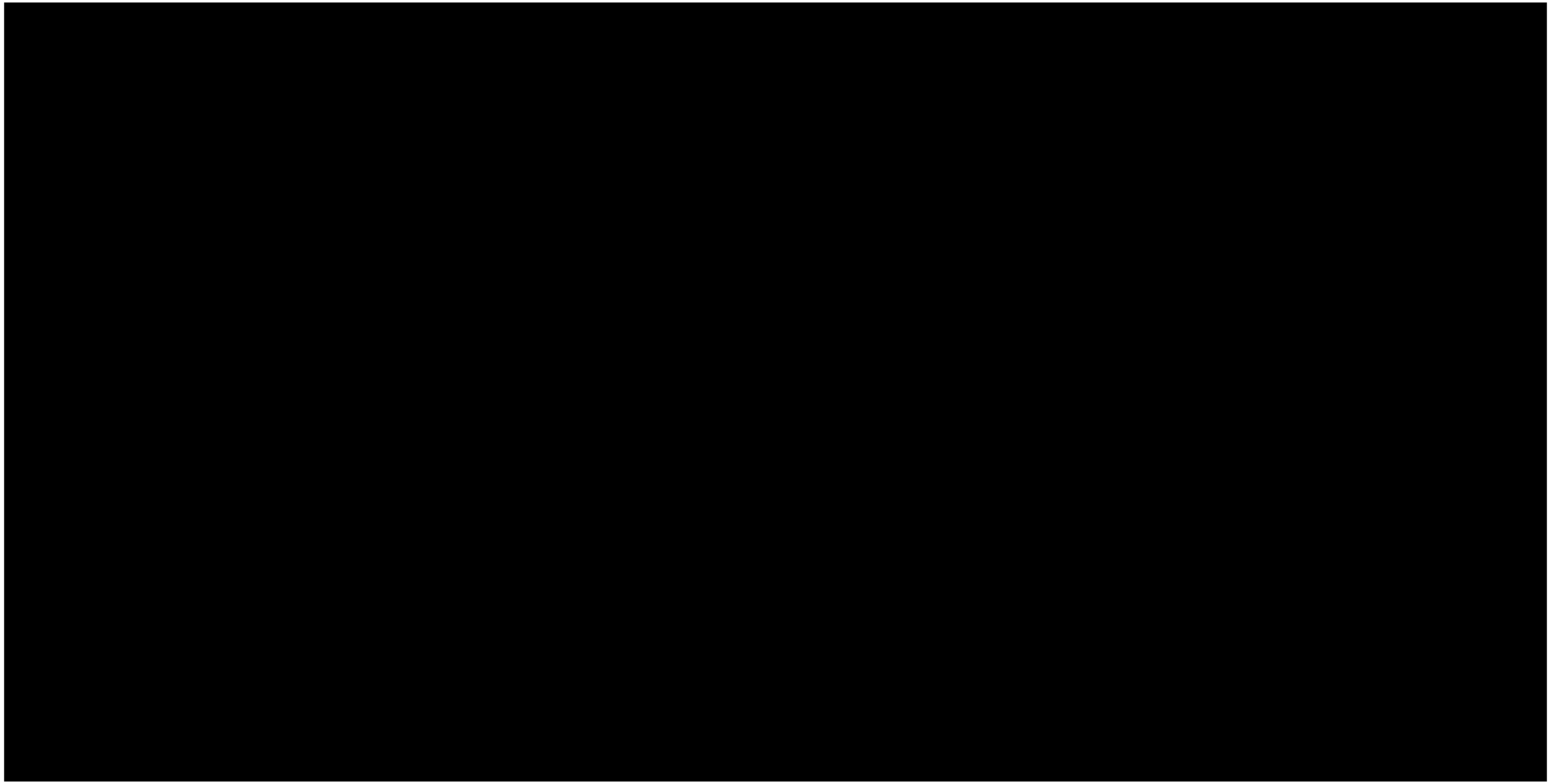
Cleaning plan



Business Processes


Budget Planning


Calendar Management

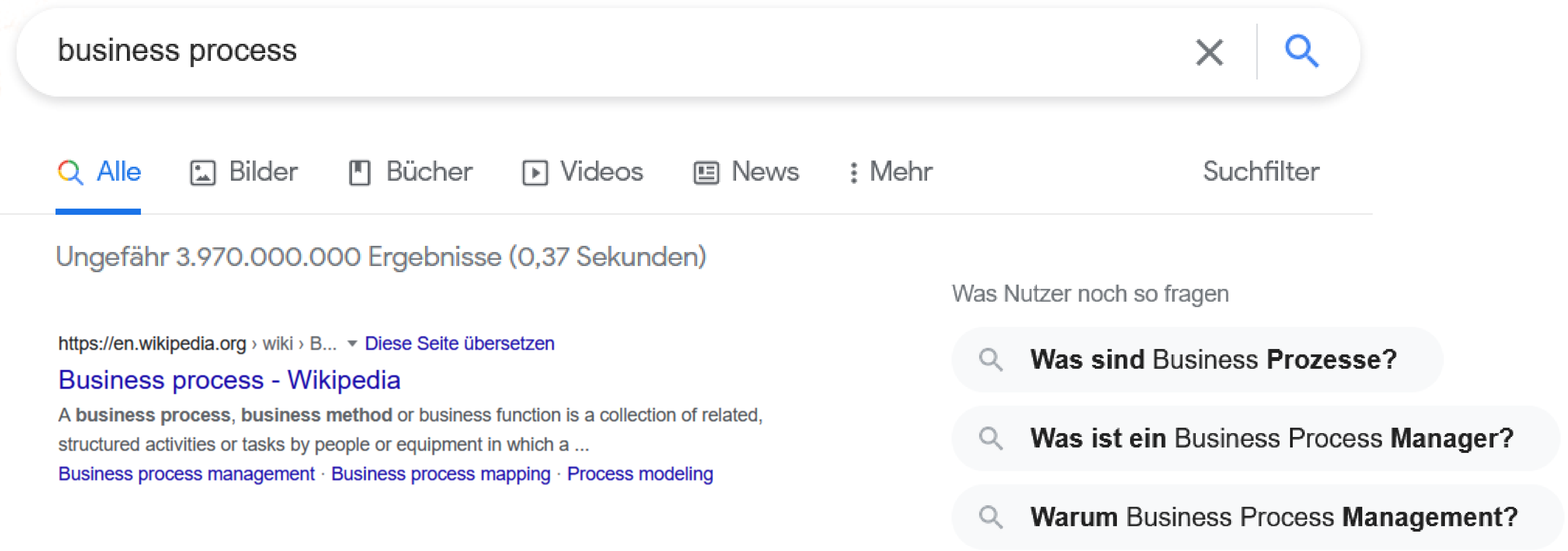
Business Process Management at Lincoln Trust



IBM (2011)

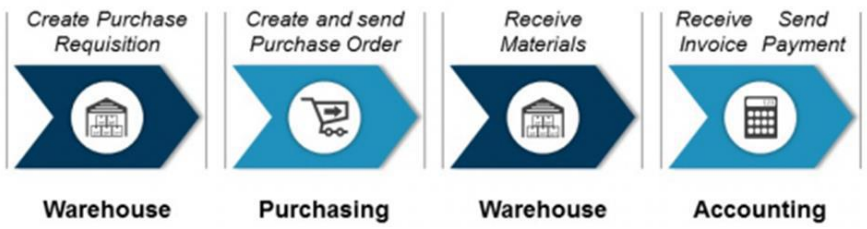
What is a Business Process?

- Querying Google with "**Business Process**", the search engine delivers about **4 billion results**



Google (2022)

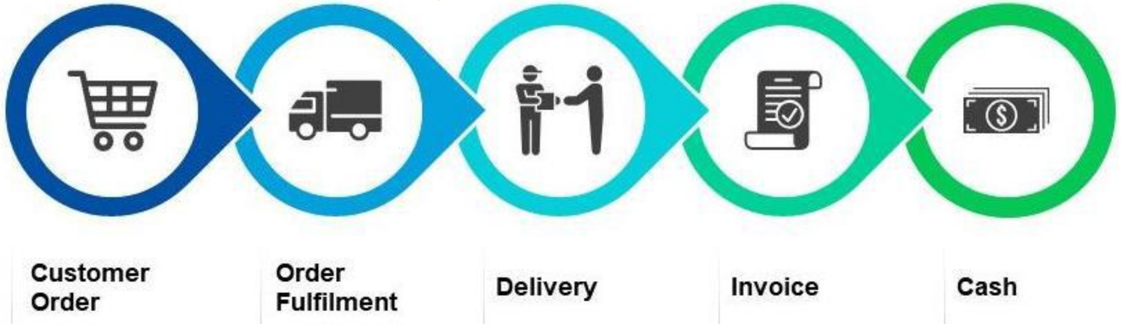
Examples for Business Processes

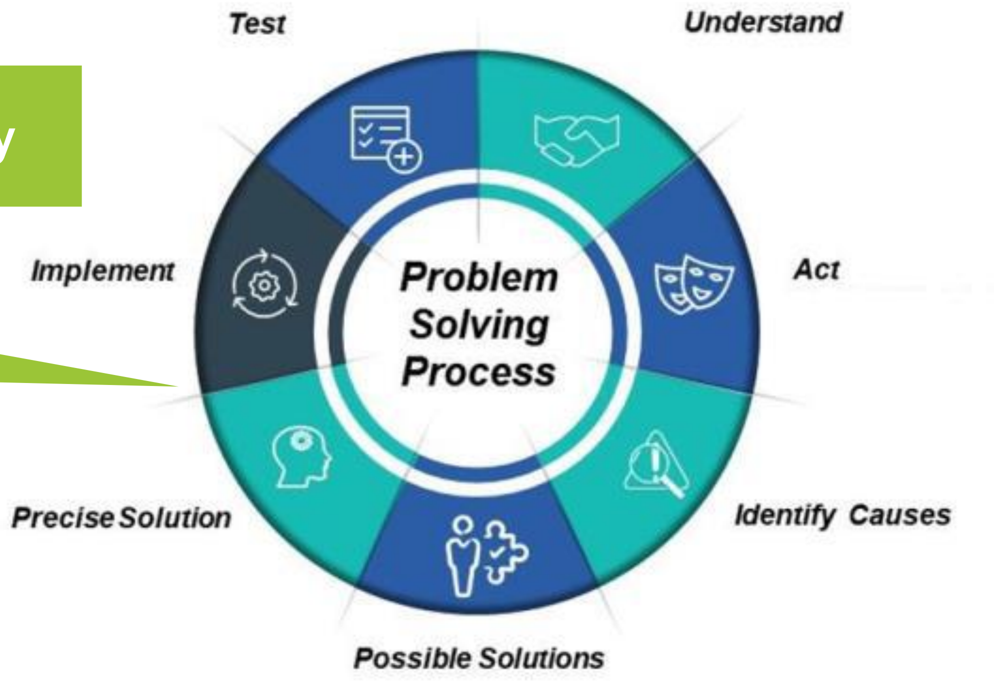


Procure-to-Pay

Order-to-Cash

Fault-to-Resolution





Application-to-Approval

Claim-to-Settlement

Elements and Structure of a Process

- Business processes are defined by a set of **”ingredients”**
  - **Actors** (internal, external // humans, organizations, software systems)
  - **Tasks and Activities** (single units of work // more coarse-grained units of work)
  - **Events** (things that happen instantaneously without resources)
  - **Decision Points** (decision affecting the way the process is executed)
  - **Physical Objects** (equipment, materials, products, paper documents)
  - **Informational Objects** (electronic documents, electronic records)

- Business Process have an **outcome** for the consumer that can be
  - **Positive** or
  - **Negative**

Dumas et al. (2018)

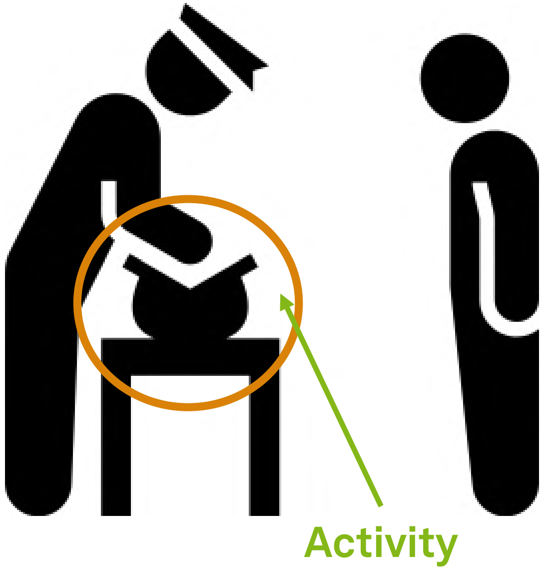
Elements and Structure of a Process

- **Actors**
  - Entities that are performing process activities and generate events
  - Can be **humans**, **organizations**, or **software systems**,
    - For example, "*Airport Security Officer*", "*Financial Officer*", etc. (roles)
    - "ERP", "CRM", etc. (software systems)

Dumas et al. (2018)

Elements and Structure of a Process

- **Activities**
  - Active elements, for example “Check Luggage”
    - **Time-consuming**
    - **Resource-demanding**
    - **State-changing**



Dumas et al. (2018)

Elements and Structure of a Process

- **Events**
  - Passive elements, for example “Sharp Object Found”
  - Stands for conditions or circumstances
  - Atomic and instantaneous

Dumas et al. (2018)

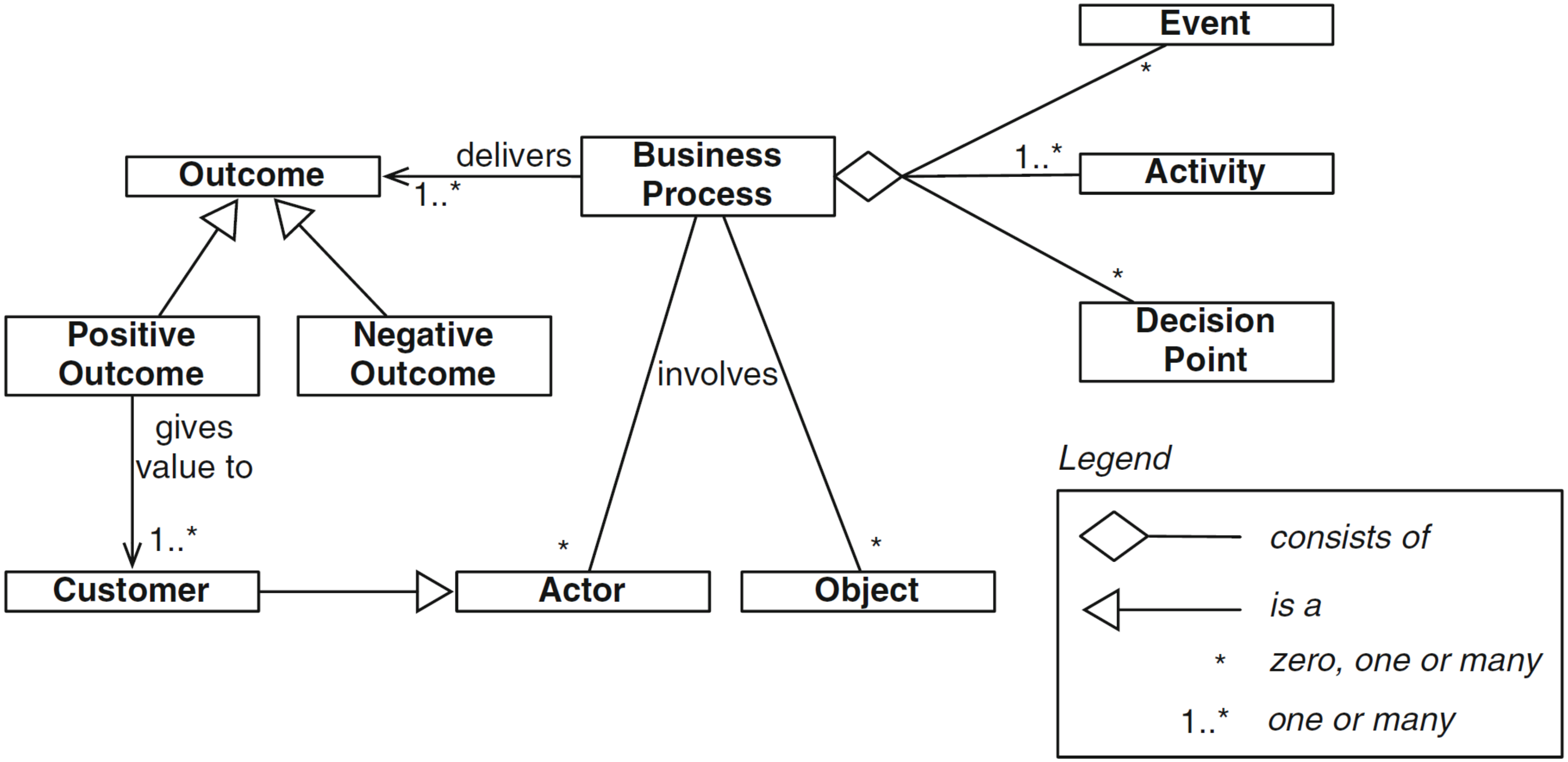
Elements and Structure of a Process

- **Decision Points**
  - Affect future process execution
  - For example, decide “whether the offense was punishable”

- **Objects**
  - For example, warrant, sales order, digital object, etc.
  - Can undergo **state changes**
  - Contain physical or electronic information

Dumas et al. (2018)

Elements and Structure of a Process Diagram



Dumas et al. (2018)

Definition of a Business Process

- A business process is
  - a **collection** of inter-related events, activities, and decision points
  - that involve a number of **actors and objects**
  - which collectively **lead to an outcome that is of value** to at least one customer
- The activities of a business process **jointly** realize a business goal

Dumas et al. (2018)

So then, what is the M in **BPM**?

An Abstract Definition of Business Process Management

- Business Process Management (BPM) is the **art and science** of overseeing how work is performed in an organization
  - to **ensure consistent outcomes** and
  - to **take advantage of improvement*opportunities**

- * improvement: one-off or continuous
  - reduce costs
  - reduce execution times
  - reducing error rates
  - **gain competitive advantage through innovation**

Dumas et al. (2018)

Definition of Business Process Management

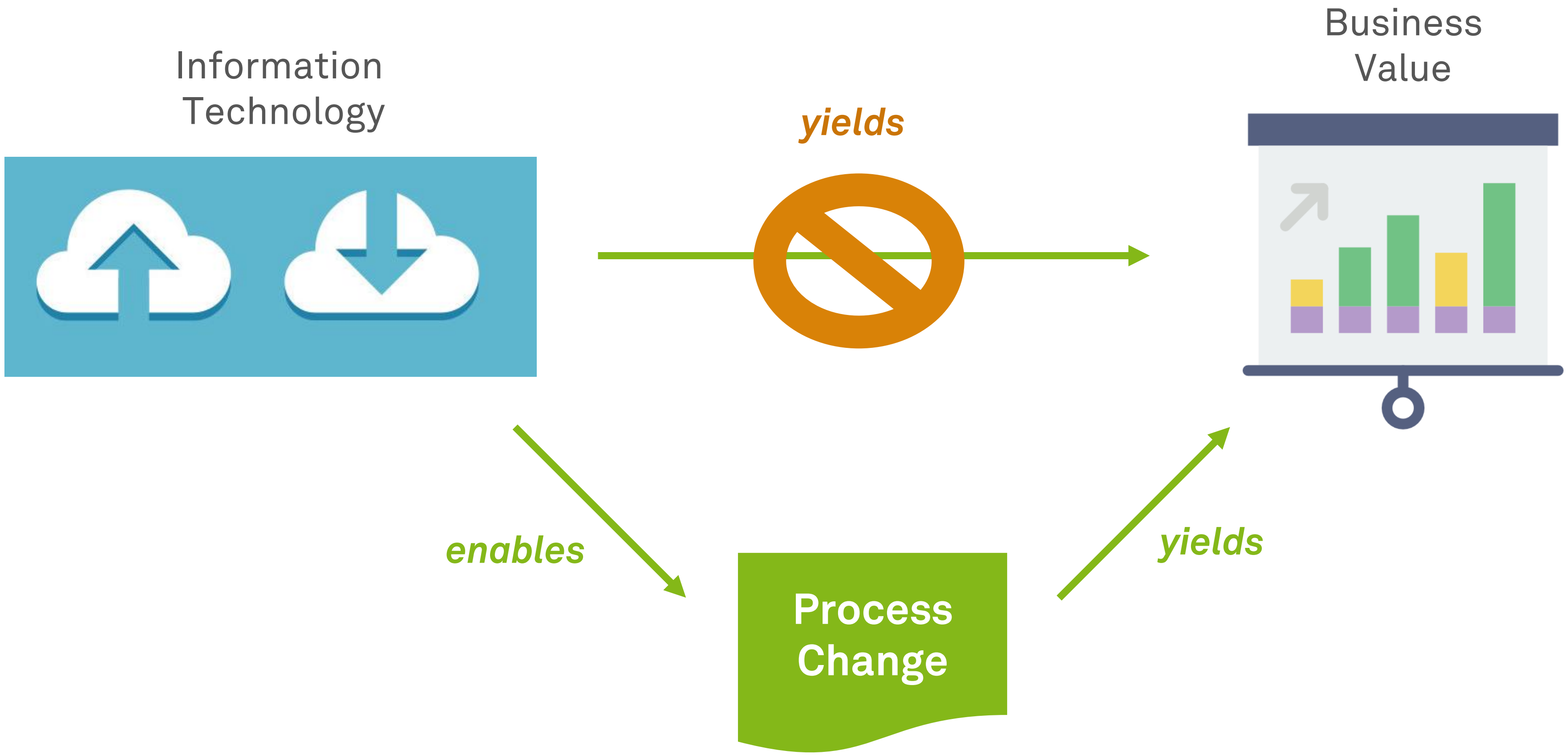
- Business Process Management (BPM) is
  - a **body of** methods, techniques, and tools
  - to **identify**, **discover**, **analyze**, **redesign**, **execute**, and **monitor** business processes
  - in order to **optimize** their performance.

- BPM **is**...
  - enabling technical and organizational improvements of processes
  - providing powerful methods for documenting processes
  - providing frameworks for introducing new processes

- BPM **is not** ...
  - the automation of manual tasks
  - re-engineering the enterprise
  - change management

Dumas et al. (2018)

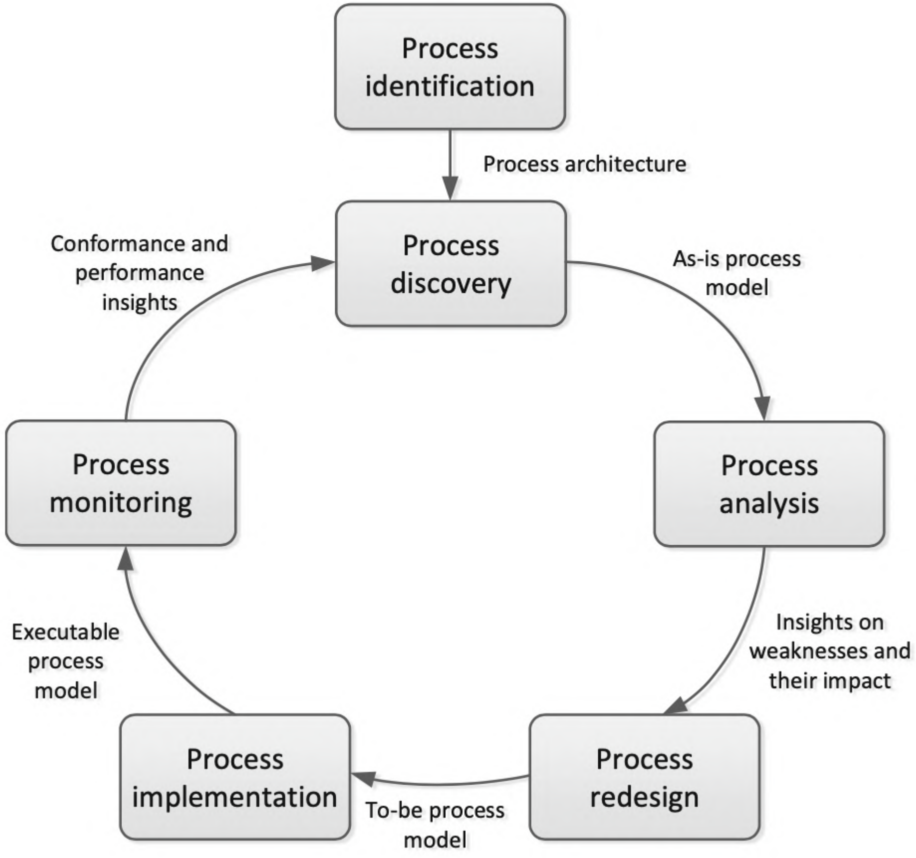
Why Do We Need BPM?



Dumas et al. (2018), Index Group (1982)

BPM Lifecycle

- Consists of **phases** related to each other
- Organized in a **cyclical structure**, showing their logical dependencies
- **No strict temporal ordering**, some phases might occur during each of the phases



Dumas et al. (2018)

BPM Six Core Elements

Strategic Alignment	Governance	Methods	Information Technology	People	Culture	Factors
Process Improvement Planning	Process Management Decision Making	Process Design & Modelling	Process Design & Modelling	Process Skills & Expertise	Responsiveness to Process Change	Capability Areas
Strategy & Process Capability Linkage	Process Roles and Responsibilities	Process Implementation & Execution	Process Implementation & Execution	Process Management Knowledge	Process Values & Beliefs	
Enterprise Process Architecture	Process Metrics & Performance Linkage	Process Monitoring & Control	Process Monitoring & Control	Process Education	Process Attitudes & Behaviors	
Process Measures	Process Related Standards	Process Improvement & Innovation	Process Improvement & Innovation	Process Collaboration	Leadership Attention to Process	
Process Customers & Stakeholders	Process Management Compliance	Process Program & Project Management	Process Program & Project Management	Process Management Leaders	Process Management Social Networks	

Rosemann & vom Brocke (2015)

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Chair of Enterprise Computing
Department of Computer Science

A Brief History of
Business Process Management

From Function Orientation to Process Orientation

Function Orientation

- Organizations seek **local** optimization and perfection of functional areas
- The aim is **to increase quality**
- Technological and organizational developments
  - Outsourcing (e.g., call centers)
  - Integrated standard software (ERP)

Process Orientation

- **Focus on processes** within the organization – not on functional areas or local optimization!
- Assume that running a business is a **continuous process**
- It makes sense to structure an organization along its **technical processes**

Becker et al. (2012)

From Function Orientation to Process Orientation

Function Orientation

- Organizatio seek **local**
  opti
  fun
- The
- Tec
  developments
  - Outsourcing (e.g., call centers)
  - Integrated standard software (ERP)

Problems

- Autonomous functional areas
- Cost increase through internal arrangements and coordination

Process Orientation

- **Focus on p cesses** within the
  org al
  are
- Ass ss is
  a **co**
- It makes sense to structure an organization along its **technical processes**

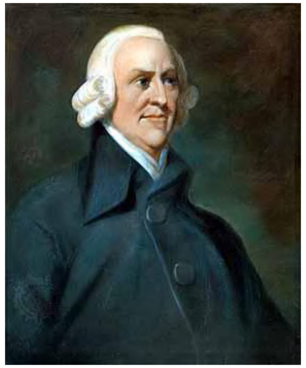
Solution

- Organizations need to move from functional orientation to **process orientation**

Becker et al. (2012)

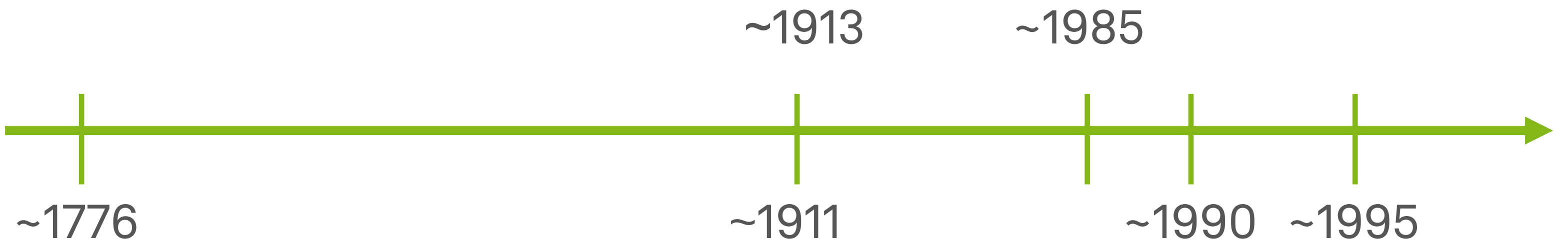
History of BPM

Management Concepts (1776)



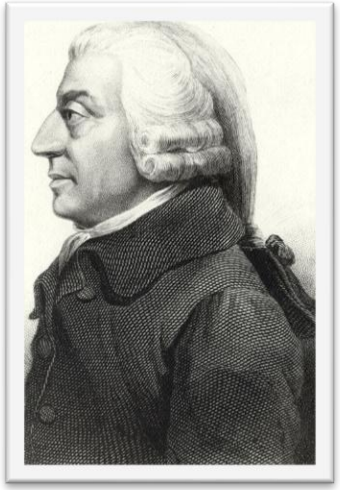
Adam Smith
1723 - 1790

- Book **"An Inquiry into the Nature and Causes of the Wealth of Nations"**
- **Specialization** of labor
- **Organized** around tasks
- **Observation** that specialized workers could produce far more output that the same number of generalists
- Gives example of pin maker

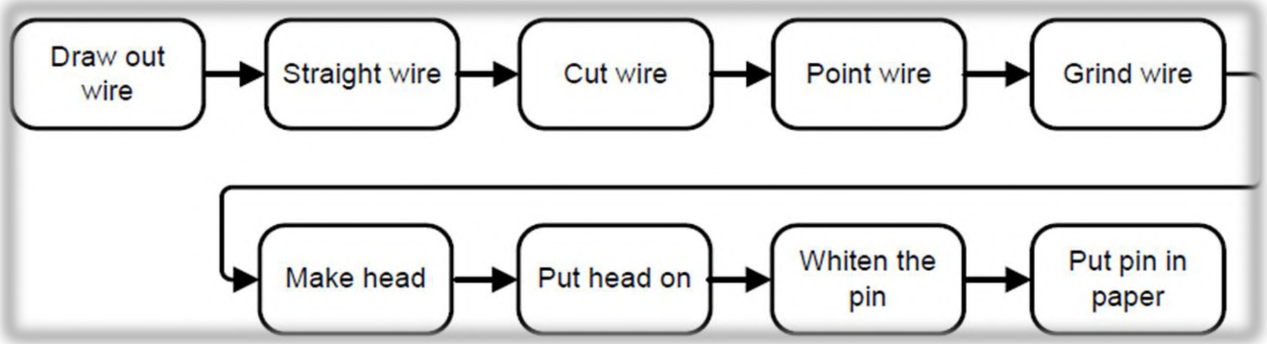


Dumas et al. (2018), Rosemann (2002)

Adam Smith: Processes and Specialization of Labor



- “To take an example, the trade of a pin-maker: But in the way in which this business is now carried on, it is divided into a number of branches:
- One man draws out the wire; another straights it;
- a third cuts it; a fourth points it; a fifth grinds it at the top for receiving the head;
- to make the head requires three operations; to put it on is a peculiar business; to whiten the pins is another; to put them into the paper;
- and the important business of making a pin is, in this manner, divided into about eighteen distinct operations.”

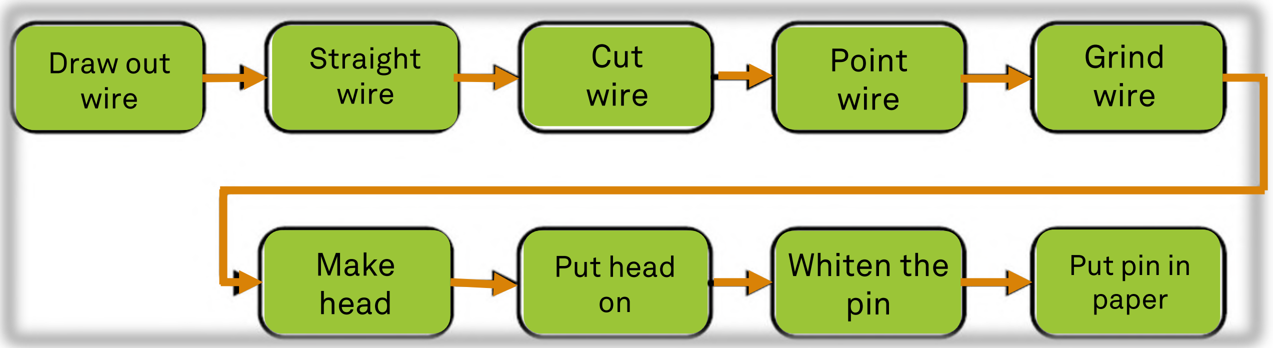


Dumas et al. (2018), Hansen et al. (2019)

Task Efficiency and Coordination Efficiency

Task Efficiency

Coordination Efficiency



Hansen et al. (2019)

History of BPM

Management Concepts (1911)

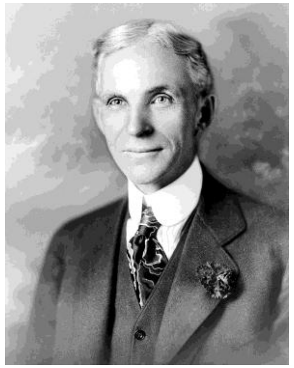
- **Scientific Management** (Taylorism)
  - Analyzes and synthesizes workflows to improve economic efficiency
  - Introduction of managers
  - **Assumption** that behavior at work can be engineered and designed according to principles of rationality and efficiency

Frederick Taylor
1856 - 1915

Dumas et al. (2018), Rosemann (2002)

History of BPM

Management Concepts (1913)



Henry Ford
1863 - 1947

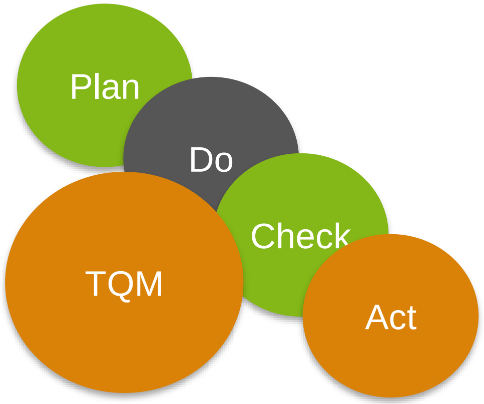
- Introduction of a moving **assembly line**
- Bring the work to the worker – not vice versa

- **Alfred Sloan**: Adapted the idea of Smith to management and created smaller divisions aiming at a division of professional labor

Dumas et al. (2018), Rosemann (2002)

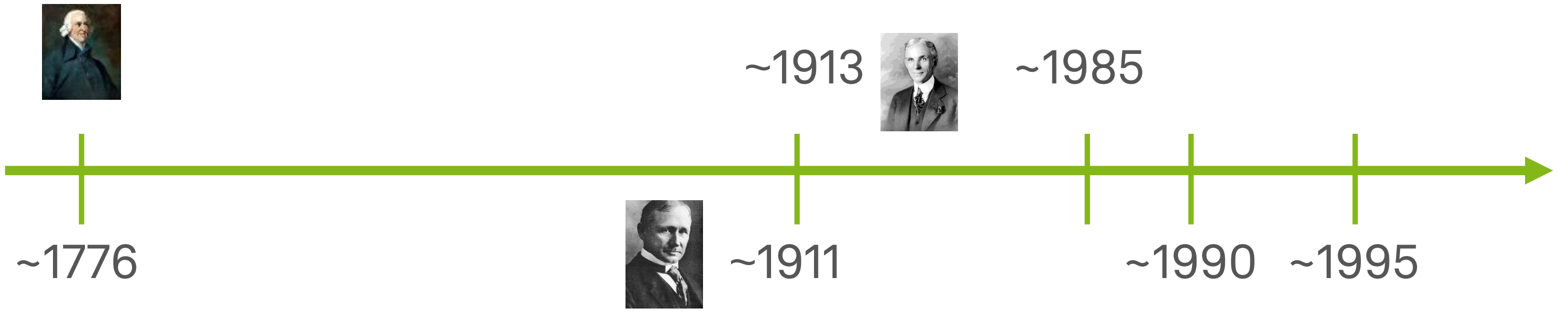
History of BPM

Management Concepts (1985)



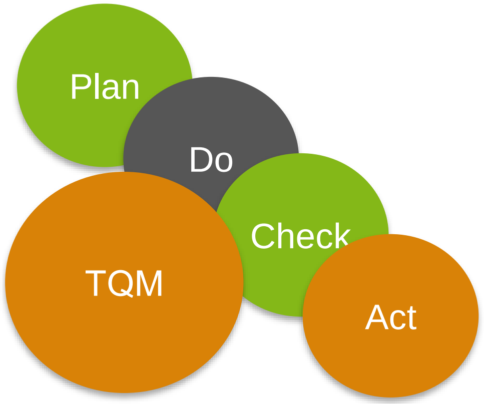
- There are many different origins of quality management
  - **Frederick Taylor** laid a foundation for quality management by empowering the efficiency movement
  - **Walter Shewhart** created a method for quality control for production (1924 firstly proposed)
  - **William Deming** brought Shewhart ideas to Japan in the 1950s

Total Quality Management
~1985



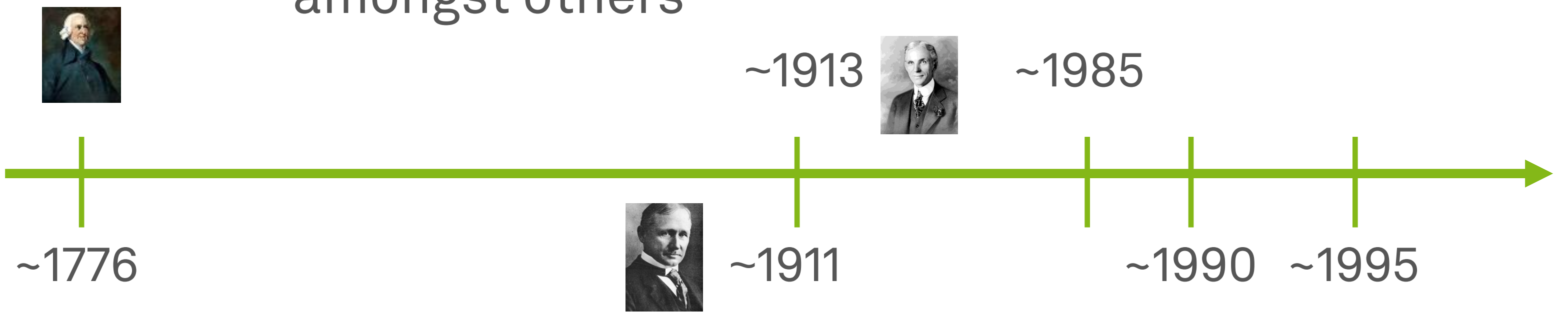
History of BPM

Total Quality Management



- Organization-wide efforts to continuously improve quality of products and services
- Necessity of ongoing improvement efforts
- There is no single popular and widely agreed-upon approach
  - Effort heavily rely on tools and techniques of quality control
  - Notable approaches include lean management and six sigma amongst others

Total Quality Management
~1985



History of BPM

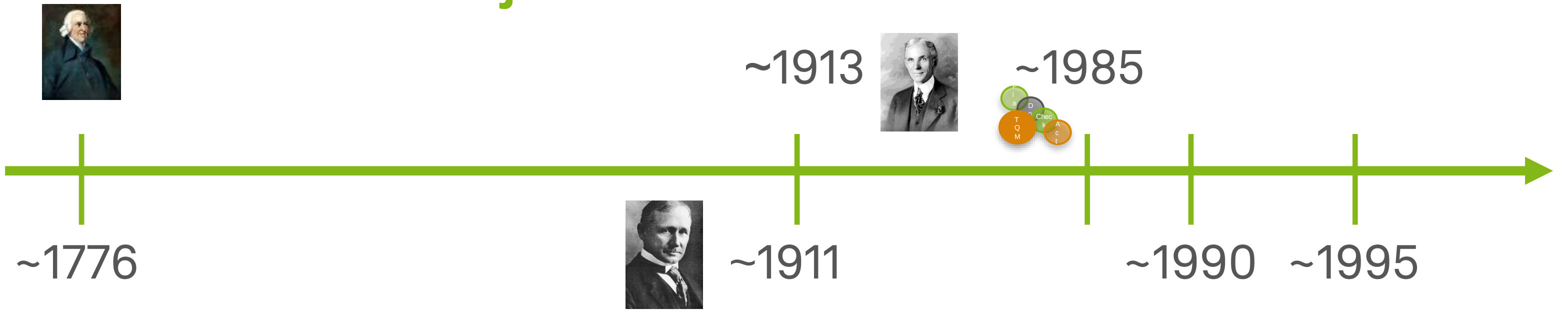
Management Concepts (1990)

Business Process Reengineering
~1990

"Sound strategy starts with having the right goal."

Michael Porter

- The basic idea of process orientation goes back to **Porter's value chain**
  - Added value is created within along the **value chain** of a business (i.e., cross-functional)
  - Decomposing the value chain exposures **transactions**, which can be assembled to **processes**
  - Porter further stresses the concept of **interoperability** across the value chain as a **major issue** within firms



Porter (2000), Hammer & Champy (1993)

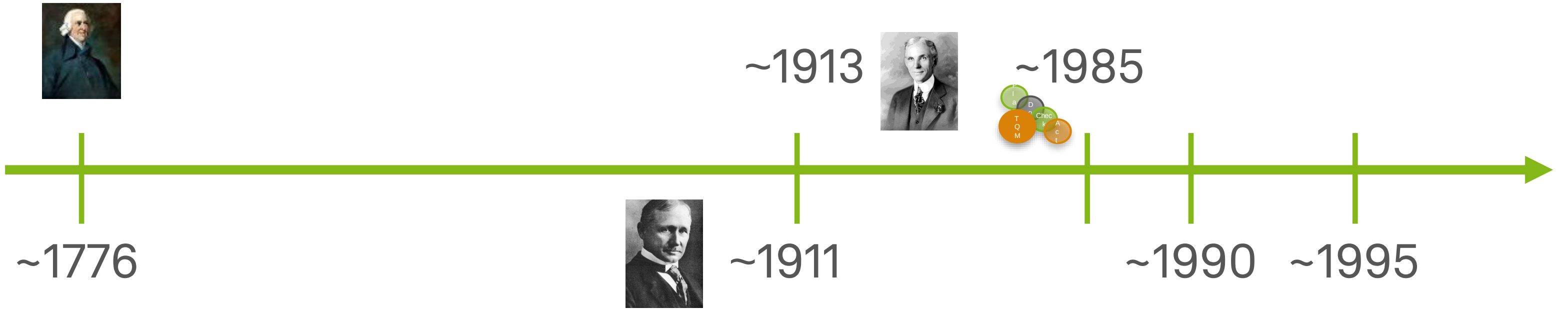
History of BPM

Management Concepts (1990)

Business Process Reengineering
~1990

- Davenport and Short, as well as Hammer and Champy and others, brought **reengineering** of processes into play
- In 1990, they popularized the use of Process Reengineering as **Business Process Reengineering** or Business Process Redesign (BPR)

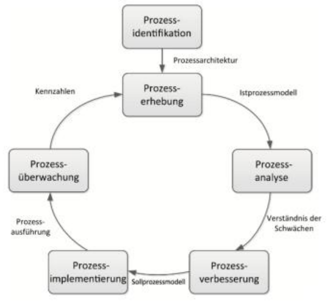
*“Reengineering means to **disregard all the assumptions and traditions** of the way business has always been done, and instead develop a **new, process-centered business organization** that achieves a **quantum leap forward** in performance.”*
Michael Hammer



Hammer & Champy (1993)

History of BPM

Management Concepts (1995+)



Business Process Management
~1995

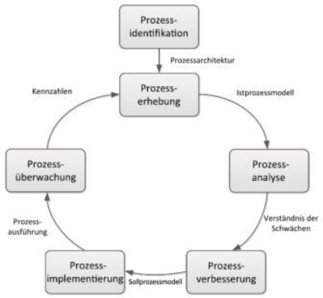
- The emergence of **Business Process Management** was caused due to **dissatisfaction and risks** with BPR
  1. Many organizations **misused** the concept of **BPR** to any possible project that did not contain processes at its core
  2. Redesign of processes has often been progressed **too radical**
  3. There was **no sufficient (IT) support** on BPR projects

Dumas et al. (2018)

History of BPM

Management Concepts (1995+)

- The rise of **process orientation**
  - **Studies** show that process-orientated organizations did better than non-process-oriented organizations
  - Different types of **IT systems emerged** made it easier to implement changes to business processes



Business Process Management
~1995

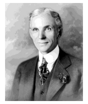
~1776

ERP

Enterprise Resource Planning

~1911

~1913



Workflow Management

~1985

EAI

Enterprise Application Integration

~1990

~1995

Dumas et al. (2018)

Learning Goals of Today

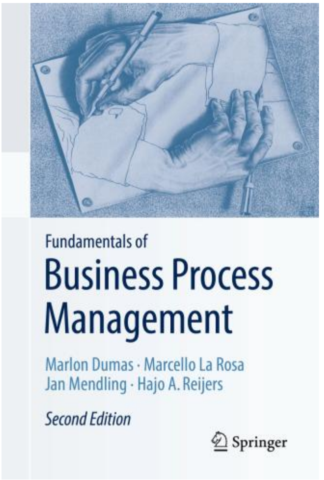
- Get to know the elements and structure of a business process
- Get an initial understanding of business process management and the business process lifecycle
- (Review (in brief) of the history of BPM)

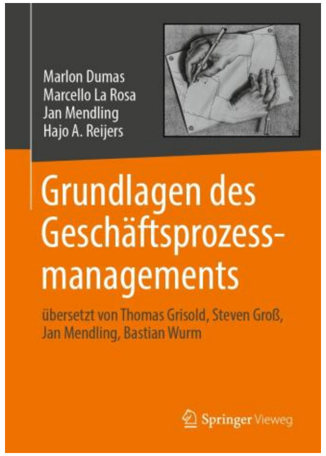
Recap Questions

- What is a business process and how is BPM defined?
- Why do we need BPM to yield business value from information technology?
- (What are the differences between function orientation and process orientation?)
- (What are the major steps in the evolution of the specialization of labor to BPM as we know it today?)

Literature

- **Section 1:** Introduction to Business Process Management





Roadmap

Introduction	10.04. Introduction	17.04. Process Identification			
Process Discovery & Modeling	24.04. Essentials of BPMN	01.05. Public Holiday	08.05. Advanced BPMN I	15.05. Advanced BPMN II	22.05. Process Discovery
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Any Questions about BPM?

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Department of Computer Science

Business Process Management

Process Identification

Roadmap

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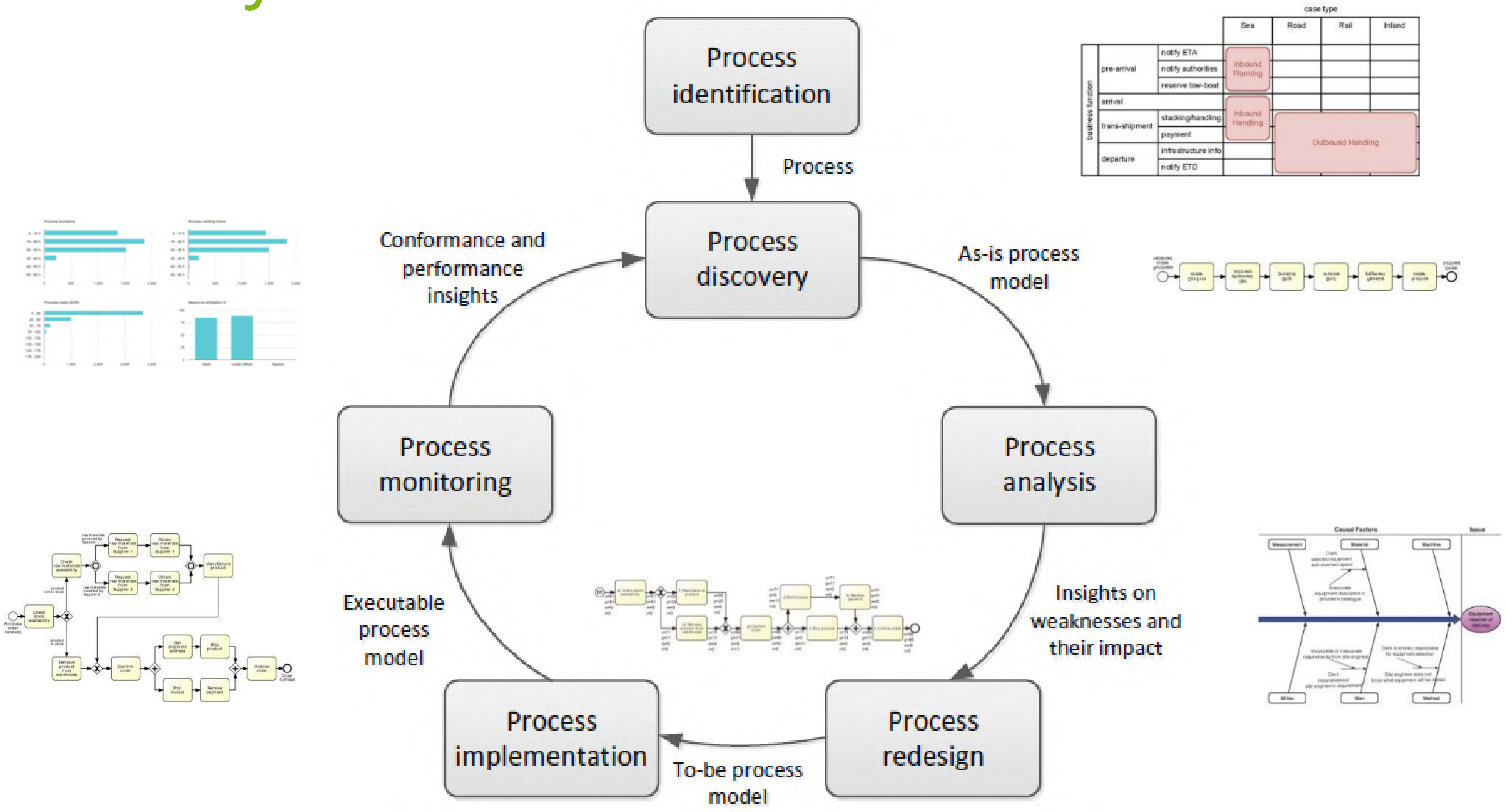
Learning Goals of Today

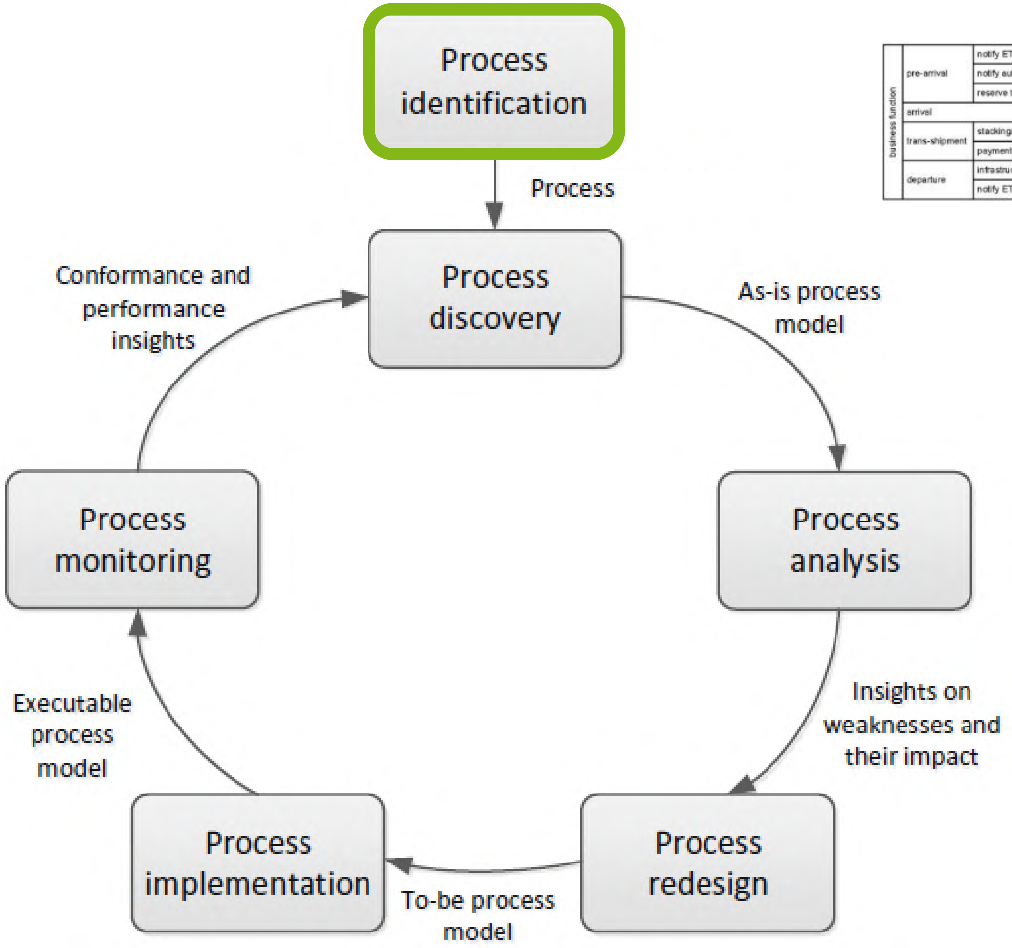
- Understand the process identification phase
- Understand process architecture definition
- Understand process selection
- Be able to work with process architectures and process portfolios

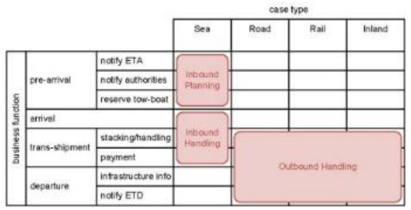
The BPM Lifecycle



Dumas et al. (2018)

The BPM Lifecycle





Dumas et al. (2018)

Overview of Process Identification




Process Identification

- ...is a **set of activities**...
- ...performed by **management, process analysts**, and **process owners**...
- ...to **establish** a process architecture (definition)...
- ...and to establish **focus** (selection) ...
- ...in order to create a working **process portfolio**.

- Can you imagine sets of processes?

- Can you imagine prioritization criteria?

Dumas et al. (2018)

Definition Phase

- **Understand** the processes and their interrelations (and the organization)
- **Define** processes and their connections
- **Determine** process scope
  - **boundaries** (horizontal and vertical) and
  - **interrelationships** (order and hierarchy)
- Caveat: There are various models to categorize processes

Davenport (1993), Dumas et al. (2018)

Selection Phase

- Establish **criteria** to prioritize the management of these processes
- **Prioritize** for modeling, redesign, automation, monitoring
- **Select** based on alignment with
  - Strategic objectives
  - Health (e.g., performance, compliance, sustainability…)
  - Feasibility to being successfully improved
  - Culture & politics
  - Risk of not improving them

Davenport (1993), Dumas et al. (2018)

Process Identification Results

- **Goal: Maximize value** of BPM initiatives
- **Output: Process Architecture** and **Process Portfolio**
- Serve as a **framework** for defining priorities and scope of subsequent BPM phases

Davenport (1993), Dumas et al. (2018)

Trade-off: Impact and Manageability

- The **larger** the process initiative, the **easier** it will become to spot opportunities for **efficiency gain**
- **But**
  - the involvement of a **large** number of staff will make effective communication **problematic**
  - keeping models of a large process **up-to-date** is more **difficult**
  - improvement projects that are related to a **large** process are more **complex**

- Rule of thumb
  - **10** to **20 main processes**
  - Distinguish **broad** and **narrow** processes

Dumas et al. (2018)

And one more thing

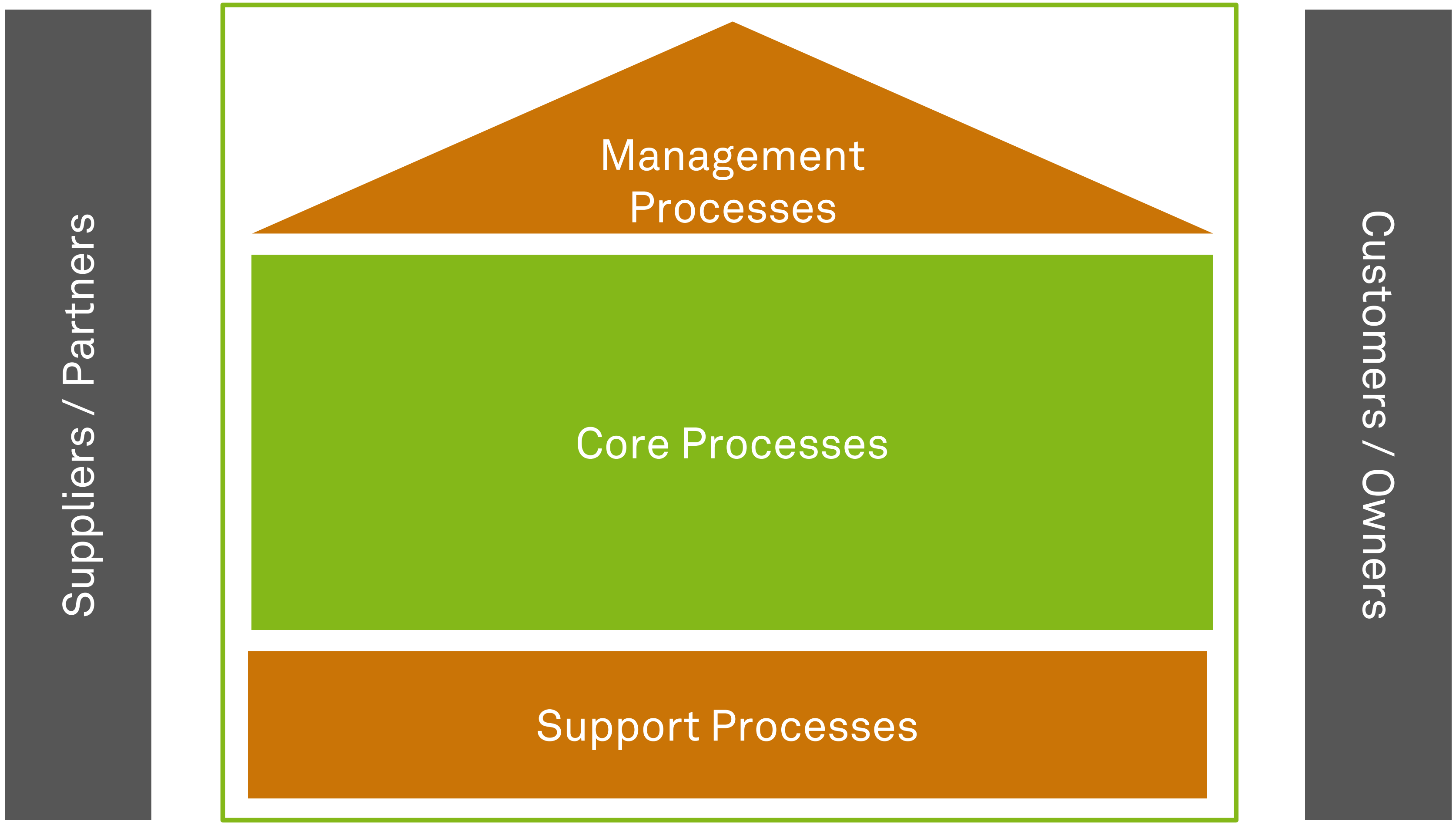
- Processes **change** over time
- Designing the perfect process architecture in **one go** is **unlikely**
- Identification should be **exploratory and iterative**
- Improvement opportunities are **time-constrained**

Dumas et al. (2018)

Building a Process Architecture and a Process Portfolio

Process Architecture

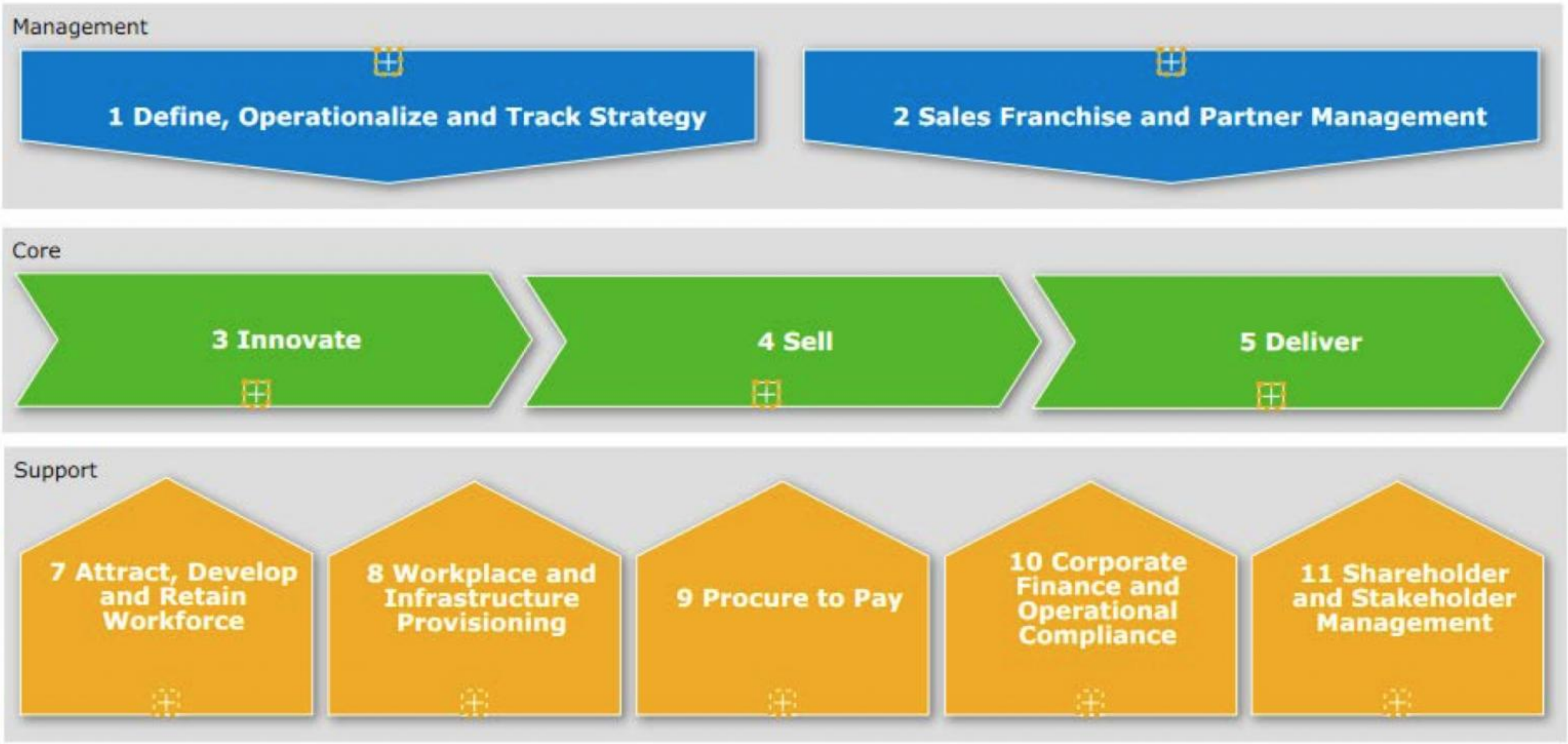
Process Categories



- What are core, support, and management processes at TU Dortmund University?

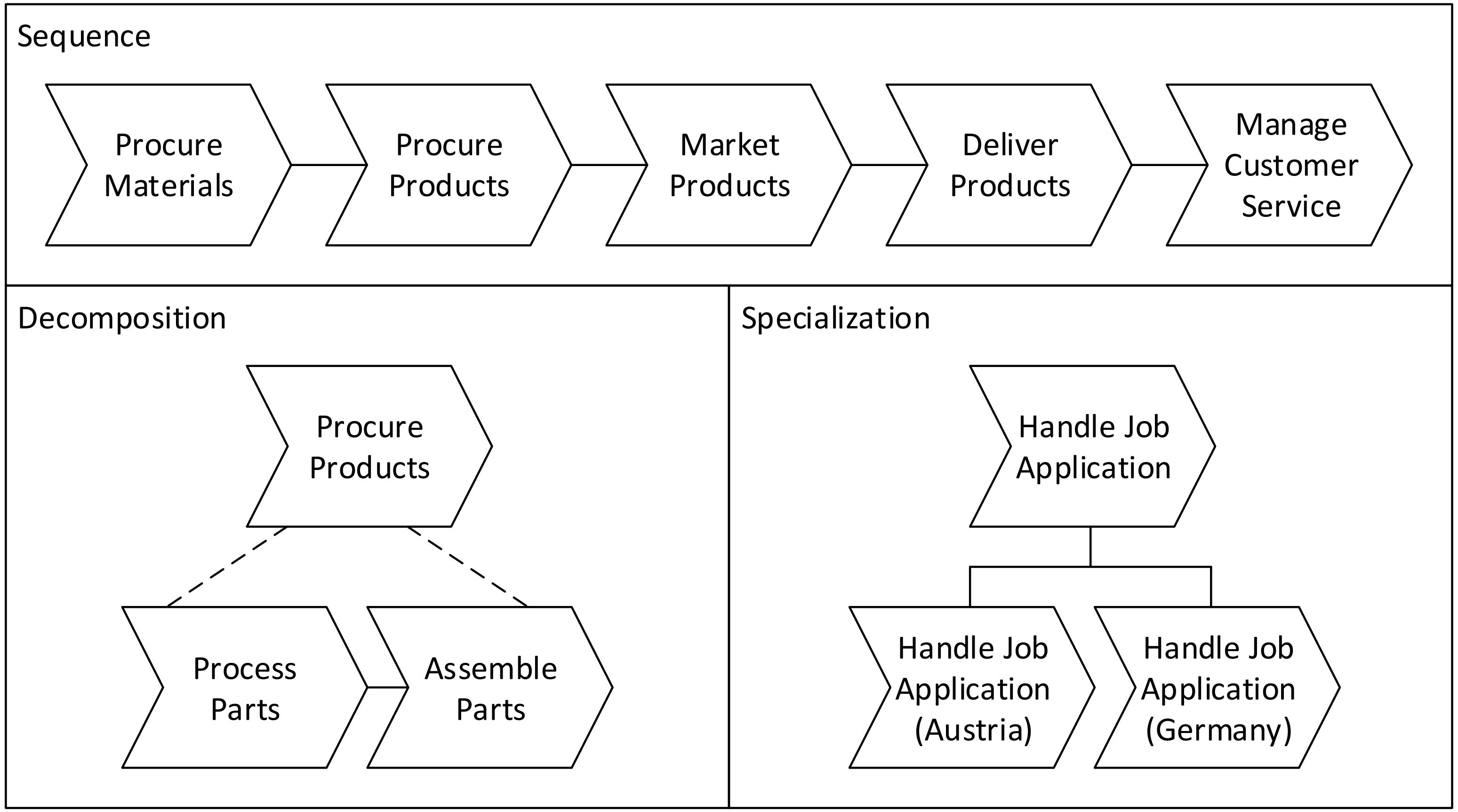
Porter (1985), Dumas et al. (2018)

Process Sets Illustrated – SAP Process Map



Reisert (2015)

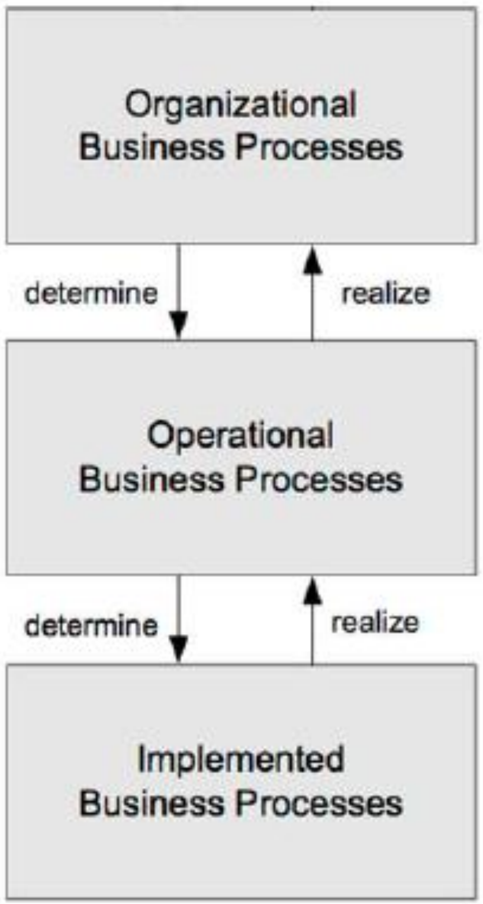
Decomposition of Business Processes (1)



Dumas et al. (2018)

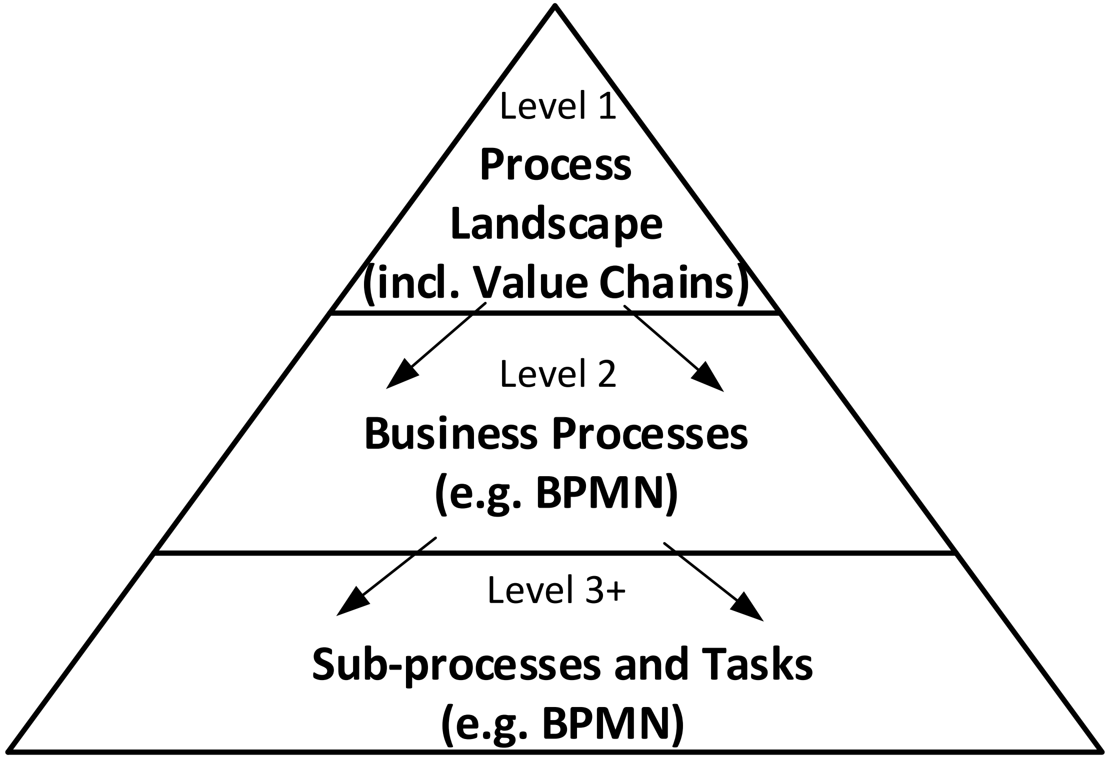
Decomposition of Business Processes (2)

- Organizational processes
  - Business processes on a **low granularity**
  - Covering the main activities of an organization
- Operational processes
  - Mapping of process activities including logical ordering
  - From the **perspective of business**
- Implemented processes
  - Include **execution information** of business processes
  - Enriched with organizational and technical aspects



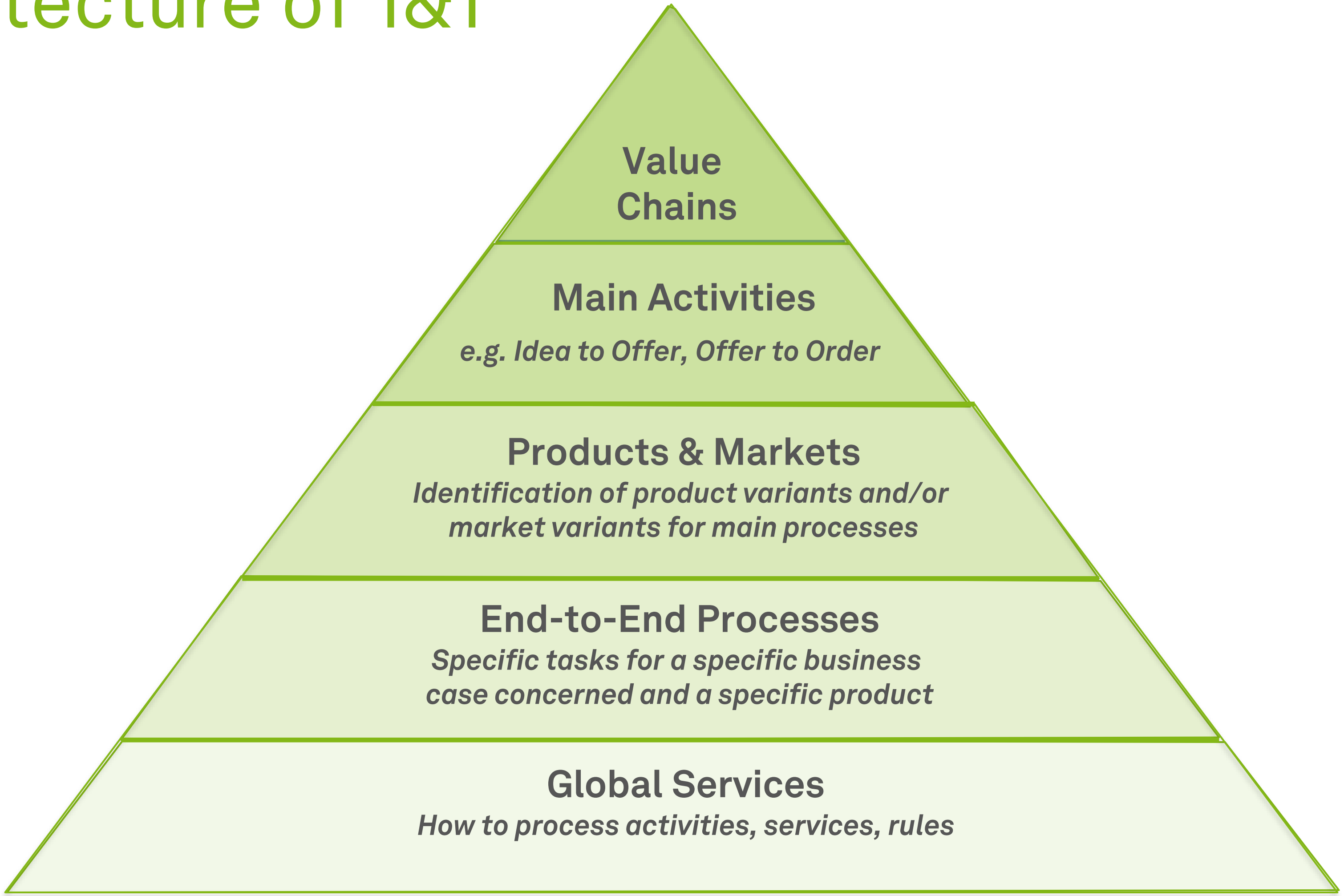
Weske (2019)

A Generic Process Architecture



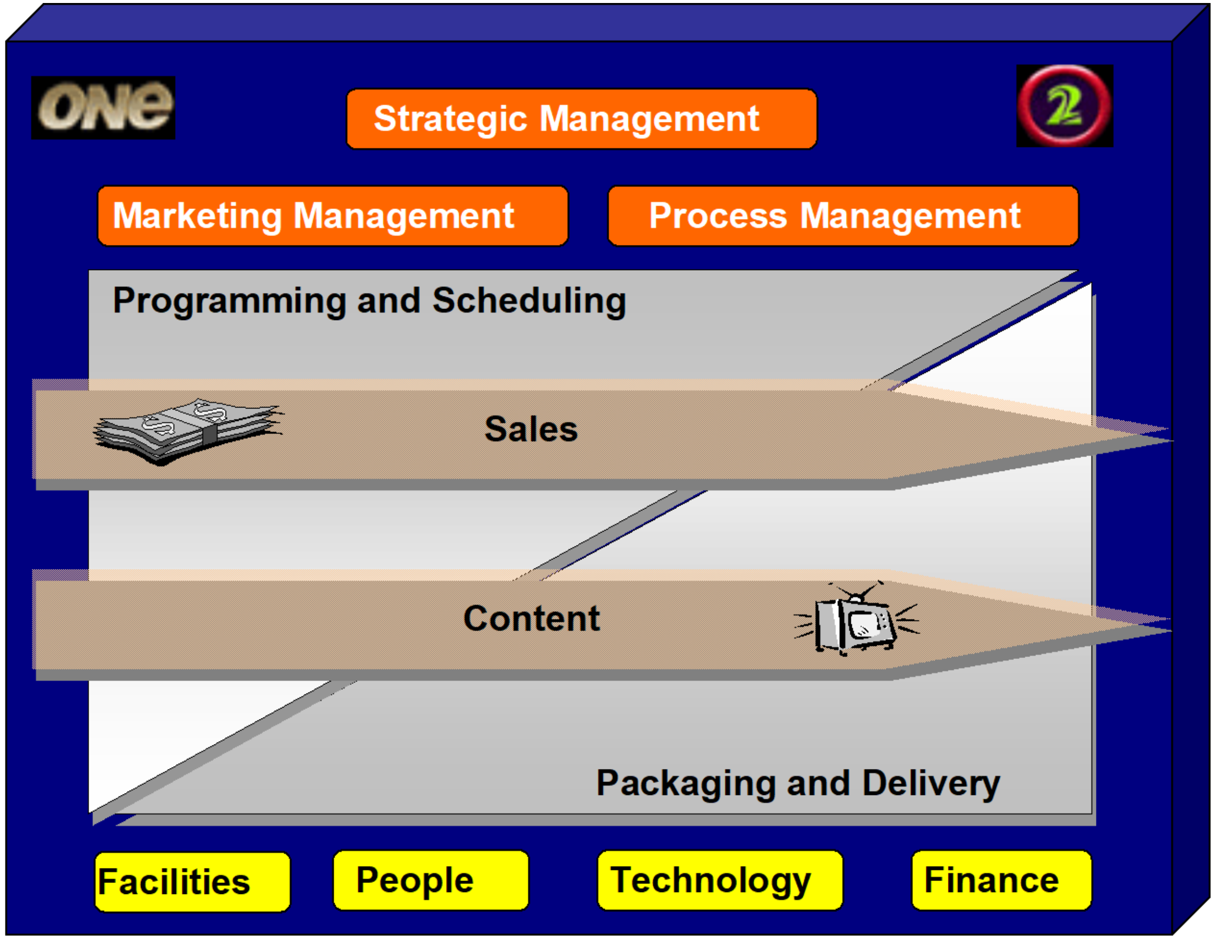
Dumas et al. (2018)

Process Architecture of 1&1



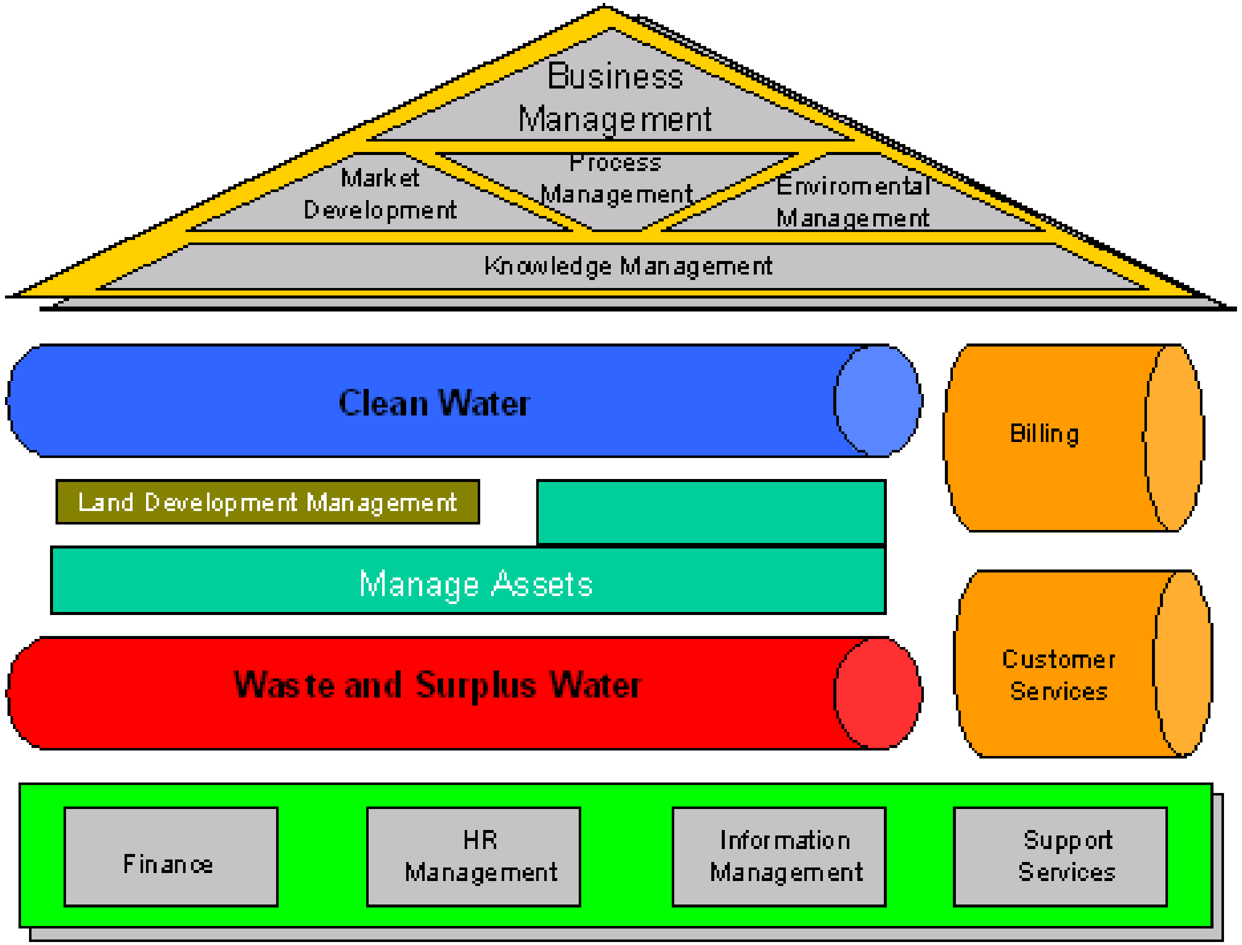
1&1 (2018)

Process Architecture of TV New Zealand



La Rosa (2013)

Process Architecture of WA Water Corporation



La Rosa (2013)

Building a Process Landscape Model (Level 1) – Overview

1. Clarify **terminology**
2. Identify **end-to-end** processes
3. For each end-to-end process, identify its **sequential processes**
4. For each business process, identify its major **management** and **support processes**
5. **Decompose and specialize** business processes
6. Compile **process profile**
7. Check **completeness** and **consistency**

Dumas et al. (2018)

Building a Process Landscape Model (1)

1. Clarify **terminology**
   - Define key terms
   - Use organizational glossary
   - Use reference models
   - Ensure that stakeholders have a consistent understanding of process landscape model
2. Identify **end-to-end** processes
   - Those processes interface with customers and suppliers
   - Goods and services that organization provides are good starting point
   - Properties help to distinguish processes, including product type, service type, channel, customer type

Dumas et al. (2018)

Building a Process Landscape Model (2)

3. For each end-to-end process, identify its **sequential** processes
   - Identify the internal, intermediate outcomes of end-to-end processes
   - Perspectives help set boundaries: product lifecycle, customer relationship, supply chain, transaction stages, change of business objects, separation
4. For each business process, identify its major **management** and **support** processes
   - What is required to execute the previously identified processes?
   - Typical support processes are management of personnel, financials, information, and materials
   - However, these can be core processes if they are integral part of business model
   - Management processes are usually generic

Dumas et al. (2018)

Building a Process Landscape Model (3)

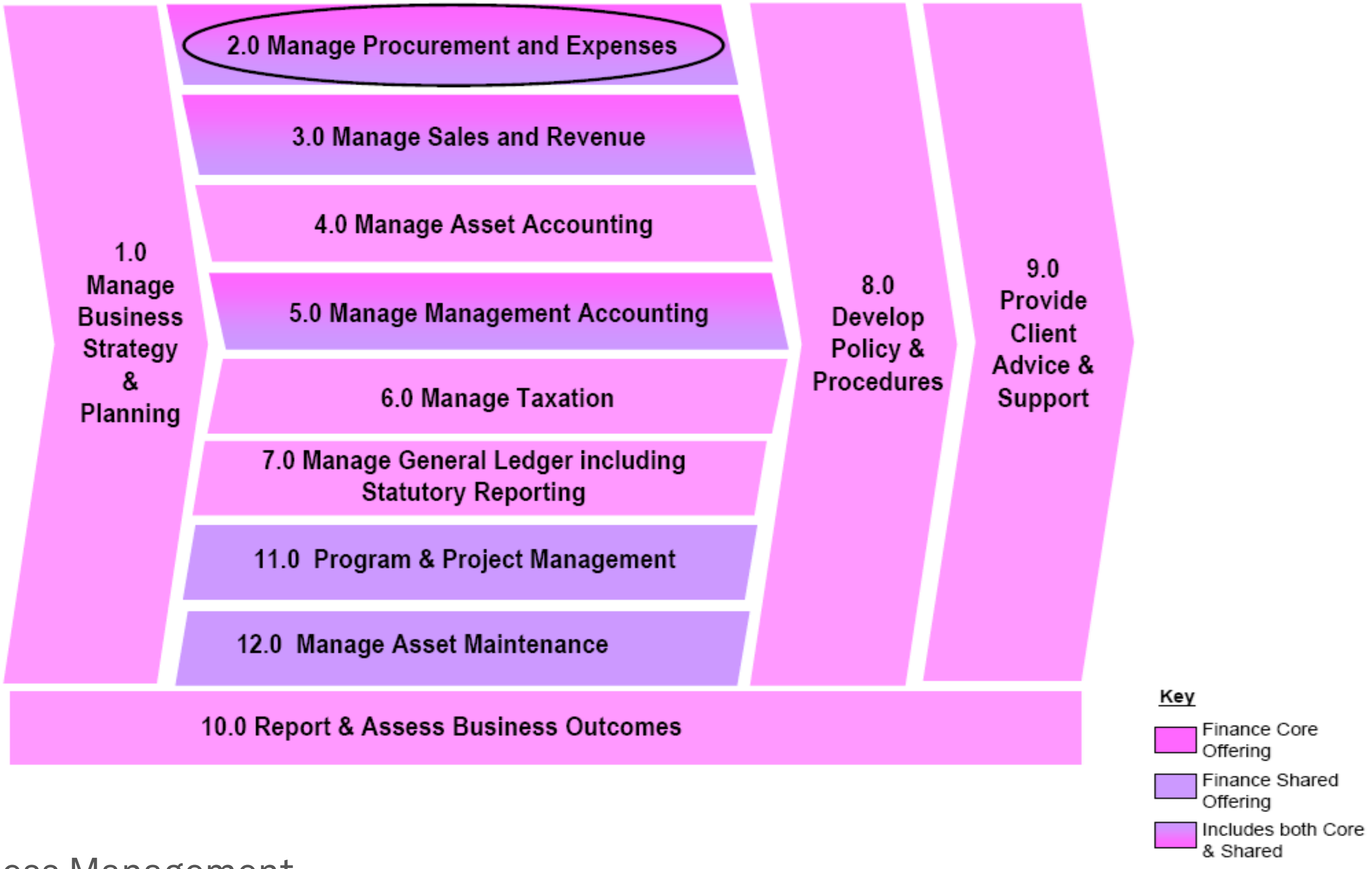
5. **Decompose** and **specialize** business processes
   - Processes of Level 1 should be further subdivided into abstract process on Level 2
   - Subdivide until processes can be managed autonomously by a single process owner
   - Considerations when this subdivision should stop: Manageability and Impact
6. Compile **process profile**
   - Each of the identified processes should be described using process profile
   - Process profile supports process documentation and communication
7. Check **completeness** and **consistency**
   - Reference models can be used to check whether all major processes are included and help to check for the consistency of terminology
   - Check whether all processes can be associated with functional units of organization chart and vice versa

Dumas et al. (2018)

Process Hierarchy Example: Queensland Shared Service Agency

Queensland Government

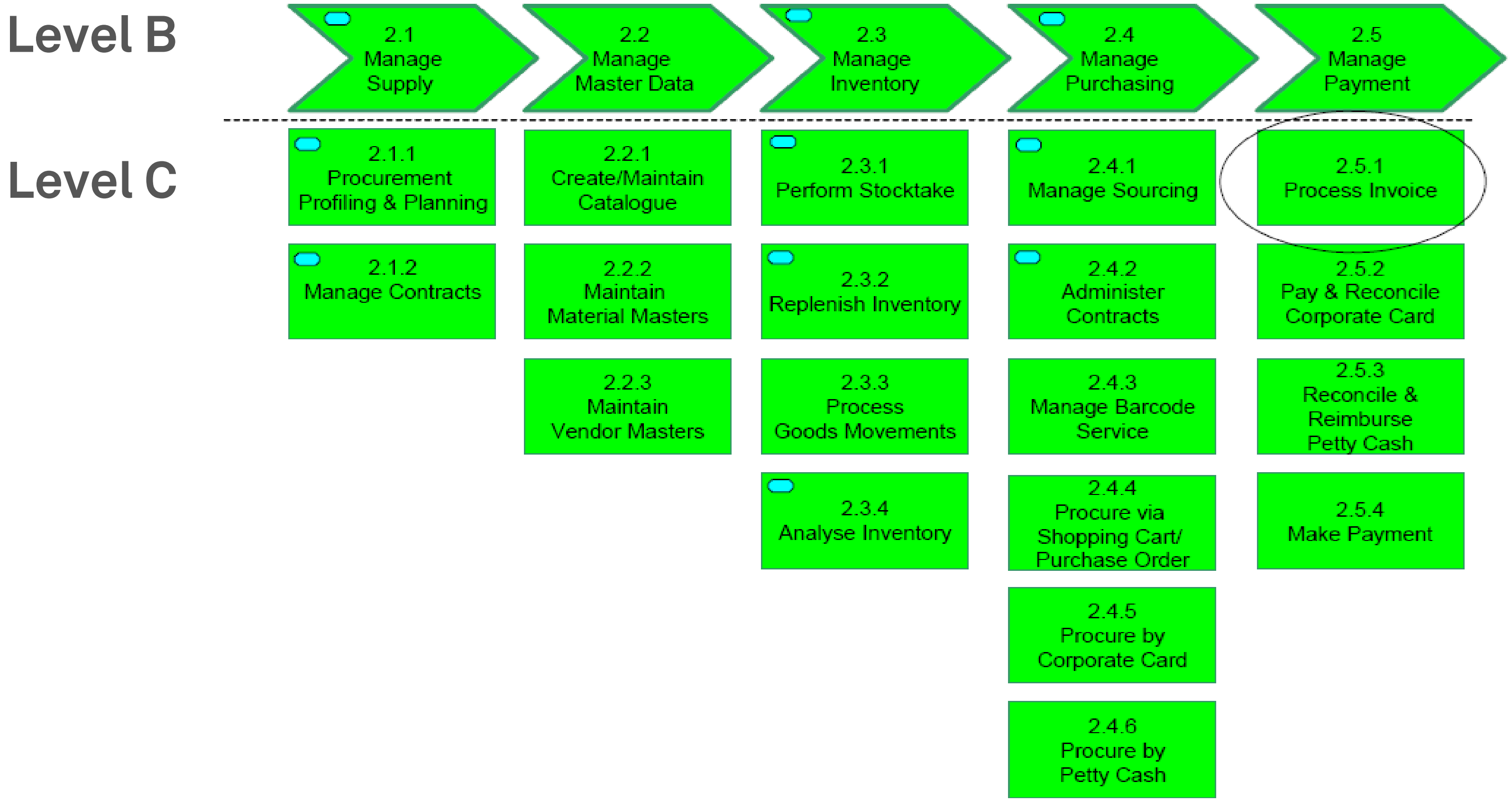
Level A



La Rosa (2013)

Process Hierarchy Example: Queensland Shared Service Agency

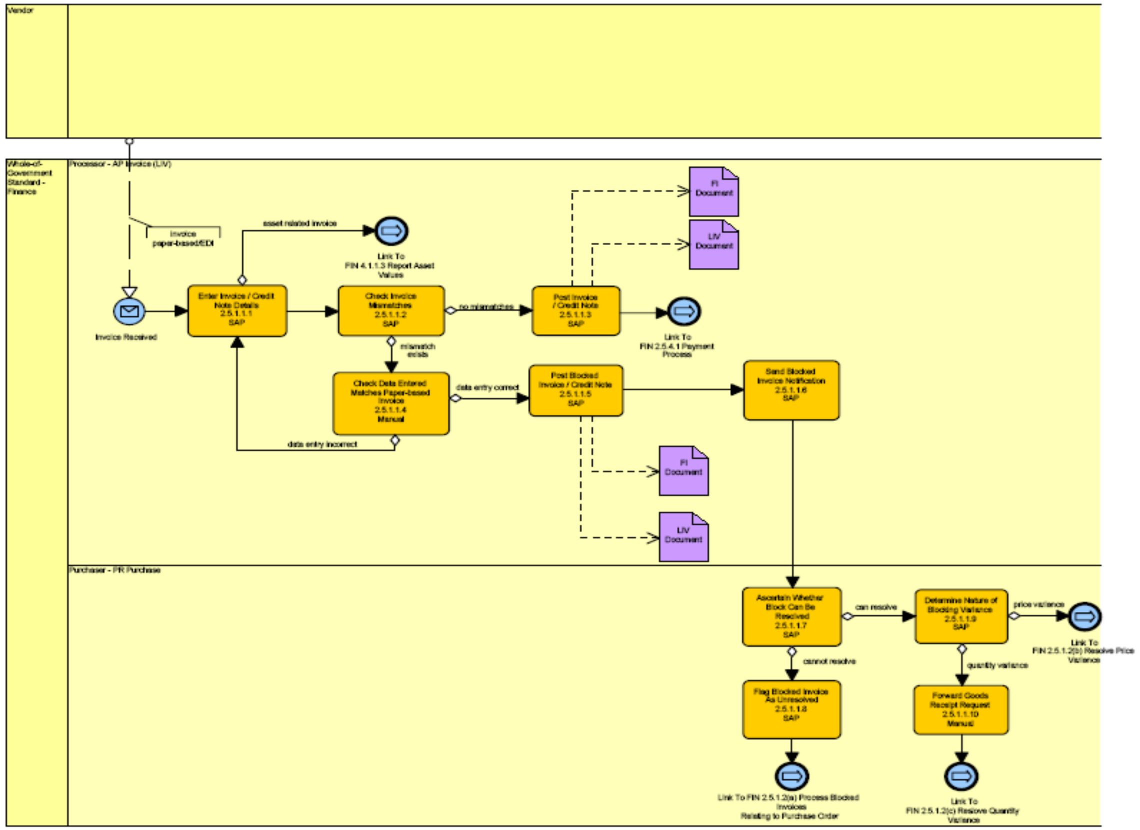
Queensland Government



La Rosa (2013)

Process Hierarchy Example: Queensland Shared Service Agency

Level D



La Rosa (2013)

Process Profile

- Accompanies a process model as a sort of fact sheet and blueprint
- Defines the boundaries (trigger, outcome) of the process
- Supports its
  - Vision
  - Process performance indicators
  - Resources, and
  - Process owner

<table>
<tr><td colspan="2">Name of Process: Procure-to-Pay</td></tr>
<tr><td colspan="2">Vision: The objective of the procurement process is to secure that the entire range of external products and services becomes available on time and is at the required level of quality.</td></tr>
<tr><td colspan="2">Process Owner: Chief Financial Officer (CFO)</td></tr>
<tr><td>Customer of process:
• Requesting unit</td><td>Expectation of customer:
• Timely, economic and complete provision</td></tr>
<tr><td colspan="2">Outcome: Delivered products or provided services for the requested unit</td></tr>
<tr><td colspan="2">Trigger: Need is identified</td></tr>
<tr><td colspan="2">First activity: Submit Request
.....
Last activity: Create Purchase Order</td></tr>
<tr><td colspan="2">Interfaces inbound: Plan-to-Procure
Interfaces outbound: Construct-to-Complete</td></tr>
<tr><td colspan="2">Required resources:
• Human resources:
Site Engineer, Clerk, Works Engineer
• Information, documents, know-how:
procurement guidelines, supplier rating, framework contract
• Work environment, materials, infrastructure:
Procurement information system</td></tr>
<tr><td colspan="2">Process Performance Measures:
• Cycle Time
• Operational Costs
• Error Rate</td></tr>
</table>

Dumas et al. (2018)

Guiding Questions for Scoping Processes

- Where does the process **fit** into the process architecture?
  - ...if a process architecture is already in place
- On what level is the **unit of analysis**?
  - i.e., end-to-end process, procedure, or operation
- What are the **previous/subsequent processes** and what are the **interfaces** to them?
- What **variants** does this process have?
- What **underlying processes** describe elements of this process in more detail?

Dumas et al. (2018)

Excursus: Techniques for Scoping Processes

- Identify relevant **stakeholders** and **objectives**
  - e.g., via a Stakeholder-Objectives Matrix
- Identify relevant **context**
  - e.g., via a SIPOC (Suppliers, Inputs, Process, Output, Customers) Diagram
- Identify relevant **guides** and **enablers**
  - e.g., via an IGOE (Input/Guides/Enablers/Outputs) Diagram
- Identify relevant **process boundaries**
  - e.g., via a Case/Function Matrix

- A combination of the above

Dumas et al. (2018)

Excursus: Identify Relevant Process Stakeholders

- **Process owner**
  - Responsible for the effective and efficient operation of the process being modeled
- **Primary process participants**
  - Those who are directly involved in the execution of the process under analysis
- **Secondary process participants**
  - Those who are directly involved in the execution of the preceding or succeeding processes

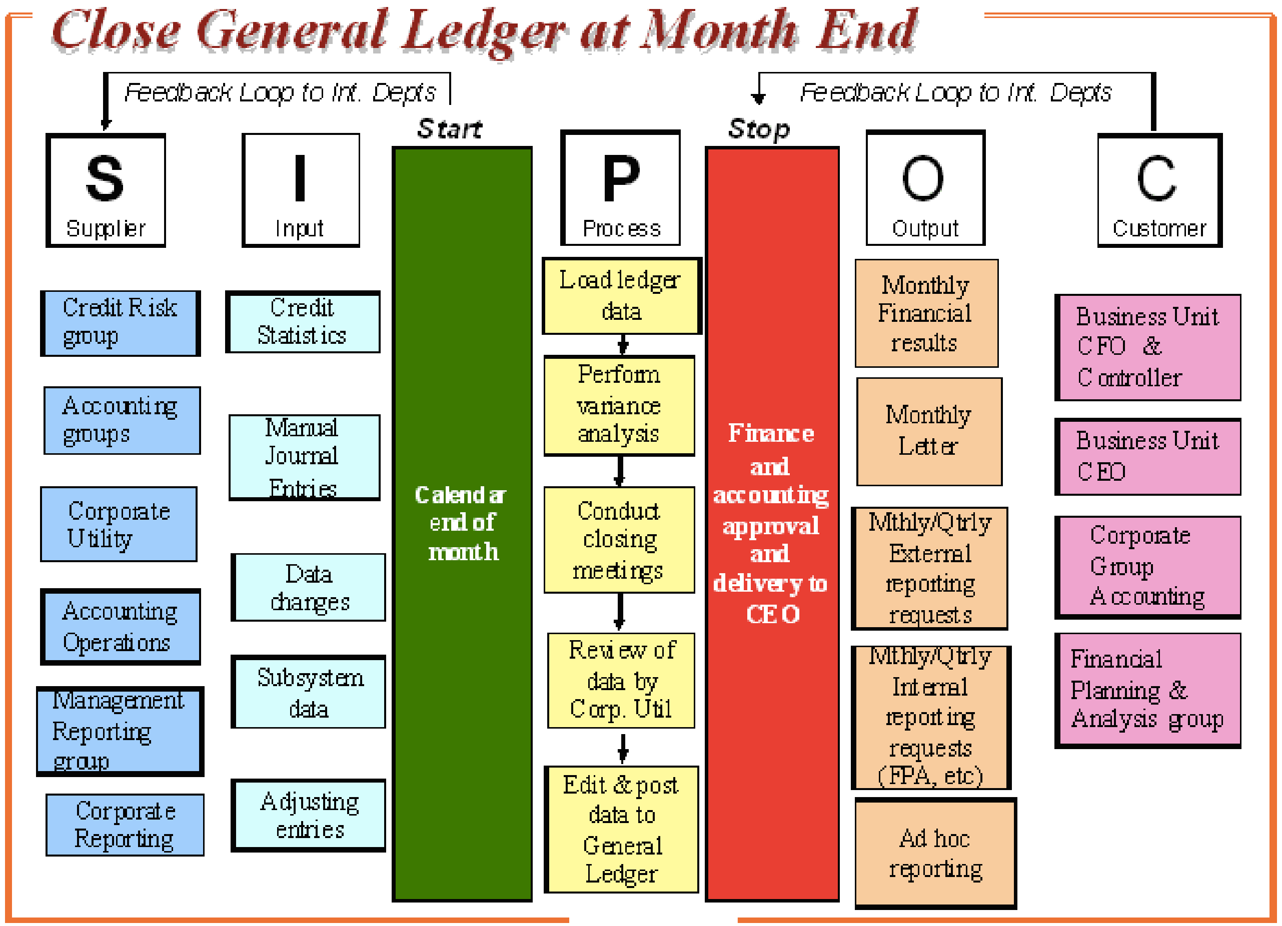
Dumas et al. (2018)

Excursus: Identify Relevant Process Objectives

- **Primary (hard)** process objectives
  - Time, cost, quality (minimize, maximize)
  - Satisfaction, compliance, flexibility, predictability
- **Secondary** process objectives
  - To purchase goods, to hire new staff members
- Back up with **appropriate process metrics**
  - e.g., cost per execution, resource utilization, waste
  - e.g., cycle time, waiting time, non-value-adding time
  - e.g., error rates, SLA violations, customer feedback
- Let involved stakeholders define their priorities

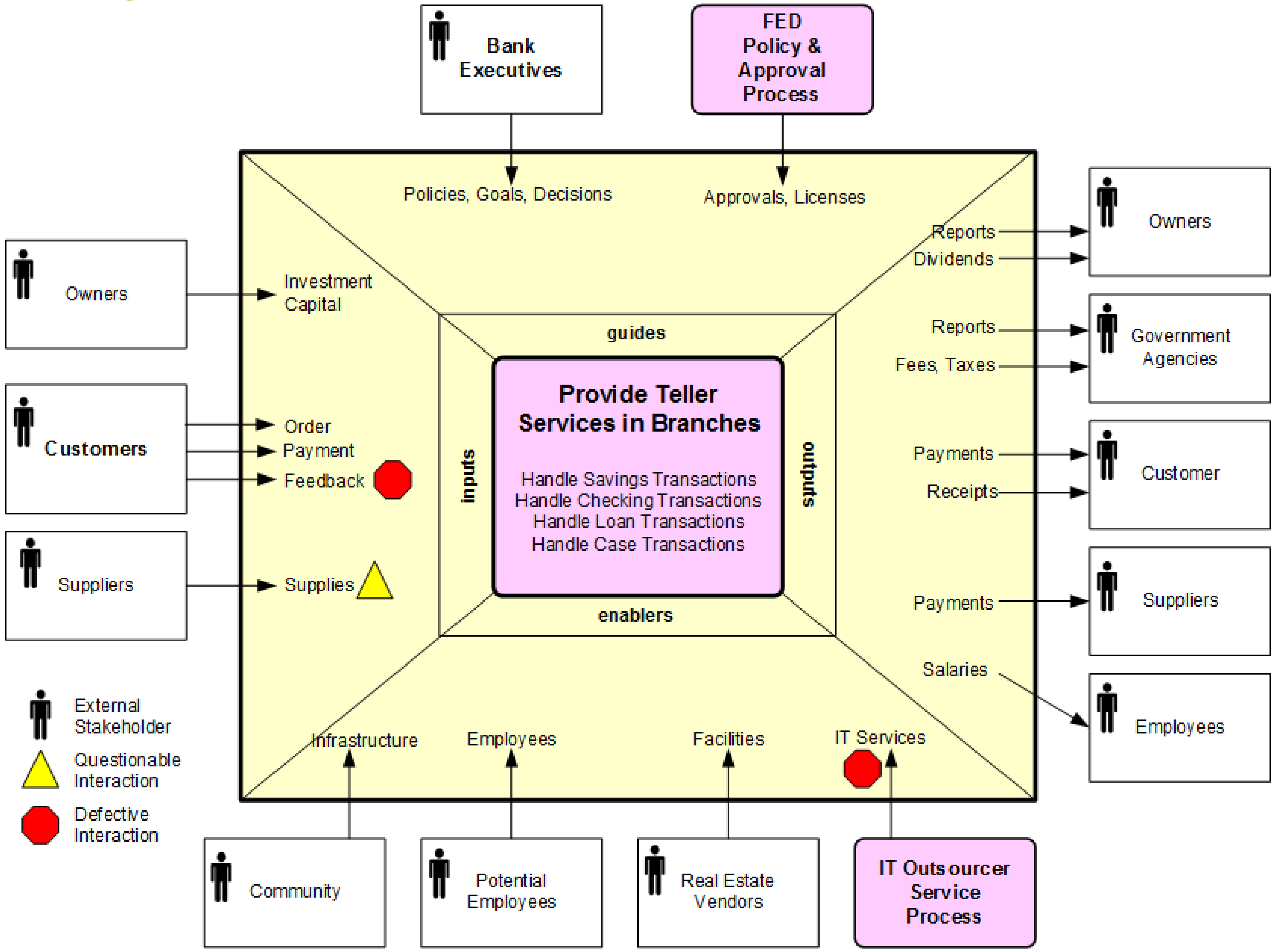
Dumas et al. (2018)

Excursus: Identify Context



American Society for Quality (ASQ) Service Quality Divison (2016)

Excursus: Identify Guides and Enablers



BPTrends (2013)

Excursus: Maturity Levels for Prioritization

- Guiding Questions
  - Does the organization cover the **expected range** of processes?
  - Are these processes **documented** and **supported**?

- Assessment is called *Appraisal*

- Example
  - Standard CMMI Appraisal Method for Process Improvement (SCAMPI)
  - Uses the Capability Maturity Model Integration (CMMI)

Dumas et al. (2013)

Excursus: CMMI Maturity Levels

- Level 1 (**Initial**)
  - Ad-hoc fashion, without any clear definition, control missing
- Level 2 (**Managed**)
  - Project planning along with project monitoring and control, measurement and analysis, process and product quality assurance
- Level 3 (**Defined**)
  - Organizations has focus on processes, process definitions available, training provided, documentation and analysis, integrated project and risk management, decision analysis and resolution
- Level 4 (**Quantitatively Managed**)
  - Process performance tracked, project management using quantitative techniques
- Level 5 (**Optimizing**)
  - Organizational performance management with causal analysis and resolution

Information Systems Audit and Control Association (ISACA) (2022)

Excursus: Reference Models

- A **reference model** is used as a template to design the process architecture

- Examples
  - Process Classification Framework (PCF)
  - Information Technology Infrastructure Library (ITIL)
  - Supply Chain Operations Reference Model (SCOR)
  - Value Reference Model (VRM)
  - Voluntary Interindustry Commerce Solutions (VICS)
  - eTOM Business Process Framework

Dumas et al. (2018)

Example: Process Classification Framework (v7.3)

American Productivity & Quality Center

- PCF from American Productivity and Quality Center (APQC)
- Industry-neutral enterprise model
- Open standard for benchmarking

Level 1 - Category — 10.0 Manage Enterprise Risk, Compliance, Remediation and Resiliency (16437)

Represents the highest level of process in the enterprise, such as Manage customer service, Supply chain, Financial organization, and Human resources.

Level 2 - Process Group — 10.1 Manage enterprise risk (17060)

Indicates the next level of processes and represents a group of processes. Perform after sales repairs, Procurement, Accounts payable, Recruit/source, and Develop sales strategy are examples of process groups.

Level 3 - Process — 10.1.4 Manage business unit and function risk (17061)

A process is the next level of decomposition after a process group. The process may include elements related to variants and rework in addition to the core elements needed to accomplish the process.

Level 4 - Activity — 10.1.4.3 Develop mitigation plans for risks (16458)

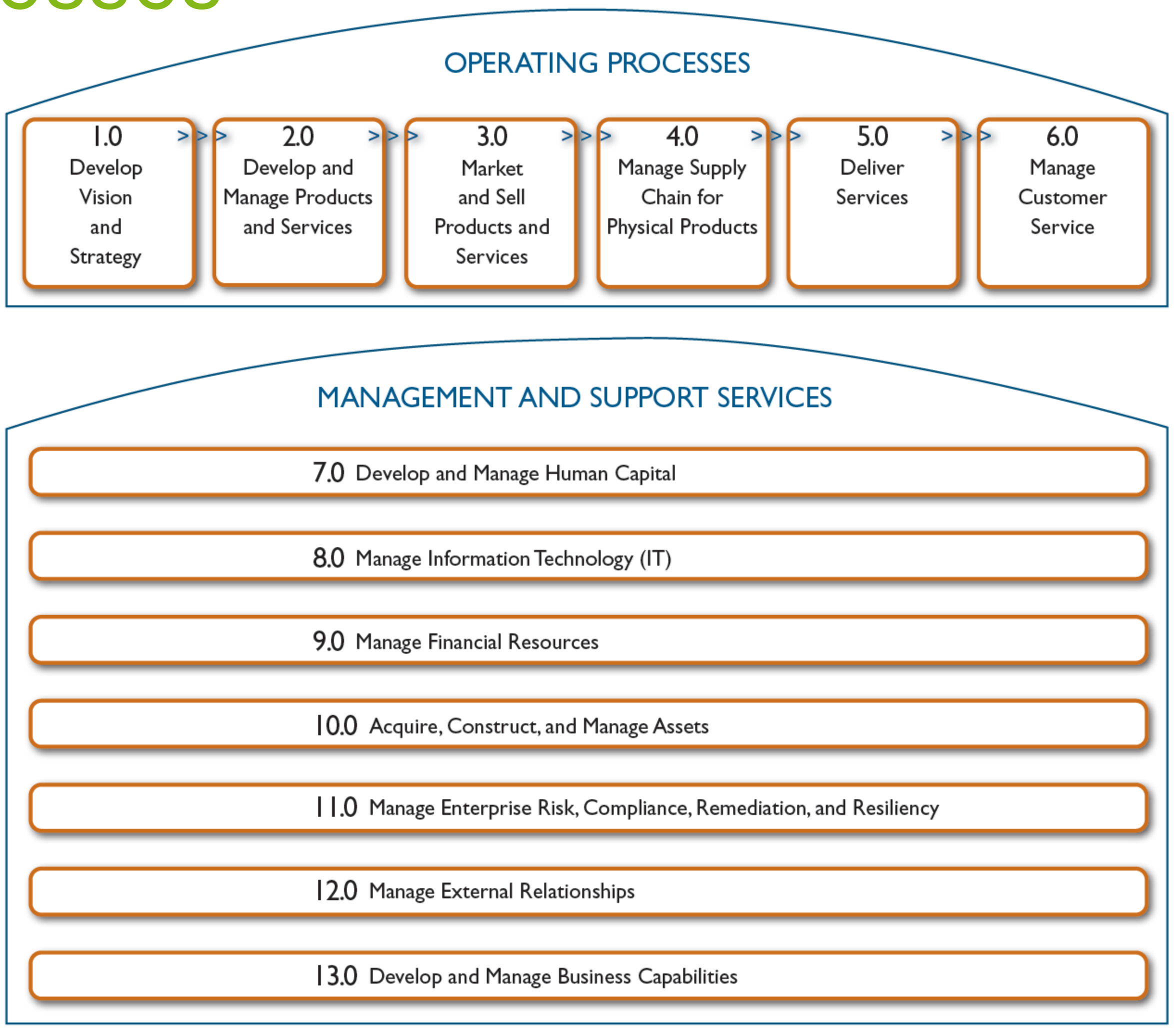
Indicates key events performed when executing a process. Examples of activities include Receive customer requests, Resolve customer complaints, and Negotiate purchasing contracts.

Level 5 - Task — 10.1.4.3.1 Assess adequacy of insurance cover (18129)

Tasks represent the next level of hierarchical decomposition after activities. Tasks are generally much more fine grained and may vary widely across industries. Examples include: Create business case and obtain funding and Design recognition and reward approaches.

APQC (2022)

APQC PCF's Core Processes



APQC (2022)

APQC PCF's Processes

APQC American Productivity & Quality Center

PCF ID	Hierarchy ID	Name	Difference Index	Change details	Metrics available?
10008	8.0	**Manage Information Technology (IT)**	3	RENAME, WAS:Manage Information Technology	Y
10563	8.1	Manage the business of information technology	2		Y
10570	8.1.1	Develop the enterprise IT strategy	1	c10609	Y
10603	8.1.1.1	Build strategic intelligence	0		N
10604	8.1.1.2	Identify long-term IT needs of the enterprise in collaboration with stakeholders	0		N
10605	8.1.1.3	Define strategic standards, guidelines, and principles	0		N
10606	8.1.1.4	Define and establish IT architecture and development standards	0		N
10607	8.1.1.5	Define strategic vendors for IT components	0		N
10608	8.1.1.6	Establish IT governance organization and processes	0		N
10609	8.1.1.7	Build strategic roadmap to develop IT capabilities in support of business objectives	1	RENAME, WAS:Build strategic plan to support business objectives	N
10571	8.1.2	Define the enterprise architecture	1	c10611	N
10611	8.1.2.1	Establish the current and future enterprise architecture definition	1	RENAME, WAS:Establish the enterprise architecture definition	N
10612	8.1.2.2	Confirm enterprise architecture maintenance approach	0		N
10613	8.1.2.3	Maintain the relevance of the enterprise architecture	0		N
10614	8.1.2.4	Act as clearinghouse for IT research and innovation	0		N

APQC (2022)

Process Selection Criteria for Prioritization

- **Strategic Importance**
  - Which processes have the greatest impact on the organization's strategic goals?
  - Consider profitability, uniqueness, or contribution to competitive advantages
- **Health (Dysfunction)**
  - Which processes are in the deepest trouble?
- **Feasibility**
  - Which process is the most susceptible to successful process management?
  - Consider culture and politics!

- Result: **Process Portfolio**

Dumas et al. (2018)

Process Performance Measures for Prioritization

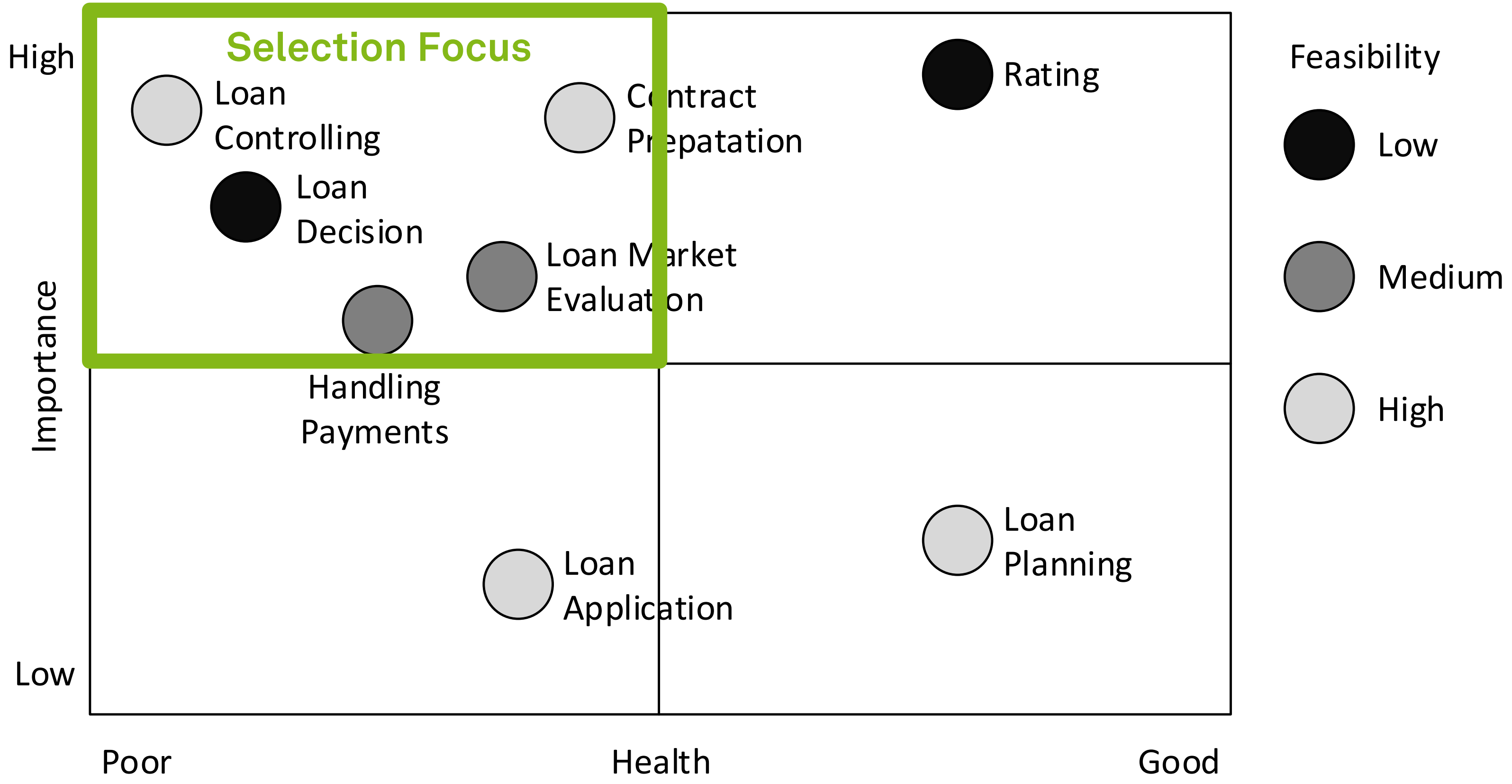
- **Time**
- **Cost**
- **Quality**
- **Flexibility**

- **Time**
- **Cost**
- **Quality**
- **Flexibility**
- **People**

- Formulate performance objectives
- Identify performance dimensions and performance measures
- Define performance target
- Result: **Process Portfolio**

Dumas et al. (2018), François et al. (2022)

Process Portfolio Example



Dumas et al. (2018)

The Hard Questions

- Does an assessment of the **importance, health,** and **feasibility** always point to the same processes to actively manage?

- Should all processes that are **not healthy**, of **strategic importance**, and **feasible** to manage be subjected to BPM initiatives?

Dumas et al. (2018)

Selection Project by Project

- Processes are identified with every **request** from a line of business
- Ensures **high relevance** for involved business unit

- **Drawbacks**
  - **Reactive** approach (-)
  - Often restricted to **discrete improvement** (-)
  - **No** conscious process **selection** approach (-)

Dumas et al. (2018)

Pitfalls of Process Identification (1)

- **Purpose** of the project is **not clear enough**
  - inappropriate scoping of the process
- **Scope** of the process is **too narrow**
  - the identified root-causes are located outside the boundaries of the process under analysis
- **Scope** of the process is **too wide**
  - process improvement project that has to be compromised in its lack of detail
- **Business process architect** has **poor facilitation skills**
  - inability to resolve emerging conflicts between the project members and stakeholders

Dumas et al. (2018)

Pitfalls of Process Identification (2)

- **Process** is **identified in isolation** to other projects due to poor portfolio management
  - redundancies and inconsistencies between projects
- Involved **project members and stakeholders** have **not** been sufficiently **informed** about the benefits of the project
  - limited participation
- Involved **project members and stakeholders** have **not** been carefully **selected**
  - very limited source of knowledge

Dumas et al. (2018)

Learning Goals of Today

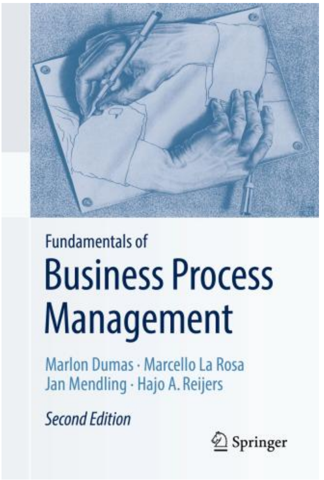
- Understand the process identification phase
- Understand process architecture definition
- Understand process selection
- Be able to work with process architectures and process portfolios

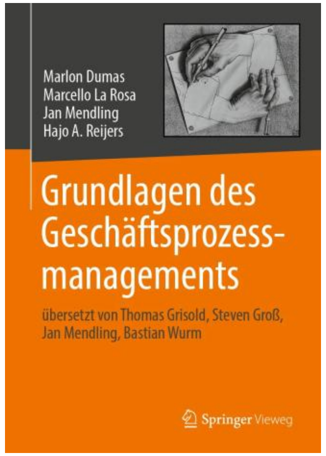
Recap Questions

- Which tasks are performed in the process identification phase and what is its outcome?
- What is the relation of a process architecture to a process landscape model?
- What is a technique for scoping business processes and what is useful for?
- What is a process portfolio and how is it related to a process architecture?

Literature

- **Section 2:** Process Identification





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Any Questions about Process Identification?

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Chair of Enterprise Computing
Department of Computer Science

Business Process Management

Essentials of Process Modeling

Roadmap

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Learning Goals of Today

- Understand the purpose of process modeling
- Understand the purpose of BPMN
- Get to know BPMN’s basic modeling constructs
- Be able to use basic BPMN

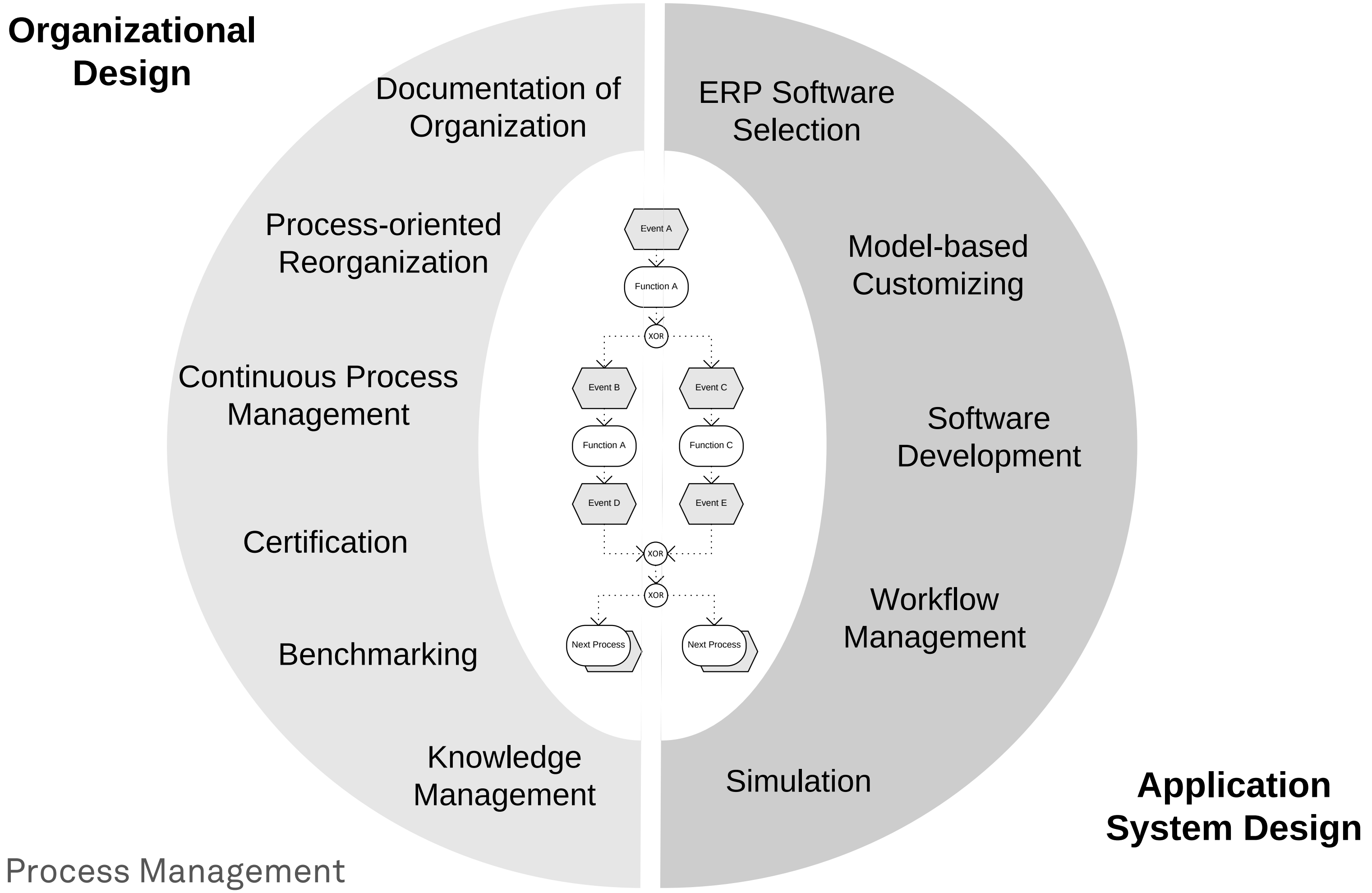
Purpose of Process Modeling




Why Process Modeling?

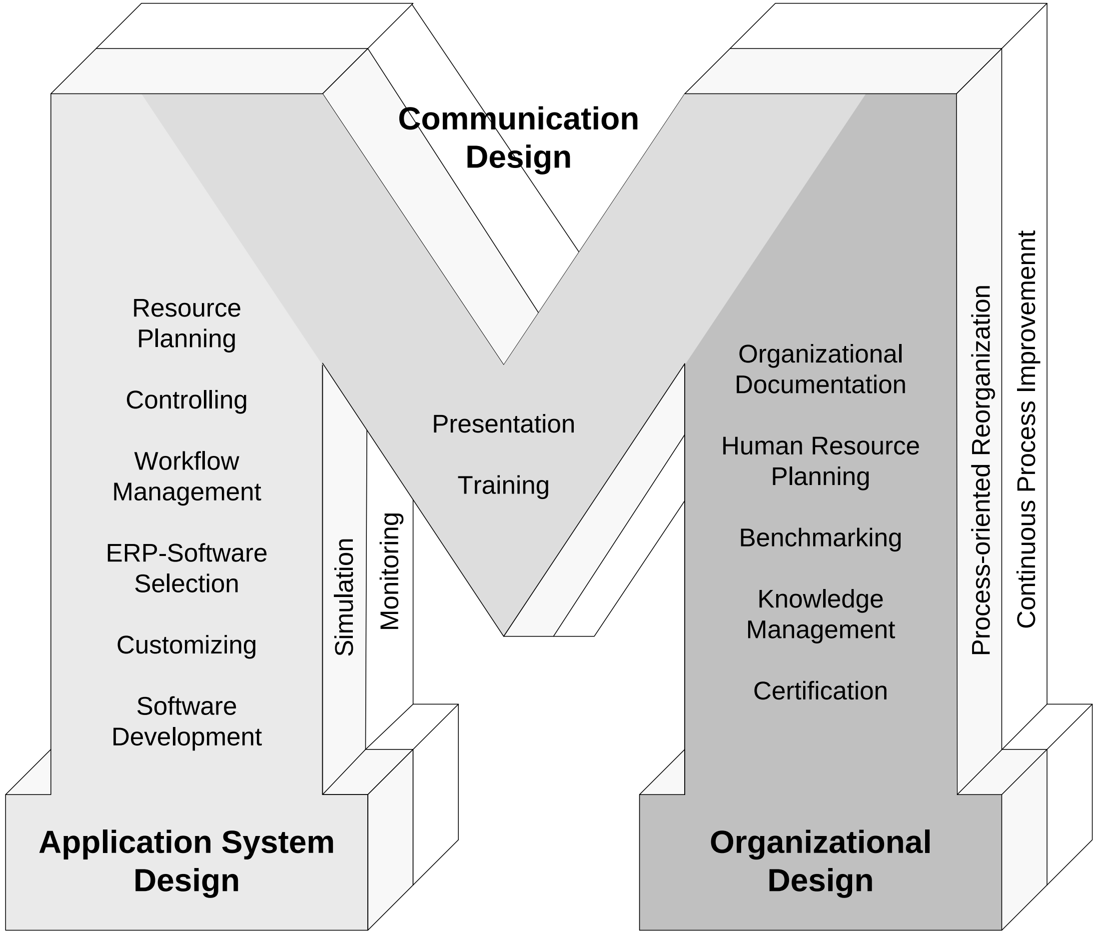
- **Understand** of the process with the people who are involved with it
- **Common language** for naming and framing an issue
- Provides a blueprint for **process automation**
- **Integrates** processes with other artefacts
  - e.g., systems, organizations, data, etc.
- Enables **walk-through**, **validation**, and **testing**
- Use as a **benchmark** for measuring improvements

Purposes of Process Modeling (1)



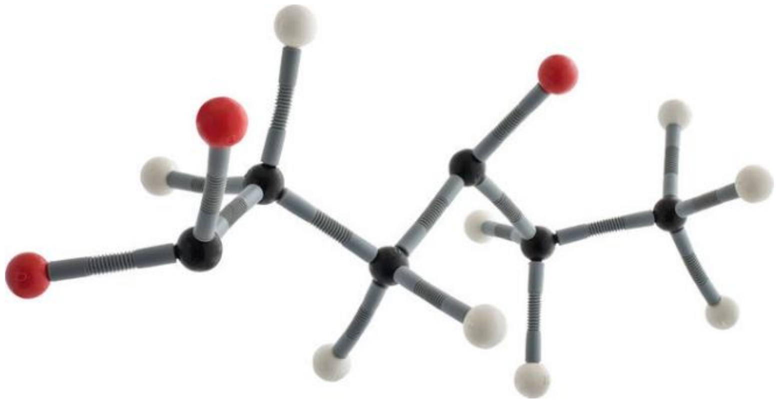
Becker et al. (2005)

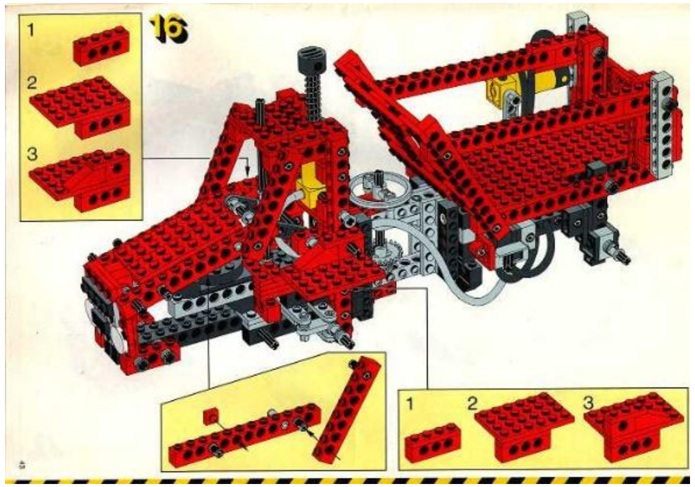
Purposes of Process Models (2)



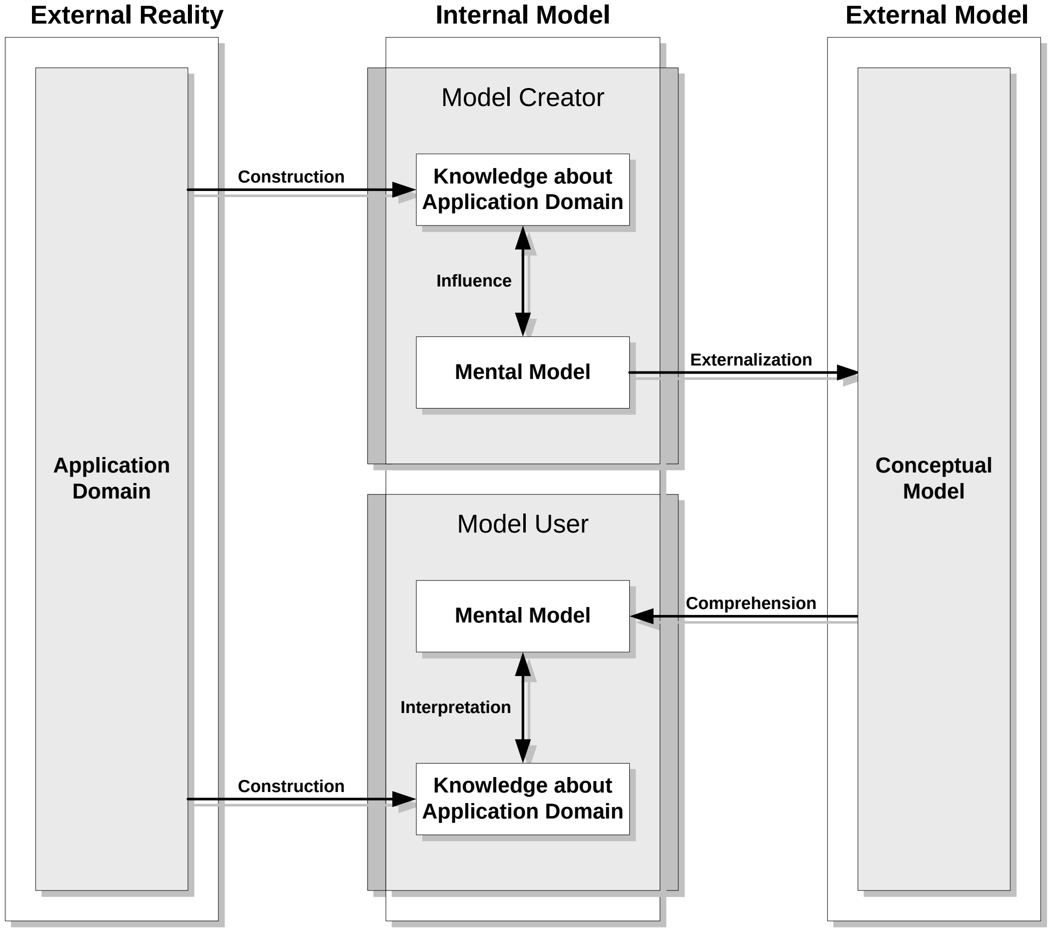
Becker et al. (2005)

What Actually is a Model?





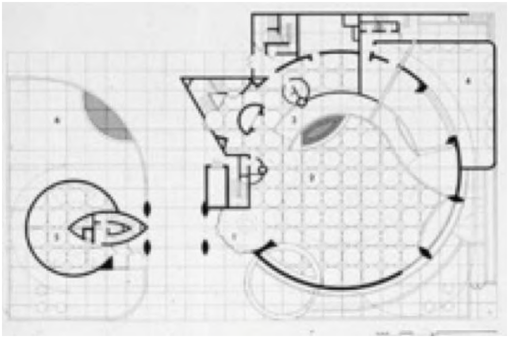
Mental Model



Janiesch (2007)

Features of a Model?

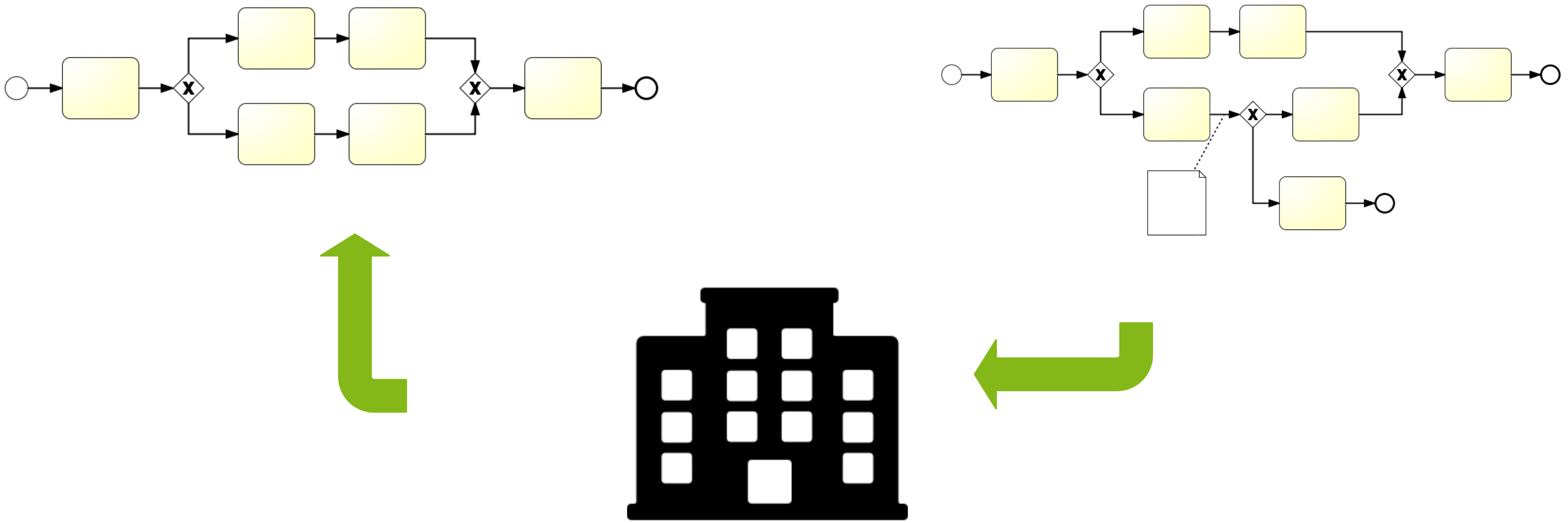
- **Mapping**
  - Models are depictions or **representation** of a natural or an artificial original, which themselves can be models again
- **Abstraction**
  - Models are **abstractions** from real world phenomena
  - Models are developed for the purpose of **reducing** overall **complexity**
- **Pragmatic purpose**
  - Models may not be clearly associated to its appropriate original
  - Models serve as a **substitute**
    - a) for certain subjects
    - b) within certain time intervals
    - c) under constraints imposed by certain operations



Stachowiak (1973), Dumas et al. (2018)

Objectives of Modeling

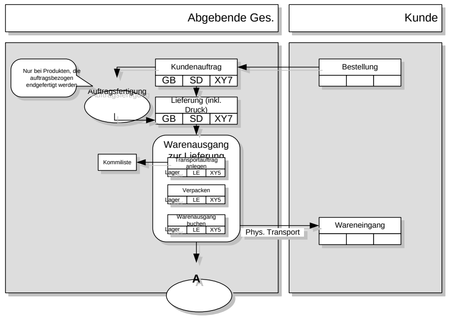
- Descriptive Modeling (as-is-processes)
- Prescriptive Modeling (to-be-processes)



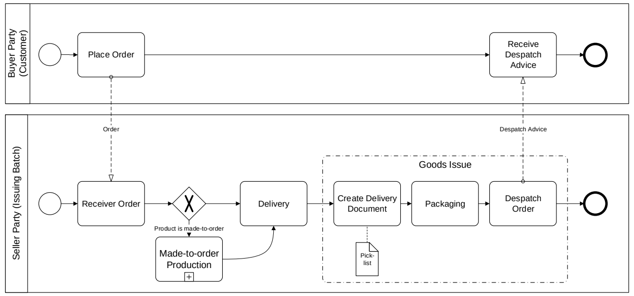
Dumas et al. (2018)

Business Process Modeling

- Business processes models are being developed in the purpose...
  - of a specific modeling **subject**,
  - for a specific target **audience**
  - with a specific modeling **purpose** in mind



confusing & complex model

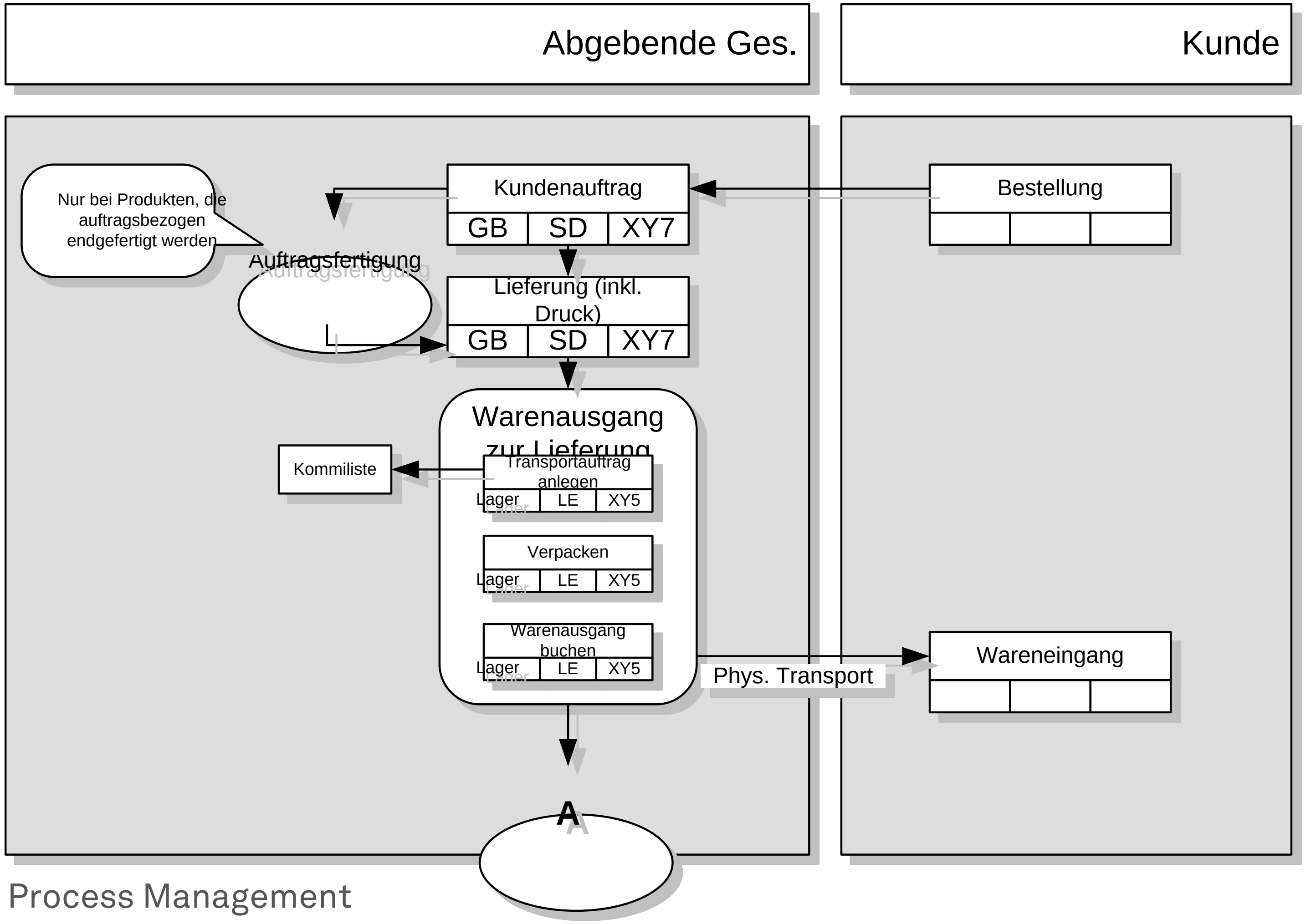


clear and easy to understand

- There is no **right** or **wrong**, but there is **relevant** and **irrelevant**

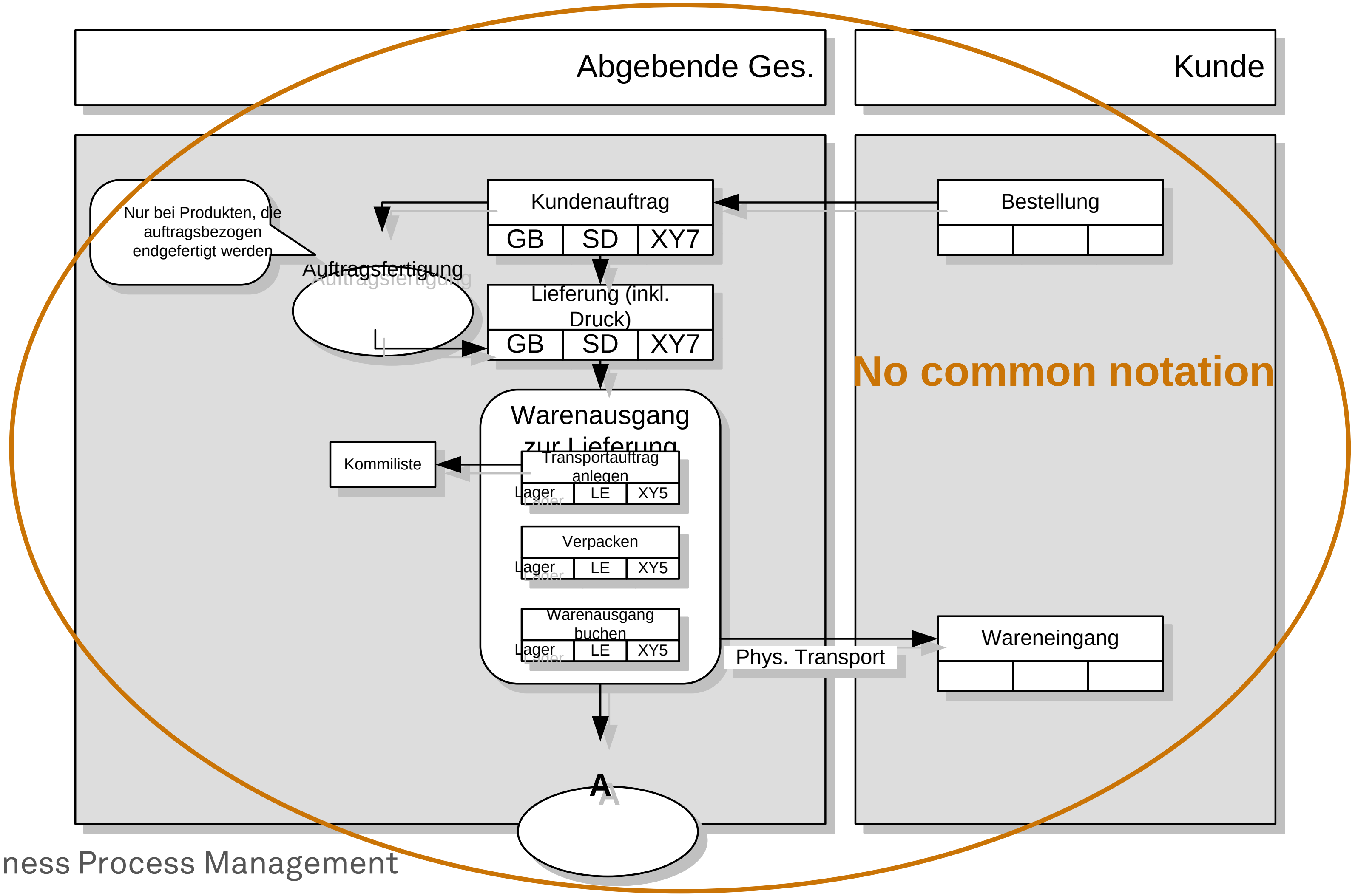
Dumas et al. (2018), Janiesch & Stein (2007)

Problems with Process Documentation



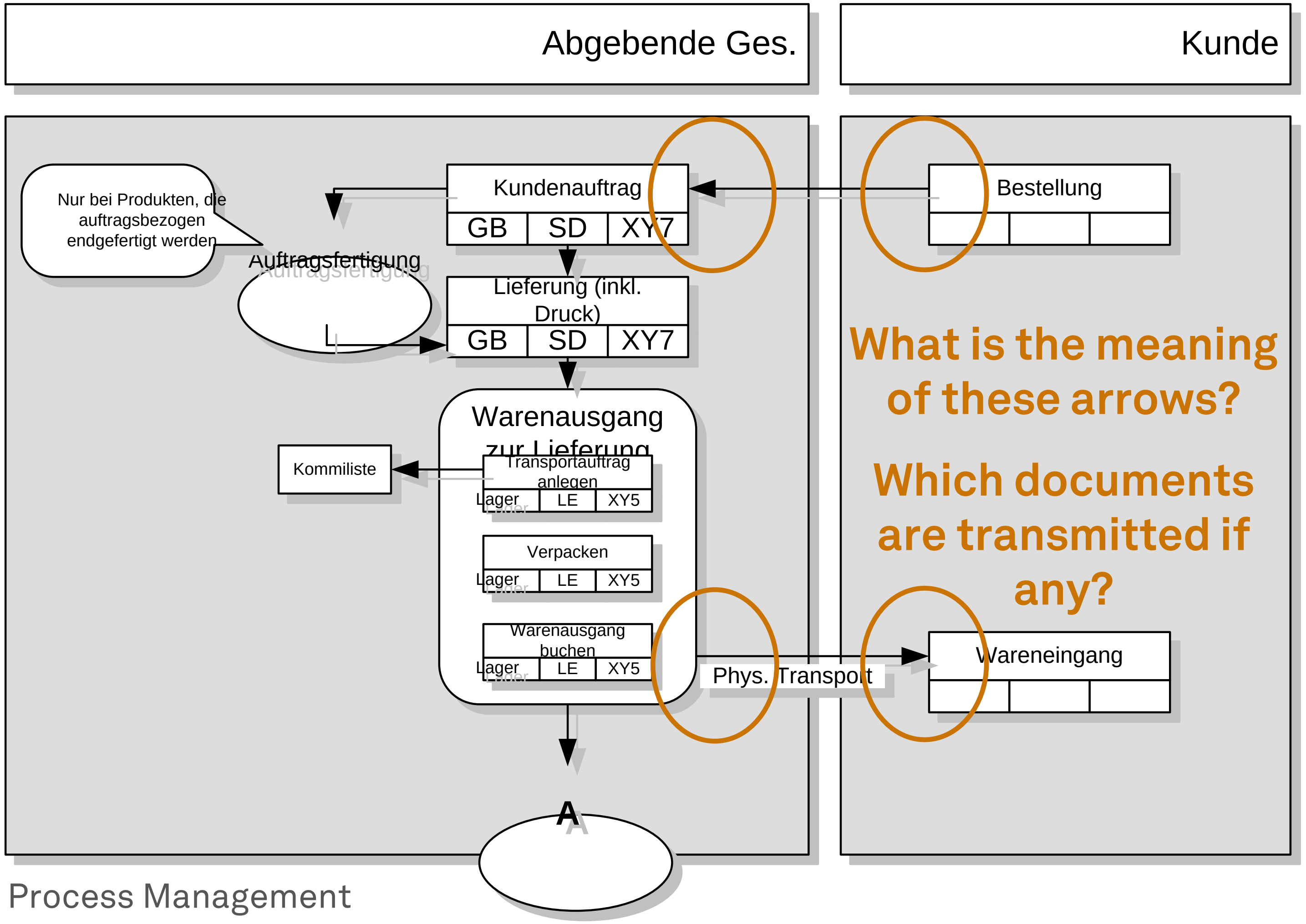
Janiesch & Stein (2007)

Problems with Process Documentation



Janiesch & Stein (2007)

Problems with Process Documentation

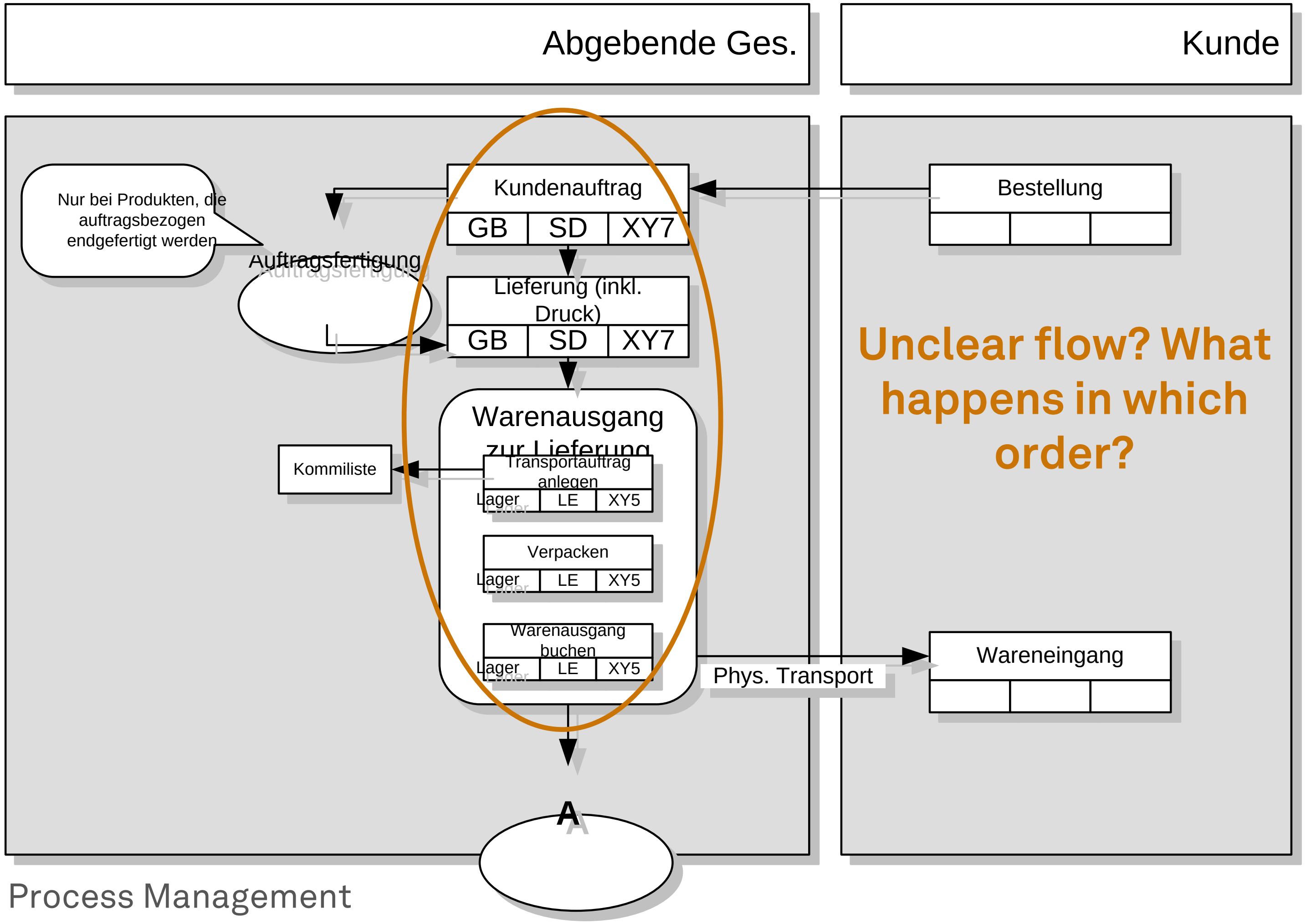


What is the meaning of these arrows?

Which documents are transmitted if any?

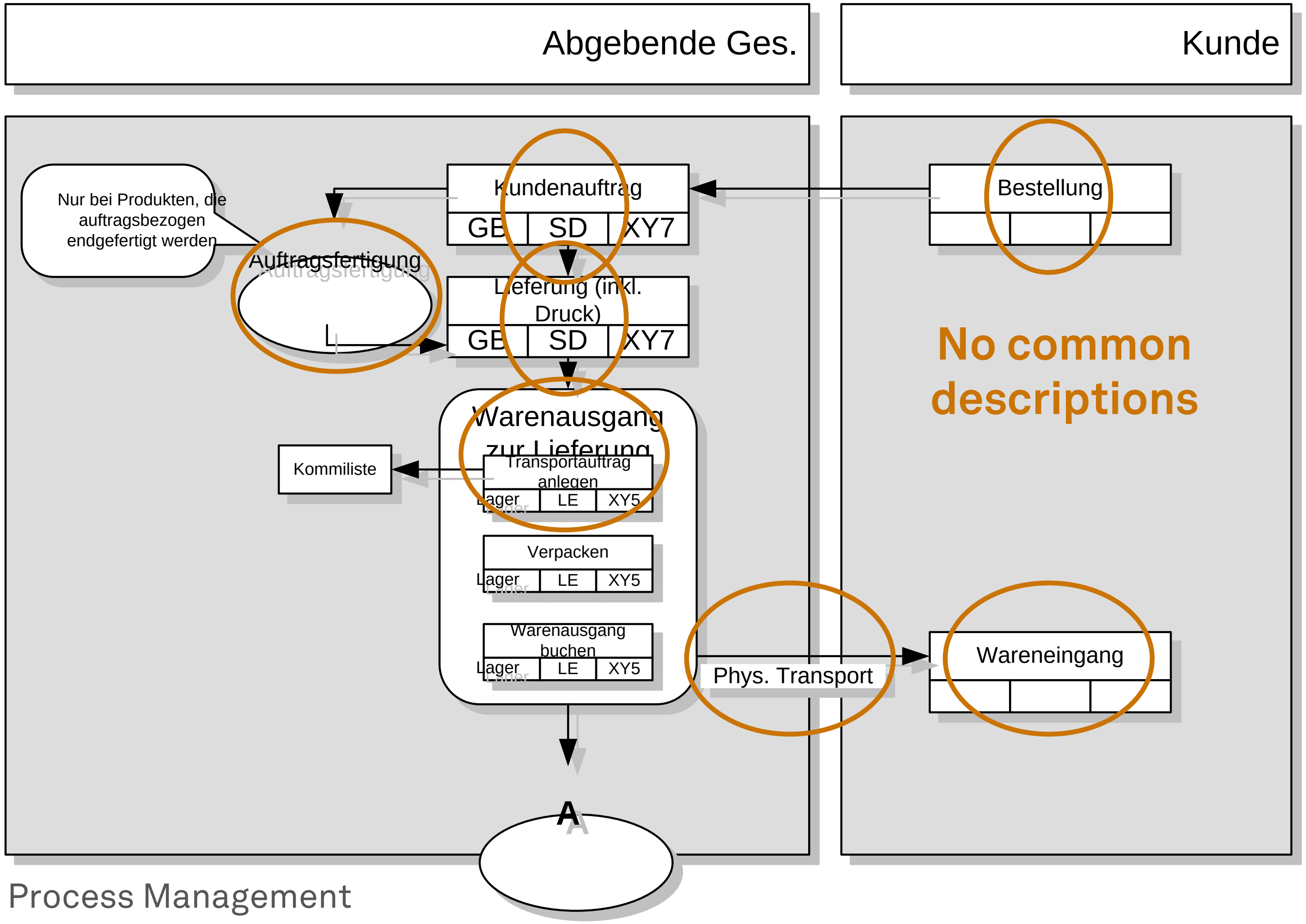
Janiesch & Stein (2007)

Problems with Process Documentation



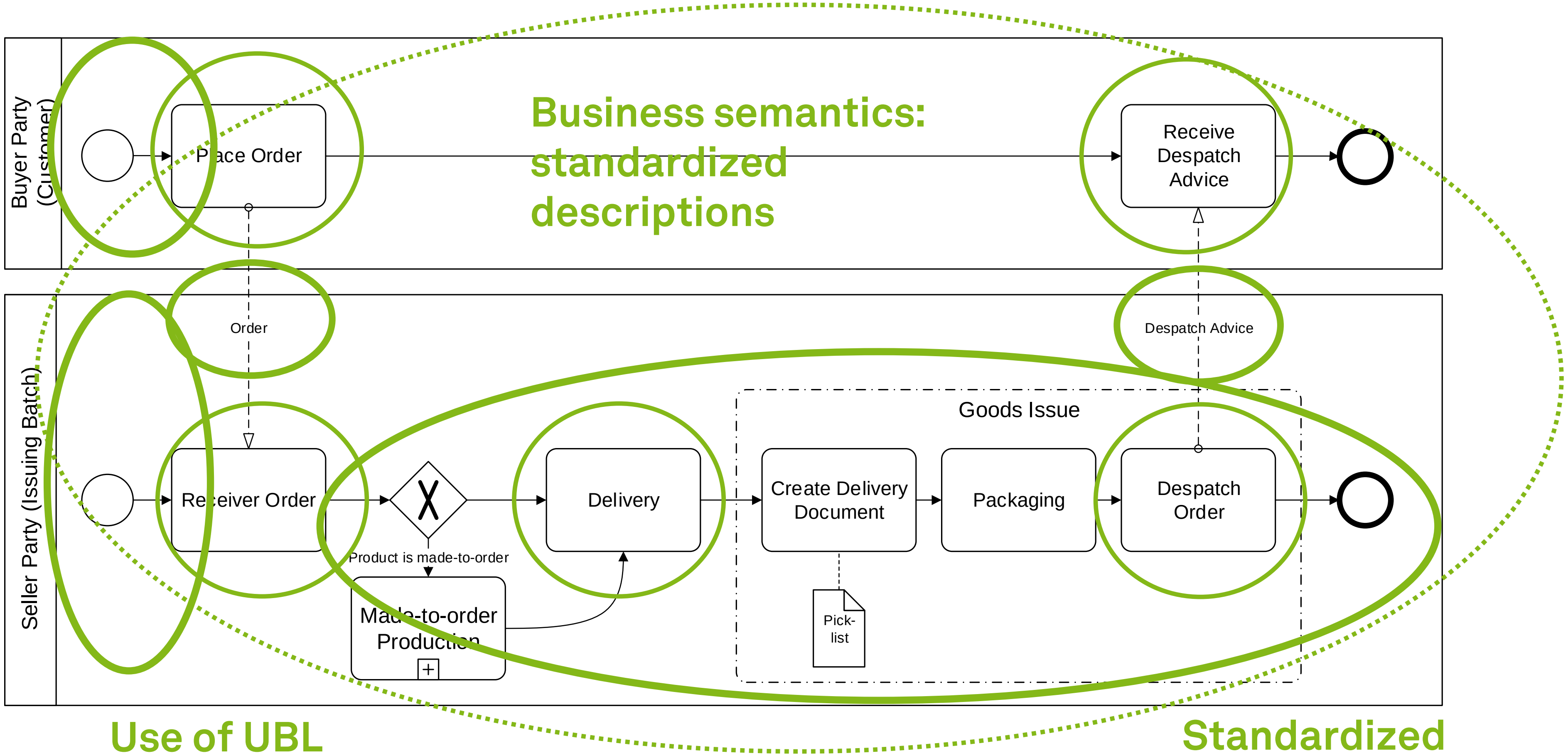
Janiesch & Stein (2007)

Problems with Process Documentation



Janiesch & Stein (2007)

Standardized Process Documentation with BPMN



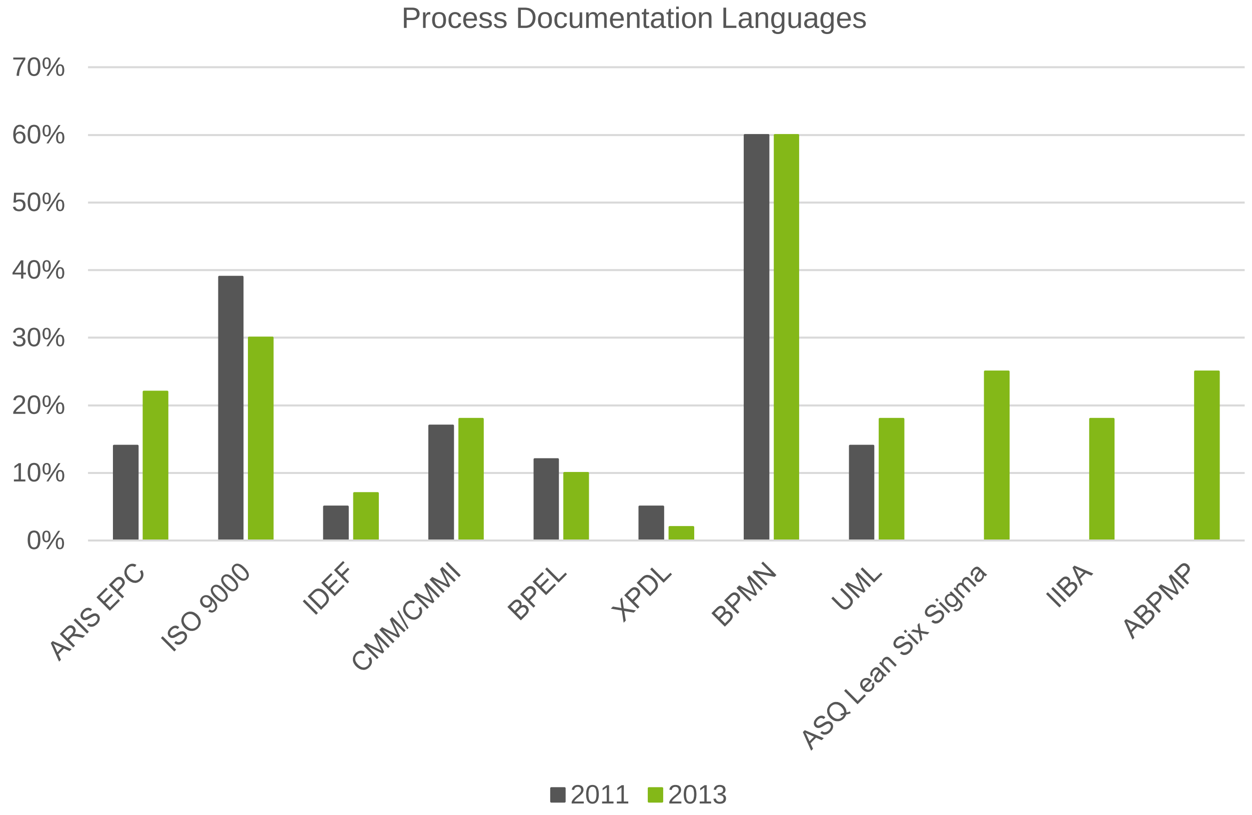
Use of UBL references processes

Standardized Notation: BPMN

Janiesch & Stein (2007)

There are many different notations for process modeling...

- ...but BPMN is the (de facto) standard



Derived from BPTrends (2014)

Business Process Model and Notation (BPMN)

Origins of BPMN

- The Business Process Management Institute (BPMI) – now merged into the OMG – developed the Business Process Modeling Language (BPML) and **realizes the need for a graphical representation**
- In 2001, the BPM Notation Working Group was formed

- Business Process Modeling Notation (BPMN 1.0/1.1/1.2)
  - May 2004: First BPMN specification was released to public
  - February 2006: BPMN specification was adopted by OMG
  - February 2008: BPMN 1.1 was released to public
  - February 2009: BPMN 1.2 was released to public

Object Management Group (2011)

Origins of BPMN

- Business Process **Model** and **Notation** (2.0/2.01/2.02)
  - January 2011: **BPMN 2.0** was released to public
    - Name changed due to changing the scope from “pure” graphical representation to a process definition language
    - Major update, focus on execution
    - V2.01 also published as ISO/IEC 19510

Object Management Group (2011)

Design Goals for BPMN (1.x)

- Must be acceptable and usable by the **business community**
- Must be able to generate **executable** processes (e.g., XPDL, BPEL) through a BPMN Model
- Should **bridge the gap** between business analysts and IT-experts

- BPMN is intended to be **methodology agnostic**
  - BPM methodologies will give guidance as to the purpose and level of detail for modeling
  - BPMN is as complex as it needs to be

BPMN 2.0 Scope

- The **primary goal** of BPMN is to provide a notation that is
  - readily **understandable** by **all business users**,
  - from the business analysts that create the initial drafts of the processes,
  - to the technical developers responsible for implementing the technology that will perform those processes,
  - and finally, to the business people who will manage and monitor those processes

Object Management Group (2011)

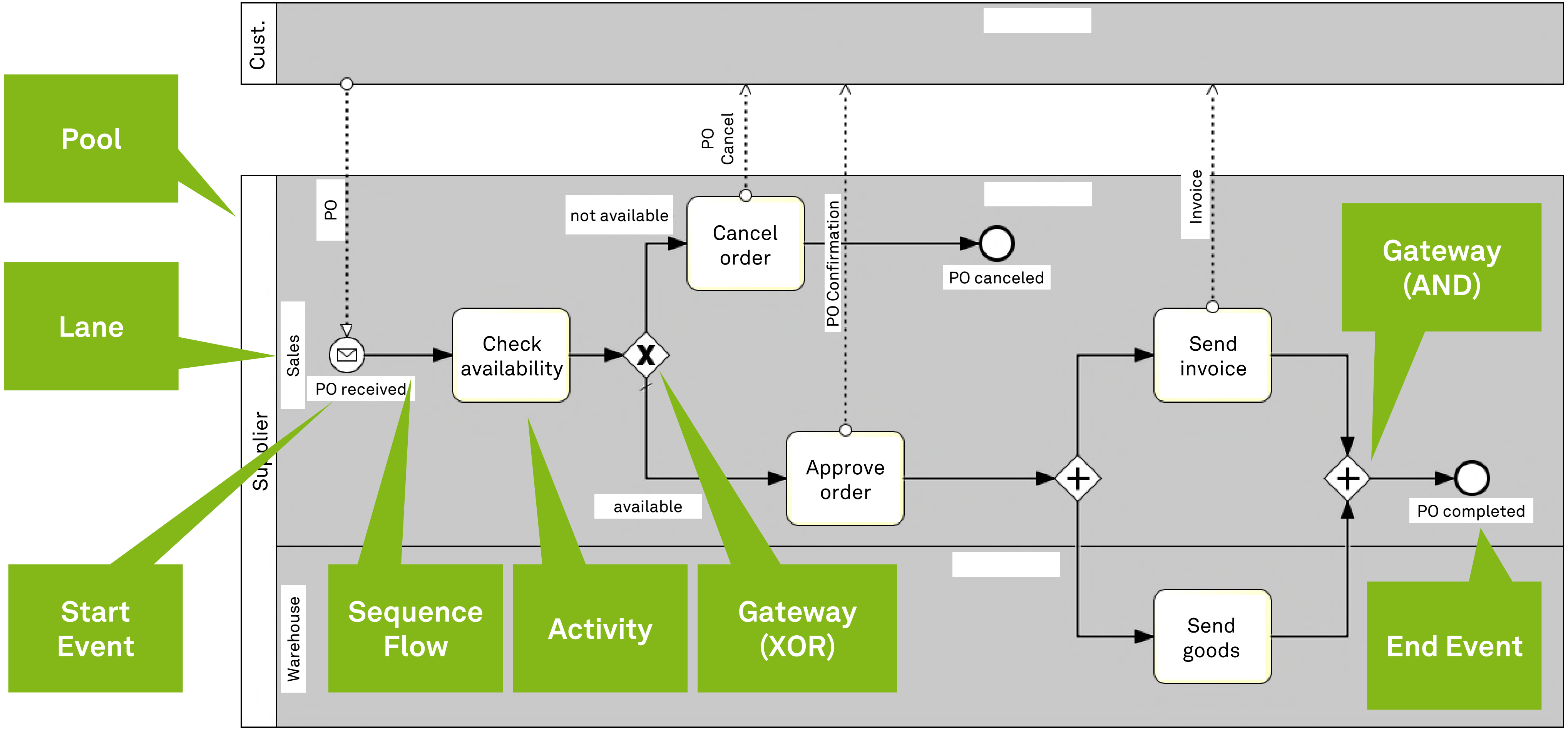
BPMN 2.0 Objects and Diagram Types

- BPMN specifies a **combination of objects**, their **graphical representation** and **supporting attributes** (preliminaries for implementation and enactment)

- Diagram Types
  - **Collaboration Diagram**
  - Choreography Diagram
  - Conversation Diagram

Object Management Group (2011)

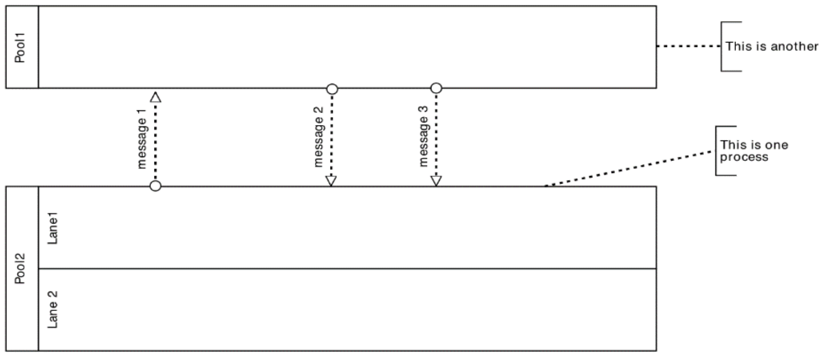
A Simple Process Model with BPMN 2.0



Dumas et al. (2018)

Structuring Elements in Process Diagrams: Pools

- A **Pool** partitions a **set of Activities** from other Pools
- A Pool is a graphical representation of an active **resource**
- Represents **independent organizational entities**
  - Customer
  - Supplier
  - etc.
- It can be a **participant**, **software**, or **hardware**
  - e.g., an ERP system
  - e.g., a manufacturing plant



Object Management Group (2011)

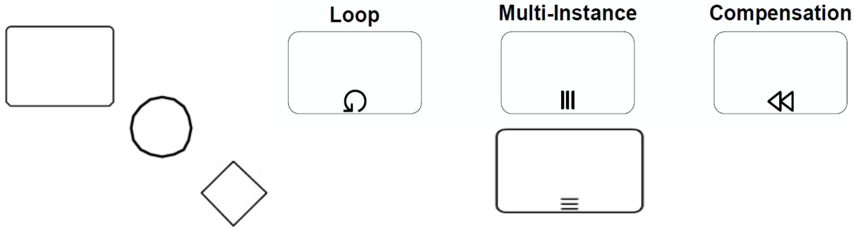
Structuring Elements in Process Diagrams: Lanes

- A **Lane** is a sub-partition within a Pool
- Lanes **extend** the **entire length** of a Pool either horizontally or vertically
- Lanes can be **nested**
- Lanes are used to **organize and categorize** Activities (typically by **roles**)
- Represent resource classes in the same organizational unit or space
  - e.g., sales department, customer care, logistics, manager, etc.

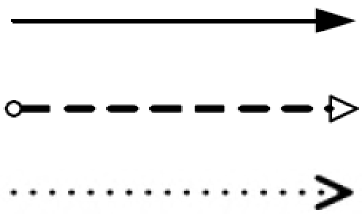
Object Management Group (2011)

Overview of Basic BPMN Objects in Process Diagrams

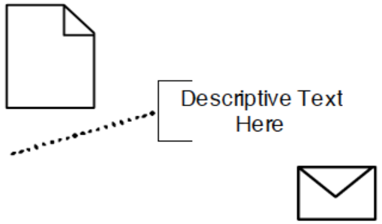
- **Flow Objects**
  - Activity
  - Event
  - Gateway



- **Connectors**
  - Sequence Flow
  - Message Flow
  - Association



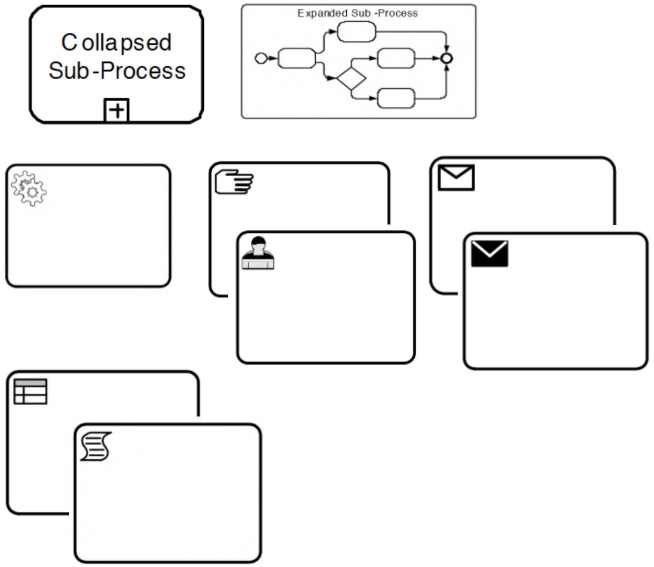
- **Artifacts**
  - Data Object
  - Text Annotation
  - Message

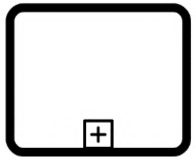


Activities and Tasks in BPMN

- Activities represent **units of work** that have a duration **performed** within a business process
  - Executable elements of BPMN process
  - Can be **atomic** (Task) or **compound** (Sub-Process)
  - Carry attributes such as **loopCharacteristics** or **maximumLoop**

- Types of Activities in a process
  - Sub-Process
  - Task
    - **Service Task**
    - Manual / User Task
    - Send / Receive Task
    - Business Rule / Script Task
  - Call Activity



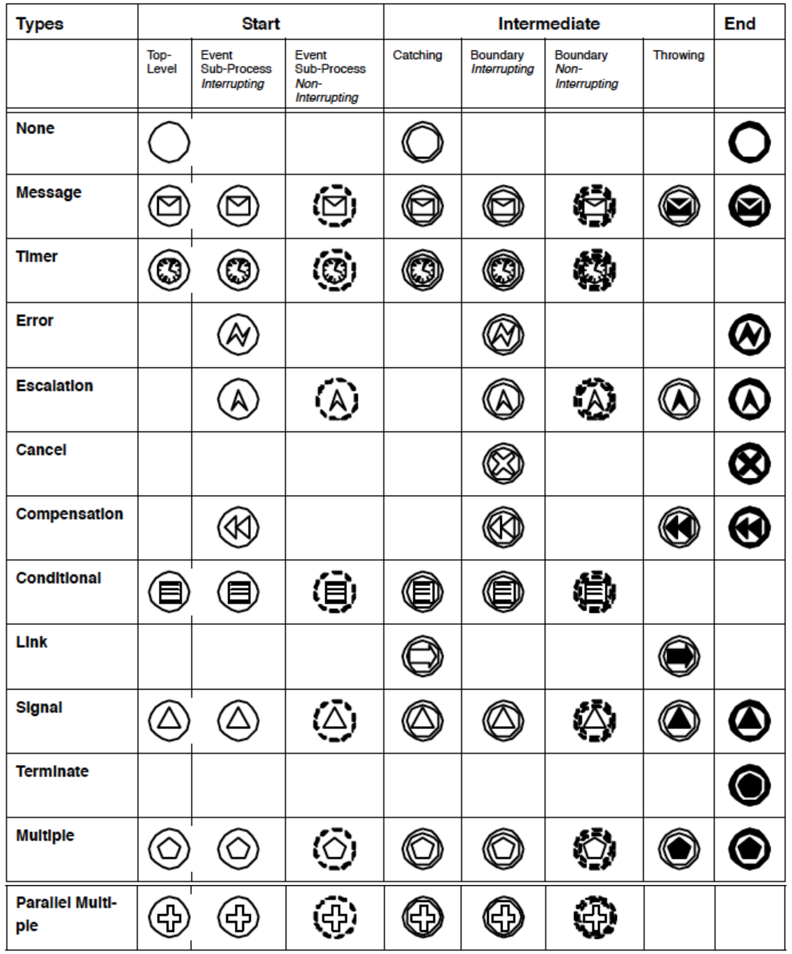


Events in BPMN

- An event is **something** that happens **instantaneously** during the course of a process
- Events **affect the flow** of a Process, usually have a cause or an impact and allow for reaction
- Types of Events
  - **Start** Events
  - **End** Events
  - **Intermediate** Events

- Types of Event behavior
  - Catch a trigger (message, timer, ...)
  - Throw a result (eventually caught by another event)

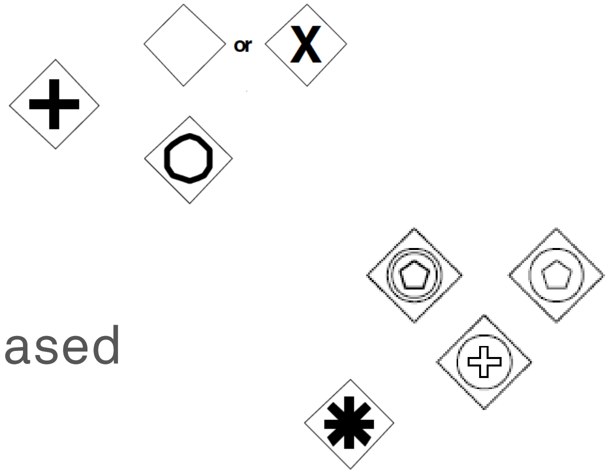
There are quite some events in BPMN



Gateways in BPMN

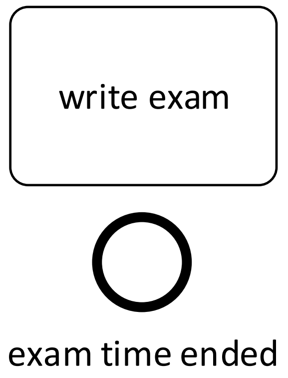
- Gateways are used to **control** how **Sequence Flows** interact as they converge and diverge within a Process
- Employ **gating mechanism** that either allows or disallows passage through the Gateway
- As Tokens arrive at a Gateway, they can be **merged** together on input and/or **split** apart on output

- Types of Gateways
  - **Exclusive**
  - **Parallel**
  - **Inclusive**
  - Event-based
  - Parallel Event-based
  - Complex

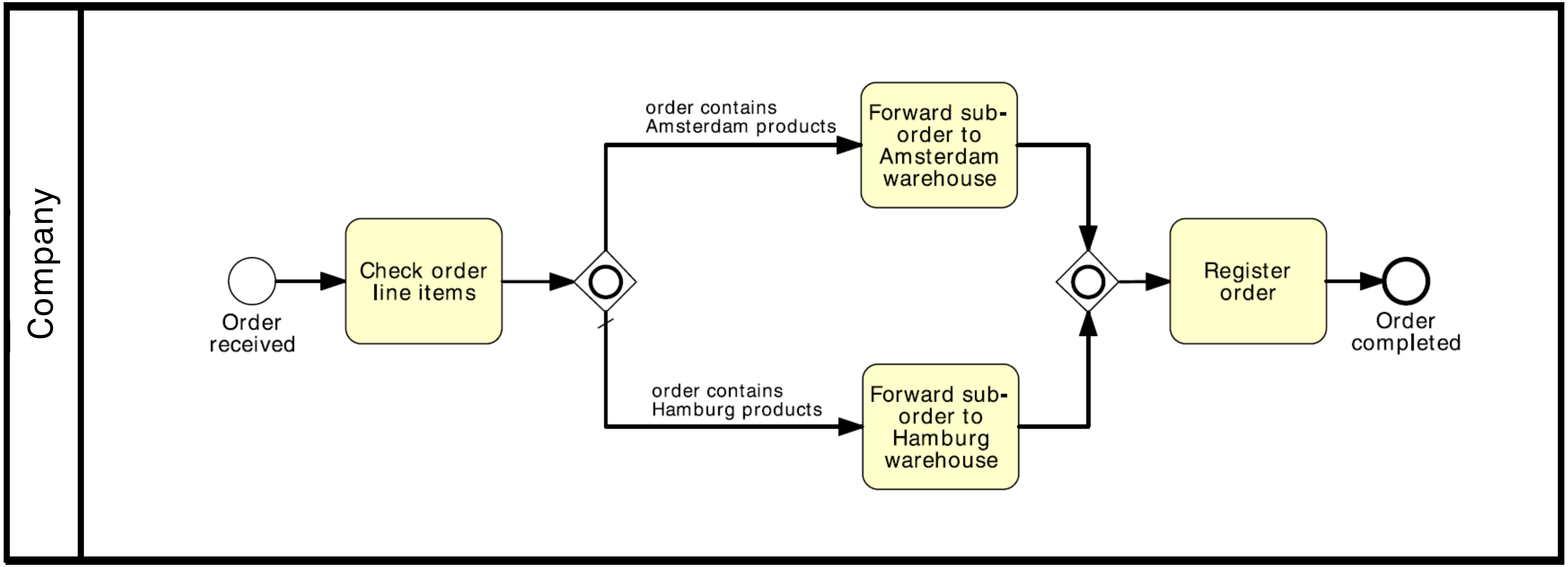


Naming Conventions

- There is **no formal restriction**, no overall naming convention
- It is **good practice** to name
  - Activities and tasks
    - **imperative verb followed by noun** (e.g., “write exam”)
  - Events
    - **Noun followed by past participle verb** (e.g., “exam time ended”)
  - Process model
    - **Nouns** (e.g., “exam marking” rather than “mark exams”)

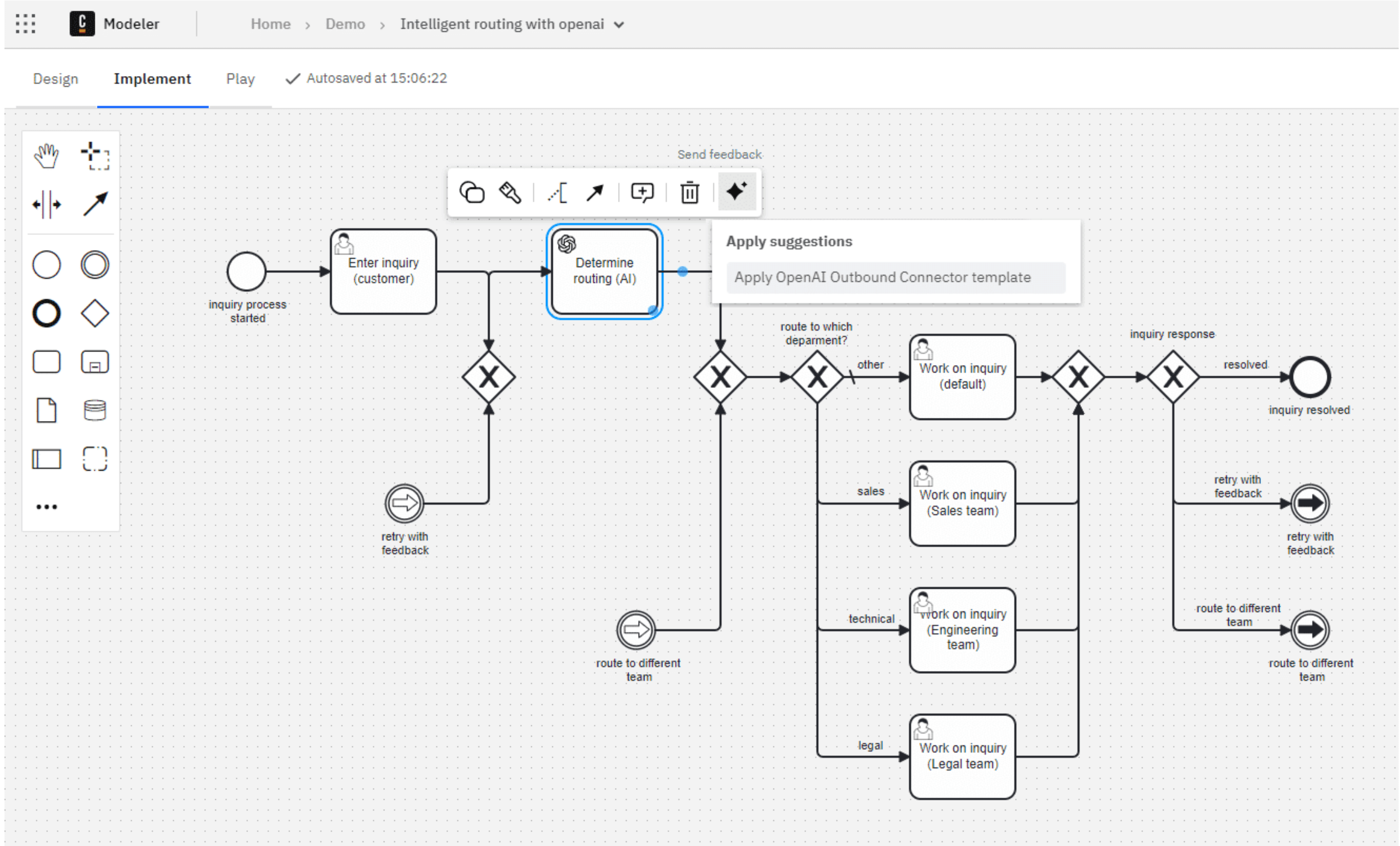


A Simple BPMN Model



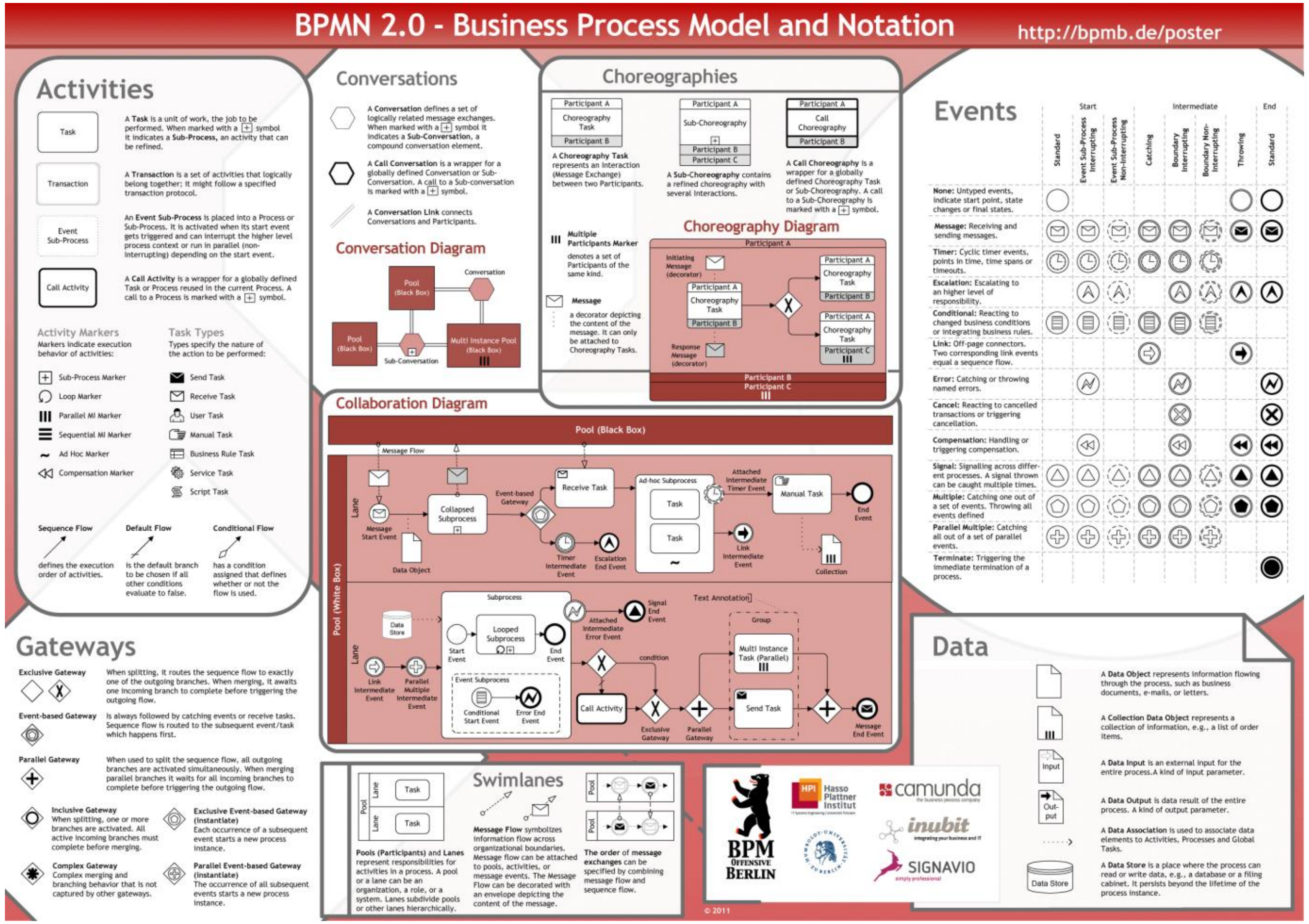
Dumas et al. (2018)

BPMN Tools - Camunda



https://camunda.com/platform/modeler/

BPMN 2.0 Poster



http://www.bpmb.de/index.php/BPMNPoster

Learning Goals of Today

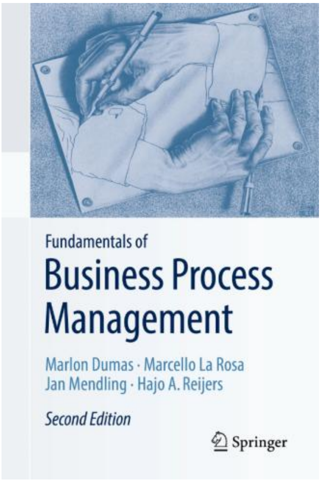
- Understand the purpose of process modeling
- Understand the purpose of BPMN
- Get to know BPMN's basic modeling constructs
- Be able to use basic BPMN

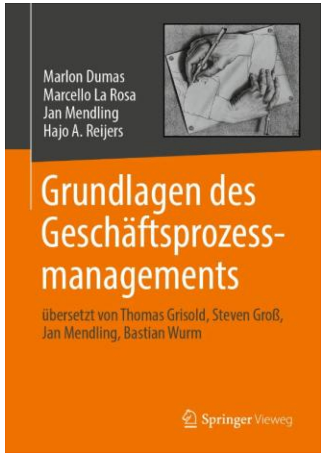
Recap Questions

- What are features of a model and why do we model processes?
- What are pools and lanes in BPMN model?
- What is the difference between activities and events in a BPMN model?
- Why do we need gateways in a BPMN model?

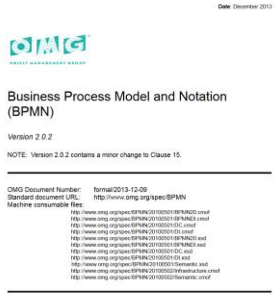
Literature

- **Section 3:** Essential Process Modeling





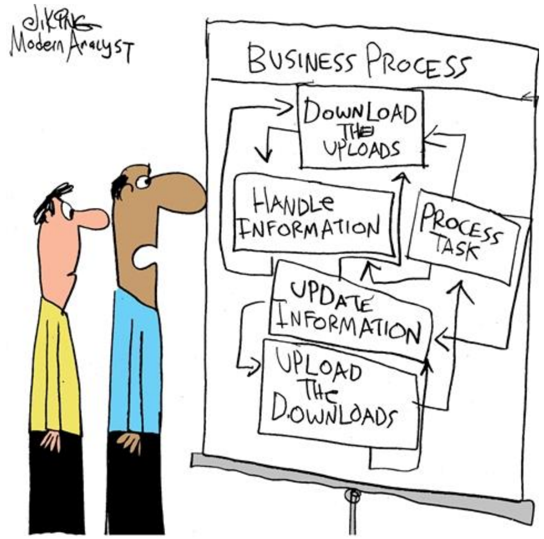
- Business Process Model and Notation (BPMN)
  - https://www.omg.org/spec/BPMN/2.0.2



Roadmap

Introduction	10.04. Introduction	17.04. Process Identification			
Process Discovery & Modeling	24.04. Essentials of BPMN	01.05. Public Holiday	08.05. Advanced BPMN I	15.05. Advanced BPMN II	22.05. Process Discovery
Process Analysis & Redesign	29.05. Public Holiday	05.06. Qualitative Analysis	12.06. Quantitative Analysis	19.06. Public Holiday	26.06.2025 Process Redesign
Application of BPM	03.07. Process Technology	10.07. Process Monitoring and Mining	15.07. BPM as an Enterprise Capability		

Any Questions about Process Modeling?



"Charlie, did you really get these details from the customer?"

modernanalyst.com (2016)

ec.cs.tu-dortmund.de

technische universität dortmund
Chair of Enterprise Computing
Department of Computer Science

Business Process Management

Advanced Process Modeling I



Roadmap

Introduction	10.04. Introduction	17.04. Process Identification			
Process Discovery & Modeling	24.04. Essentials of BPMN	01.05. Public Holiday	08.05. Advanced BPMN I	15.05. Advanced BPMN II	22.05. Process Discovery
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Learning Goals of Today

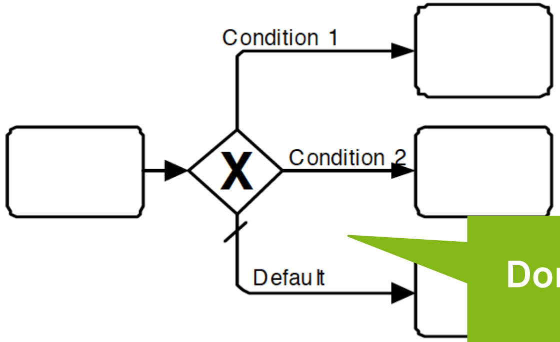
- Better understand the syntactics and semantics of BPMN
- Understand branching and merging in BPMN
- Understand decomposition, resources, and collaborations in BPMN
- Be able to use advanced BPMN

Branching & Merging



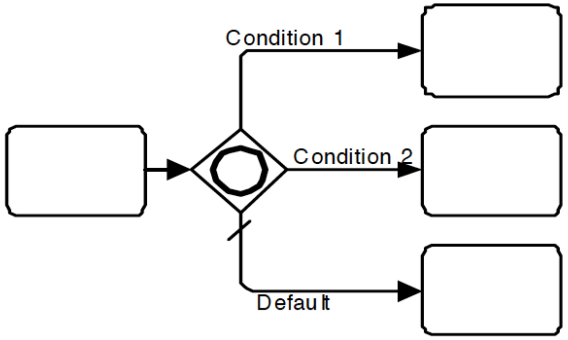

Gateways in Action

- Exclusive Decision

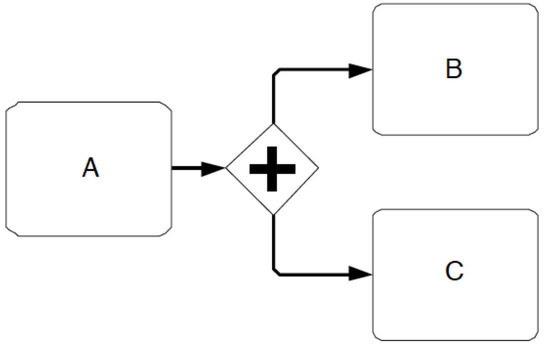


Don’t forget to specify conditions!

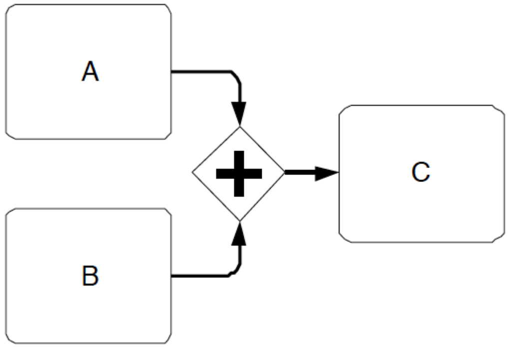
- Inclusive Decision



- Parallel Split

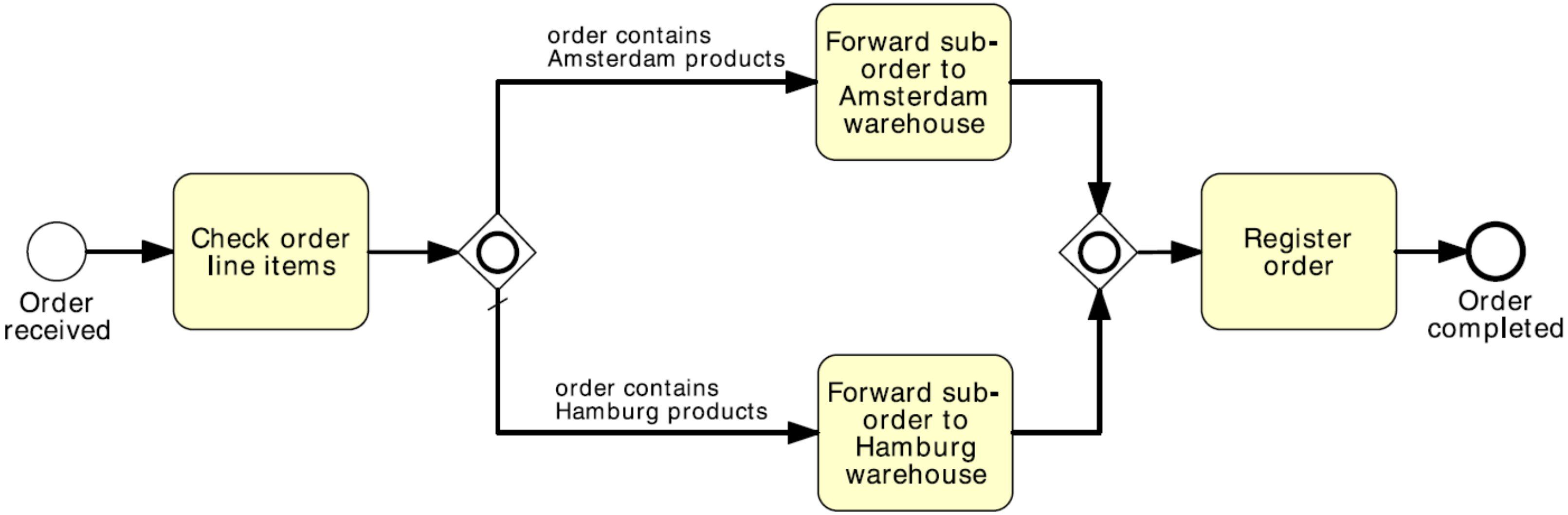


- Synchronizing Merge



Do We Really Need an Inclusive OR?

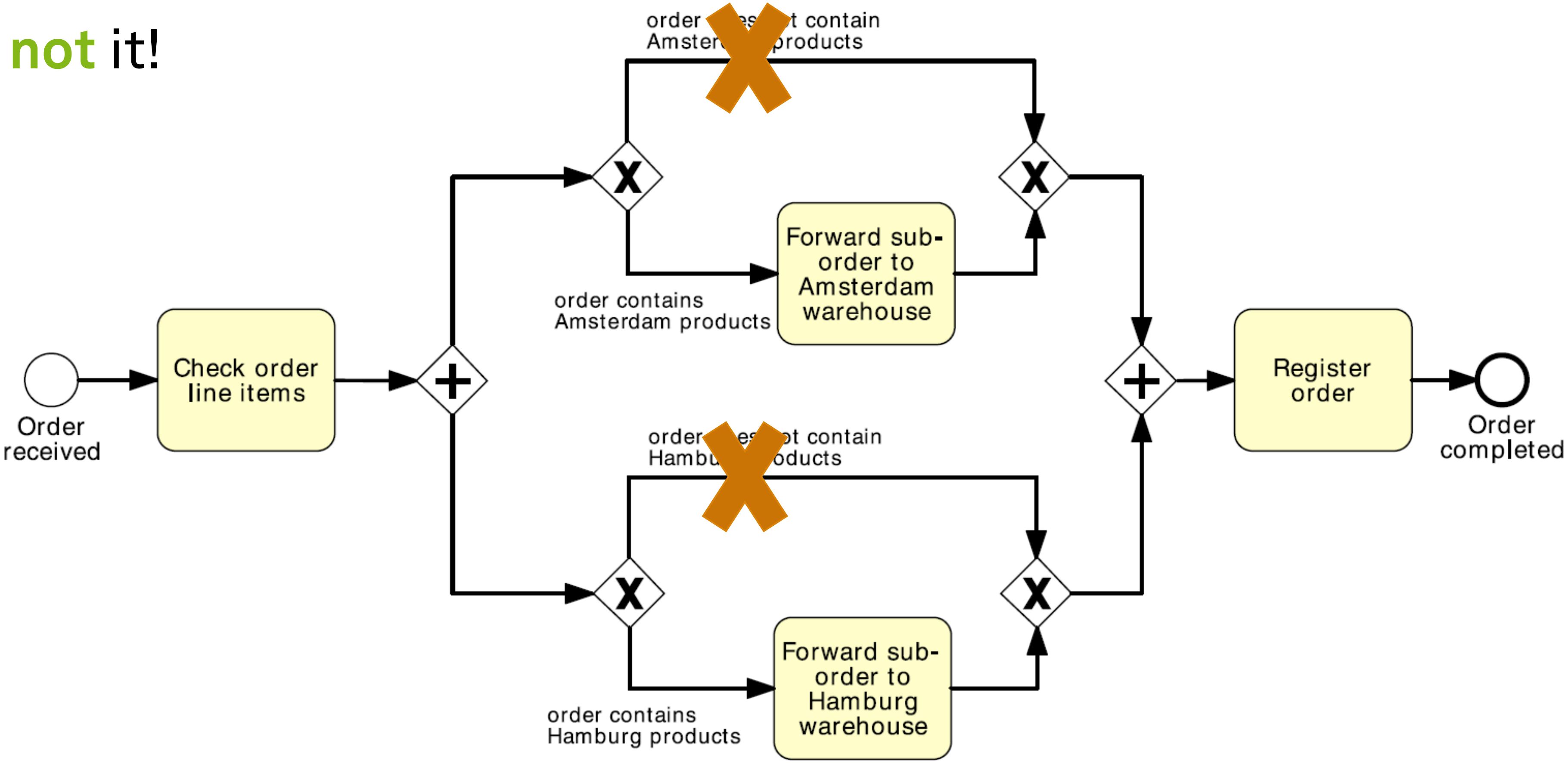
- Imagine this process
- You can visit **one or both** branches
- Can you model it **without an OR gateway**?



Dumas et al. (2018)

A Solution Without OR?

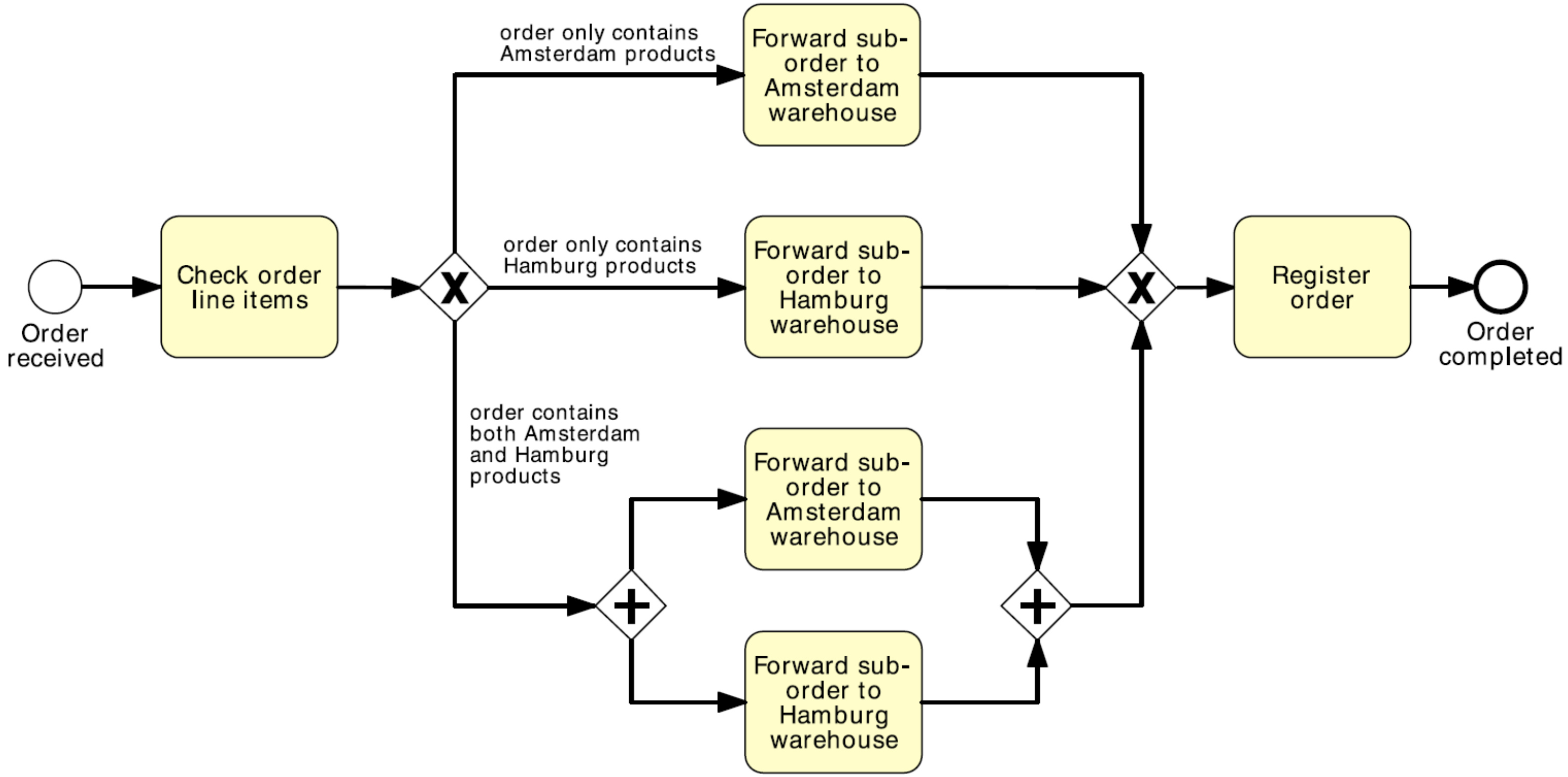
- This is **not** it!



Dumas et al. (2018)

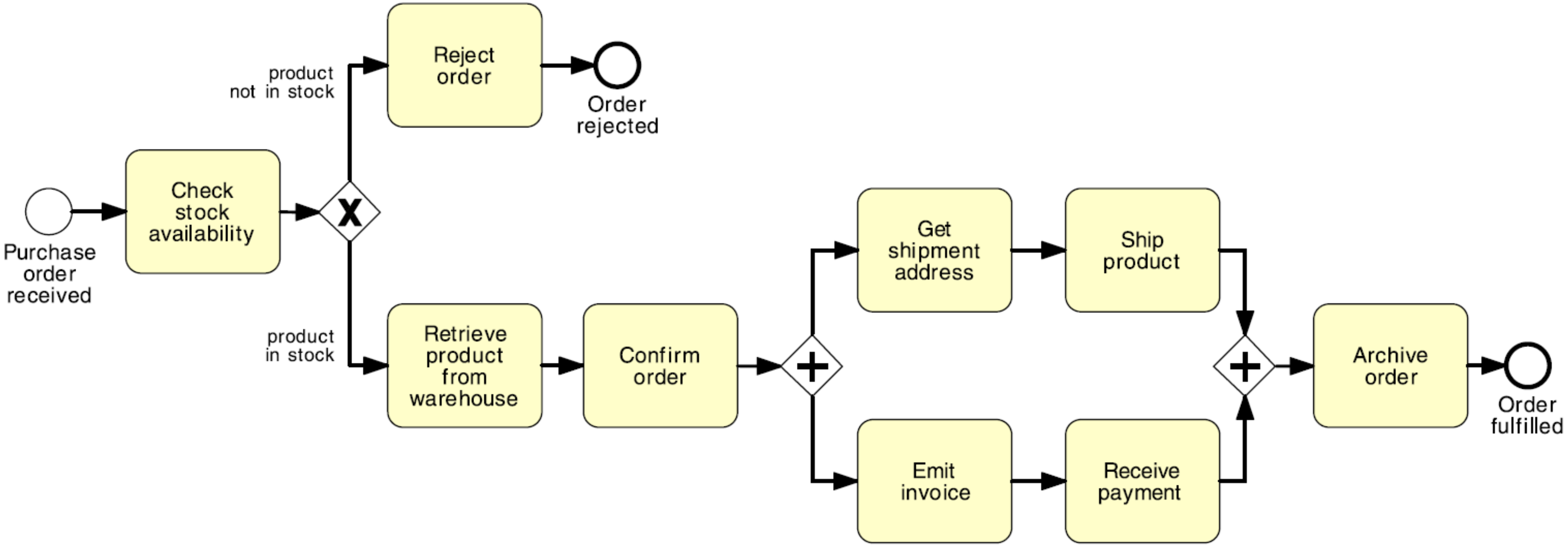
A Solution Without OR?

- **Yes!** This is it!
- But it is very cumbersome for 3+ branches!



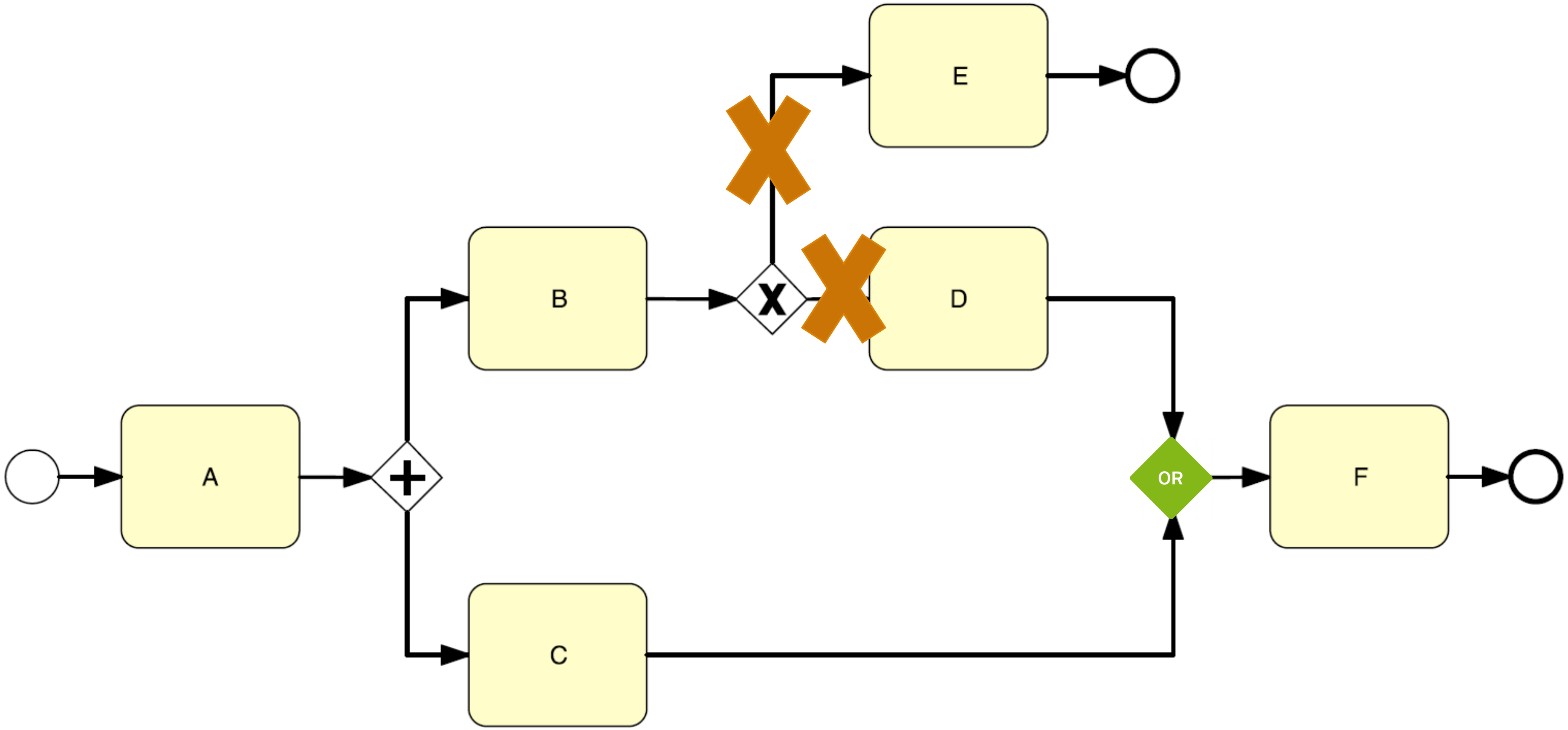
Dumas et al. (2018)

Token Flow and Implicit Termination



Dumas et al. (2018)

How Do We Fix This?

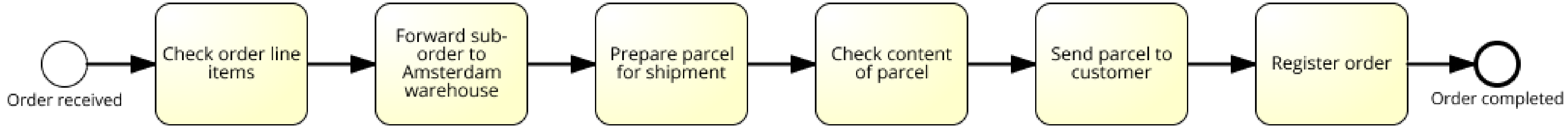


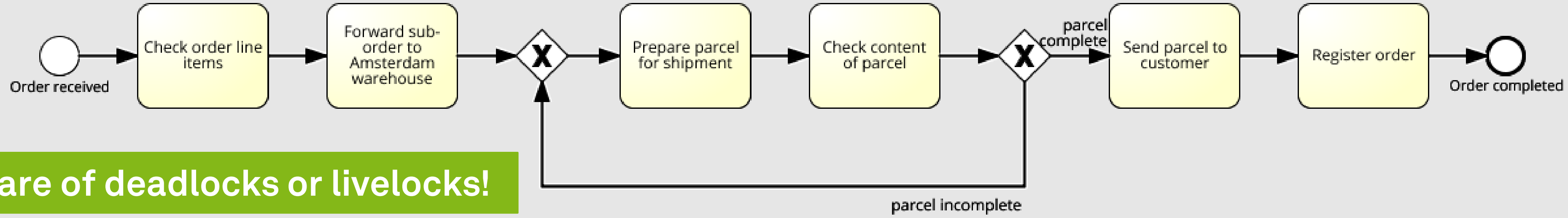
Dumas et al. (2018)

Rework Blocks

- Processes might need to **rework** one or several activities
- Imagine the following process model

- How do we **model** rework?
  - Activity “Check content of parcel” failed
  - Activity “Prepare parcel for shipment” needs to be reworked

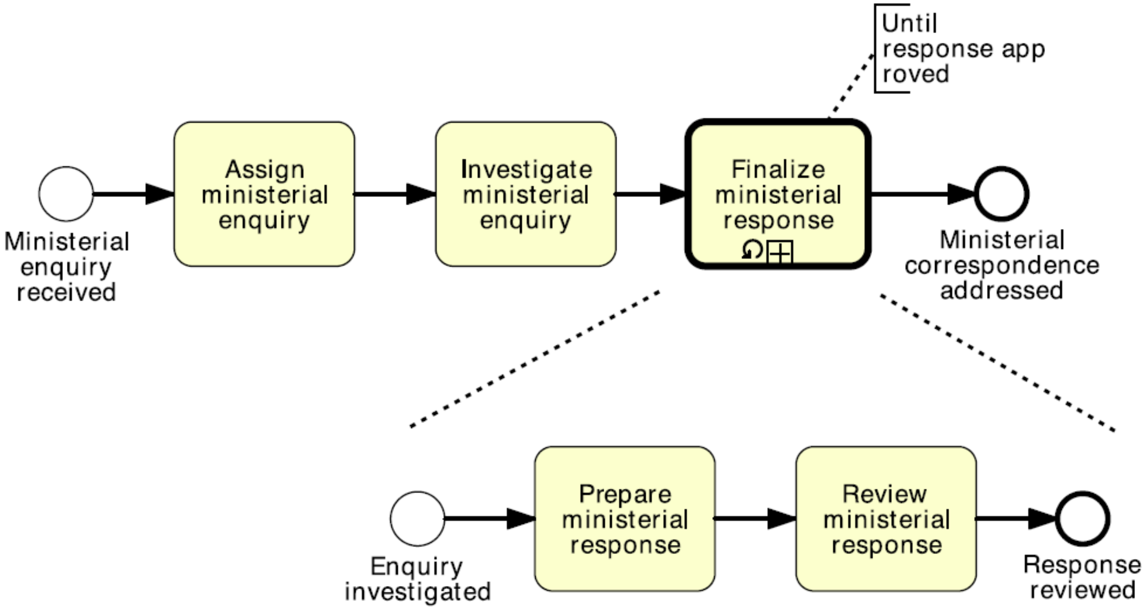




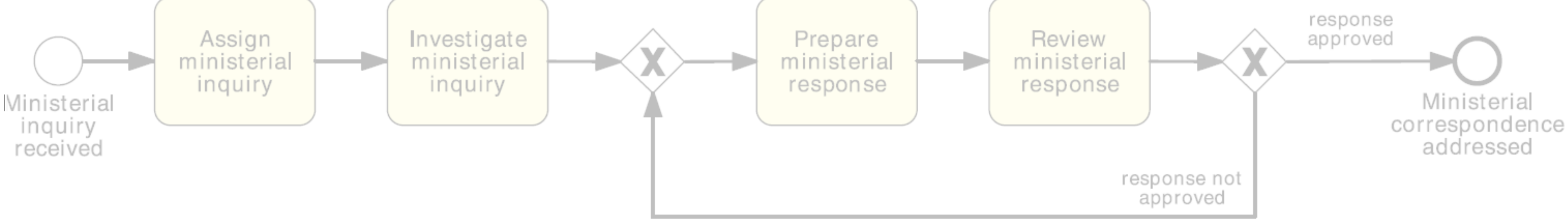
Be aware of deadlocks or livelocks!

Dumas et al. (2018)

Loop Activities

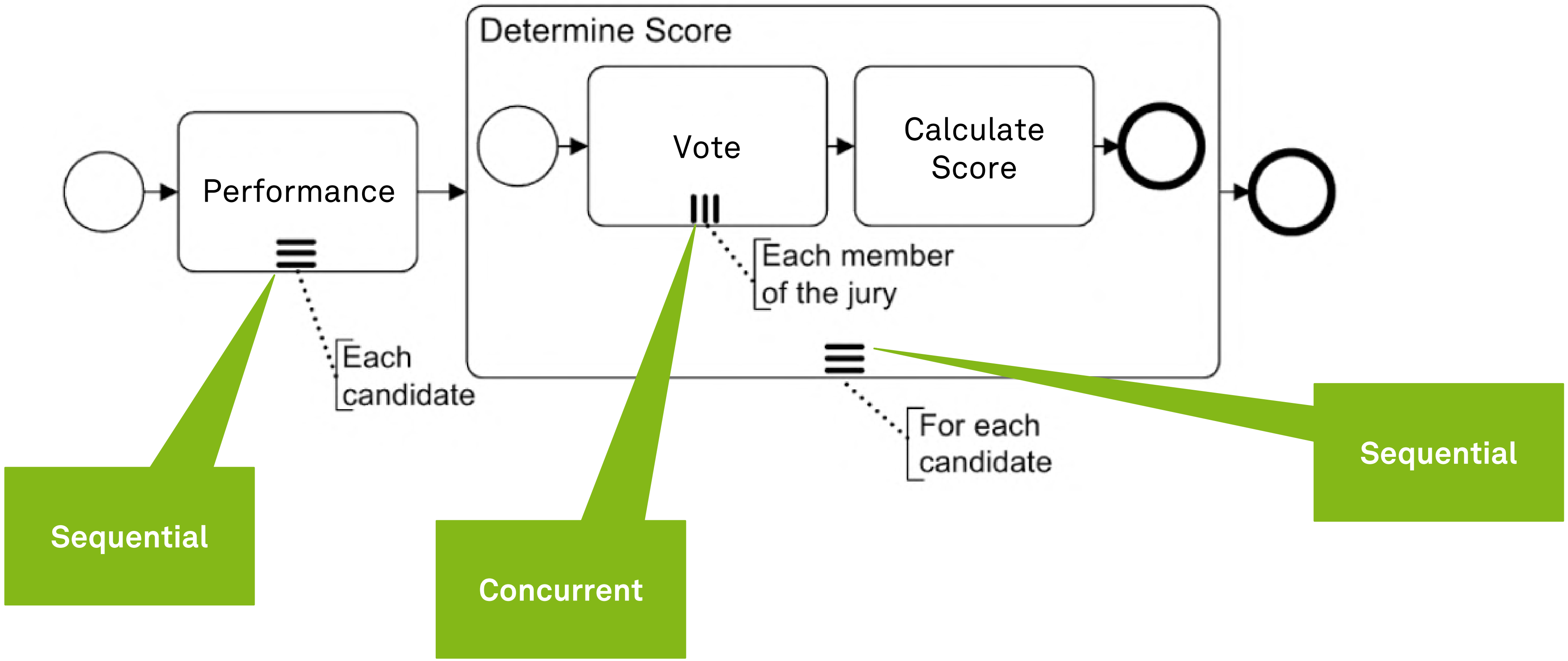


- For loops like these
  - **use XOR**,
  - not OR
  - **never AND**!



Dumas et al. (2018)

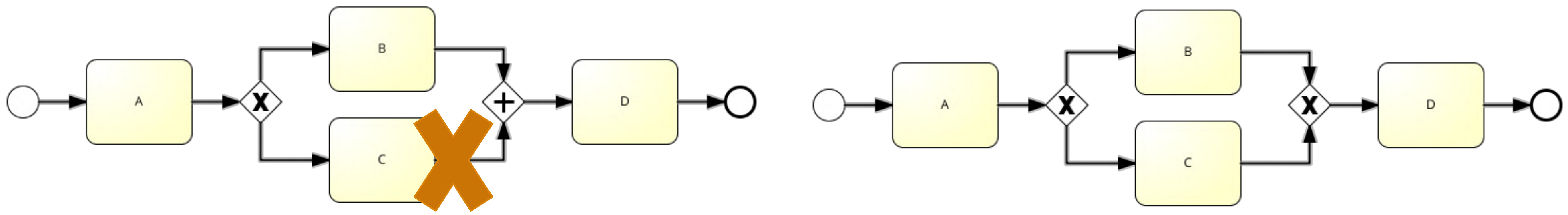
Multi-Instance Activities



Allweyer (2015)

BPMN Use – Syntax

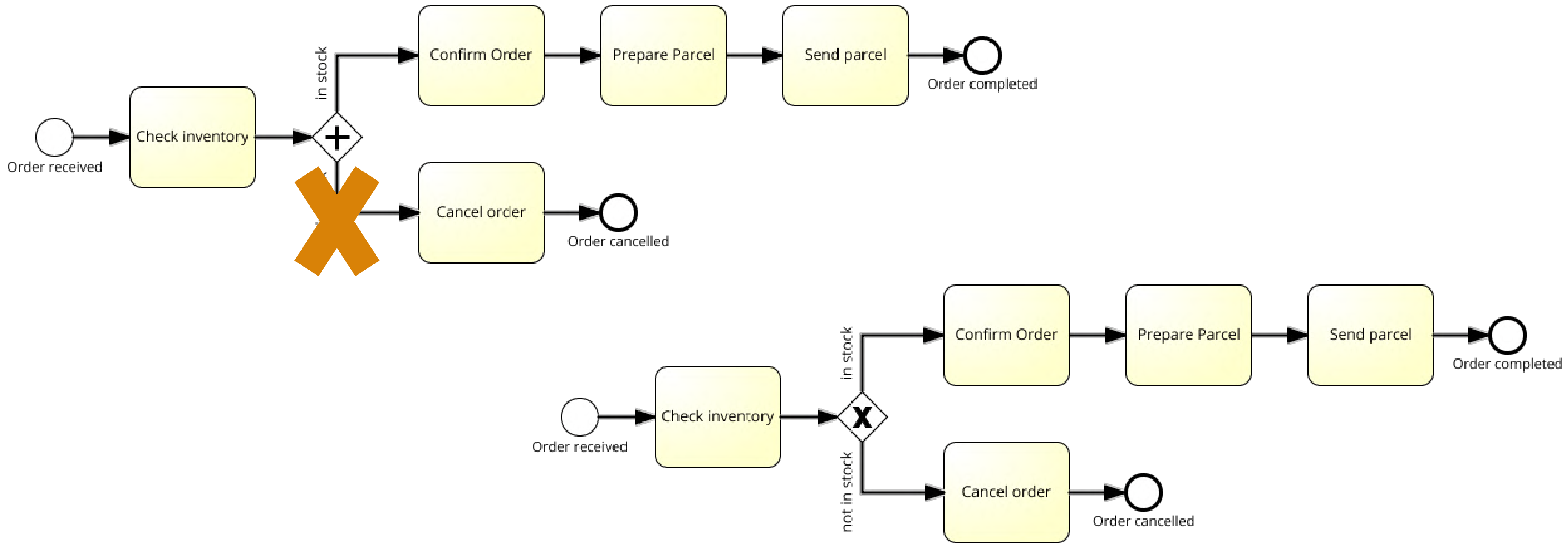
- Includes constructs and the **set of rules** to combine these constructs
- In linguistics, you determine the **structure of sentences** in a given language



La Rosa (2013)

BPMN Use – Semantics

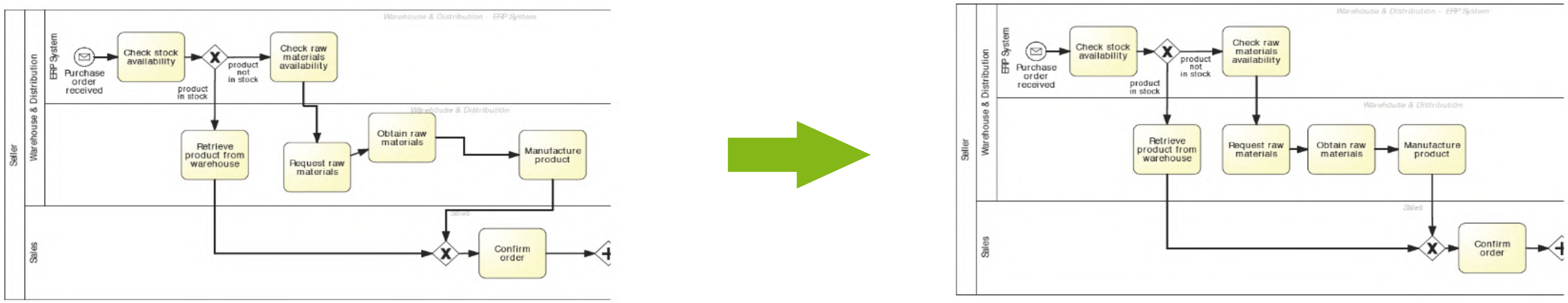
- Focusses on the study of **meaning**, and the study of **relations** between different elements.



La Rosa (2013)

BPMN Use – Pragmatics

- Modeling using BPMN is from **top-left to bottom right**
- Do not cross arrows unless it is unavoidable

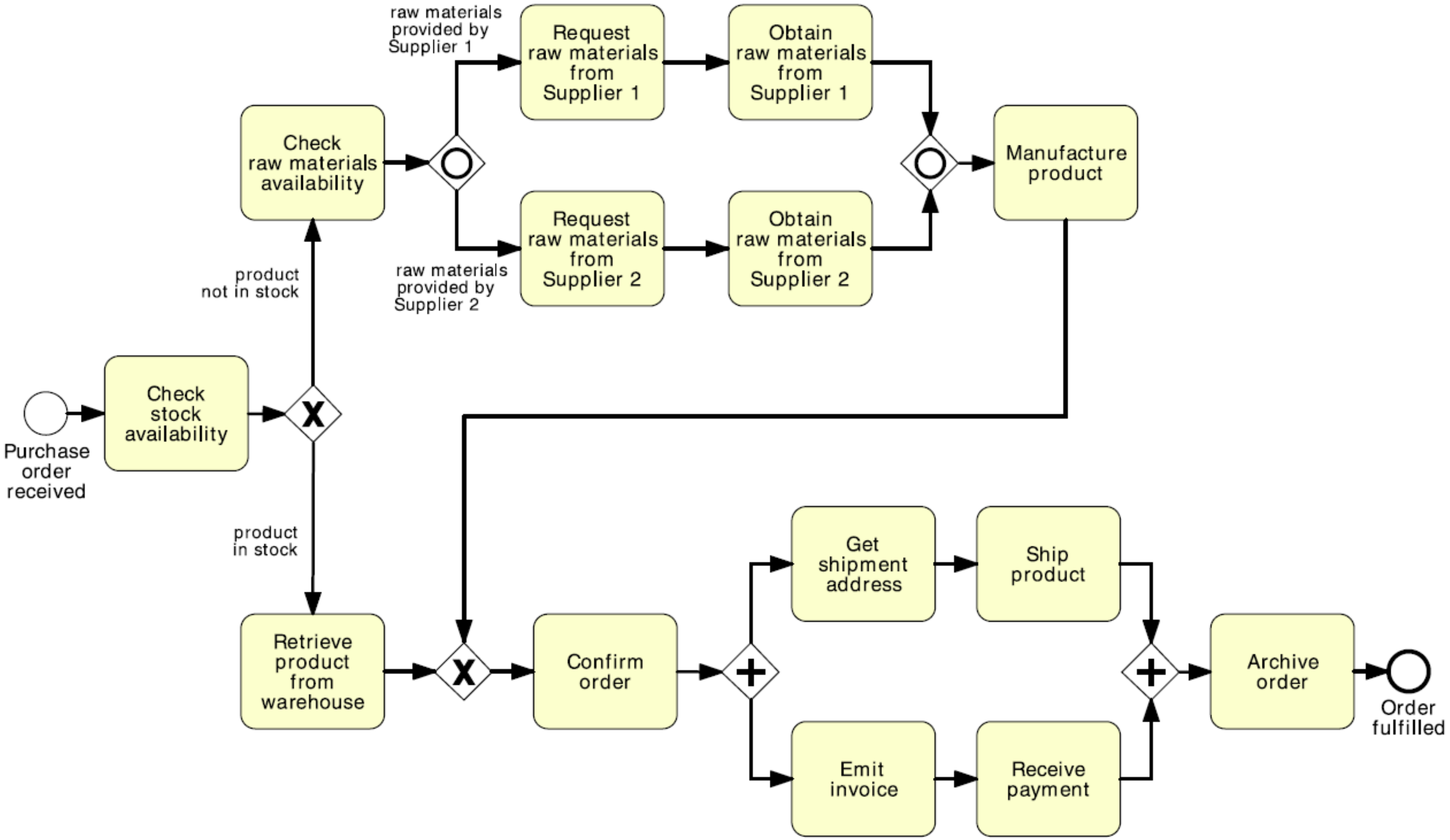


La Rosa (2013)

Decomposition, Resources, and Collaborations




Process Decomposition

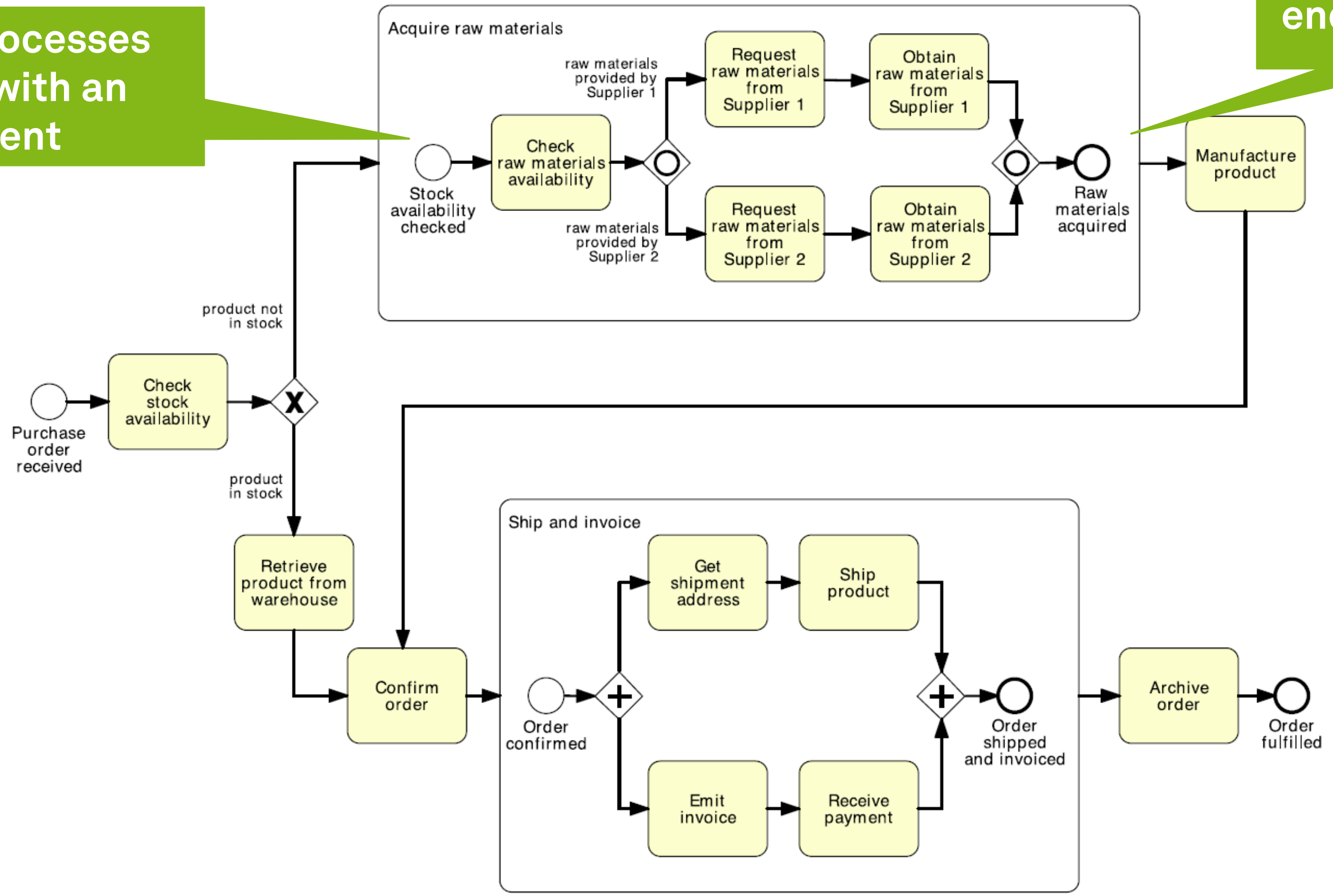


Dumas et al. (2018)

Identify Sub-processes

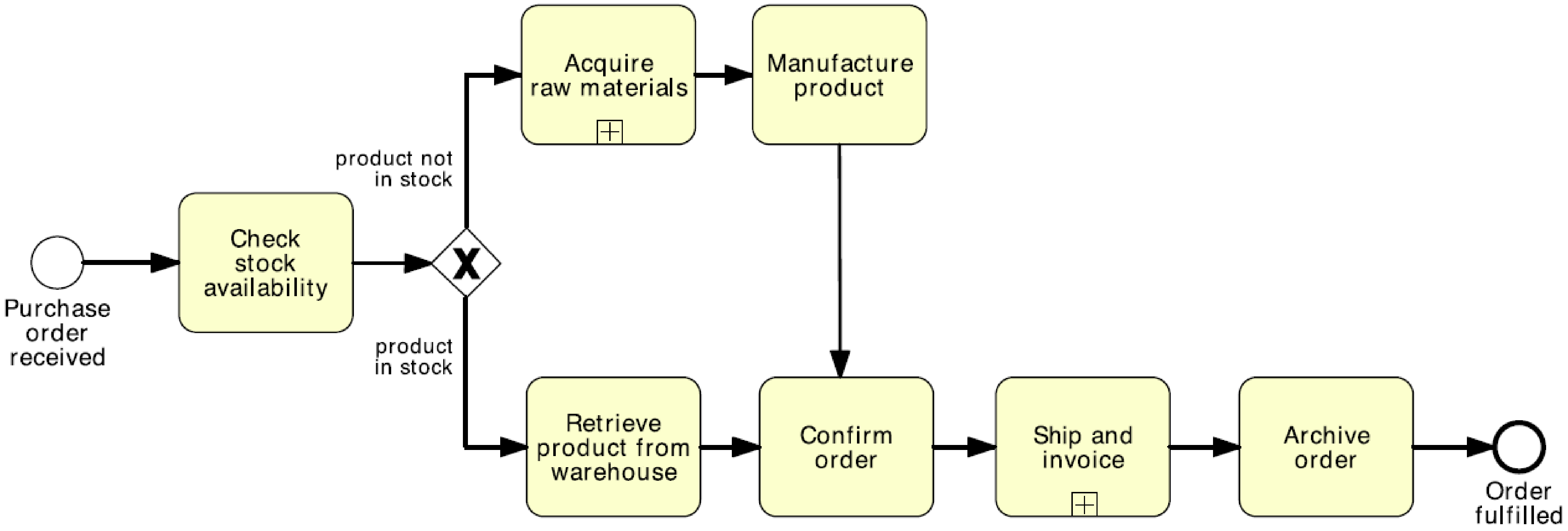
Sub-processes start with an event

Sub-processes end with an event



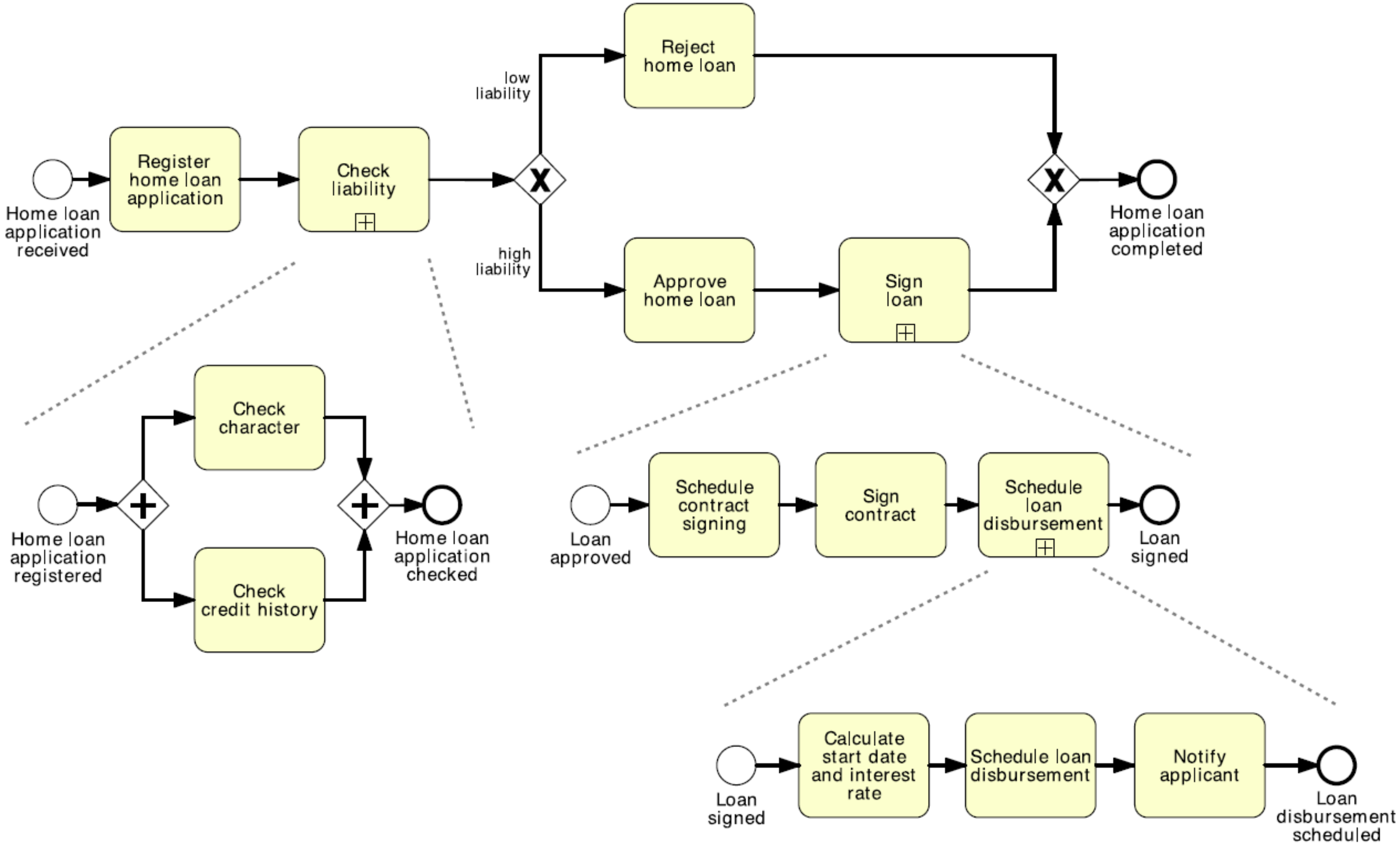
Dumas et al. (2018)

Process with Collapsed Activities



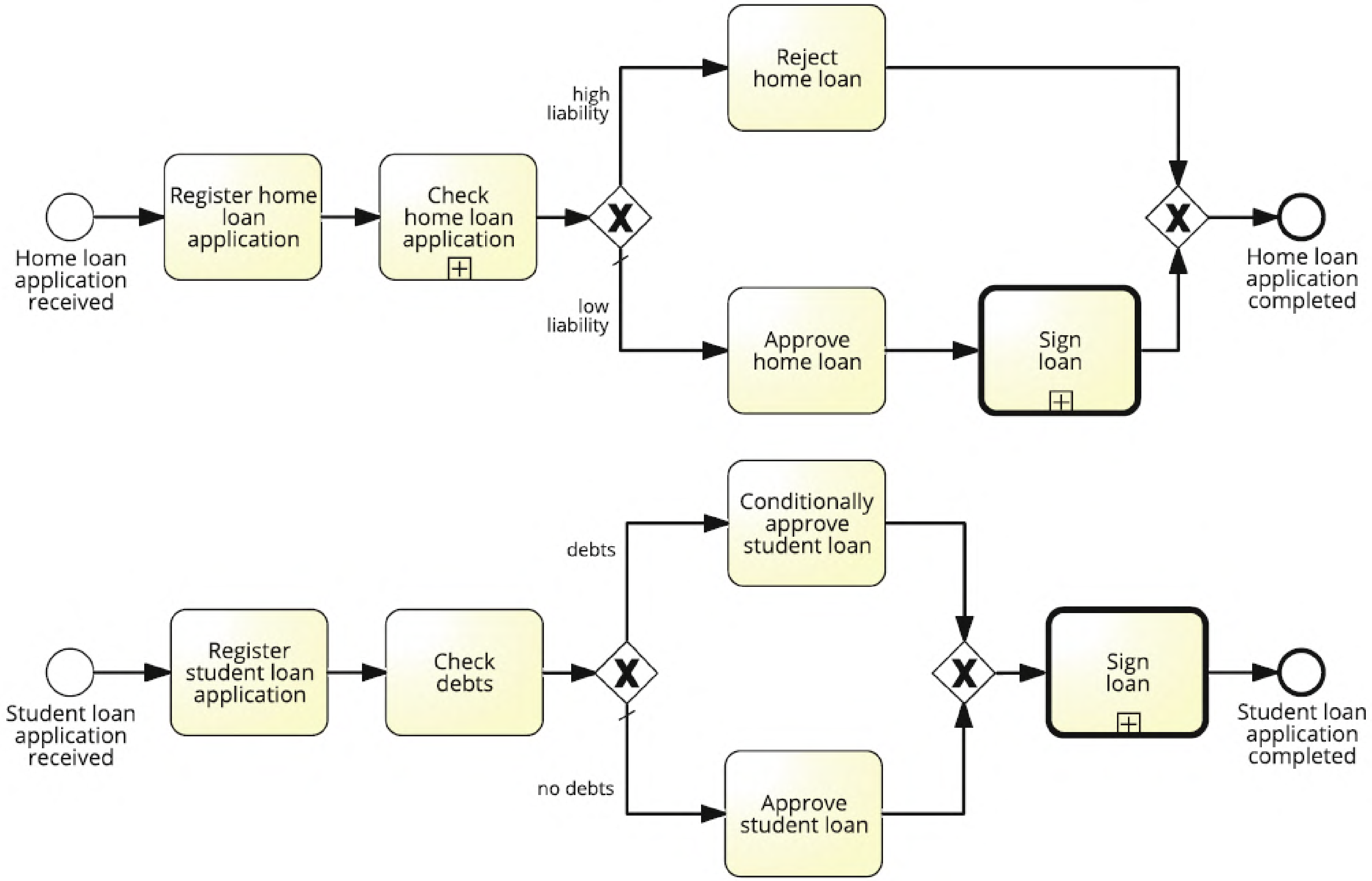
Dumas et al. (2018)

Hierarchically Decomposed Process



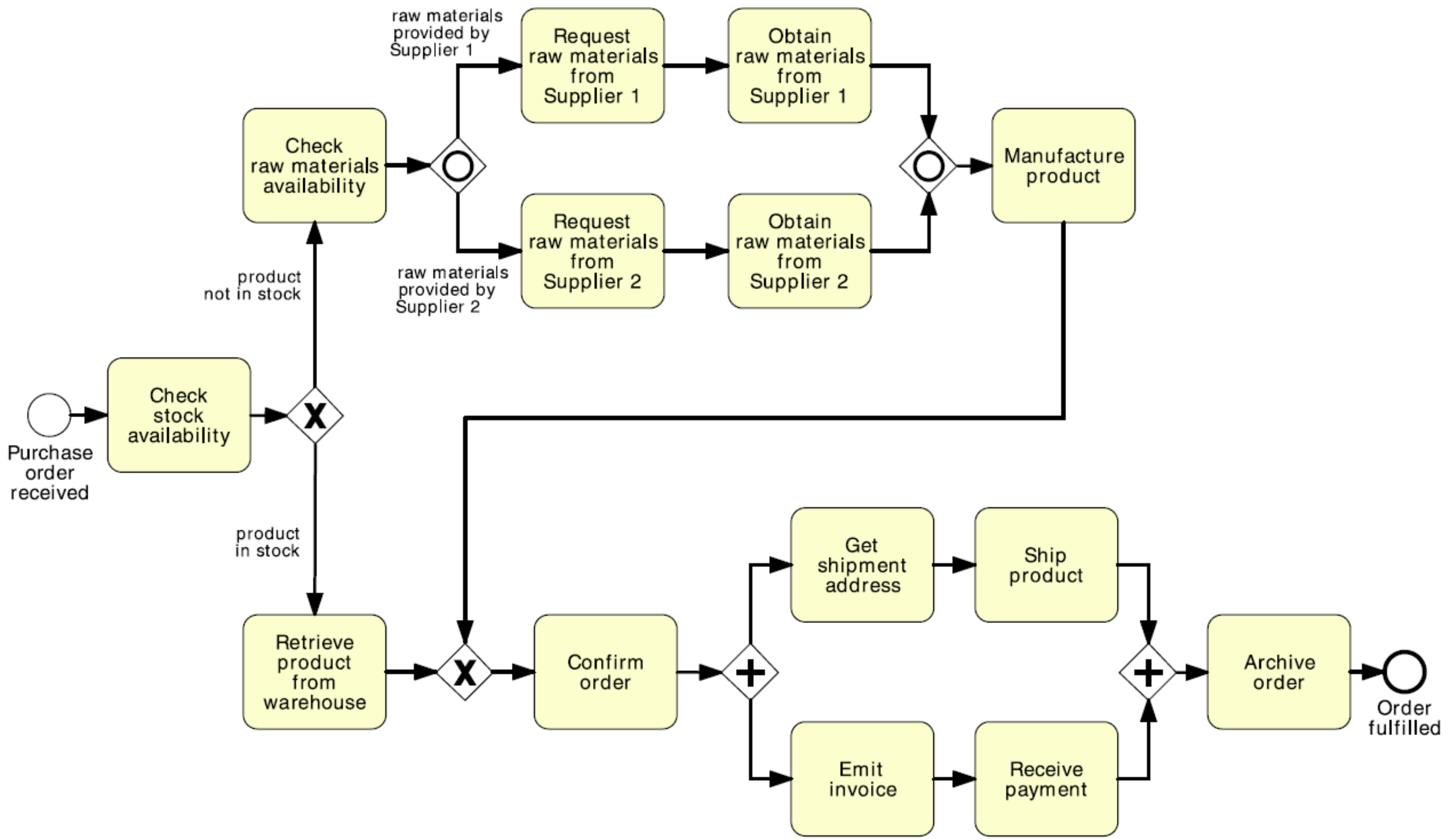
Dumas et al. (2018)

Call Activities



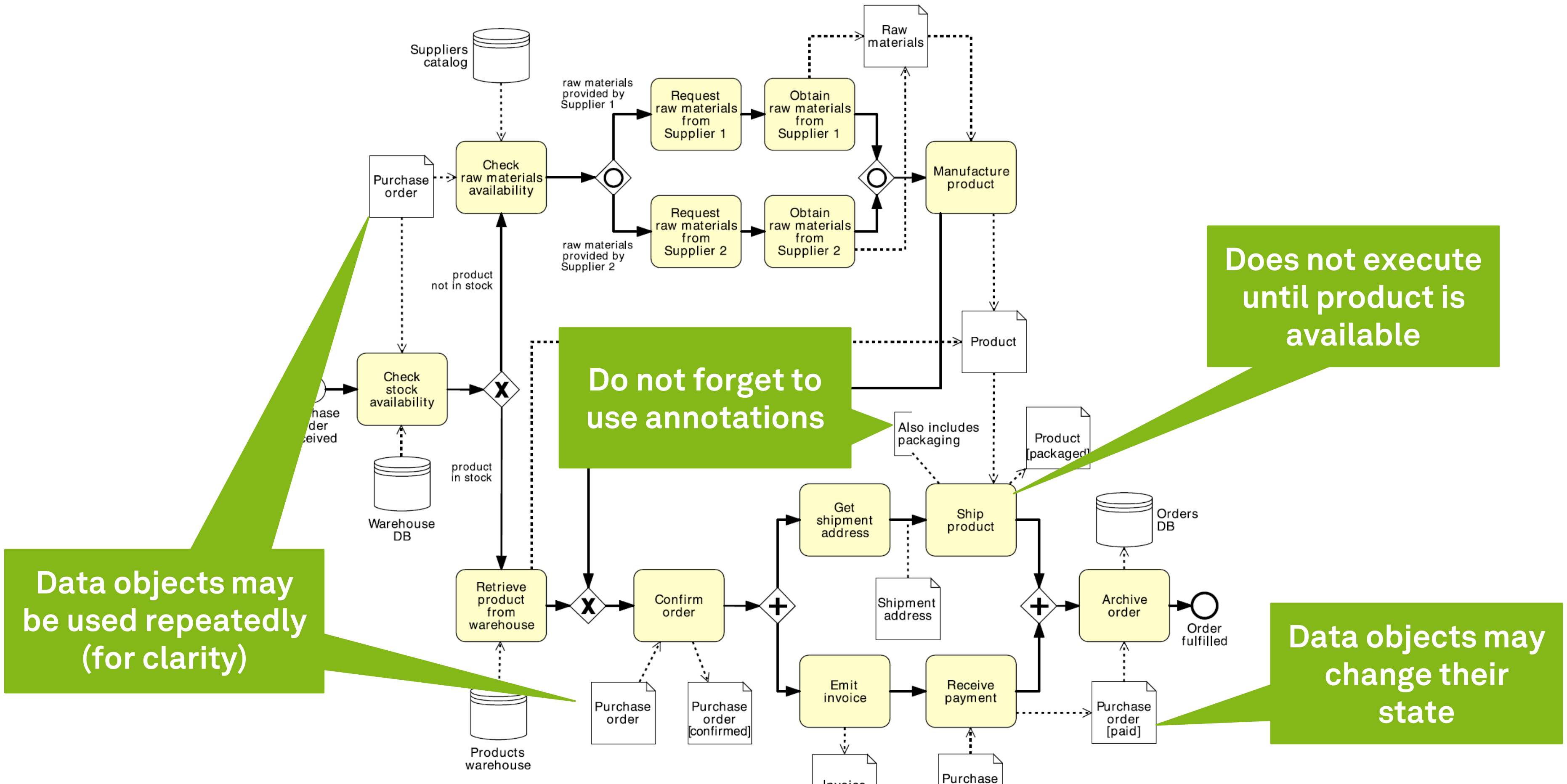
Dumas et al. (2018)

"But what about documents and stuff?"



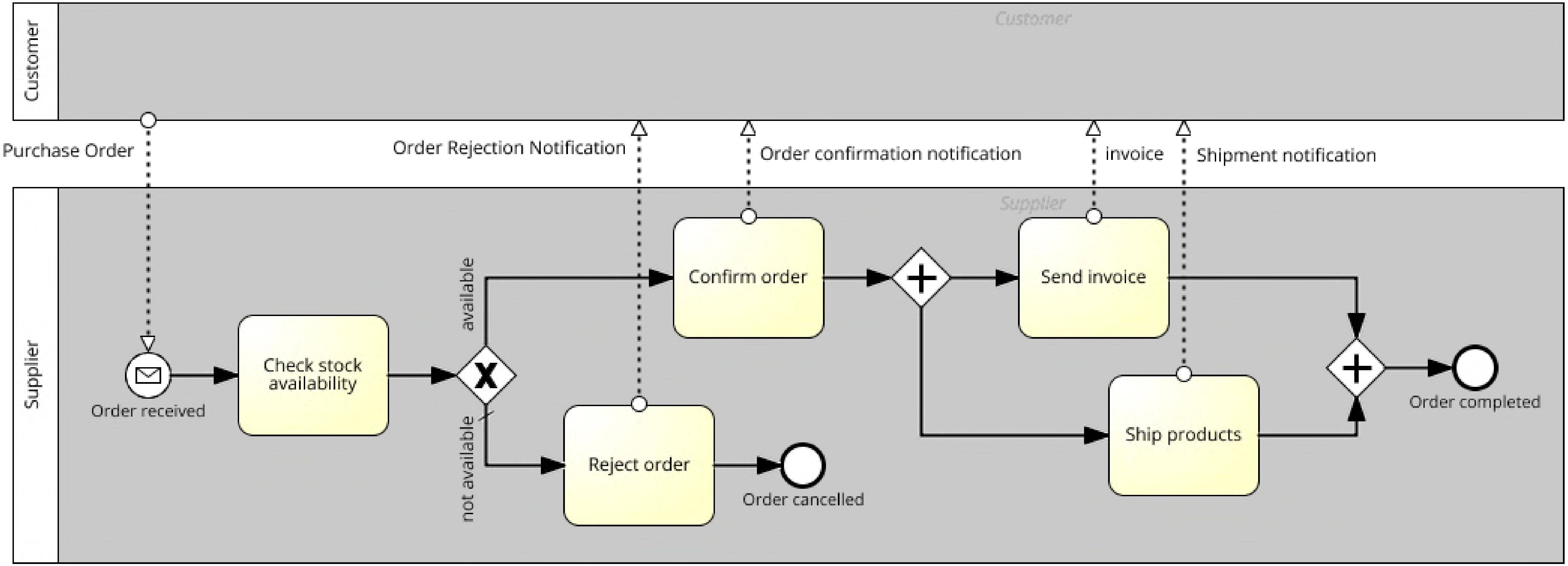
Dumas et al. (2018)

Passive Resources: The Data Perspective



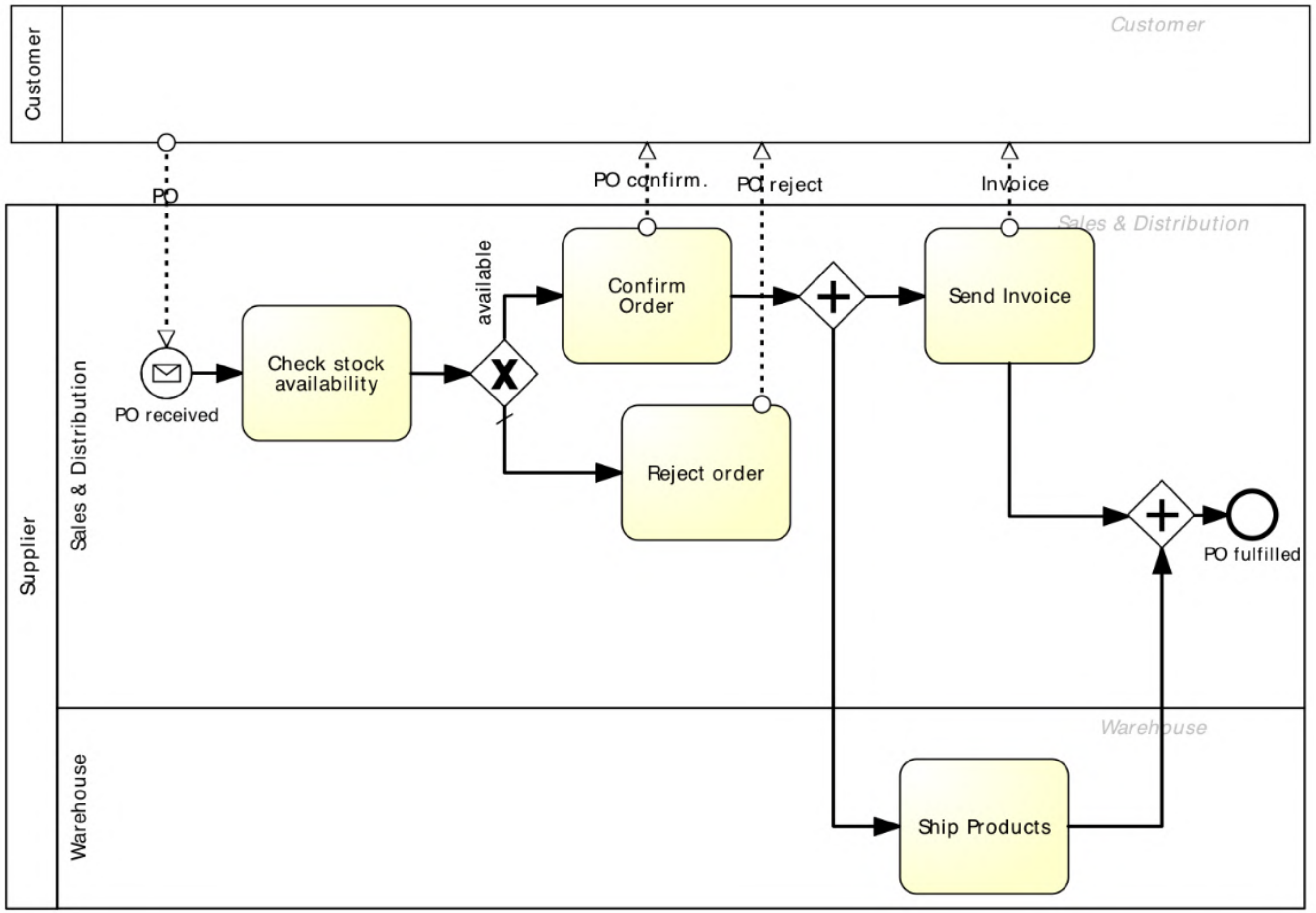
Dumas et al. (2018)

Active Resources: The Organizational Perspective



Dumas et al. (2018)

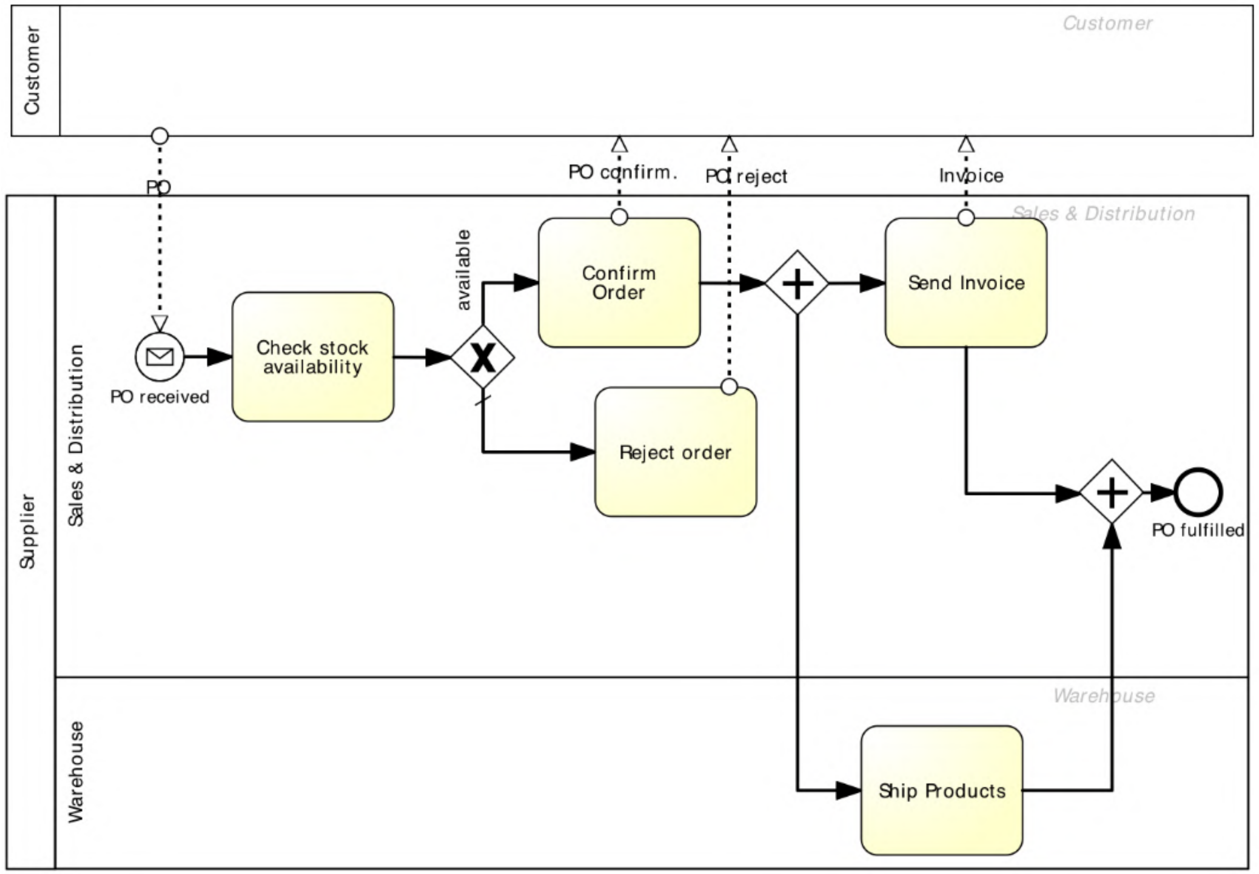
Order Management Process with Pools and Lanes



Dumas et al. (2018)

Modeling a Pool – Black Box vs. White Box

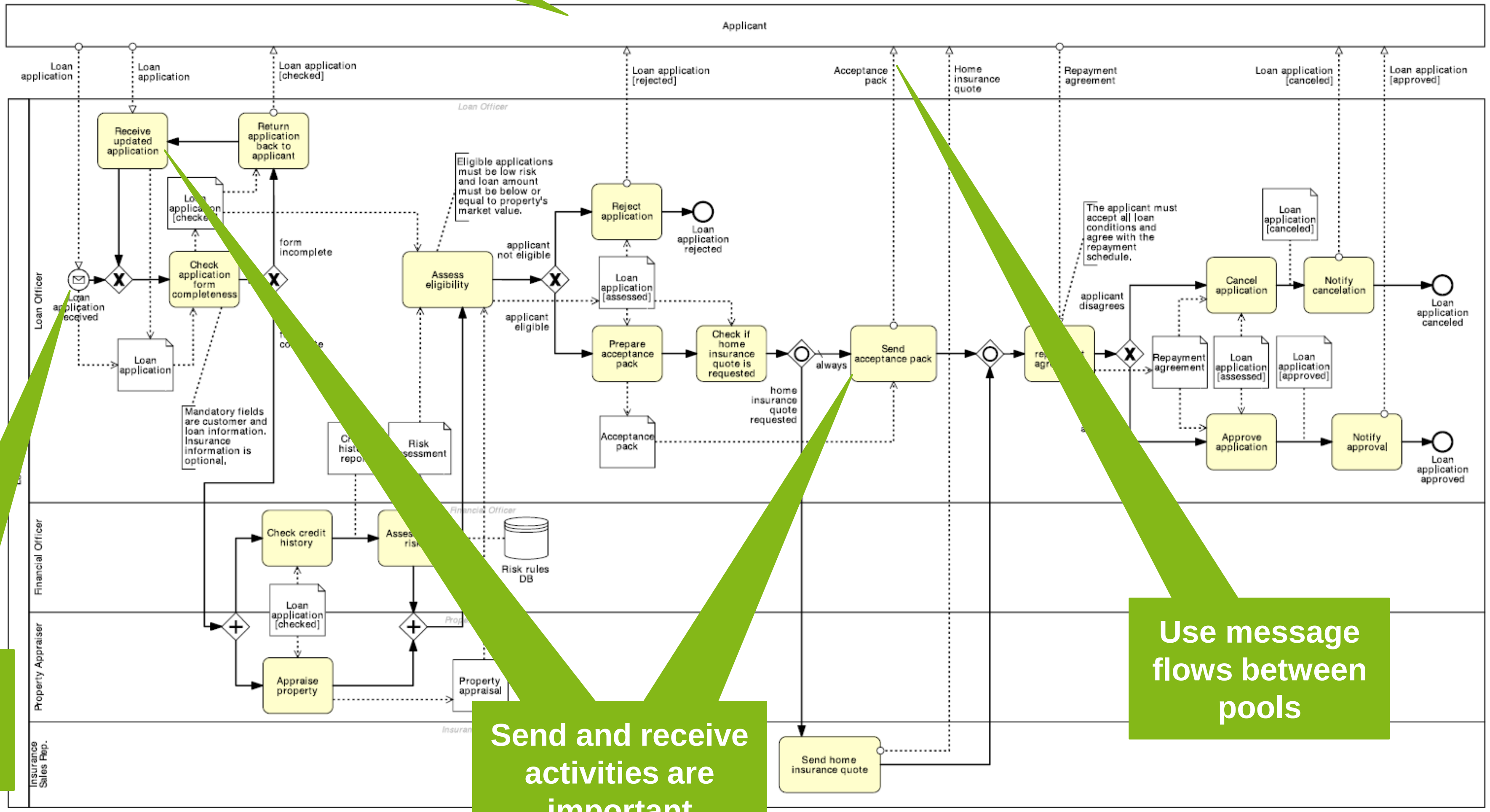
- When do we use a **black box** and when do we use a **white box**?



Dumas et al. (2018)

Collaborations

Private processes are collapsed pools



If the start is a message use a message event

Send and receive activities are important

Use message flows between pools

Dumas et al. (2018)

Learning Goals of Today

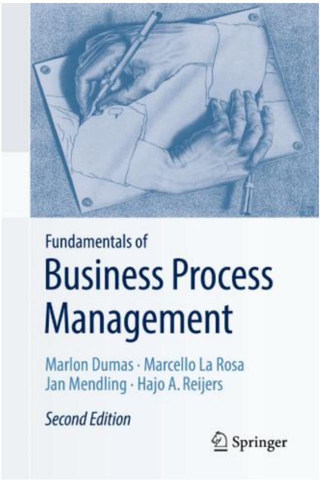
- Better understand the syntactics and semantics of BPMN
- Understand branching and merging in BPMN
- Understand decomposition, resources, and collaborations in BPMN
- Be able to use advanced BPMN

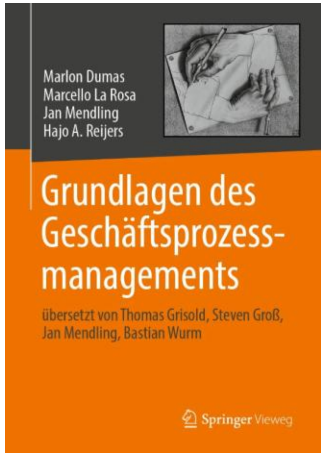
Recap Questions

- Can we model everything without the inclusive-OR construct in BPMN?
- What is a loop, how do we model them in BPMN, and why can they cause deadlocks or even livelocks?
- What is process decomposition and its relation to private pools in BPMN?

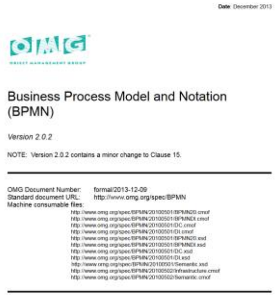
Literature

- **Section 3**: Essential Process Modeling
- **Section 4**: Advanced Process Modeling





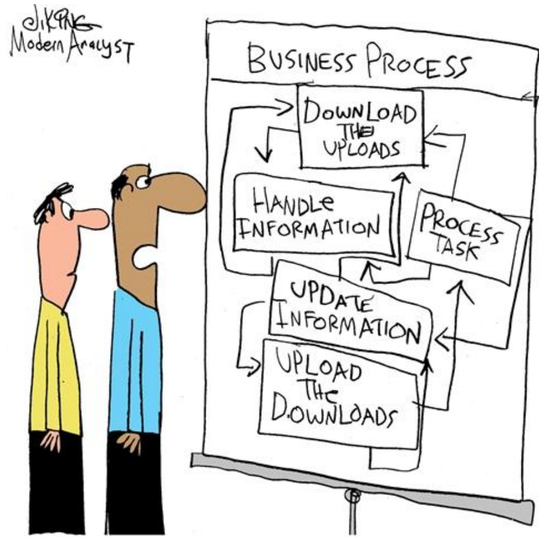
- Business Process Model and Notation (BPMN)
  - https://www.omg.org/spec/BPMN/2.0.2



Roadmap

Introduction	10.04. Introduction	17.04. Process Identification			
Process Discovery & Modeling	24.04. Essentials of BPMN	01.05. Public Holiday	08.05. Advanced BPMN I	15.05. Advanced BPMN II	22.05. Process Discovery
Process Analysis & Redesign	29.05. Public Holiday	05.06. Qualitative Analysis	12.06. Quantitative Analysis	19.06. Public Holiday	26.06.2025 Process Redesign
Application of BPM	03.07. Process Technology	10.07. Process Monitoring and Mining	15.07. BPM as an Enterprise Capability		

Any Questions about Process Modeling?



"Charlie, did you really get these details from the customer?"

modernanalyst.com (2016)

ec.cs.tu-dortmund.de

technische universität
dortmund

Chair of Enterprise Computing
Department of Computer Science

Business Process Management

Advanced Process Modeling II



Roadmap

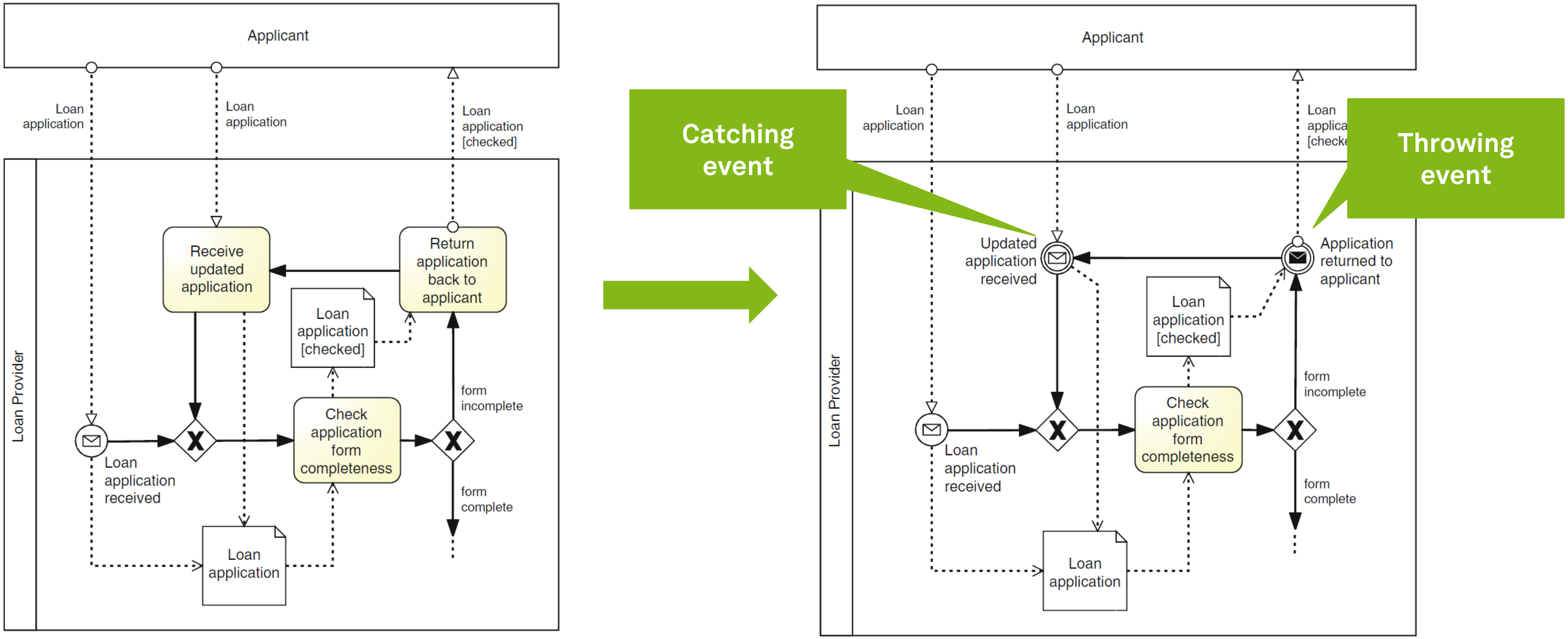
Introduction	10.04. Introduction	17.04. Process Identification			
Process Discovery & Modeling	24.04. Essentials of BPMN	01.05. Public Holiday	08.05. Advanced BPMN I	15.05. Advanced BPMN II	22.05. Process Discovery
Process Analysis & Redesign	29.05. Public Holiday	05.06. Qualitative Analysis	12.06. Quantitative Analysis	19.06. Public Holiday	26.06.2025 Process Redesign
Application of BPM	03.07. Process Technology	10.07. Process Monitoring and Mining	15.07. BPM as an Enterprise Capability		

Learning Goals of Today

- Understand throwing and catching events in BPMN
- Understand exception handling in BPMN
- Understand and know workflow patterns
- Be able to use basically all of BPMN

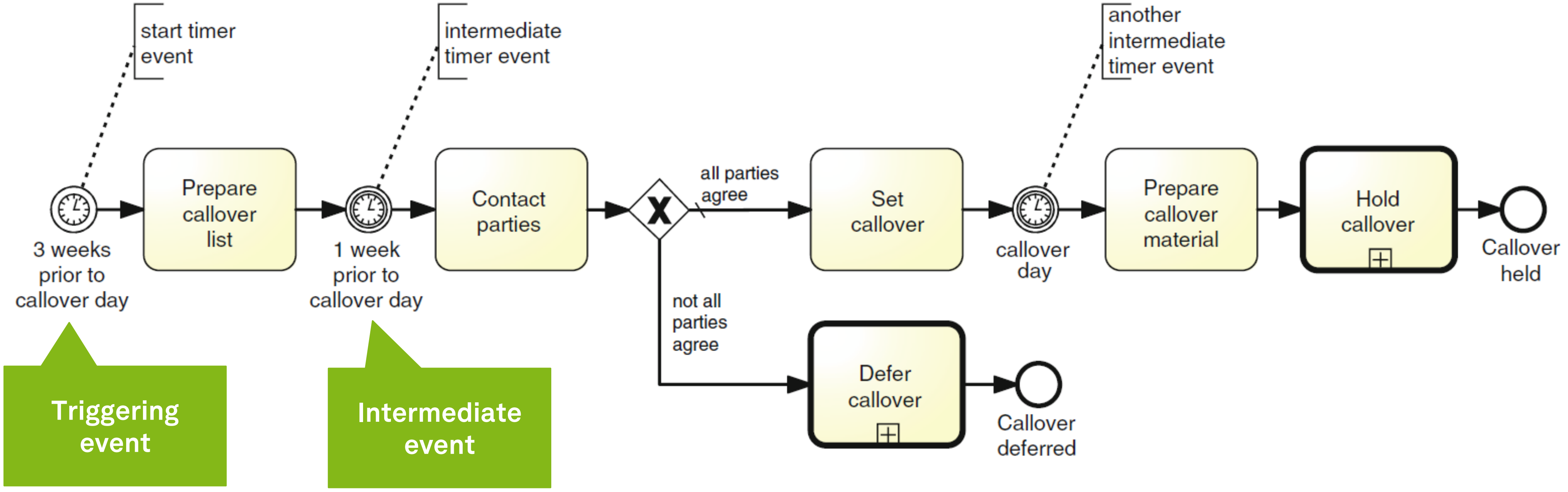
Event Handling

Message Events



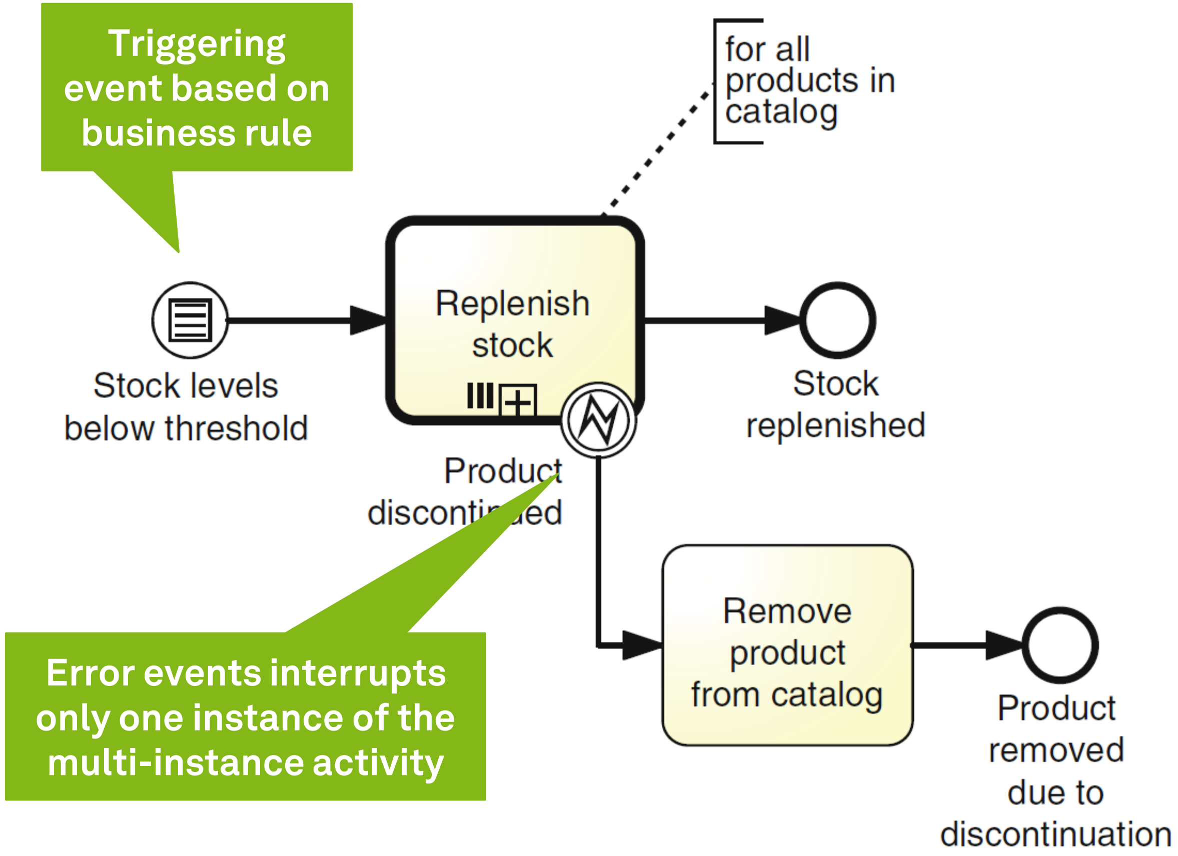
Dumas et al. (2018)

Temporal Events



Dumas et al. (2018)

Conditional Events

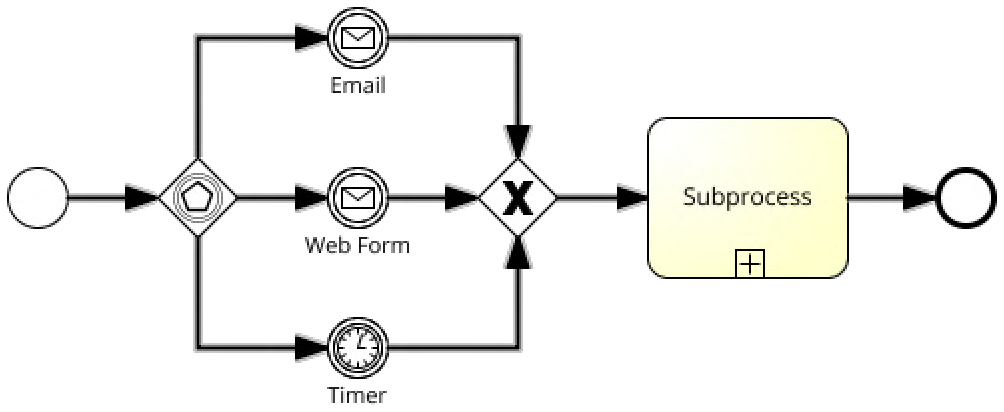


Dumas et al. (2018)

Data-based vs. Event-based XOR-Split

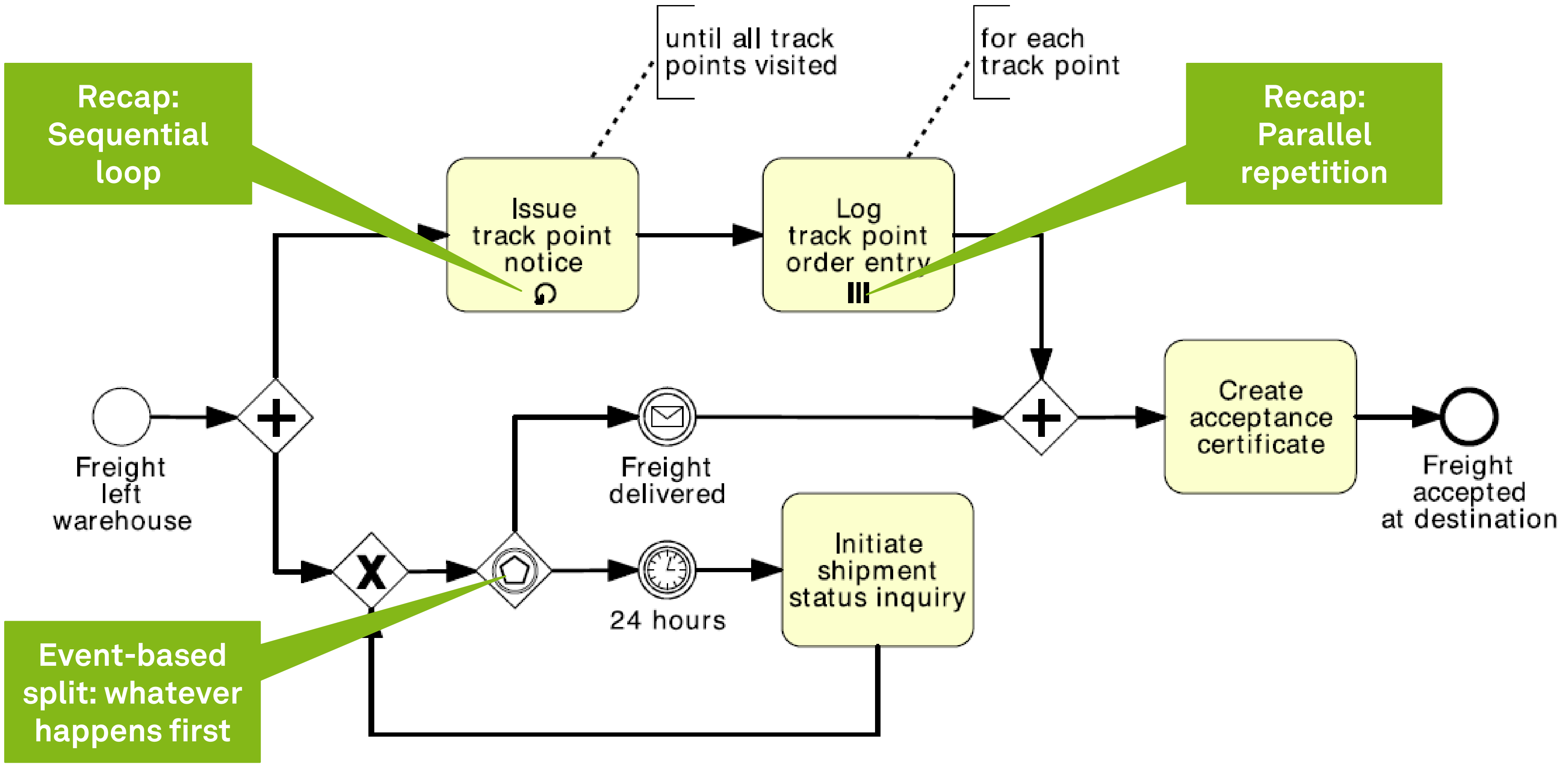
- A branch in an XOR gateway is chosen based on expressions evaluated over available data
- The decision is made **without any delay** (not time consuming)
- However, there are cases in which the decision must be **delayed** until something happens
  - Approval of management
  - Customer decision
  - ...

- In this case, BPMN provides an **event-based decision gateway**



Dumas et al. (2018)

Racing Events

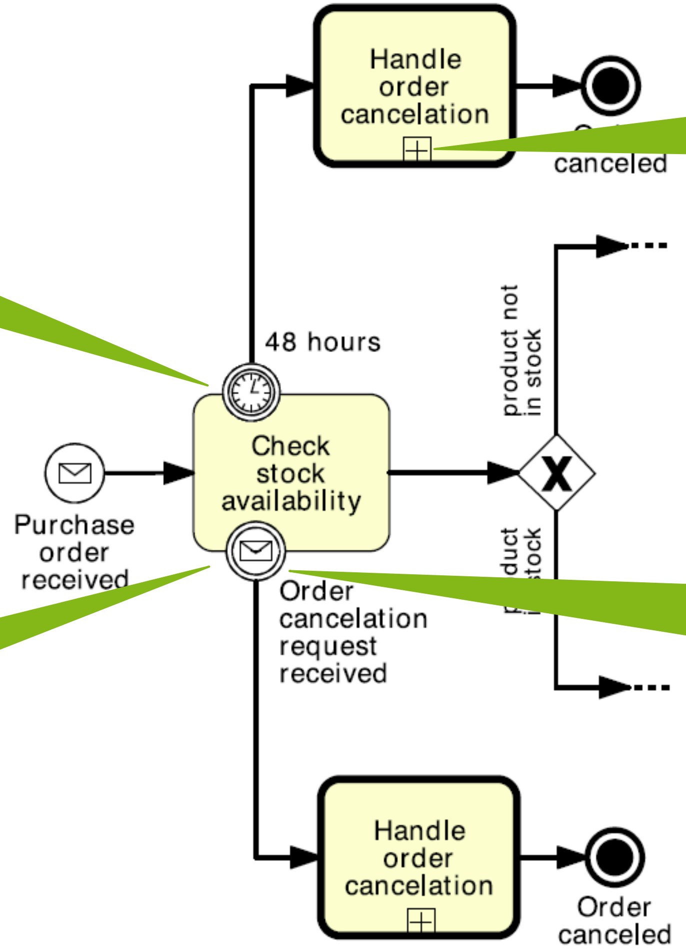


Dumas et al. (2018)

Boundary Events

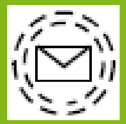
Boundary event cancels after 48 ours

Boundary event cancels on message receipt



If the event does not only interrupt one activity, it can also be placed at the sub-process level

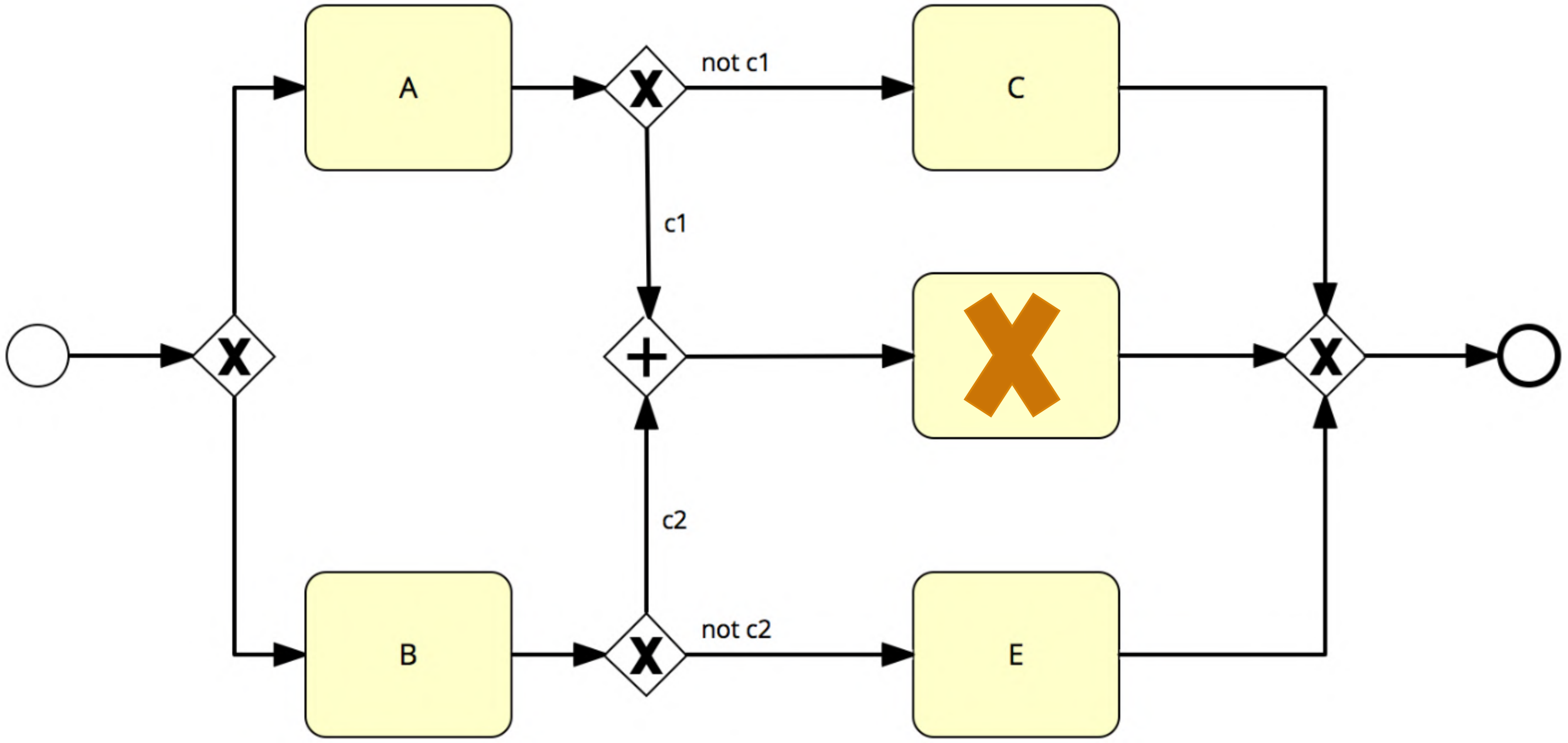
Alternative option: non-interrupting event



Dumas et al. (2018)

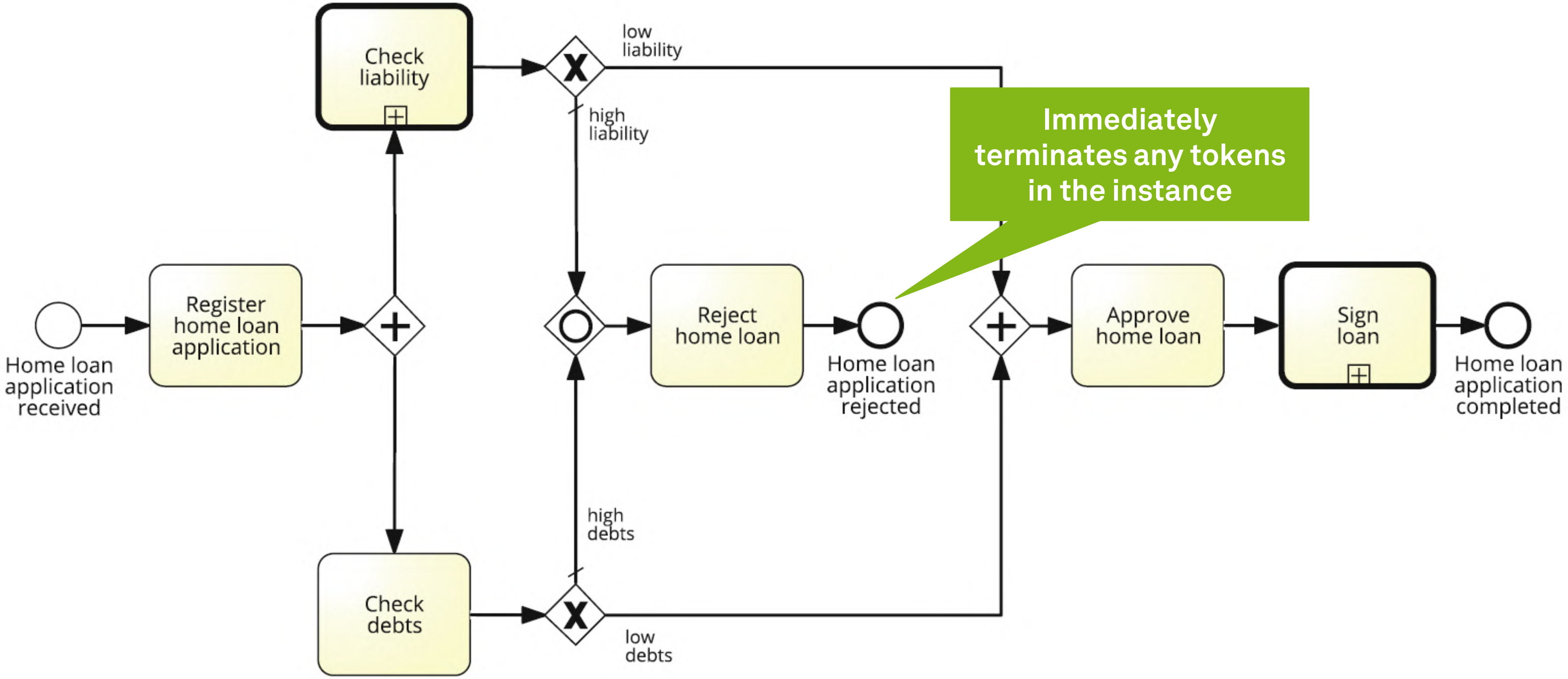
Exception Handling

Instance Termination: Correct or Incorrect?



La Rosa (2013)

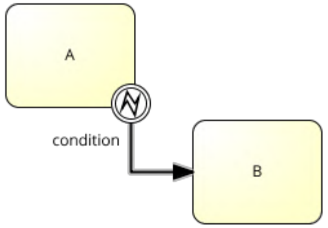
Process Termination



Dumas et al. (2018)

Exception Handling

- Exception handling often means stopping a sub-process and performing a special activity

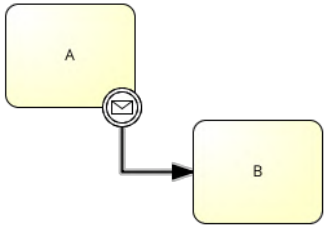


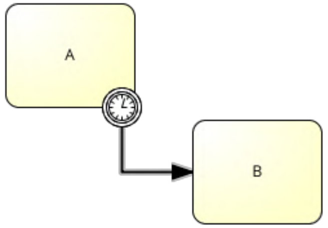
Internal Exception

An error **inside** the activity occurred. The execution of the activity must be interrupted. Handled with **Error** event.

External Exception

An error **outside** the activity occurred. The execution of the activity must be interrupted. Handled with **Message** event.



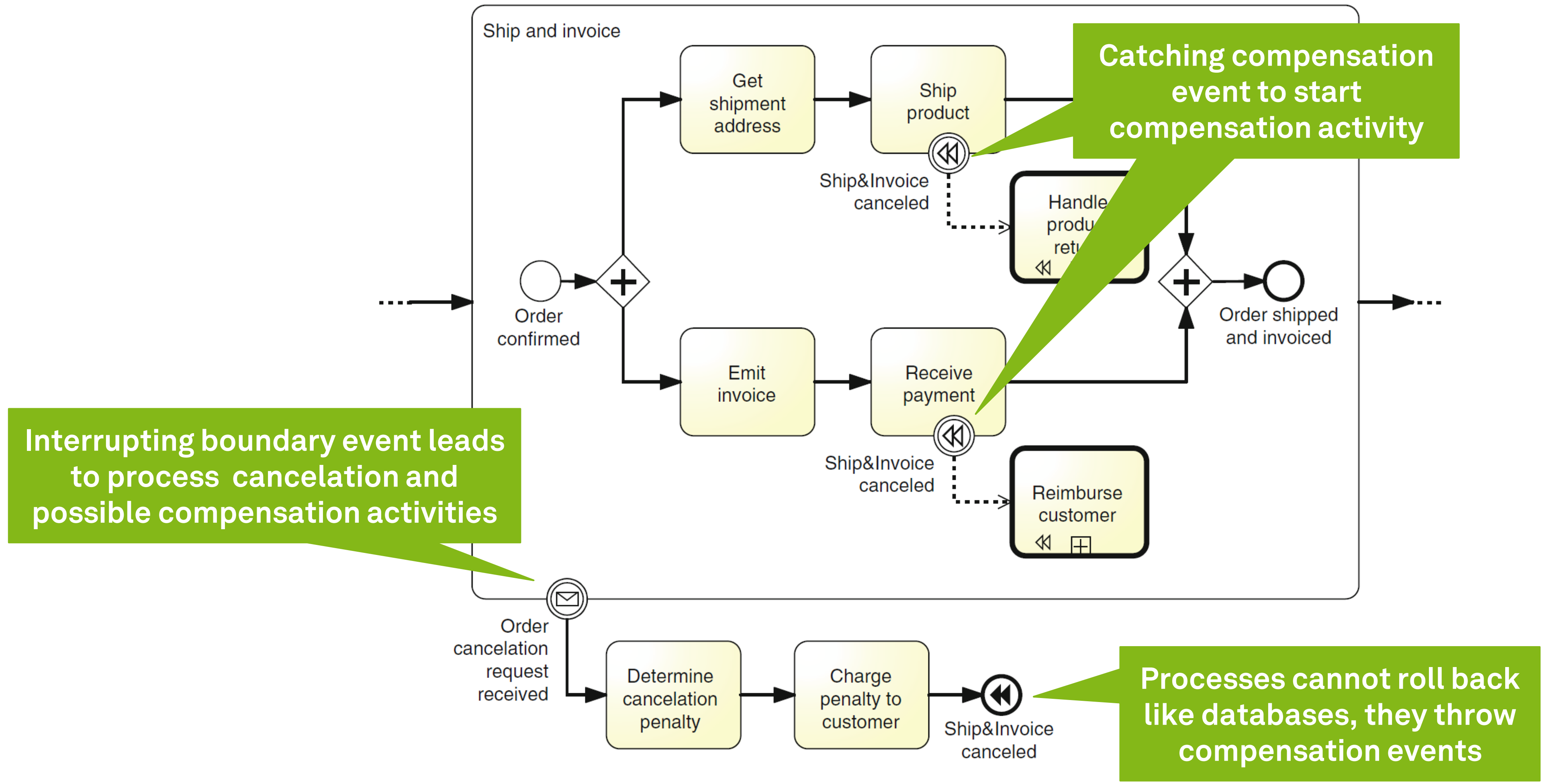


Timeout Exception

An activity lasts **too long** and must be interrupted. Handled with the **Timer** event.

La Rosa (2013)

Compensation Events and Activities



Dumas et al. (2018)

Complex Exception Handling Scenario

Process continues from here...

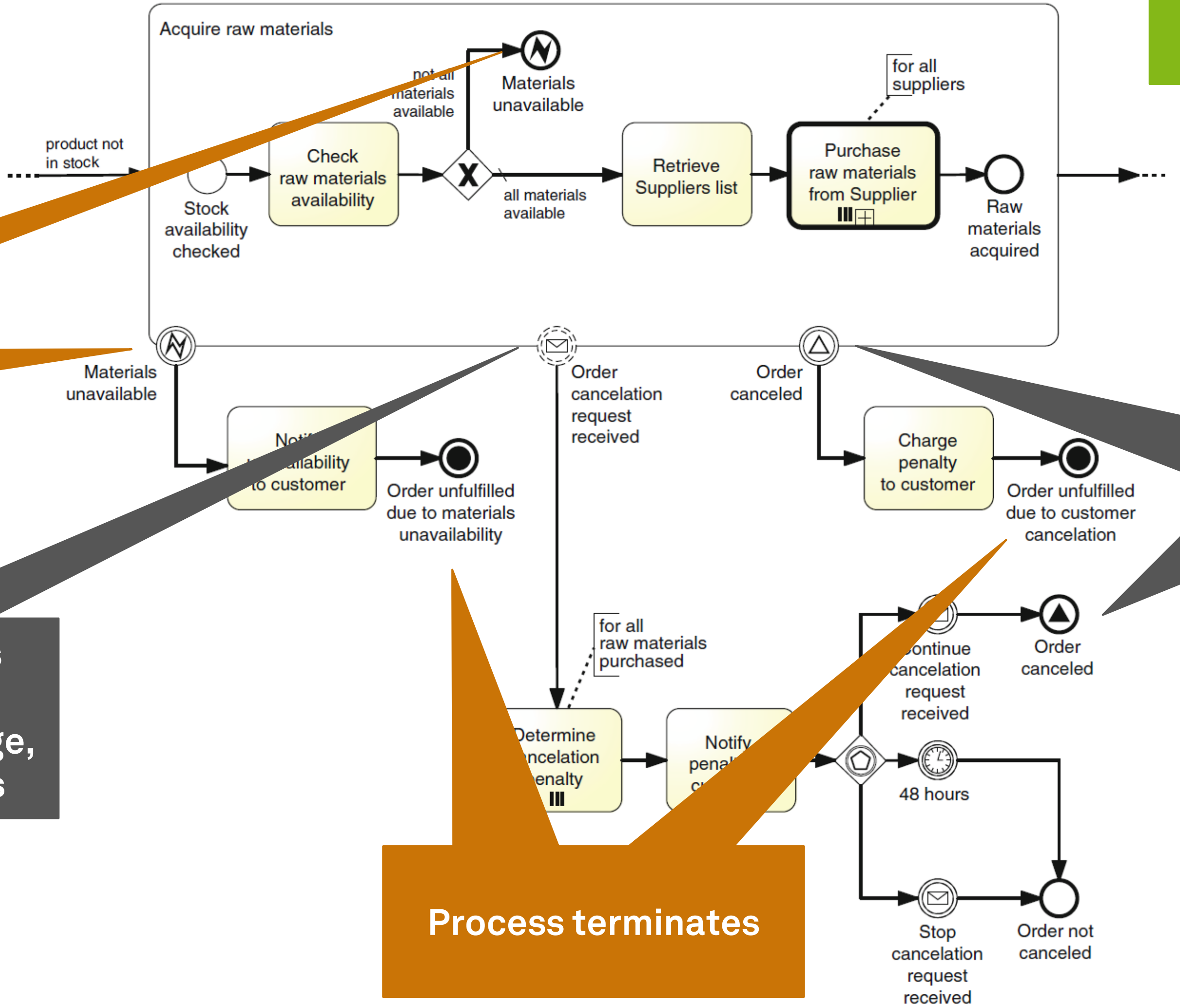
Internal exception if no materials are available

Customer cancels order via non-interrupting message, process continues

Process terminates

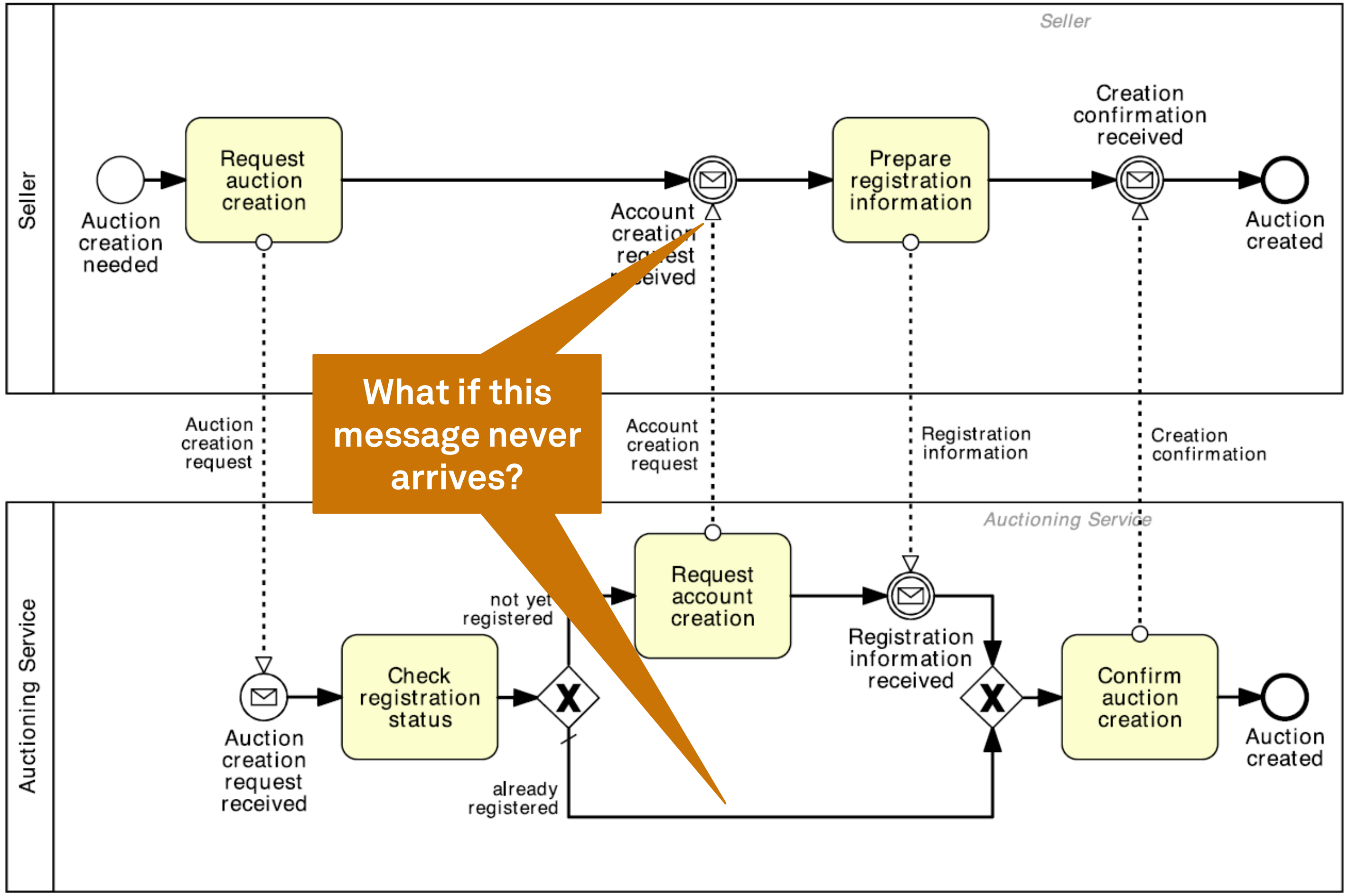
Signal event broadcasts order cancellation

Process continues here if completed successfully



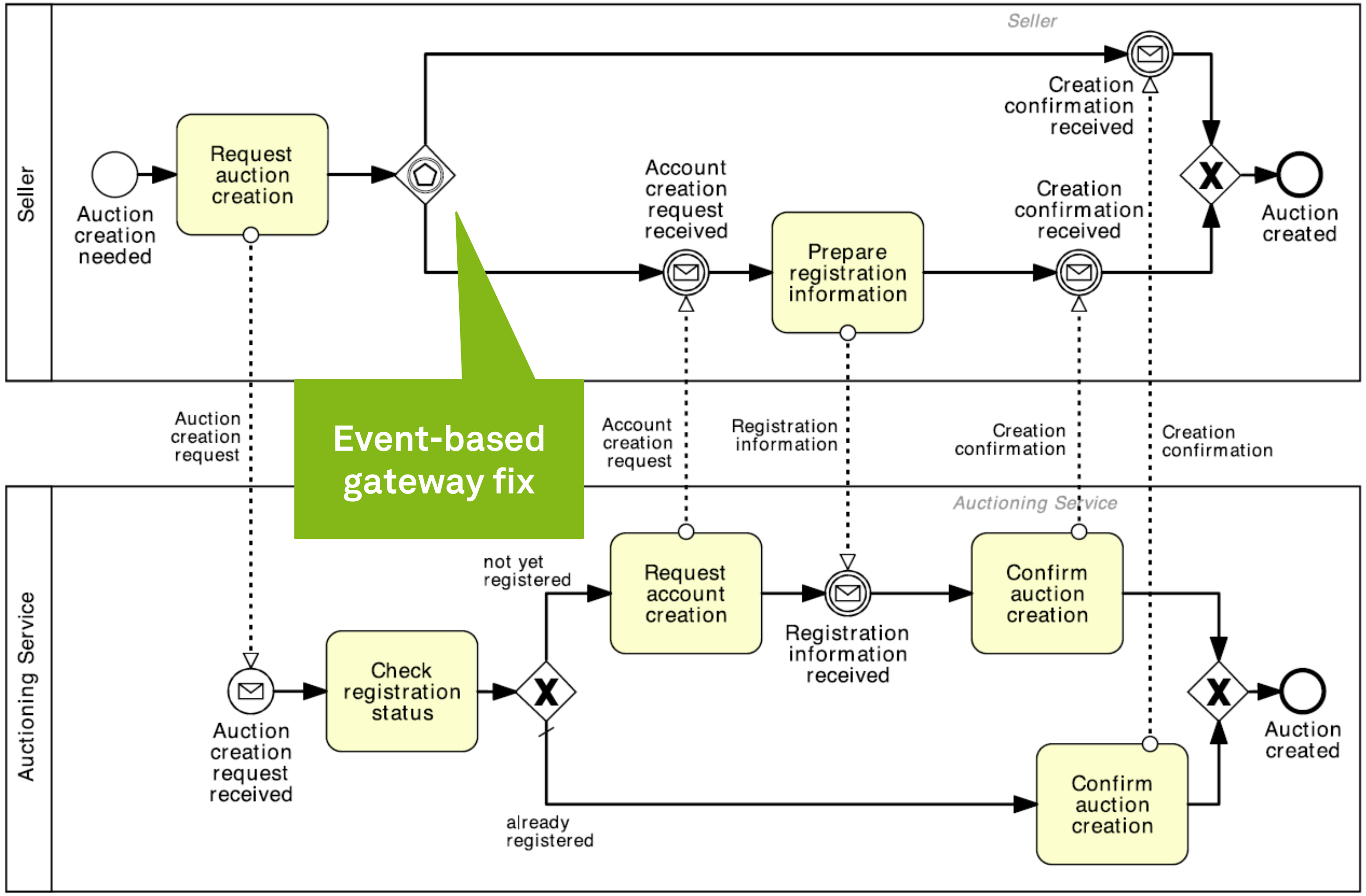
Dumas et al. (2018)

Variants and Collaborations: What's Wrong Here?



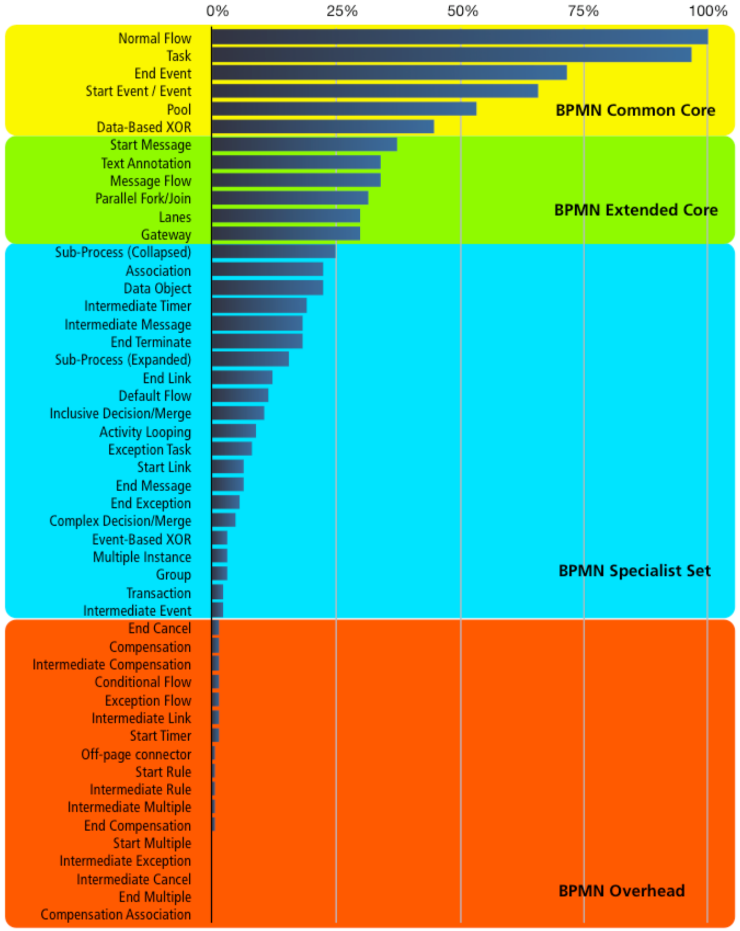
Dumas et al. (2018)

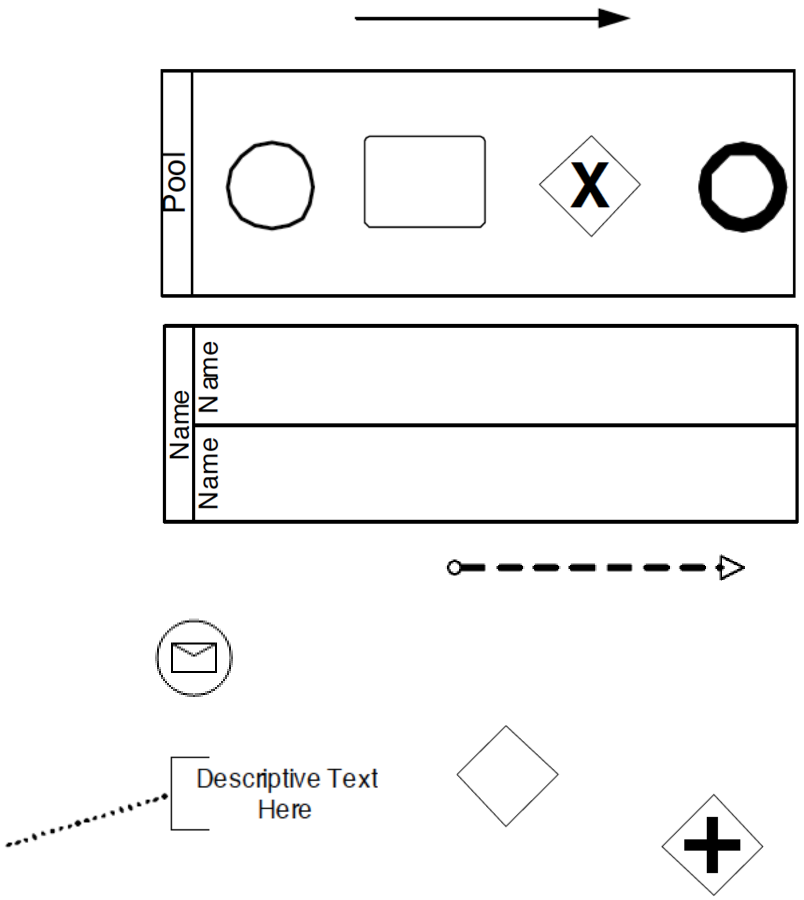
Variants and Collaborations: What's Wrong Here?



Dumas et al. (2018)

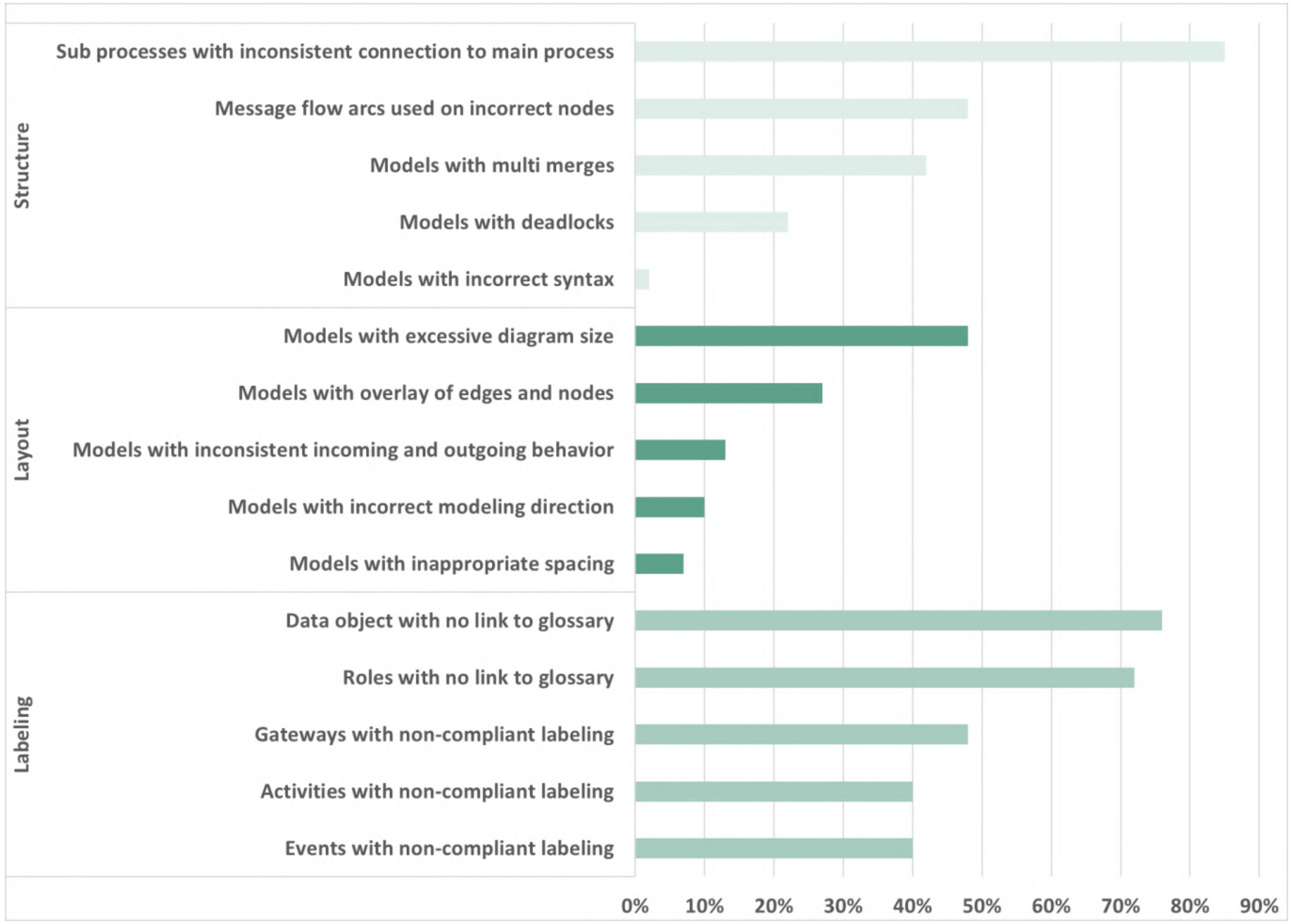
BPMN has over 100 symbols, but...





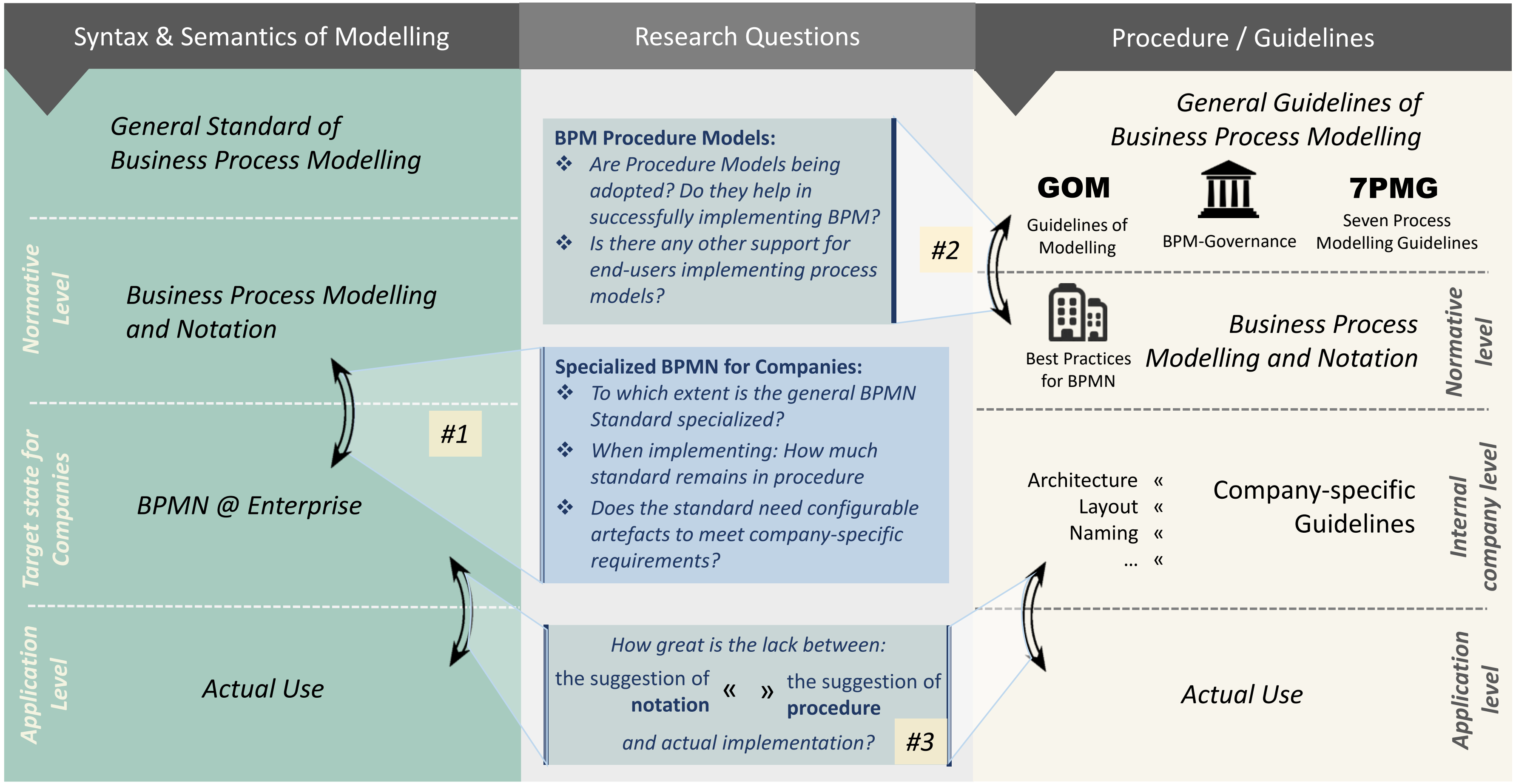
zur Mühlen, Recker (2008)

Quality Issues of BPMN Models



Derived from Leopold (2015)

BPMN Use – Levels of Abstraction



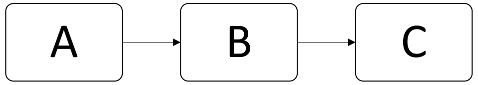
Imgrund (2015)

Workflow Patterns

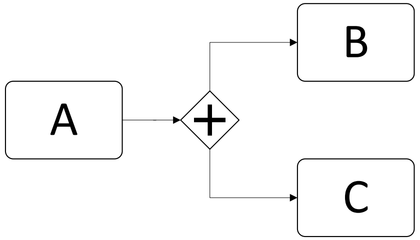



Control Flow Patterns

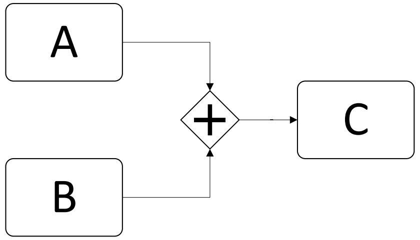
1. Sequence



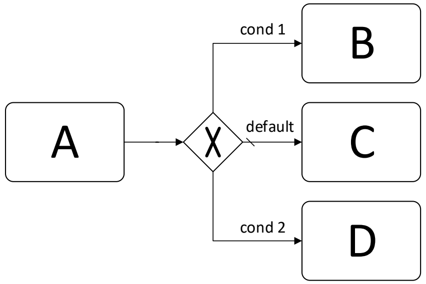
2. Parallel Split



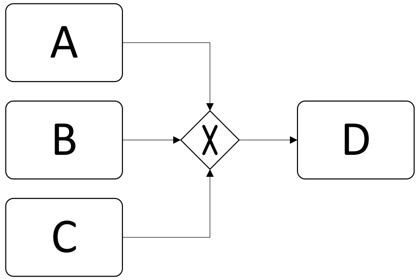
3. Synchronization



4. Exclusive Choice



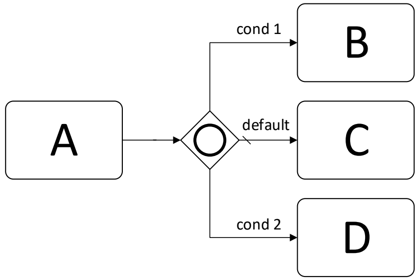
5. Simple Merge



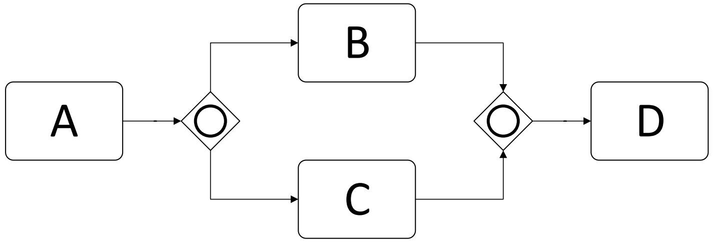
http://www.workflowpatterns.com/
http://www.ariscommunity.com/users/sstein/2010-07-20-bpmn-2-workflow-patterns

Advanced Branching and Synchronization Patterns

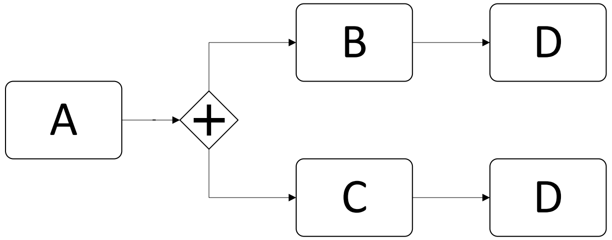
6. Multi-Choice



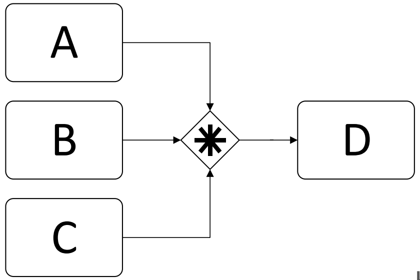
7. Structured Synchronizing Merge



8. Multi-Merge



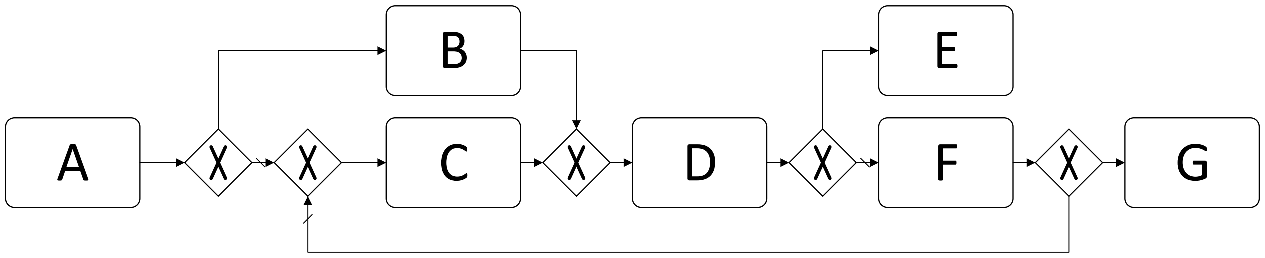
9. Structured Discriminator



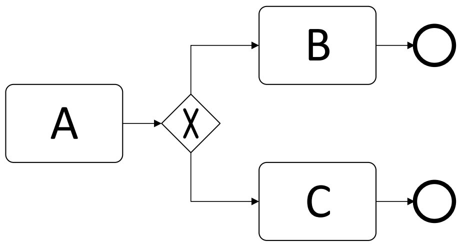
http://www.workflowpatterns.com/
http://www.ariscommunity.com/users/sstein/2010-07-20-bpmn-2-workflow-patterns

Iteration Patterns and Termination Patterns

10. Arbitrary Cycles



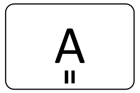
11. Implicit Termination



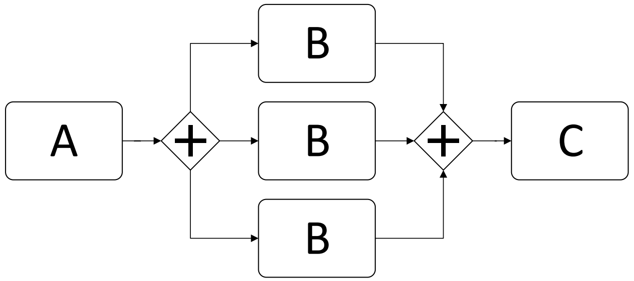
http://www.workflowpatterns.com/
http://www.ariscommunity.com/users/sstein/2010-07-20-bpmn-2-workflow-patterns

Multiple Instance Patterns

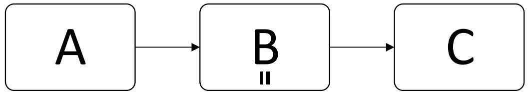
12. Multiple Instances without Synchronization



13. Multiple Instances with a Priori Design-Time Knowledge



14. Multiple Instances with a Priori Run-Time Knowledge



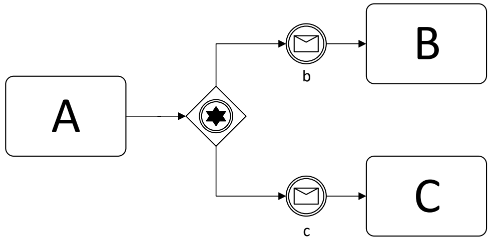
15. Multiple Instances without a Priori Run-Time Knowledge

No support in BPMN, workaround loop activity and event sub-processes

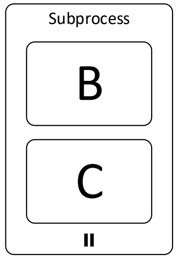
http://www.workflowpatterns.com/
http://www.ariscommunity.com/users/sstein/2010-07-20-bpmn-2-workflow-patterns

State-based Patterns

16. Deferred Choice



17. Interleaved Parallel Routing



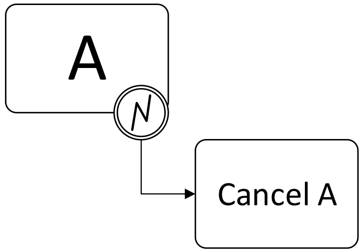
18. Milestone

No support in BPMN

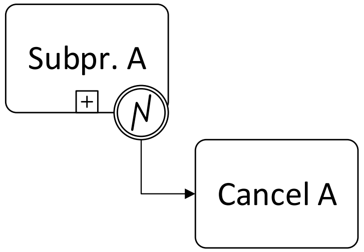
http://www.workflowpatterns.com/
http://www.ariscommunity.com/users/sstein/2010-07-20-bpmn-2-workflow-patterns

Cancellation and Force Completion Patterns

19. Cancel Task



20. Cancel Case

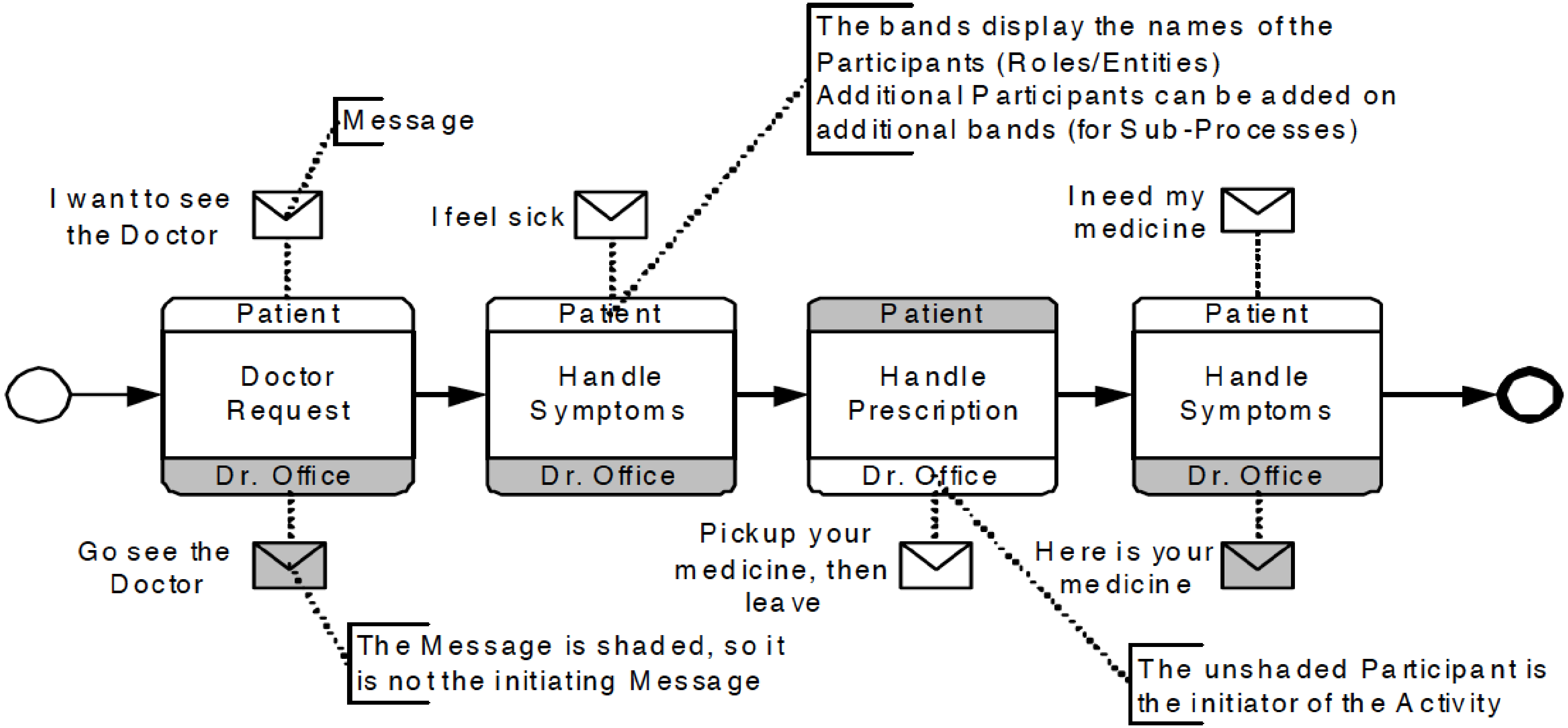


http://www.workflowpatterns.com/
http://www.ariscommunity.com/users/sstein/2010-07-20-bpmn-2-workflow-patterns

Even More BPMN!

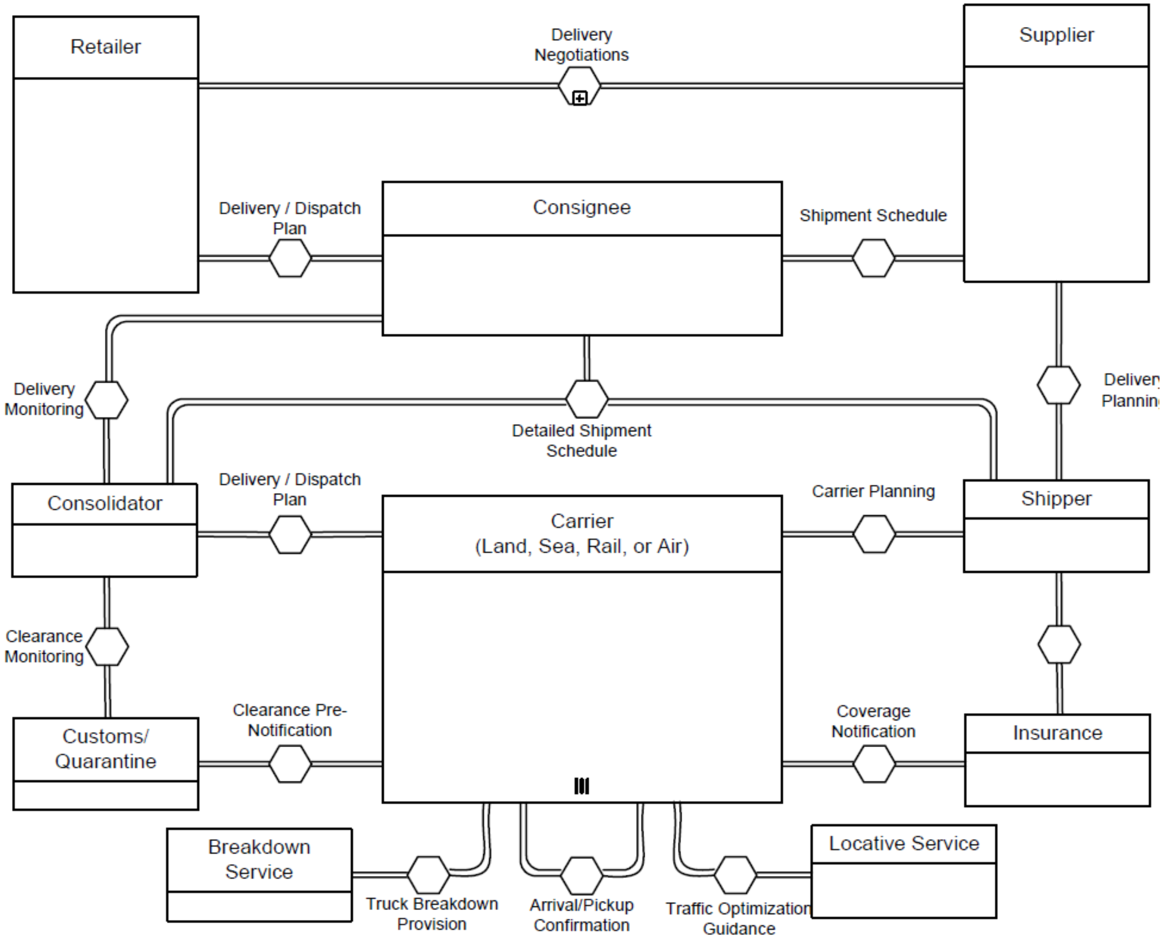


BPMN Choreography Example



Object Management Group (2011)

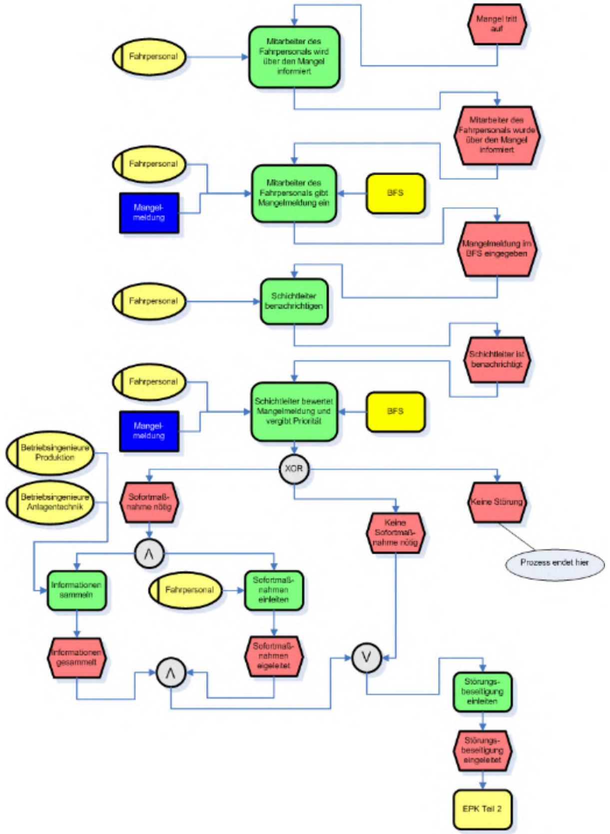
BPMN Conversation Example



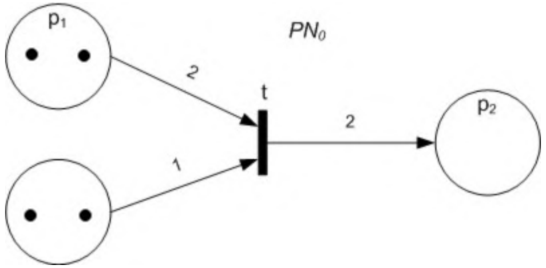
Object Management Group (2011)

Good to know: Alternatives to BPMN

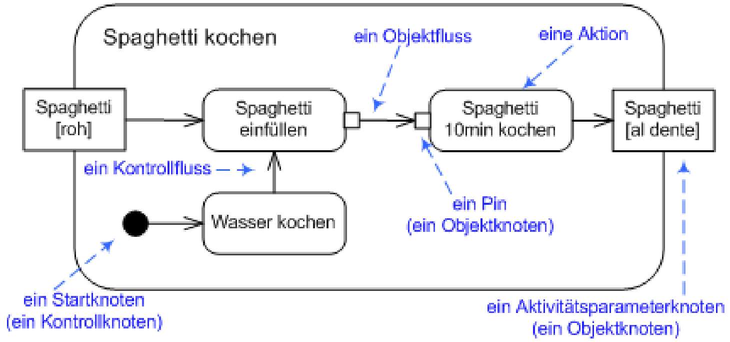
- Event-driven Process Chains



- Petri Nets



- UML 2.0 Activity Diagrams



Learning Goals of Today

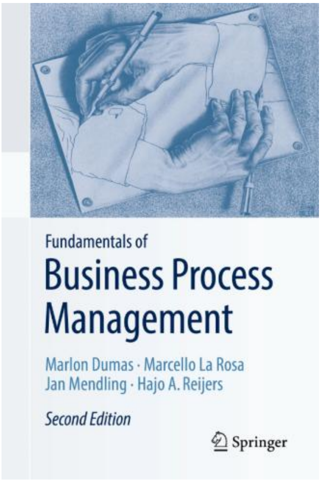
- Understand throwing and catching events in BPMN
- Understand exception handling in BPMN
- Understand and know workflow patterns
- Be able to use basically all of BPMN

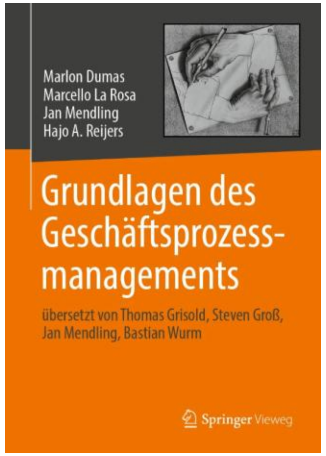
Recap Questions

- What are racing events in BPMN and how do we model them?
- What are boundary events in BPMN and how do we model them?
- How do I model internal and external exceptions in BPMN and what is a compensation event and how do I use it in BPMN?
- What is a workflow pattern?

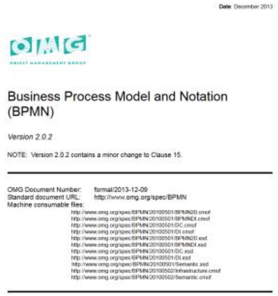
Literature

- **Section 4**: Advanced Process Modeling





- Business Process Model and Notation (BPMN)
  - https://www.omg.org/spec/BPMN/2.0.2


Date: December 2013

OMG

Business Process Model and Notation (BPMN)

Version 2.0.2

NOTE: Version 2.0.2 contains a minor change to Clause 15.

OMG Document Number: formal/2013-12-09
Standard document URL: http://www.omg.org/spec/BPMN
Machine consumable files:

Roadmap

Introduction	10.04. Introduction	17.04. Process Identification			
Process Discovery & Modeling	24.04. Essentials of BPMN	01.05. Public Holiday	08.05. Advanced BPMN I	15.05. Advanced BPMN II	22.05. Process Discovery
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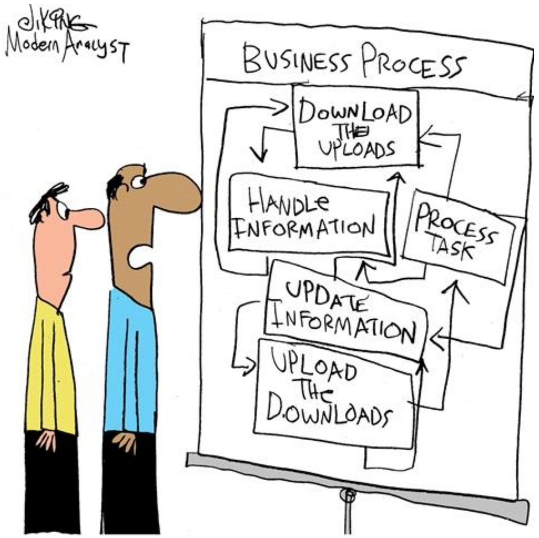
Any Questions about Process Modeling?



"Charlie, did you really get these details from the customer?"

modernanalyst.com (2016)

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Business Process Management

Process Discovery



Roadmap

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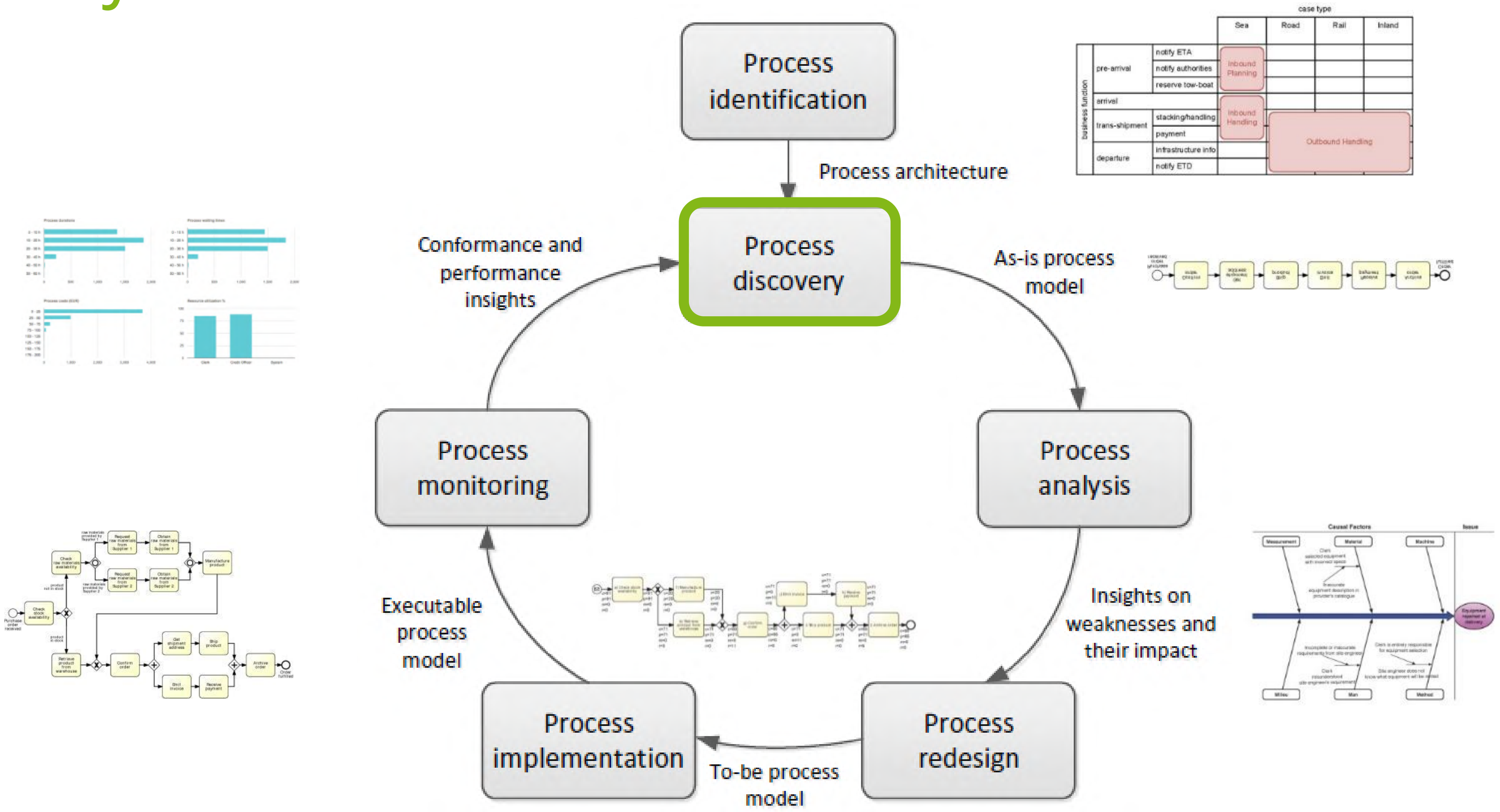
Learning Goals of Today

- Understand the challenges of process discovery
- Understand and be able to choose process discovery methods
- Be able to use process modeling during process discovery

BPM Lifecycle



Dumas et al. (2018)

Process Discovery is...

- ...the act of **gathering information** about an existing process and ...
- ...**organizing it** in terms of an as-is process model

→ **Process discovery is a much broader activity than just modeling a process**

Dumas et al. (2018), Soroco (2022)

Process Discovery Phases

1. **Define the setting**
   – Assemble a team in a company that will be responsible for working on the process
2. **Gather information**
   – Build an understanding of the process with discovery methods
3. **Conduct the modeling task**
   – Create the process model(s)
4. **Assure process model quality**
   – Guarantee that the resulting process models meet different quality criteria

Dumas et al. (2018)

Define the Setting: Process Discovery

Process Discovery: Who is involved?

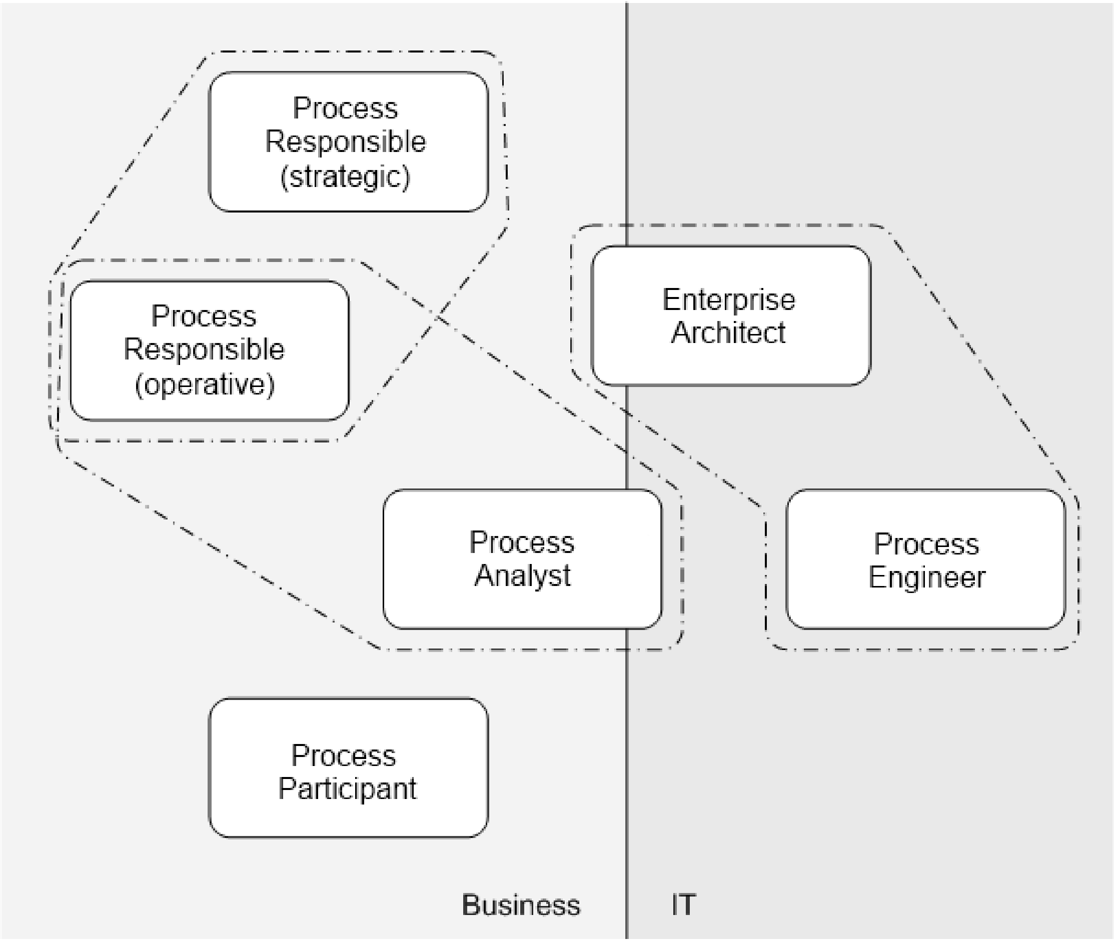
Process Analyst

Domain Expert

- **Process owner** needs to secure commitment and involvement of both

Dumas et al. (2018)

Involvement of Business and IT



Mendling (2013)

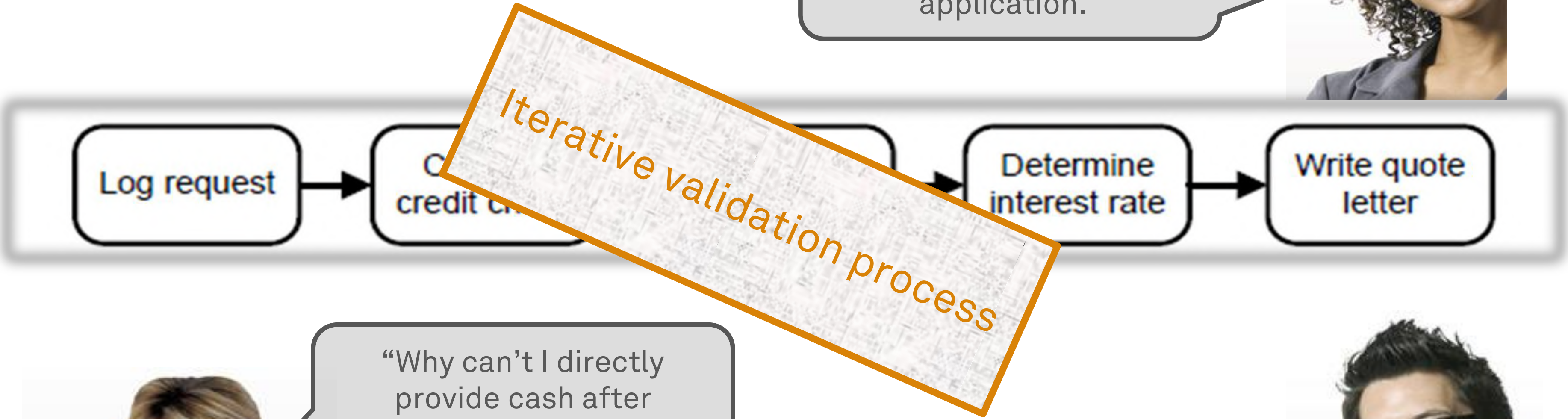
Strength and Weaknesses

Aspect	Process Analyst	Domain Expert
Modeling Skills	Strong	Limited
Process Knowledge	Limited	Strong

Dumas et al. (2018)

Challenge of Fragmented Process Knowledge

“I make a photocopy before handing over the application.”

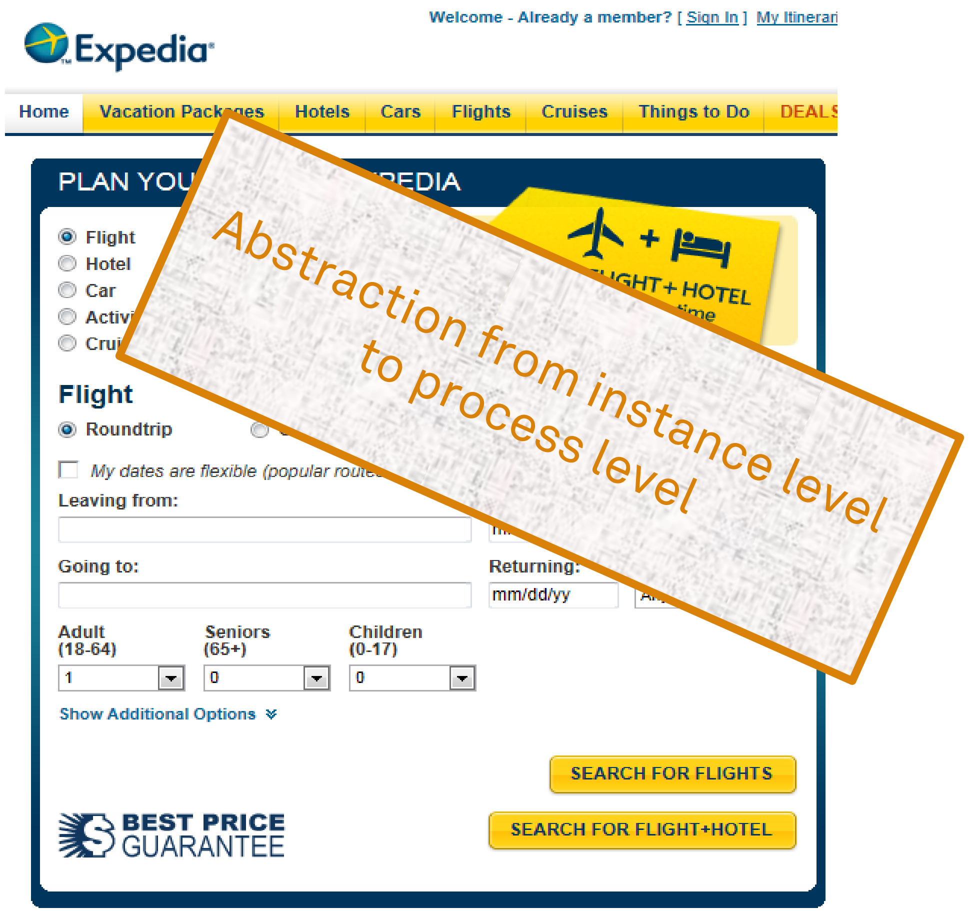


“Why can’t I directly provide cash after approval?”

“We bundle refinancing to get better interest rates.”

La Rosa (2013)

Challenge of Instance/ Case Level Thinking



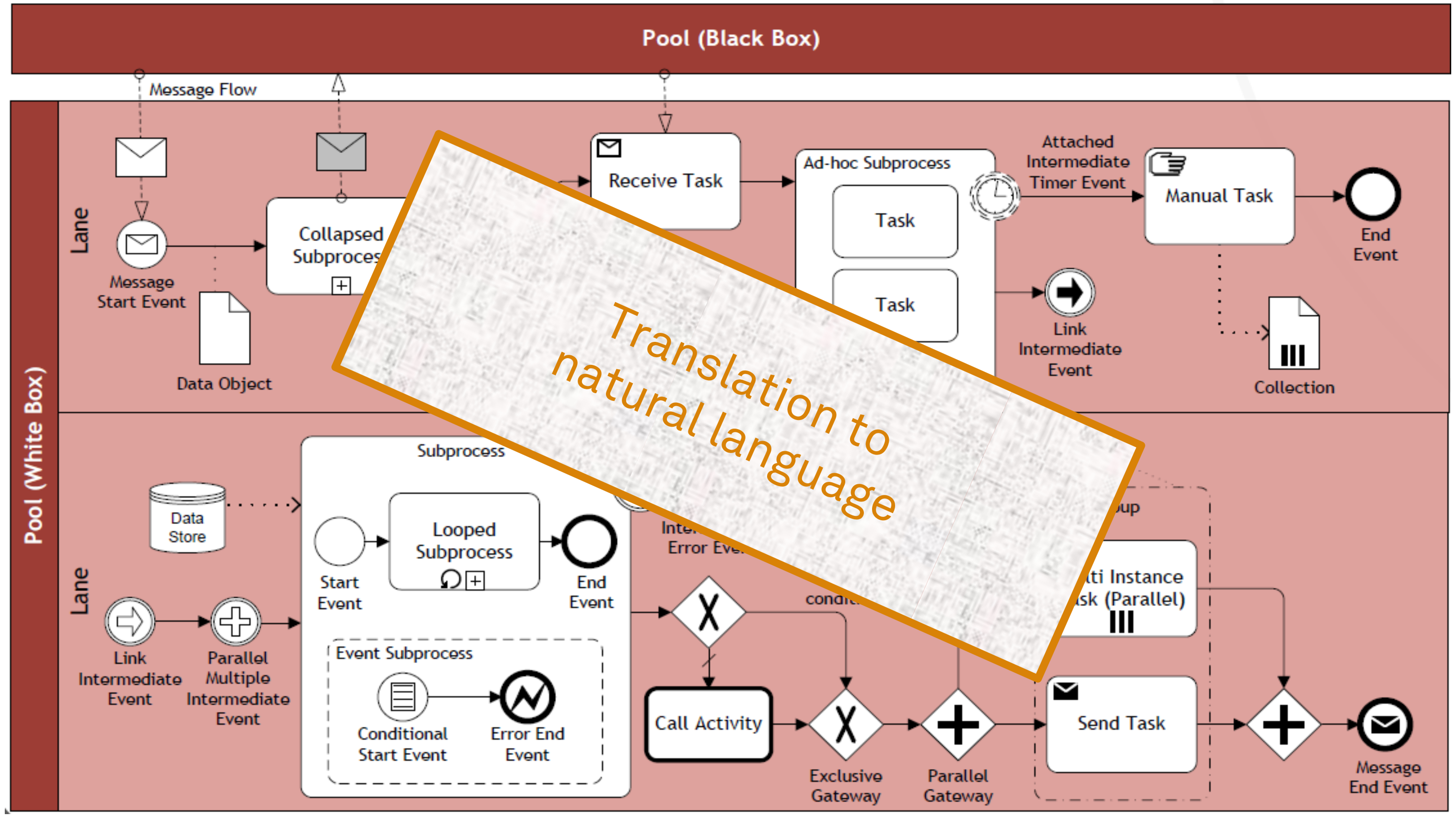
“Every trip is different.”

“You cannot really compare. Our customers go to different places in different seasons using different modes of transportation.”

“We can never do anything exactly in the same way. There are so many special conditions.”

La Rosa (2013)

Challenge of Unfamiliarity with Process Modeling Methods



La Rosa (2013)

Profile of an Expert Process Analyst

- Five Factor Model
  - openness (appreciating art, emotion, and adventure)
  - conscientiousness (tendency to self-discipline, achievement, and planning)
  - extraversion (being positive, energetic, and seeking company)
  - agreeableness (being compassionate and cooperative)
  - neuroticism (being anxious, depressed, and vulnerable)

Dumas et al. (2018)

Profile of an Expert Process Analyst

- Five Factor Model for Analysts
  - openness (appreciating art, emotion, and adventure)
  - **conscientiousness** (tendency to self-discipline, achievement, and planning)
  - **extraversion** (being positive, energetic, and seeking company)
  - agreeableness (being compassionate and cooperative)
  - neuroticism (being anxious, depressed, and vulnerable)

Dumas et al. (2018)

The Problem with Process Discovery

- Process discovery in general belongs to the category of **ill-defined problems**

- This means in the beginning of a process discovery project, it is **not** exactly **clear**
  - who of the **domain experts** must be contacted,
  - which **documentation** can be used, and
  - which **agenda** the different stakeholders might have in mind

Dumas et al. (2018)

Process Discovery Guidelines

- **Get the right people on board**
  - When talking to a participant make sure his supervisor is involved
- **Have a set of working hypotheses**
  - Prepare questions and assumption for workshops to structure the process at different levels of detail
- **Identify patterns**
  - Gather facts on alternatives and conditions, independence and concurrency of activities to sketch process models
- **Pay attention to model quality**
  - Presentation is important
  - Use the right level of abstraction and the right set of elements

Dumas et al. (2018)

Gather Information: Process Discovery Methods




Process Discovery Methods

- **Evidence-based**
  - Document analysis
  - Observation
  - Automated process discovery

- **Dialog-based**
  - Interview-based
  - Workshop-based

Dumas et al. (2018)

Document Analysis

- Documents point to existing **roles**, **activities**, and **business objects** and their **relations**
  - Process descriptions (ideal scenario) ☺
  - Internal policies
  - Organization charts & employment plans
  - Quality certificate reports
  - Glossaries and handbooks
  - Forms
  - Work instructions

- Potential Issues
  - Document **organization might not be process-oriented**
  - Document **granularity might not fit**
  - Documents **might not show reality**

- This method is helpful **before** talking to domain experts

Dumas et al. (2018)

Observation

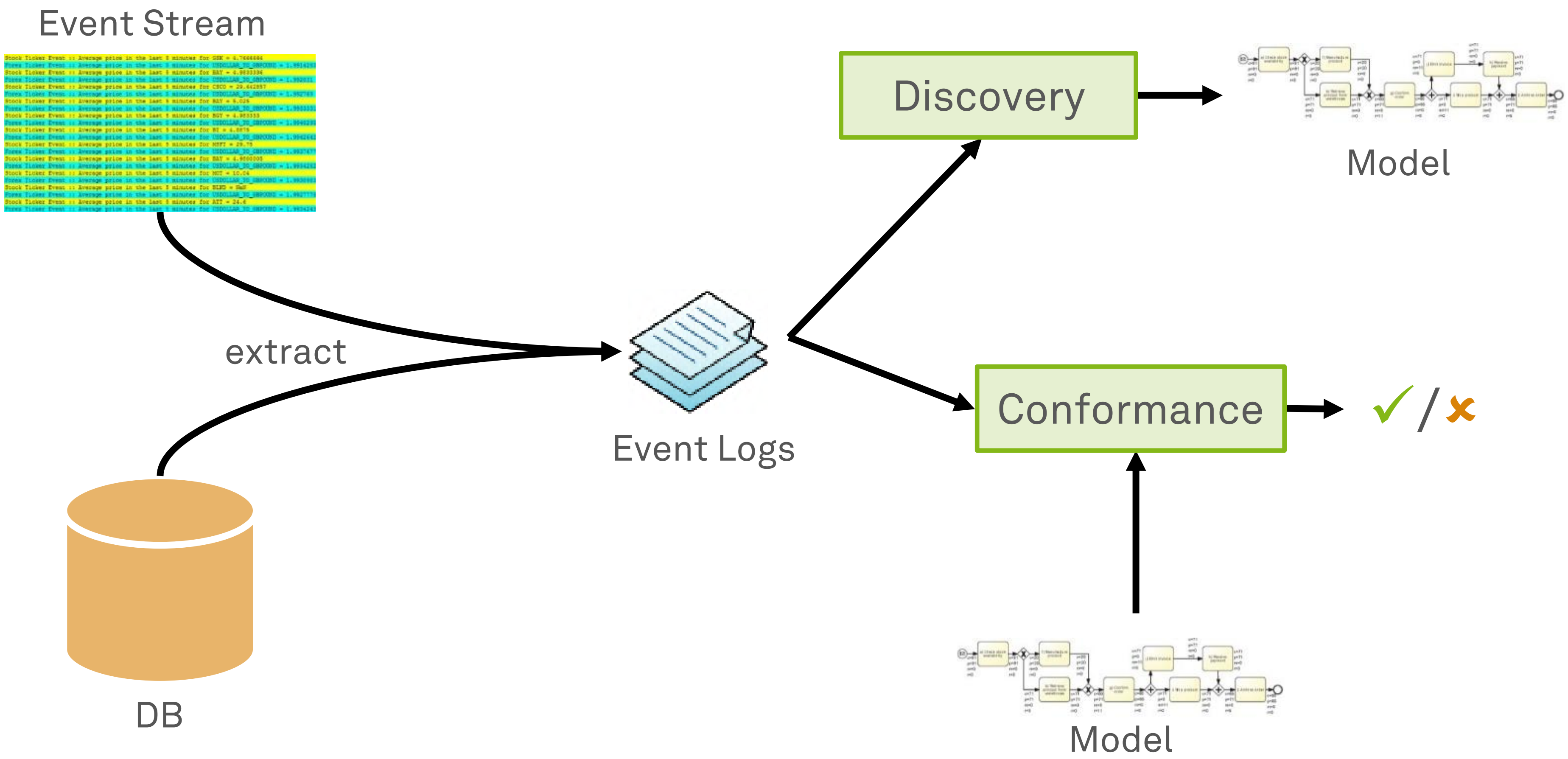
- Follow directly the processing of individual cases
  - **Active role**
    - play a specific role, e.g. customer
  - **Passive role**
    - observe participants and their environment (shadowing)
- Trace business objects in the course of their lifecycle
- Inspect the work environment

- Potential Issues
  - Active role: **cases only**, no big picture
  - Passive role: participants' **bias** when being observed

- This method reveals how a process is conducted in reality as of today

Dumas et al. (2018)

Automated Process Discovery



La Rosa (2013)

Automated Process Discovery

- Event logs capture actual execution
  - **Objective** representation of the process
- **Rich set** of process-related information beyond the tasks
  - e.g., timestamps, resources, business objects
- **Not limited** to one system
  - End-to-end-process representation

- Potential Issues
  - (Quality) event logs may not be **not always available**
  - Event logs **may be fragmented** (granularity, missing values)
  - Event logs **only represent the interaction with the system** (see manual tasks)

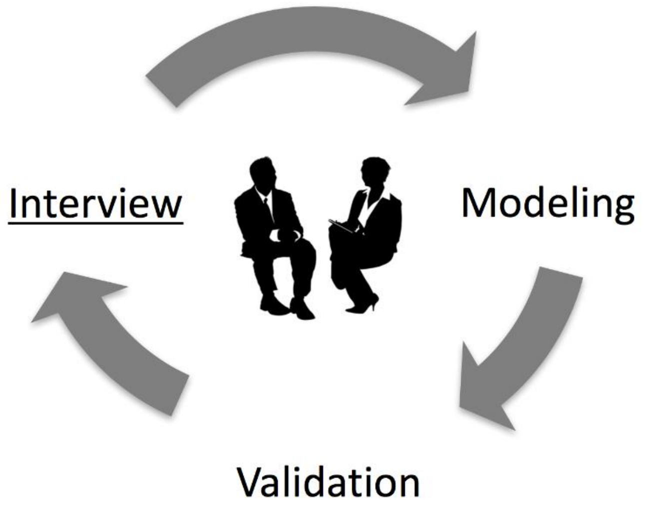
- This method reveals how a process is executed with BPM systems

Dumas et al. (2018)

Interviews

- **Procedure**
  - Interview – Model – Validate // Iterate
- **Strategies**
  - **Backward** (upstream) vs. **forward** (downstream) perspective
  - **Structured** vs. **unstructured** (free-form) interviews
    - Recommendation: ¾ to ¼ ratio
  - “**Sunny day**“ vs. “**rainy day**“
- Remember the **mental model** (e.g., shared terminology)

- This method reveals a rich picture of the process and its participants



Dumas et al. (2018)

Workshop

- Gathers **all key stakeholders** together
- Participants interact to create **shared understanding**
- Typically, one **process analyst** (facilitator), multiple **domain experts** (modeler, scribe), **process owner** may also attend
- **Model** is directly created during the workshop (typically a separate role – tool operator)
- Usually 3 to 5 half-day sessions

- Potential Issues
  - Team opinion **may not be the sum** of individual opinions
  - **Not everybody dares to speak up**

- This method reveals a rich and consolidated picture of the process and its participants

Dumas et al. (2018)

Can I apply any method in any organization?

Company Culture

- Before starting with process discovery, understand the **culture and the sentiment** of an organization
- Companies can preach and practice an **open culture** in which all employees are encouraged to utter their ideas and their criticism
  - They can benefit a lot from workshops as participants are likely to present their ideas freely
- **Strictly hierarchical** organizations can be close-mouthed
  - It is necessary to take special care that every participant gets an equal share of attention
  - Ideas and critique are often held back

Dumas et al. (2018)

Method Results

Aspect	Evidence	Interview	Workshop
Objectivity	high	medium-high	medium-high
Richness	medium	high	high
Time consumption	low-medium	medium	medium
Immediacy of feedback	low	high	high

Dumas et al. (2018)

Method Strengths and Weaknesses

Technique	Strength	Weakness
Document Analysis	✓ Structured information ✓ Independent from availability of stakeholders	x Outdated material x Wrong level of abstraction
Observation	✓ Context-rich insight into process	x Potentially intrusive x Stakeholders likely to behave differently x Only few cases and not all processes can be observed
Automatic Discovery	✓ Extensive set of cases ✓ Objective data	x Potential issue with data quality and level of abstraction x Data may not be available or be available only in part x Data extraction and preparation is time-consuming
Interview	✓ Context-rich insights	x Requires time of process stakeholders x Time-consuming: several iterations required before sign-off
Workshop	✓ Context-rich insights ✓ Direct resolution of conflicting views	x Synchronous availability of several stakeholders x Time-consuming: multiple sessions typically required

Dumas et al. (2018)

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Conduct the Modeling Task: Process Modeling in Process Discovery

Modeling the Process

1. Identify the **process boundaries**
2. Identify **activities** and **events**
3. Identify **resources** and their **handoffs**
4. Identify the **control flow**
5. Identify **additional elements**

Dumas et al. (2018)

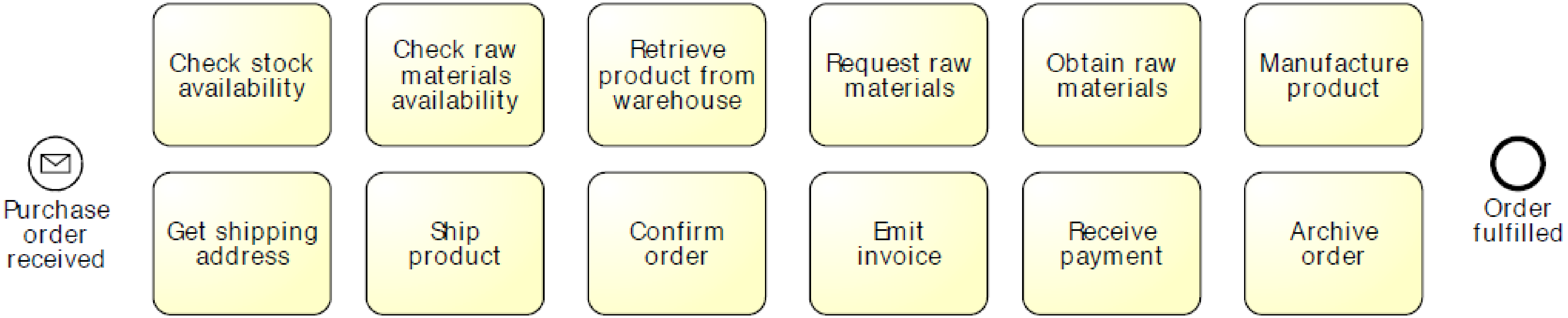
1) Identify the Process Boundaries

- What are the **process triggers**?
  – Manual, timer, message, …
- What are the possible **outcomes** ?
  – Positive, negative
- Which **perspective** do we assume?
  – Internal, external, collaborative
- What **business objects** (artifacts) are required to the process?
  – Input, output

Dumas et al. (2018)

2) Identify Activities and Events

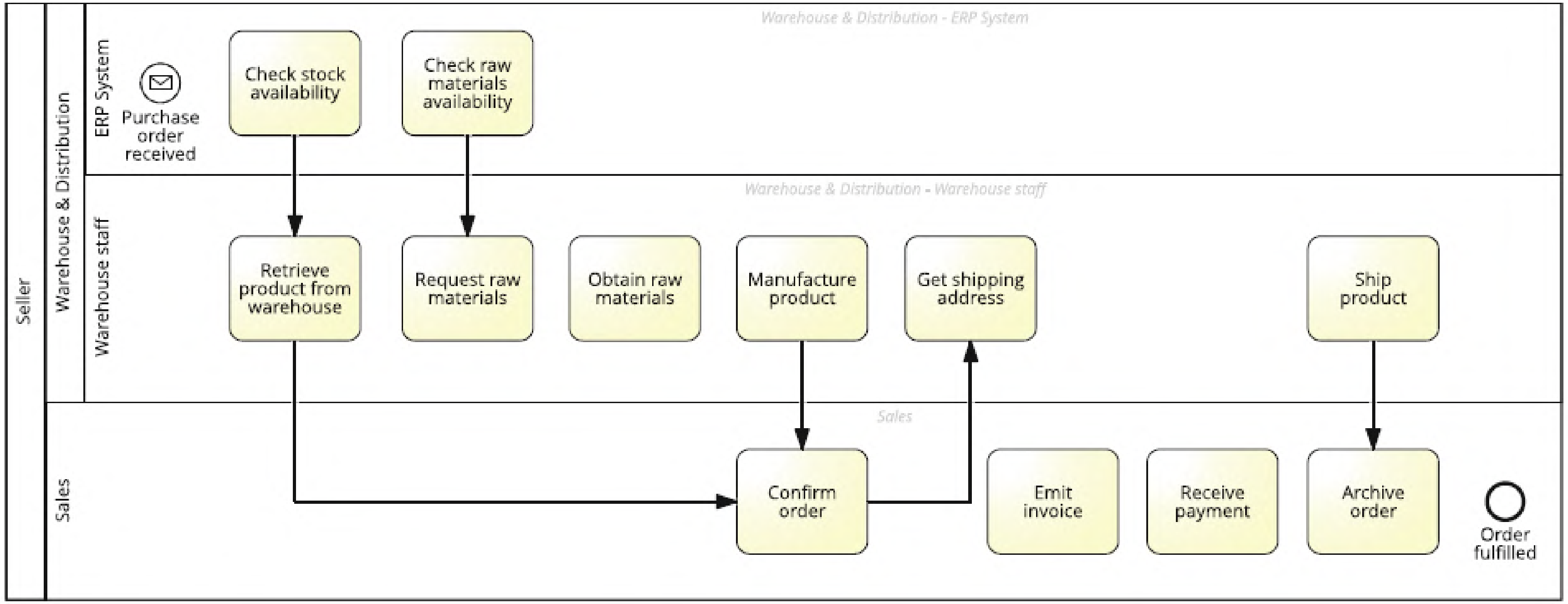
- Which are the **main activities**?
- What are **significant events**?



Dumas et al. (2018)

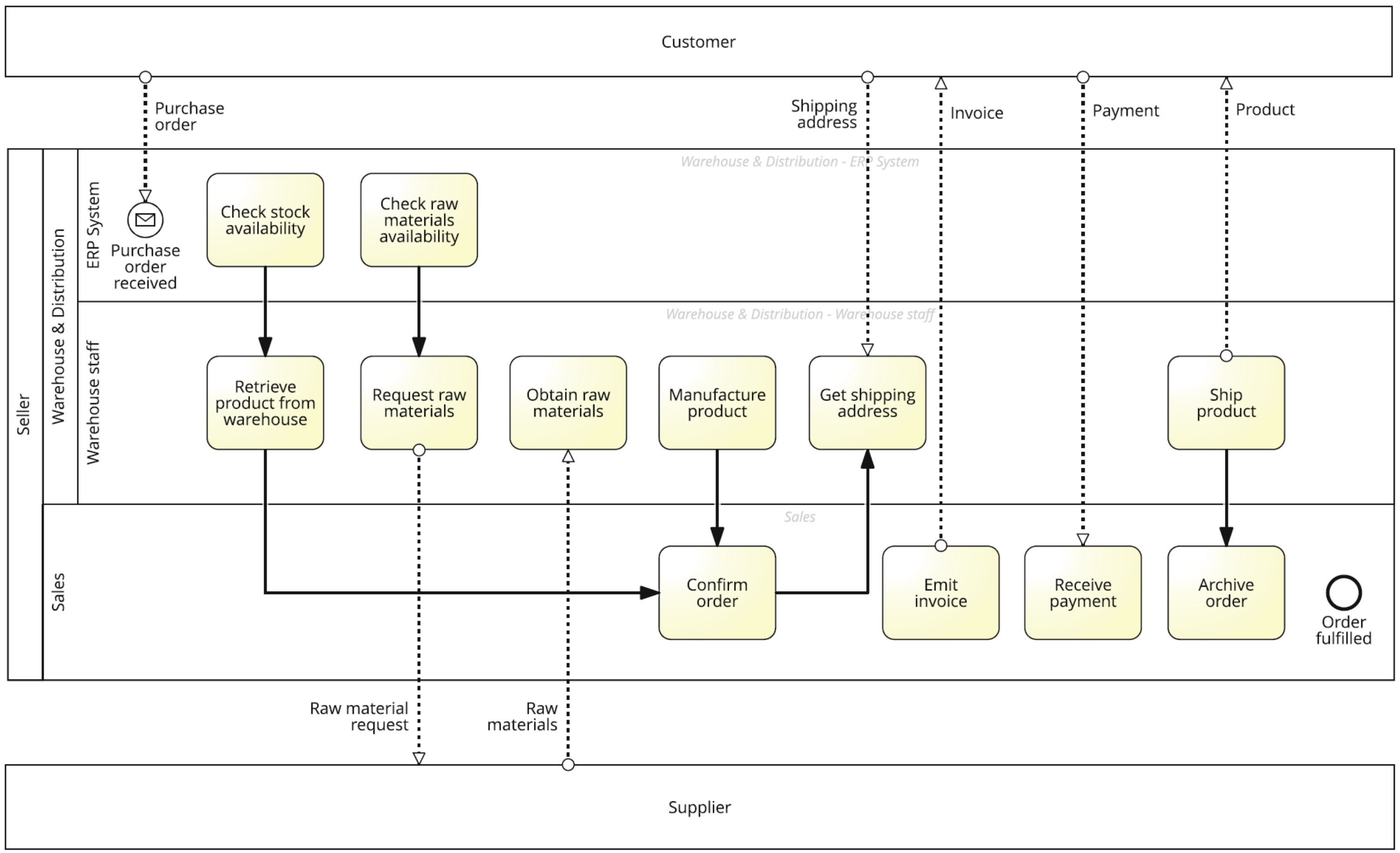
3) Identify Resources and their Handoffs

- Who is **responsible**?
- Where do we find **handoffs**?



Dumas et al. (2018)

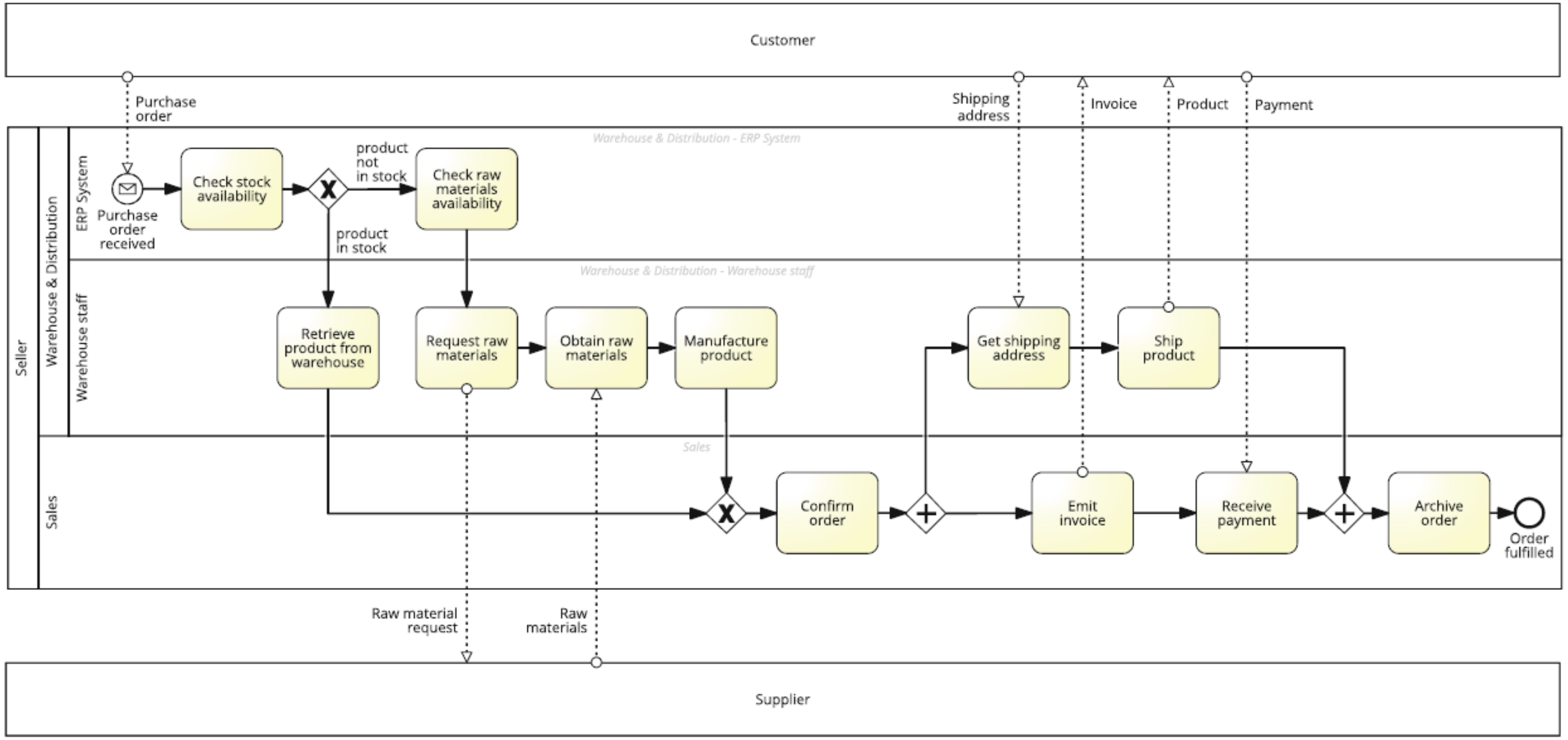
3) Identify Resources and their Handoffs



Dumas et al. (2018)

4) Identify the Control Flow

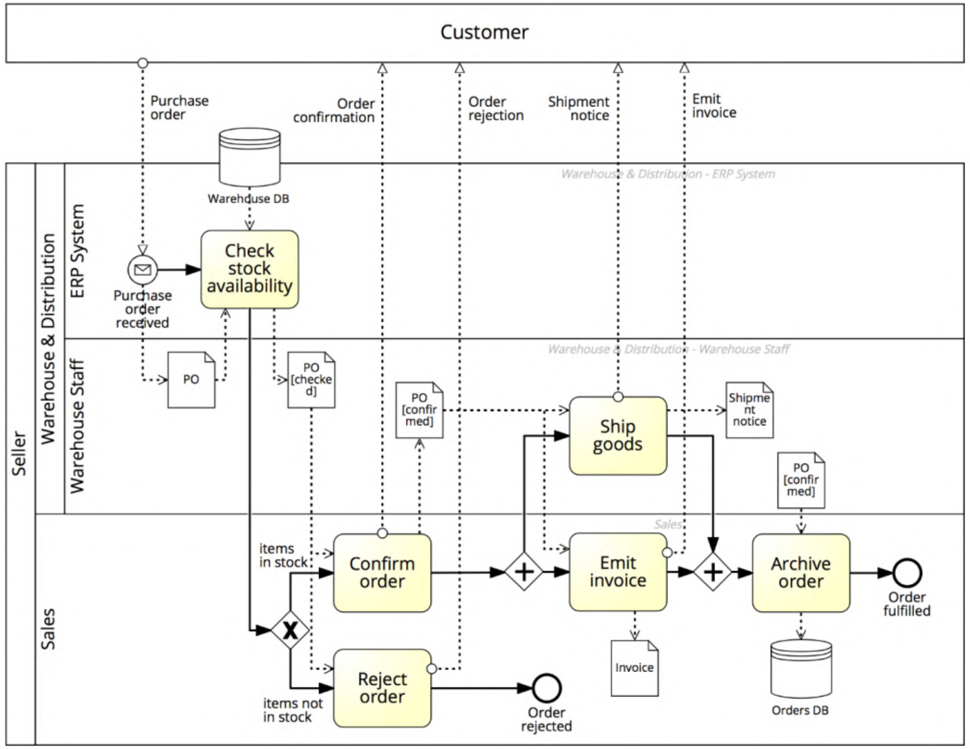
- **When** and **why** are activities executed?



Dumas et al. (2018)

5) Identify Additional Elements

- What are involved **business objects**?
  - Data objects
  - Data stores
- What are involved **exception handlers**?
  - Boundary events
  - Exception flows
  - Compensation handlers
- What **data** is necessary for **execution**?
- Add risk or cost data
- **When do we stop modeling?**



Dumas et al. (2018)

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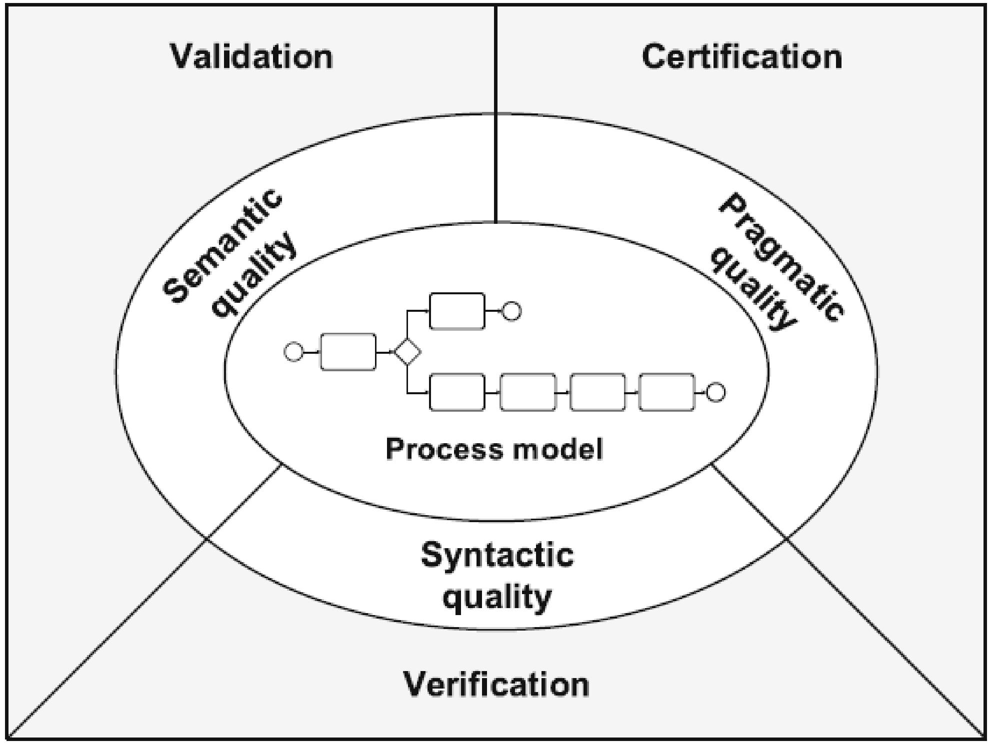
Assure Process Model Quality: Process Models in Process Discovery

Process Model Quality Assurance

Validity
Completeness



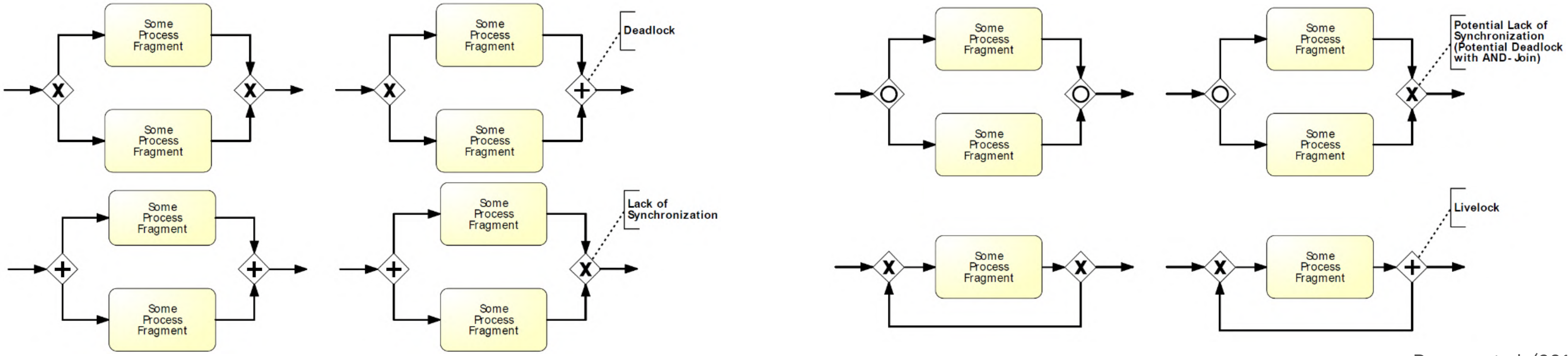
Understandability
Maintainability
Learning

Structural correctness
Behavioral correctness

Dumas et al. (2018)

Verification: Examples of Behavioral Anomalies

- Sound Model
  - **Option to complete** (any instance must complete)
  - **Proper completion** (each remaining token must be in a different end event)
  - **No dead activities** (any activity can be executed)



Dumas et al. (2018)

Process Modeling Guidelines and Conventions

- Objectives
  - Safeguard **model consistency** and improve **standardization** and reuse
  - Reduce the **dependency** on individual process analysts
  - Facilitate **access** to models by non-modeling experts

- Restrictions
  - **Vocabulary** (avoid certain elements, e.g. subprocesses)
  - **Structure** (limit the structure of the model, e.g. by restricting layers)
  - **Semantics** (avoid element meanings, e.g. limiting boundary events to business faults)
  - **Appearance** (restrict the model appearance, e.g. labels, layout, and notation

Dumas et al. (2018)

Seven Process Modeling Guidelines (7PMG)

G1 Use as few elements in the model as possible

G2 Minimize the routing paths per element

G3 Use one start and one end event

G4 Model as structured as possible

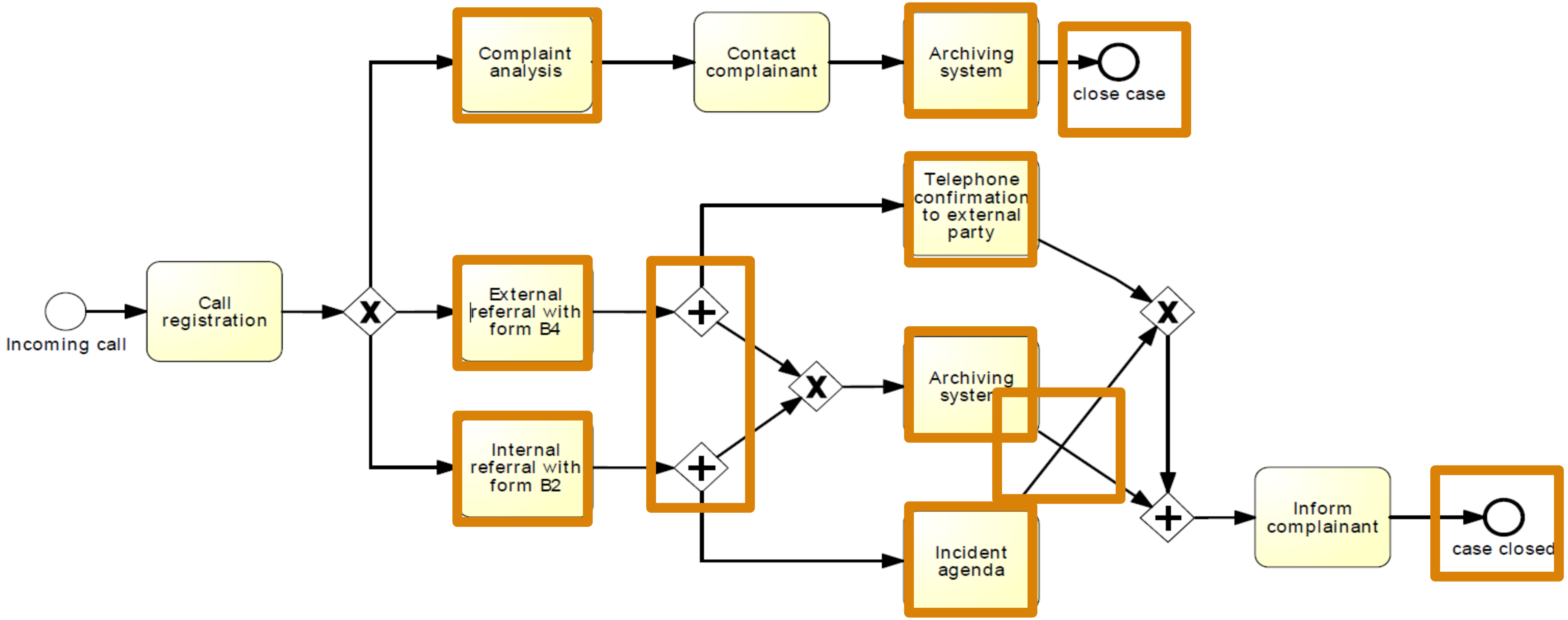
G5 Avoid, where possible, OR routing elements

G6 Use verb-object activity labels

G7 Decompose a model with more than 30 elements

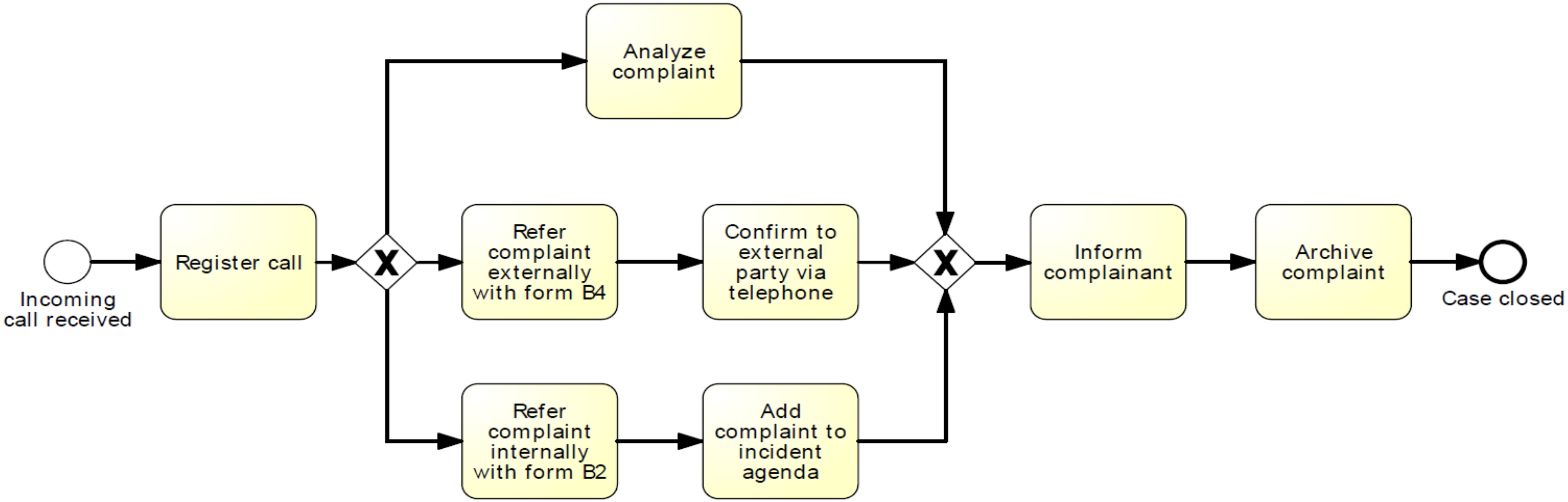
Dumas et al. (2018)

Bad Example



Dumas et al. (2018)

Good Example



Dumas et al. (2018)

Learning Goals of Today

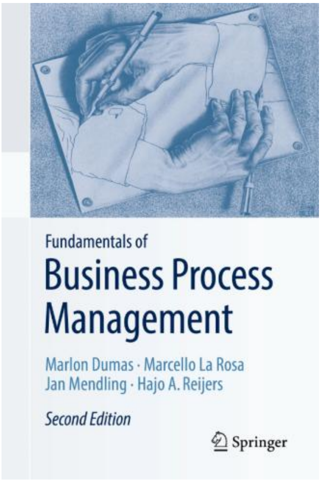
- Understand the challenges of process discovery
- Understand and be able to choose process discovery methods
- Be able to use process modeling during process discovery

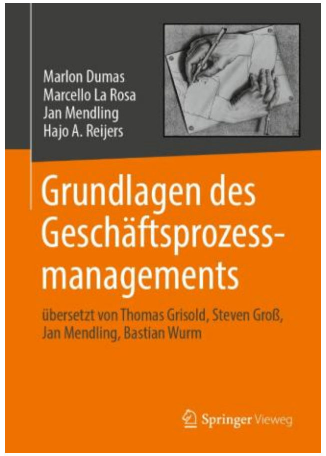
Recap Questions

- What is process discovery and what are its four phases about?
- Which evidence-based and dialogue-based process discovery methods do you know, what are their benefits and drawbacks, and when can they be applied?
- Which dimensions for process model quality assurance are you aware of?

Literature

- **Section 4:** Process Discovery





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Any Questions about Process Discovery?

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Qualitative Process Analysis



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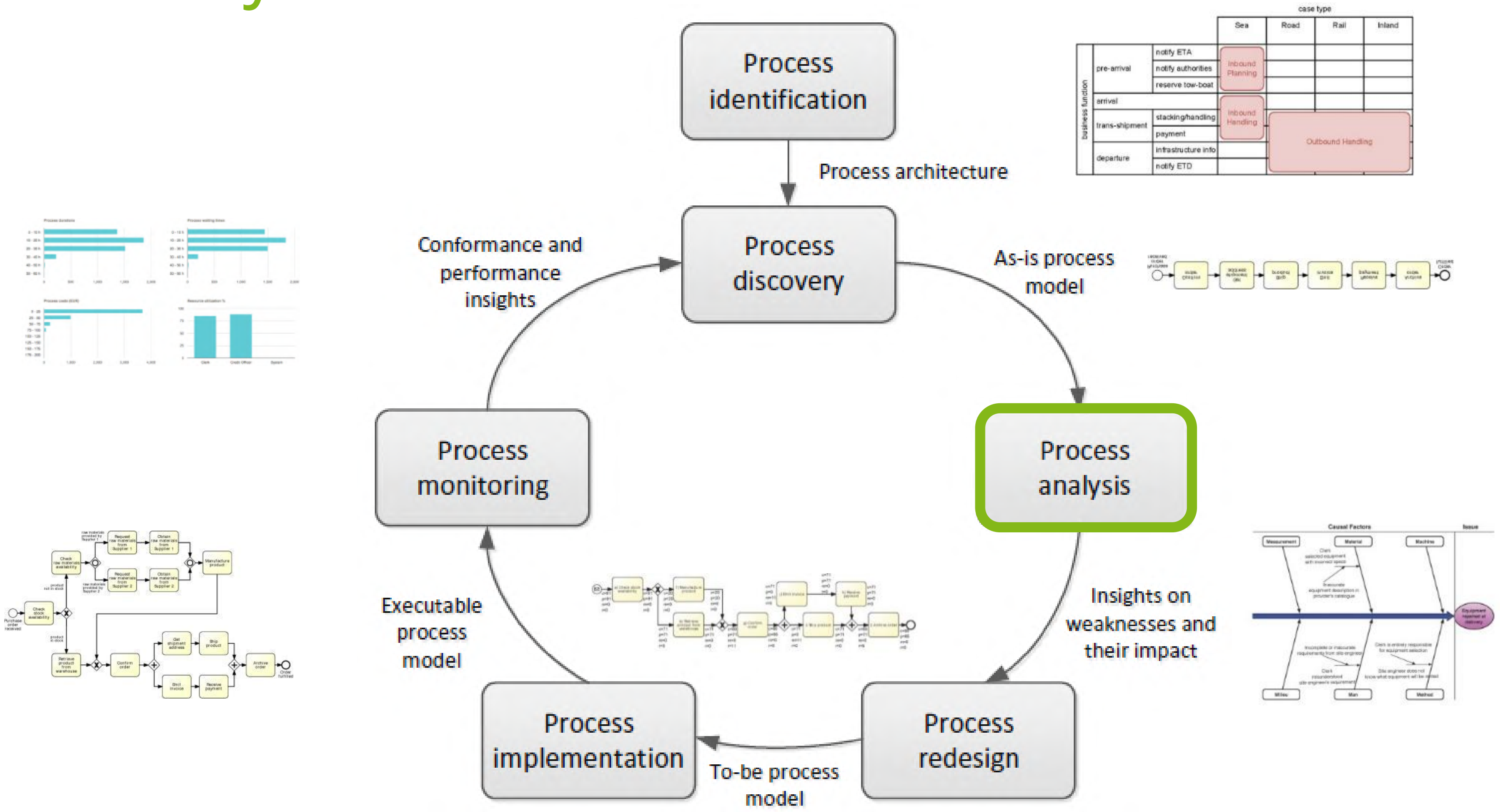
Learning Goals of Today

- Understand and be able to use value-added analysis
- Understand and be able to use waste analysis
- Understand and be able to use stakeholder analysis and issue documentation
- Understand and be able to use root cause analysis

The BPM Lifecycle



Dumas et al. (2018)

Process Analysis

“Quality is free, but only to those who are willing to pay heavily for it.”

Tom DeMarco (1940–)

- Analyzing Processes is an **art** and a **science**
- **Qualitative analyses** use a range of **principles and techniques**
- **Quantitative analyses** use **measures** to judge a process
- There is no single way of producing good results

Dumas et al. (2018)

A Selection of Process Analysis Techniques

- **Qualitative** analysis
  - Value-added Analysis
  - Waste Analysis
  - Stakeholder Analysis
  - Root cause Analysis

- **Quantitative** Analysis
  - Flow Analysis
  - Queueing Analysis
  - Process Simulation

Dumas et al. (2018)

Purpose of Qualitative Process Analysis

- Identify and eliminate **unnecessary steps in a process**
  - Valued-added analysis
  - Waste analysis

- Identify, document, understand, and prioritize **issues**
  - Stakeholder analysis and issue documentation
  - Root cause analysis
  - Pareto charts, PICK charts

Dumas et al. (2018)

Value-added Analysis

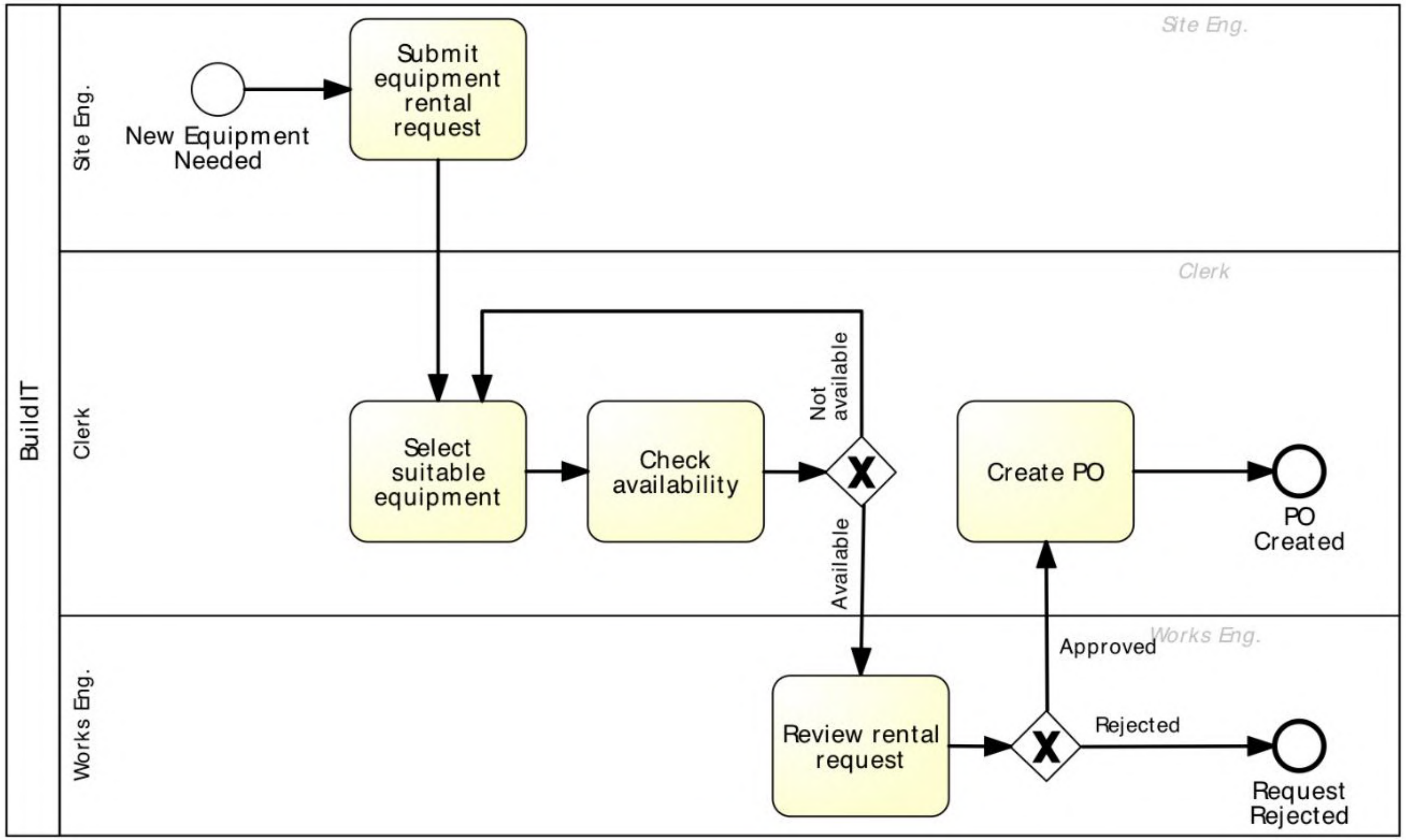
   

Value-added Analysis

- Technique to **identify unnecessary steps** in a process and eliminate them
- **Decompose** the process into steps and identify customer & positive outcomes
  - Analyze documentation
    - e.g., process models, checklists
  - Analyze tacit knowledge
    - e.g., through observation and interviews
- **Classify** each step into
  - **Value-adding (VA):** Produces value or satisfaction to the customer
  - **Business value-adding (BVA):** Necessary or useful for the business to run smoothly, or required due to the regulatory environment, e.g. checks, controls
  - **Non-value-adding (NVA):** Everything else including handovers, delays, and rework

Dumas et al. (2018)

Value-added Analysis Example



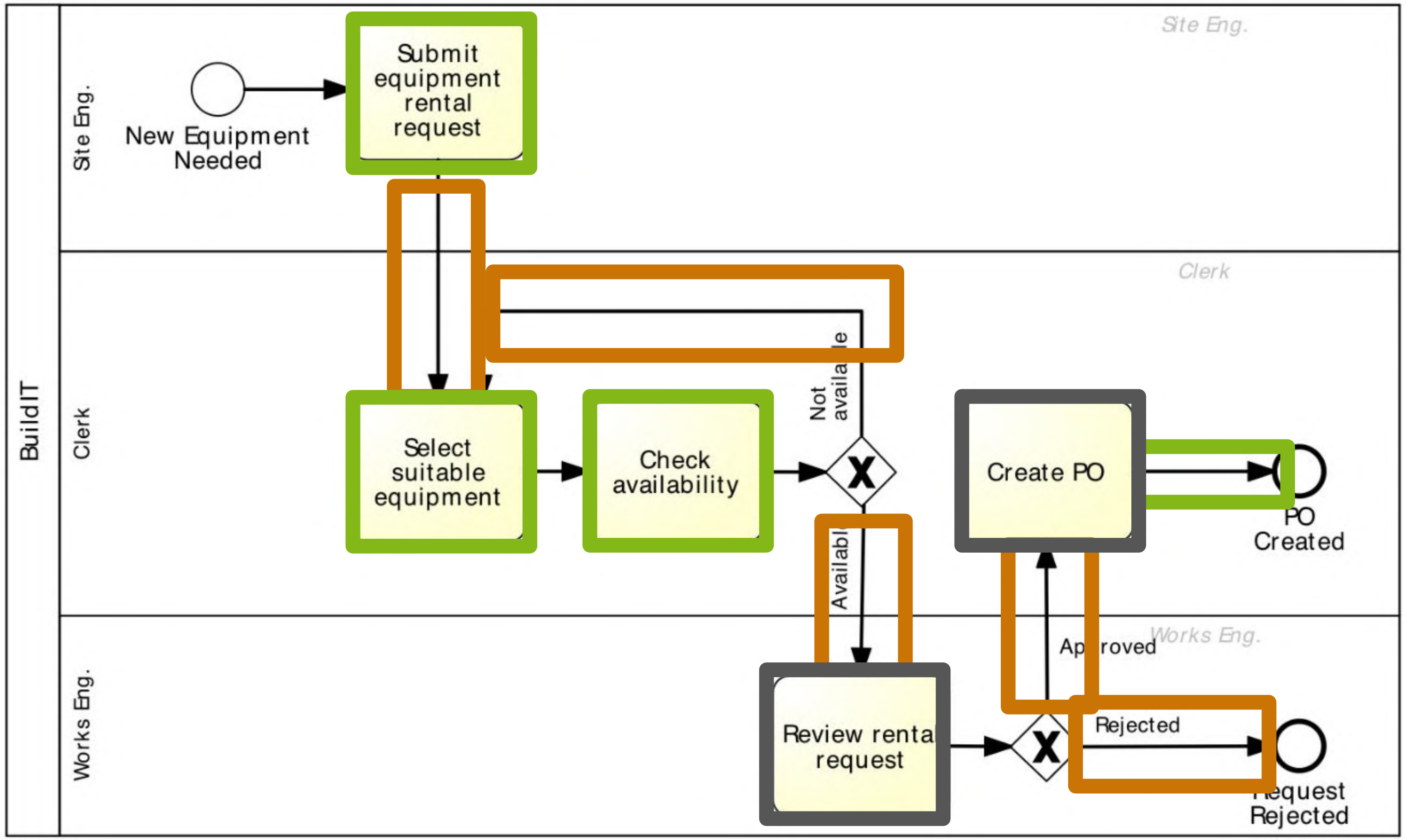
Dumas et al. (2018)

VA

BVA

NVA

Value-added Analysis Example



Dumas et al. (2018)

Guidelines for Value-added Analysis

- **Handoff steps** between process participants are **non-value adding**
- **Control steps** are generally **business-value adding** (an excessive amount may be NVA)
- Creating a **formal document** in general is typically **non-value adding**
- **Steps** imposed by **accounting or legal** requirements are generally **business-value adding** (beware of hearsay)
- Differentiate between **passing** information (NVA) and **creating** and **making** information **available** (VA)
- **Classification is to some extent subjective and depends on context!**
(That is why it is an art!)

Dumas et al. (2018)

Guidelines for Elimination

- **Minimize and Eliminate NVA steps**
  - Use automation (BPMS)
  - Reconsider participants (e.g., clerk)
  - Consider process variants (e.g., low value cases without approval)

- **Minimizing and Eliminating BVA steps is a trade-off**
  - Consider the value for the company w.r.t. goals and strategy
  - Consider regulations

- More about this when we discuss process redesign

Dumas et al. (2018)

Waste Analysis

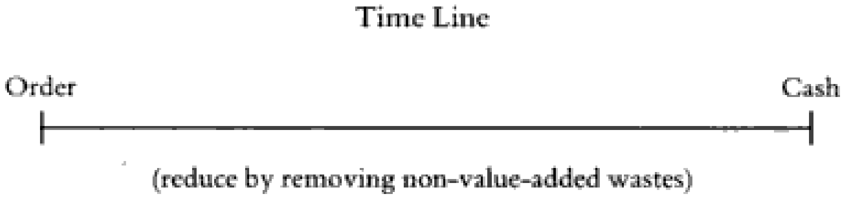



Waste Analysis

"All we are doing is looking at the timeline,
from the moment the customer gives us an order
to the point when we collect the cash.

And we are reducing that timeline
by removing the non-value-added wastes."

Taiichi Ōhno (1912–1990)



Ohno (1988)

7+1 Sources of Waste

- **Transportation** (send, receive)
- **Motion** (drop-off, pick-up, go to)
- **Inventory** (large work-in-progress)
- **Waiting** (idle time between tasks)
- **Defects** (compensation to fix defects)
- **Overprocessing** (performing what is not needed)
- **Overproduction** (unnecessary cases)

- **Skills** (resource underutilization, idle resources)

- **Move**
  - Transportation & motion
- **Hold**
  - Inventory (+ *overprovisioning*) & waiting
- **Overdo**
  - Defects, overprocessing, overproduction

Ohno (1988), Liker (2004), Six Sigma, Dumas et al. (2018)

Move

- **Transportation**
  - Physical transport of materials from A to B
  - Physical transport of documents
  - Handoffs between participants
  - → Electronic Data Interchange (EDI) has automated some, lanes in pools help to identify further waste
- **Motion**
  - Physical movement of process participants from A to B
  - Application switch in digitalized processes
  - → Robotic Process Automation (RPA) can help to reduce some of this waste

Dumas et al. (2018)

Hold

- **Inventory**
  – Physical hold of more stock than necessary to perform a task
  – Work-in-process
  – *(Resource/skill overprovisioning – more capacity than necessary)*
  → See Quantitative Process Analysis
- **Waiting**
  – Change-over times
  – Idle times, suspend times
  – Resource availability
  – **Transportation is a constant source of waiting waste**
  → See Quantitative Process Analysis

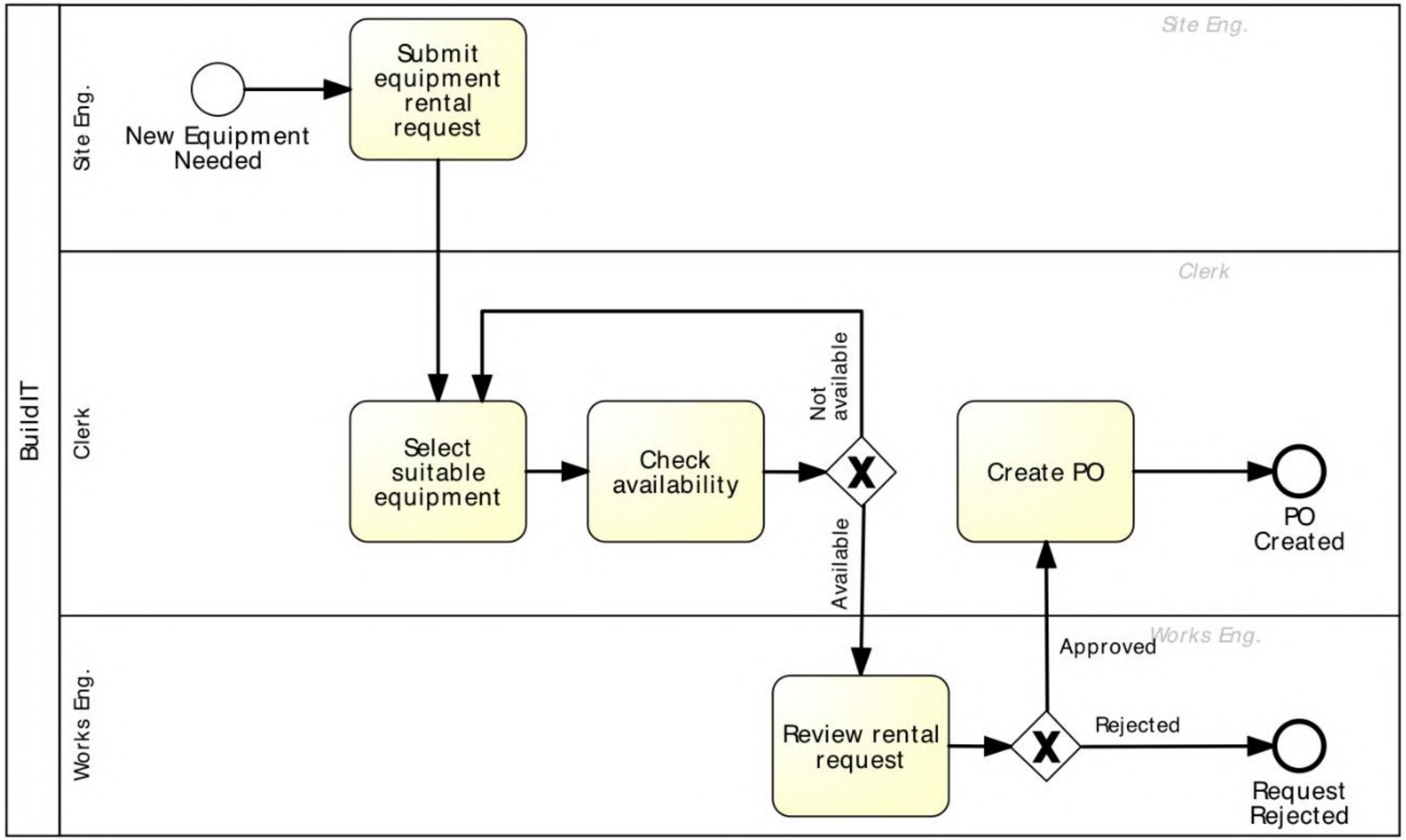
Dumas et al. (2018)

Overdo

- **Defect**
  - Correct, repair, or compensate for issues
  - Results in rework (loops)
  - → See Process Redesign
- **Overprocessing**
  - Unnecessary tasks or unnecessary perfectionism
  - → See Process Redesign
- **Overproduction**
  - Successful process instances that were not necessary (e.g., unrequired approval)
  - Failed process instance that should not have been initialized (e.g., invalid application)
  - → See Process Redesign

Dumas et al. (2018)

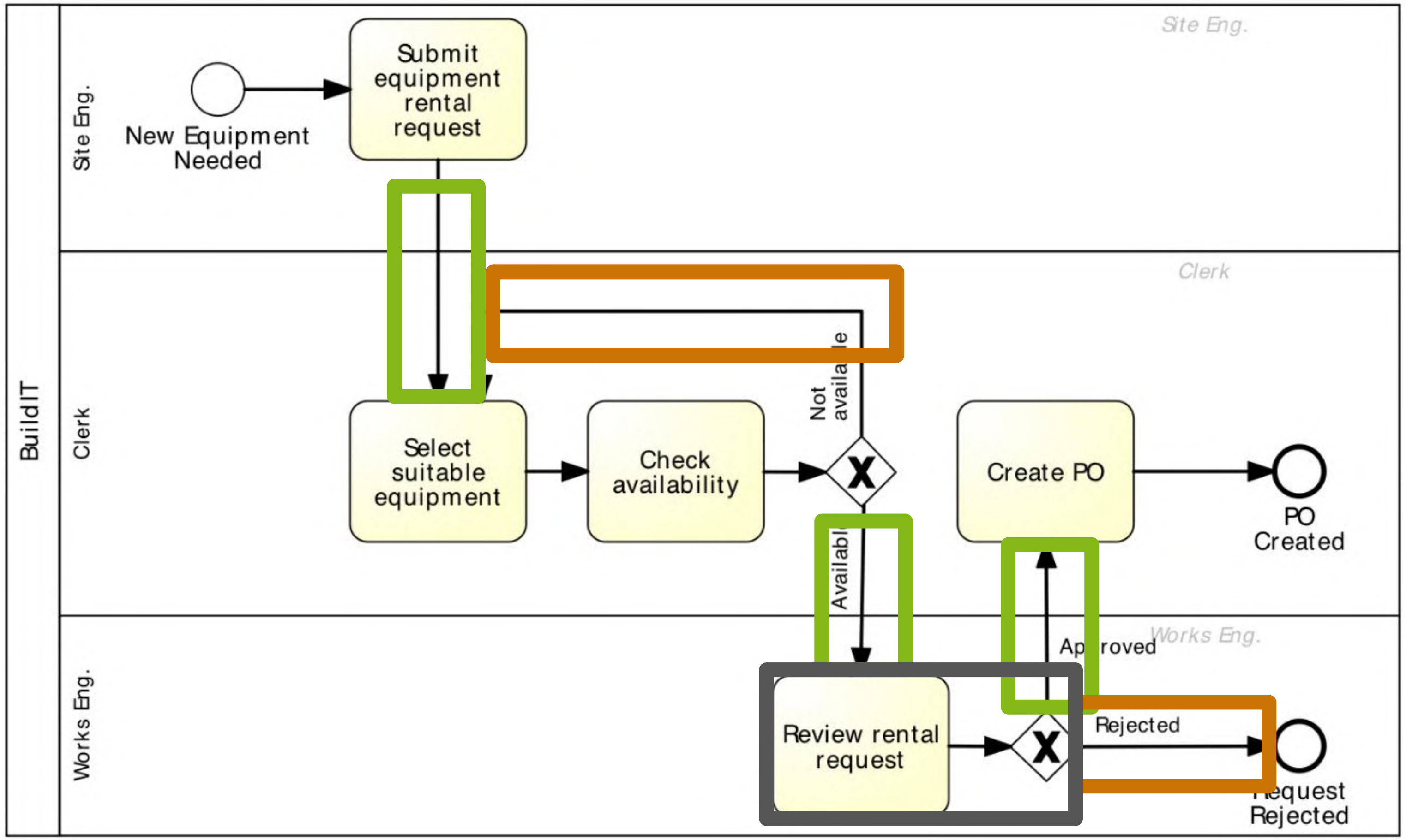
Value-added Analysis Example



Dumas et al. (2018)

Move
Hold
Overdo

Value-added Analysis Example



Dumas et al. (2018)

Stakeholder Analysis & Issue Documentation




Stakeholder Analysis

- Gathering data from **multiple sources** by interviewing stakeholders of different types and reconciling their viewpoints
  → See Process Discovery
- There are typically **five categories of stakeholders**
  - **Customers** of the process
  - **Process participants**
  - **Process owner** and operational managers who supervise the process participants
  - **External parties**, e.g., suppliers, sub-contractors
  - **Sponsor** of the process improvement effort and other executive managers who have a stake in the performance of the process

Dumas et al. (2018)

Stakeholder Concerns

- **Customers**
  - Slow cycle time, defects, lack of transparency, or lack of traceability (inability to observe the current process status)
- **Process participants**
  - Resource utilization, working under stress, defects arising from handoffs and wastes
- **Process owner**
  - Performance (cycle times, processing times, common defects and wastes, compliance)
- **External parties**
  - Steady or growing stream of work from the process, ability to plan ahead, meet SLA
- **Sponsor**
  - Strategic alignment of the process, contribution to key performance indicators

Dumas et al. (2018)

Issue Documentation

- **Issue register**
  - Categorize identified **issues and their impact** as part of as-is process modelling
  - Can contains **issues** and **factors**

Name	Description	Qualitative Impact	Assumptions	Quantitative Impact
Rejected equipment	Site engineers reject delivered equipment due to non-conformance to their specifications	Disruption to schedules. Employee stress and frustration	• We rent 3000 pieces of equipment p.a. • Each time an equipment is rejected due to an internal mistake, we are billed the cost of one day of rental, that is 100 €. • 5% of them are rejected due to an internal mistake	3.000 × 0.05 × 100 € = **15.000** € p.a.

Dumas et al. (2018)

Issue Documentation

- **Issue register tables** can have the following columns
  - Issue number
  - Issue name
  - Description/ explanation
  - Priority
  - Assumptions/ input data
  - Impact: qualitative (cost) and quantitative (nuisance)
  - Caused by/ causes
  - Possible solution

Dumas et al. (2018)

Impact Assessment

- A small **subset of issues** in the issue register are likely responsible for the **largest share of impact**
- For a given issue, a small **subset of factors** behind this issue are likely responsible for the **largest share of occurrences** of this issue
- Techniques necessary to **prioritize** a collection of issues or factors behind an issue
  - Pareto analysis: 80-20 principle
  - Growth-share matrix
  - ...

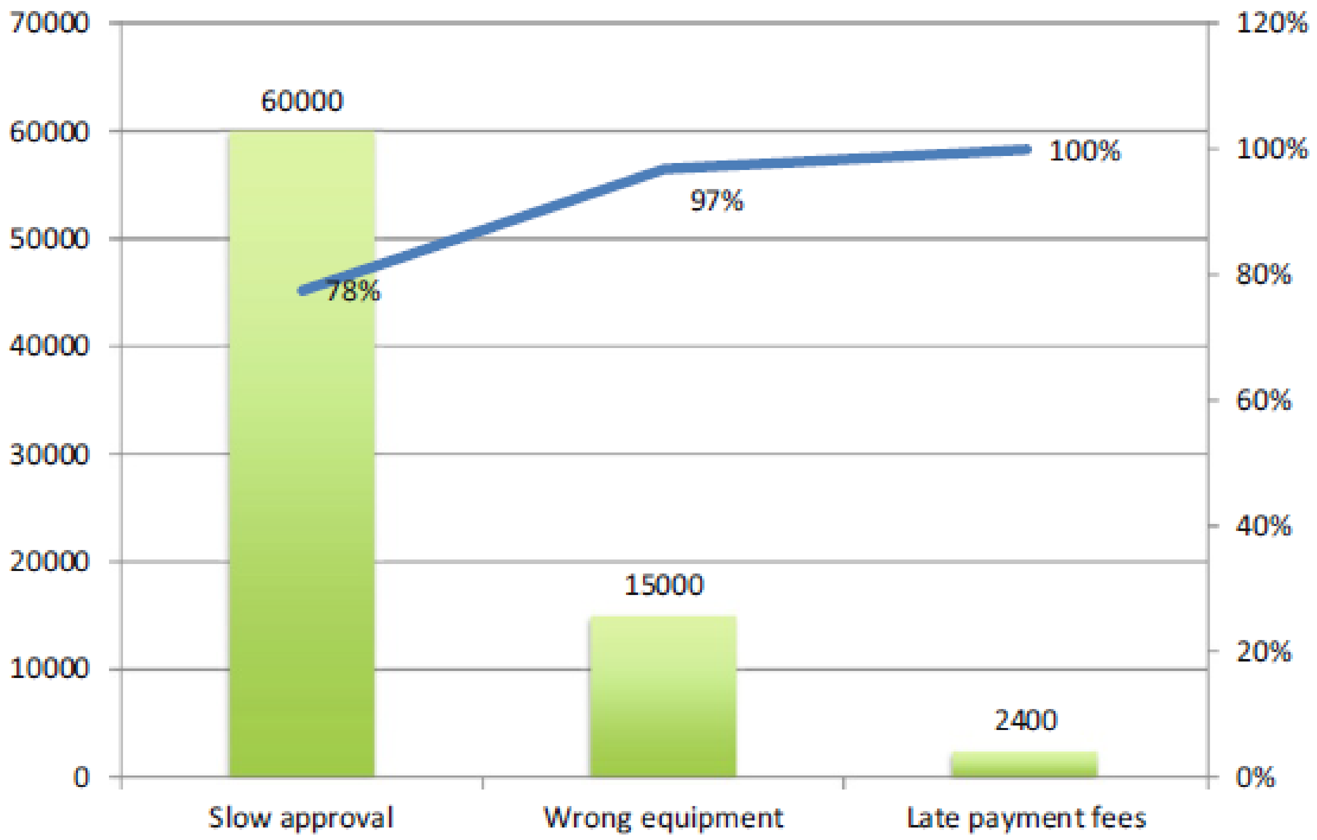
Dumas et al. (2018)

Impact Assessment Process – Pareto Analysis

- Few problems are responsible for large parts of the deficiencies
- **Define** the **effect** to be analyzed and the **measure** via which this effect will be quantified
- Identify all **relevant issues** that contribute to the effect to be analyzed
- **Quantify each issue** according to the chosen measure
  - Can be done on the basis of the issue register, i.e. the quantitative impact column
- Sort the issues according to the chosen measure (from highest to lowest impact) and draw a so-called **Pareto chart**

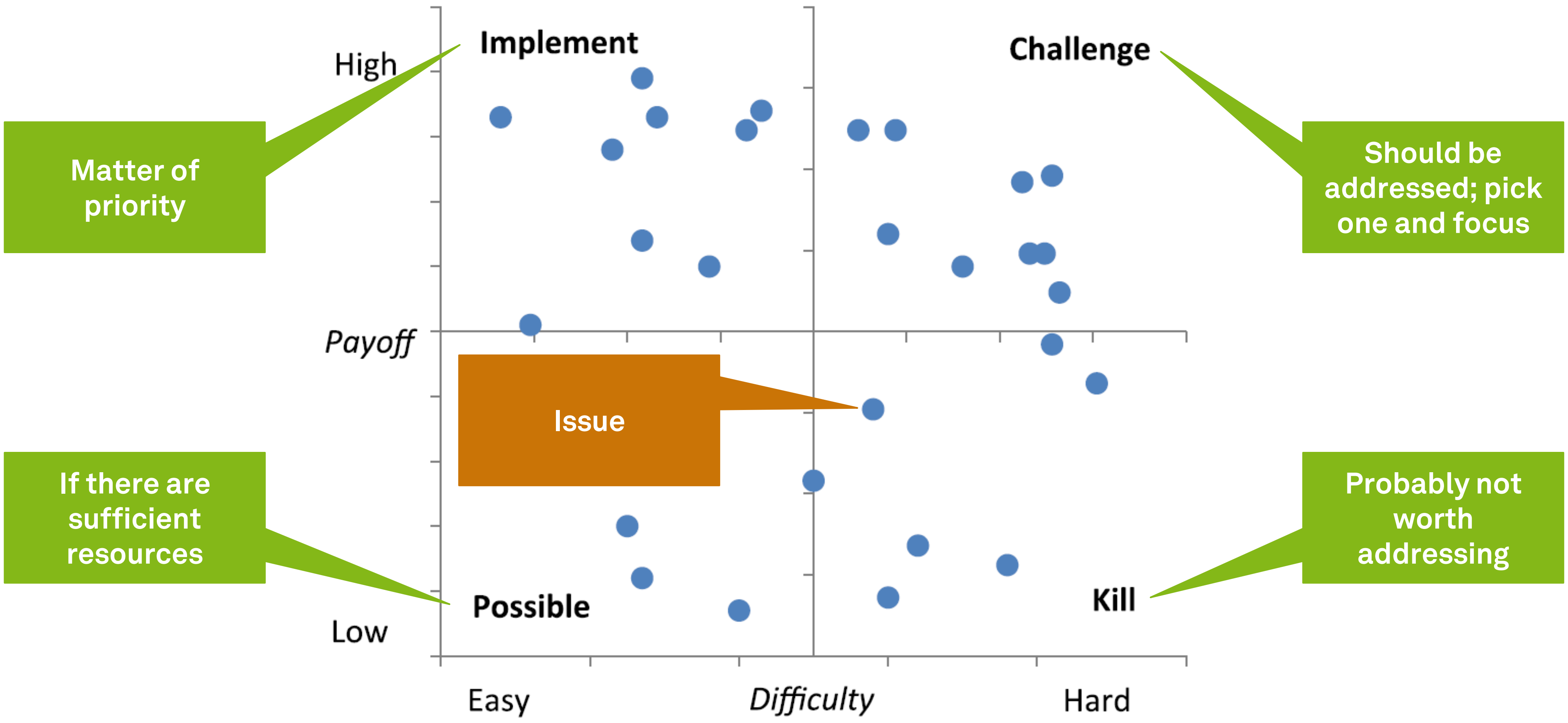
Dumas et al. (2018)

Pareto Chart



Dumas et al. (2018)

PICK Chart



Dumas et al. (2018)

Root Cause Analysis

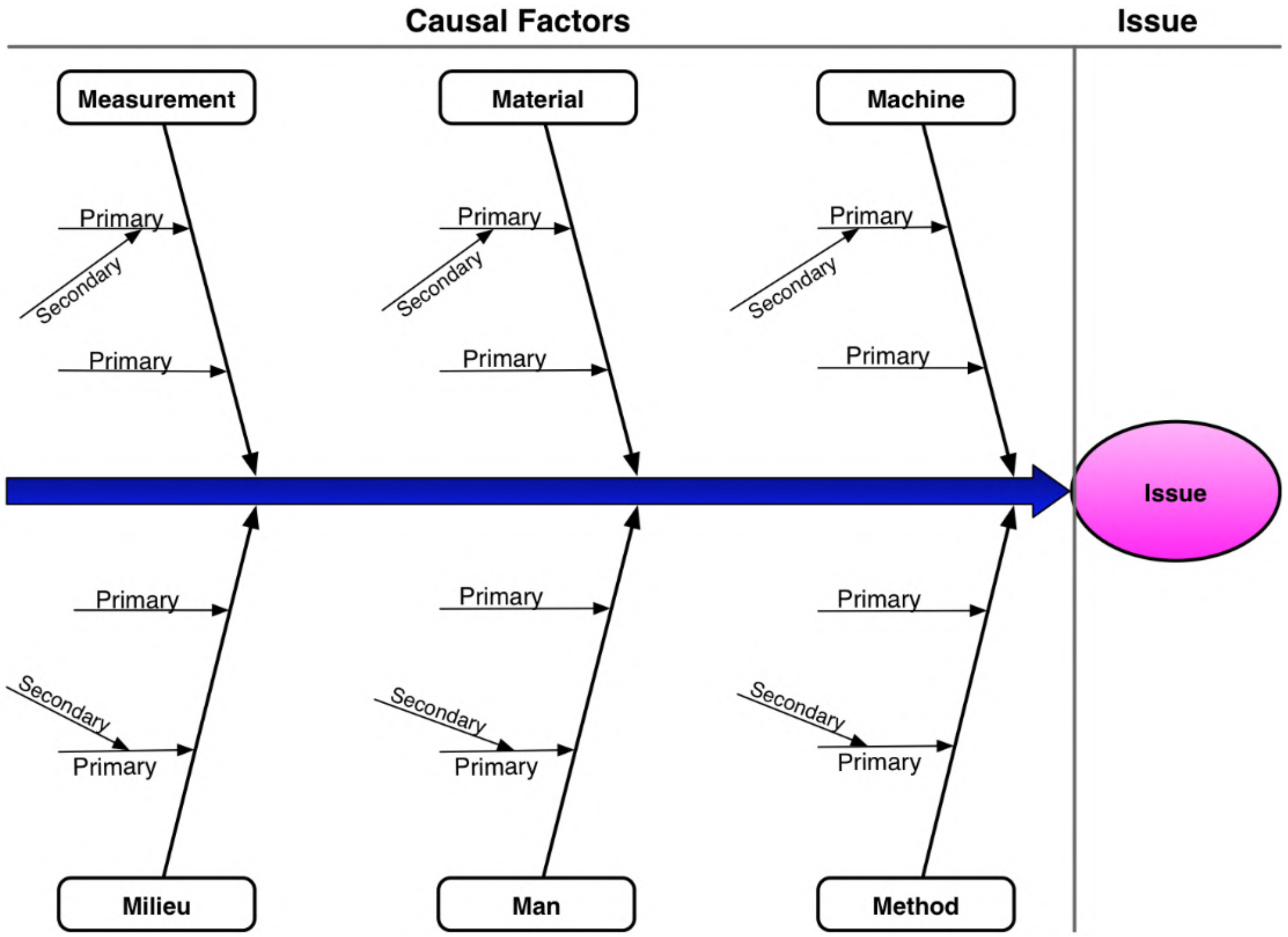



Root Cause Analysis

- Root cause analysis is a **family of techniques**
- Goal is to **understand root causes** of problems why a process does not have a better performance
- Distinguishes
  - **Causal factors** (root cause)
  - **Contributing factors** (enabler, multiplier)
- Analysis are usually performed as **interviews** or **workshops**
  - Be aware of multiple perspectives/ perceptions on one issue
- Results are documented as
  - **Cause-Effect diagrams**
  - **Why-Why diagrams**

Dumas et al. (2018)

Cause-Effect Diagram



Dumas et al. (2018)

The 6 M (1/3)

- **Machine** (technology)
  - **Lack of functionality** in application systems
  - **Redundant storage** of data across systems, leading e.g. to double data entry and data inconsistencies
  - **Low performance** of IT of network systems, leading e.g. to low response times for process participants
  - **Poor user interface** design, leading e.g. to process participants not realizing that some data are missing or available
  - **Lack of integration** between multiple systems within the enterprise or with external systems

Dumas et al. (2018)

The 6 M (2/3)

- **Method** (process)
  - **Unclear, unsuitable or inconsistent assignment** of decision-making and processing responsibilities to process participants
  - **Lack of empowerment** of process participants, leading to process participants not being able to make decisions
  - **Excessive empowerment** may lead to process participants having too much discretion and causing losses to the business
  - **Lack of** timely **communication** between process participants
- **Material**
  - **Incorrect** data leading to a wrong decision being made during the execution of a process

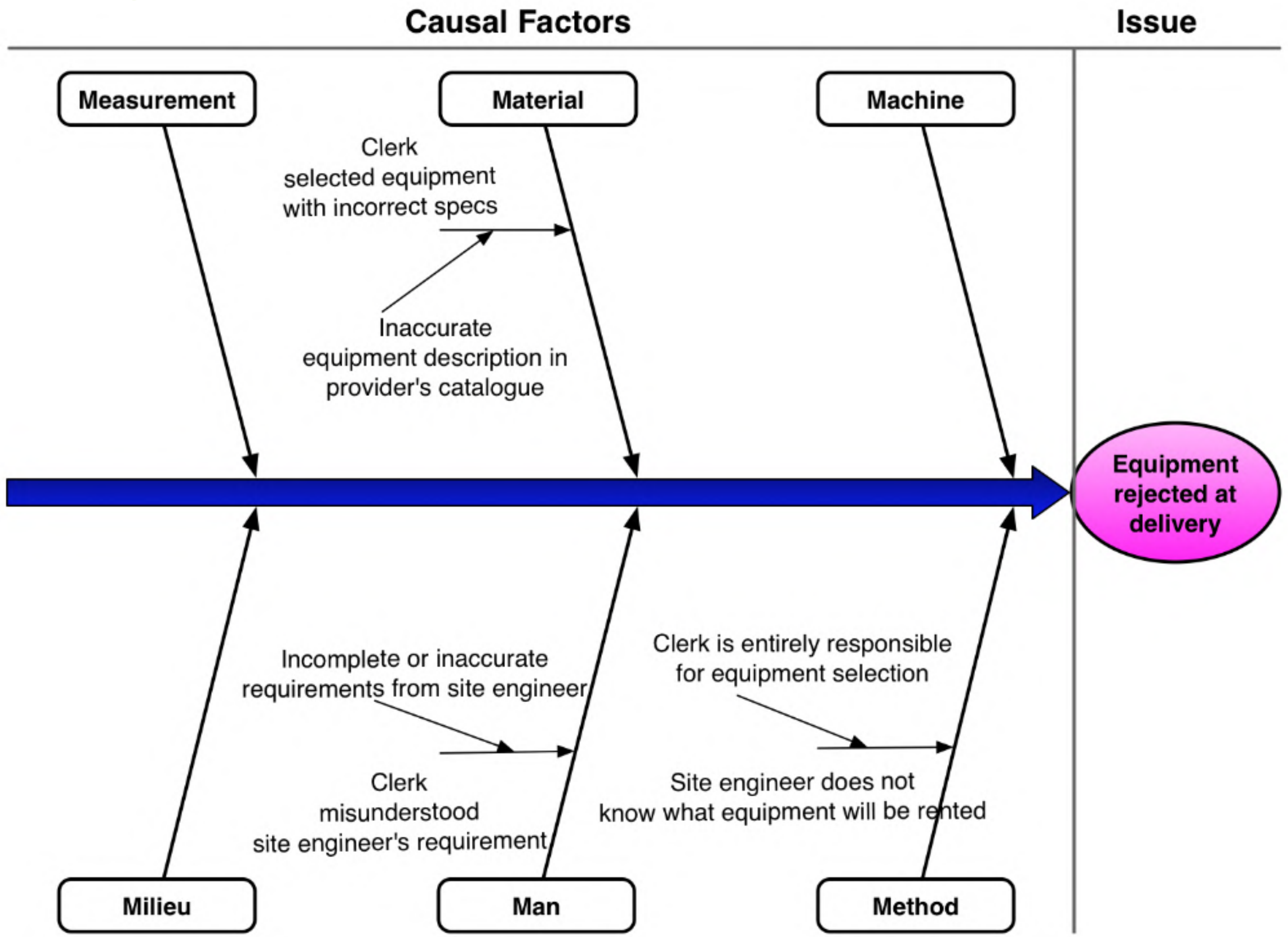
Dumas et al. (2018)

The 6 M (3/3)

- **Man**
  - **Lack of training** and clear instructions for process participants
  - **Lack of incentive system** to motivate process participants sufficiently
  - **Expecting** too much from process participants
  - **Inadequate recruitment** of process participants
- **Measurement**
  - **Mistakes in measurements** or calculations made during the process
- **Milieu**
  - Factors stemming from the **environment** in which the process is executed (customer, suppliers or other external actors)
  - Distinction between global, external, and internal factors

Dumas et al. (2018)

Cause-Effect Diagram – Example



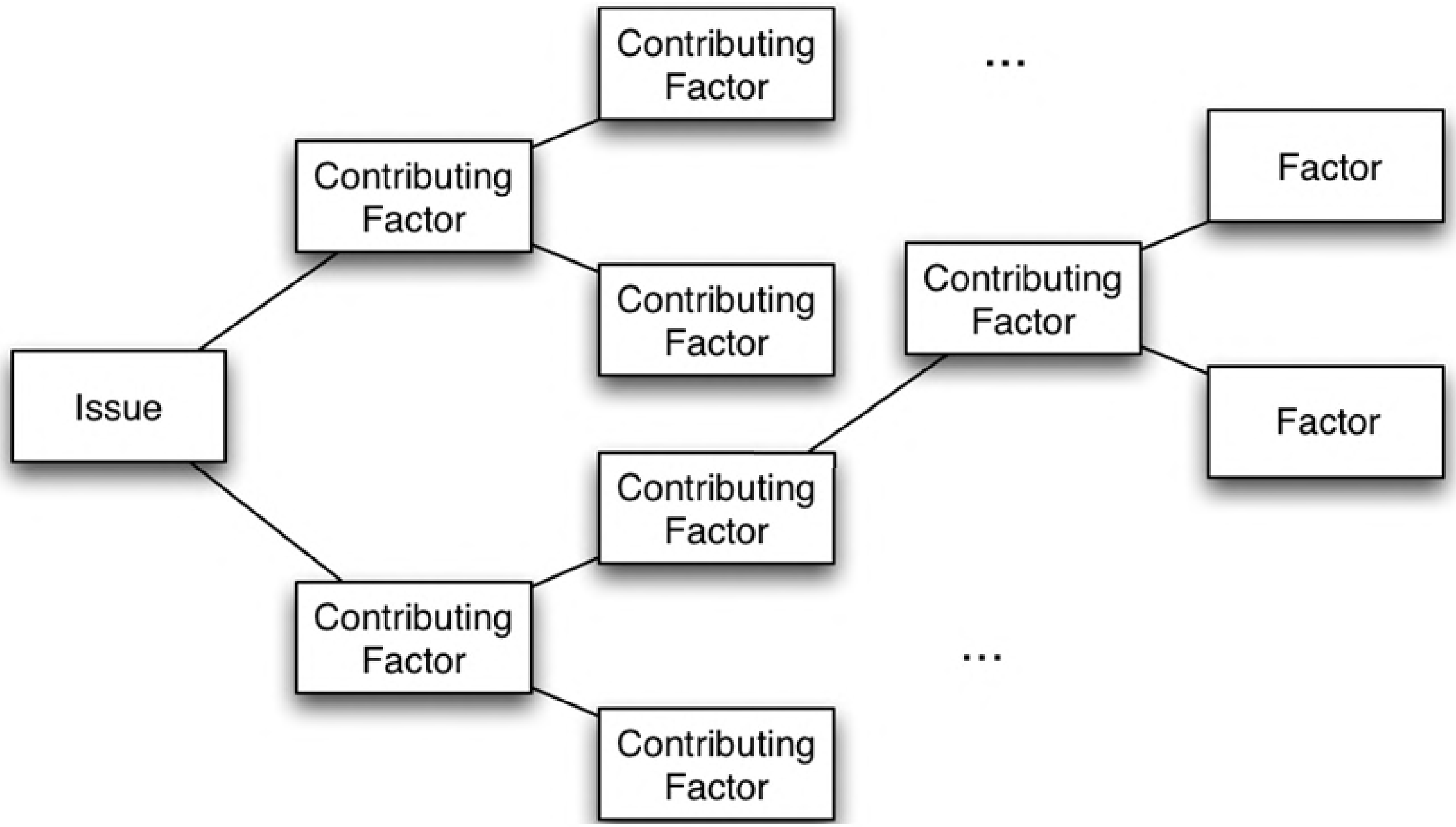
Dumas et al. (2018)

Why-Why Questions

- Recursively ask the question **“Why?”**
  - “Why does this issue happen?”
  - “What is/are the main sub-issue(s) leading to this issue?”
- After asking **five times** you should arrive at the root-cause
- At level 3 to 4 on should focus on **issues that can be solved**
- **Leaves** should be **fundamental** factors (atomic, cannot be explained otherwise)

Dumas et al. (2018)

Why-Why Diagram



Dumas et al. (2013)

Why-Why Diagram - Example

- **Issue** "Site engineers sometimes reject delivered equipment"
  why?
  - "Wrong equipment is delivered"
    why?
    - "equipment descriptions in supplier's catalog not accurate"
    - "miscommunication between site engineer and clerk"
      why?
      - "site engineer provides only brief/ inaccurate description of what they want"
      - "site engineer does not (always) see the supplier catalogs when making a request and does not communicate with the supplier"
        why?
        - » site engineer generally **does not have Internet connectivity**

Dumas et al. (2018)

Qualitative Process Analysis

- **This is not the end**

- There are tons of qualitative techniques which could be used for process analysis
  - Six Sigma techniques
  - Causal factor charts (capture context)
  - Theory of constraints (identify bottlenecks)
  - ...

Dumas et al. (2013)

Learning Goals of Today

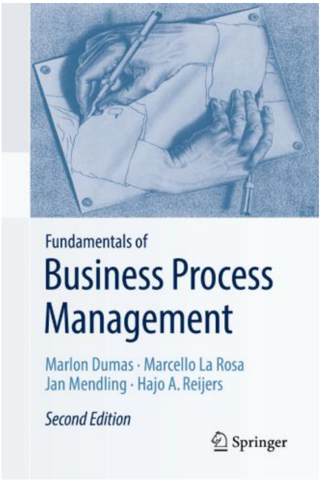
- Understand and be able to use value-added analysis
- Understand and be able to use waste analysis
- Understand and be able to use stakeholder analysis and issue documentation
- Understand and be able to use root cause analysis

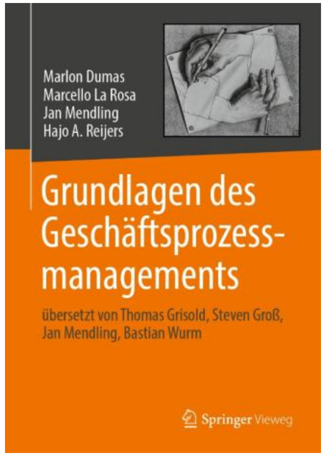
Recap Questions

- Why is process analysis an art and a science?
- What are different techniques of qualitative process analysis and what are their key benefits?
- Can you give an own example for the application of each qualitative process analysis technique?

Literature

- **Section 6:** Qualitative Process Analysis

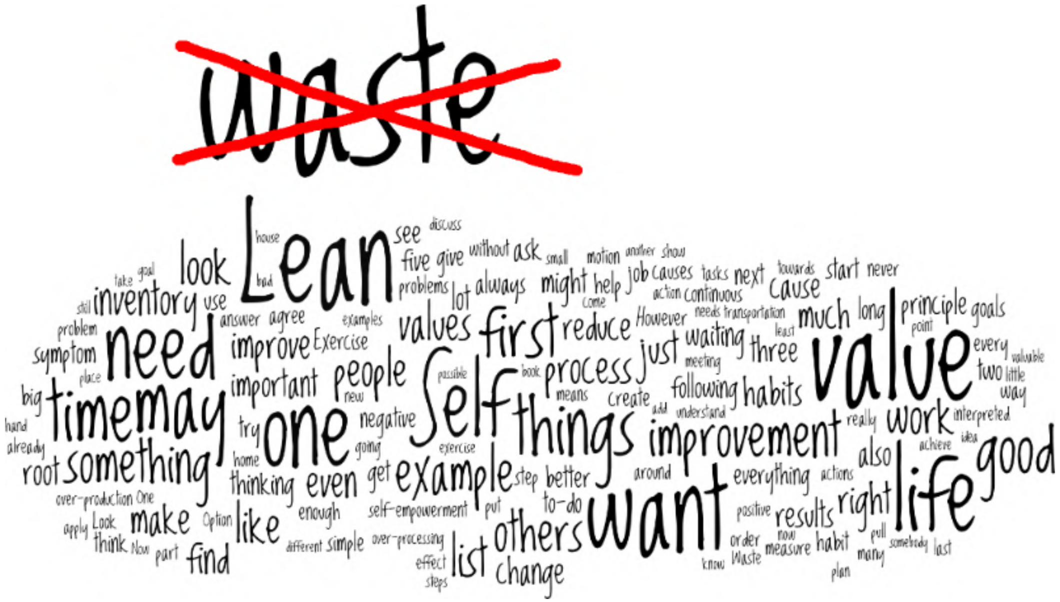




Roadmap

Introduction	10.04. Introduction	17.04. Process Identification			
Process Discovery & Modeling	24.04. Essentials of BPMN	01.05. Public Holiday	08.05. Advanced BPMN I	15.05. Advanced BPMN II	22.05. Process Discovery
Process Analysis & Redesign	29.05. Public Holiday	05.06. Qualitative Analysis	12.06. Quantitative Analysis	19.06. Public Holiday	26.06.2025 Process Redesign
Application of BPM	03.07. Process Technology	10.07. Process Monitoring and Mining	15.07. BPM as an Enterprise Capability		

Any Questions about Qualitative Process Analysis?



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dortmund

Chair of Enterprise Computing
Department of Computer Science

Business Process Management

Quantitative Process Analysis



Roadmap

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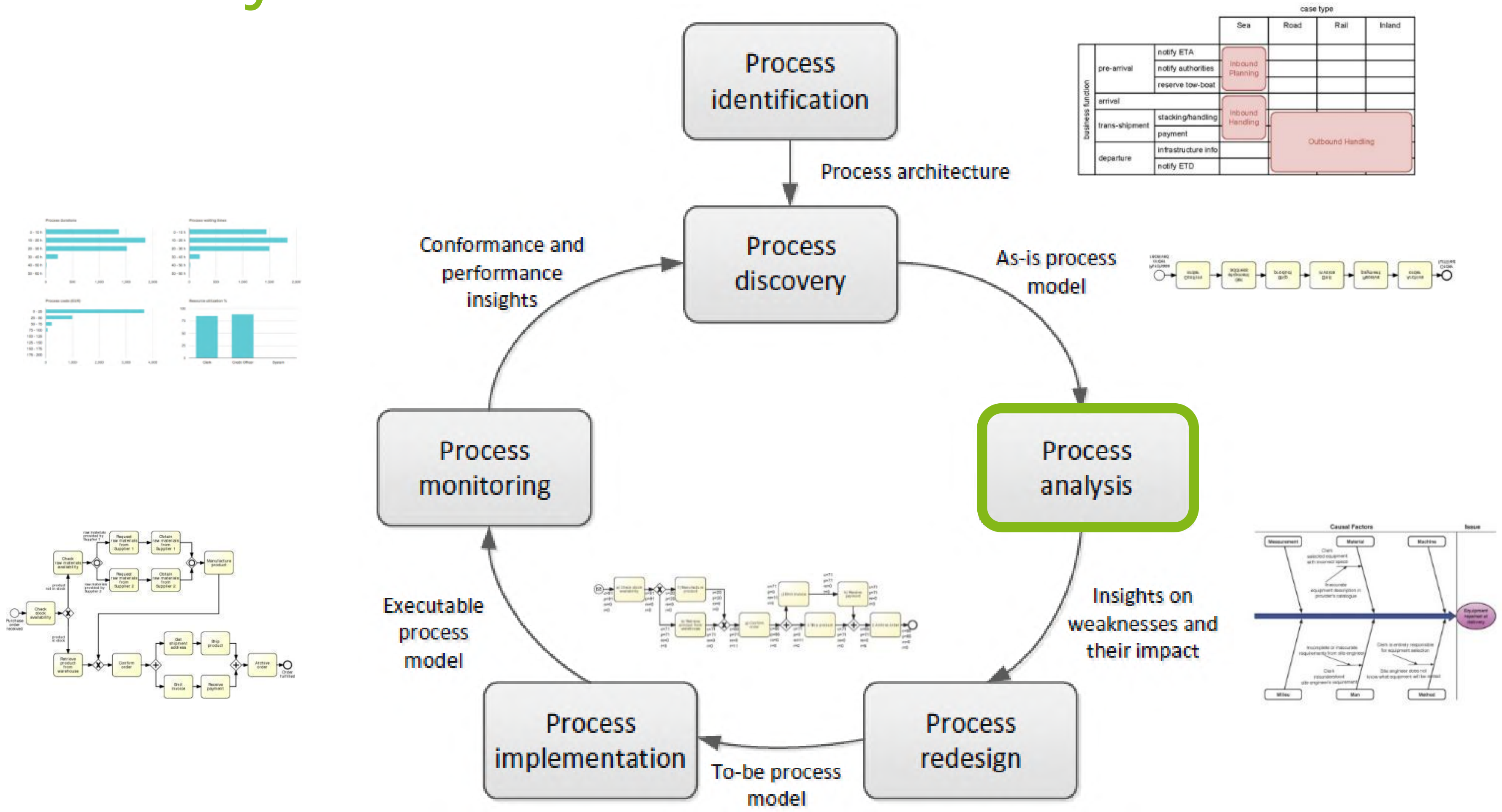
Learning Goals of Today

- Understand process performance measures
- Understand and be able to use flow analysis
- Understand and be able to use queueing analysis
- Understand process simulation

The BPM Lifecycle



Dumas et al. (2018)

Process Analysis

“Quality is free, but only to those who are willing to pay heavily for it.”

Tom DeMarco (1940–)

- Analyzing Processes is an **art** and a **science**
- **Qualitative analyses** use a range of **principles and techniques**
- **Quantitative analyses** use **measures** to judge a process
- There is no single way of producing good results

Dumas et al. (2018)

A Selection of Process Analysis Techniques

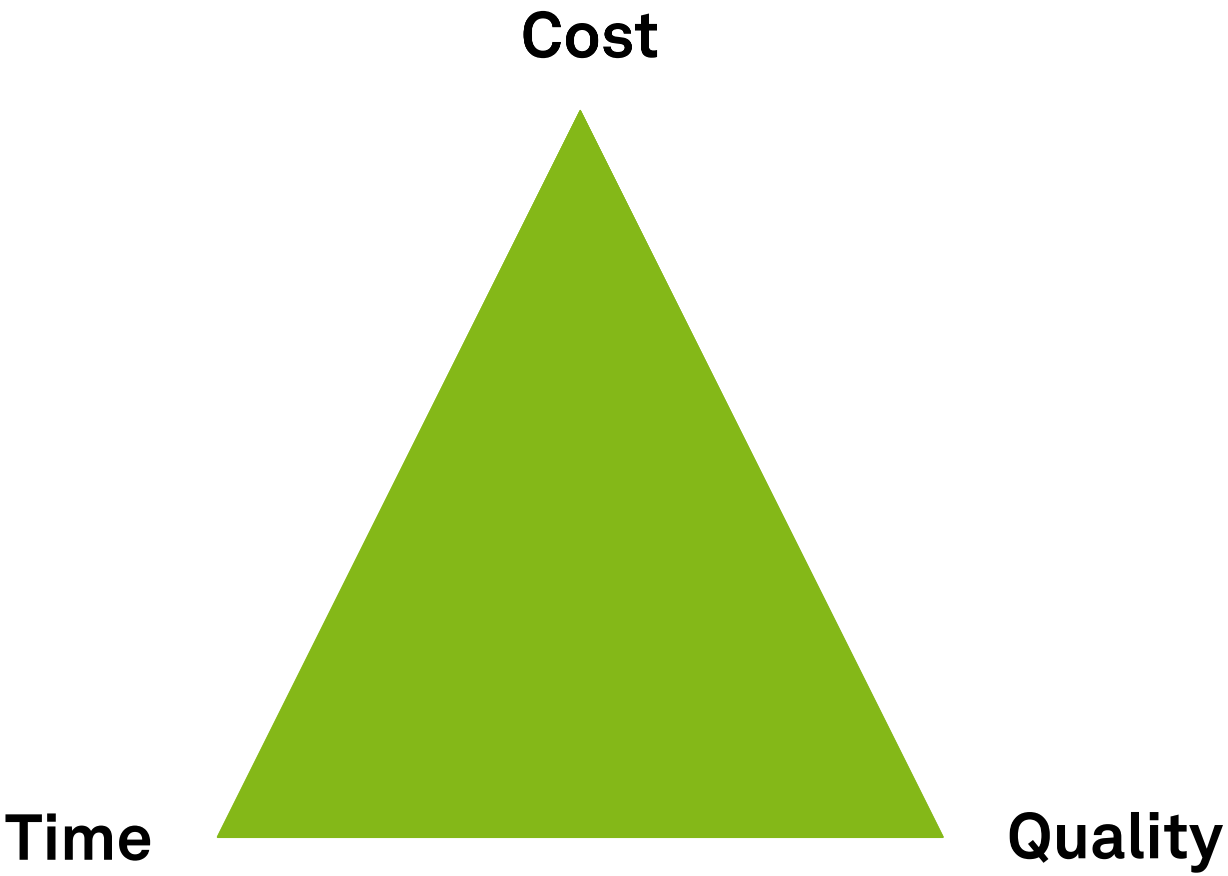
- **Qualitative** analysis
  - Value-added Analysis
  - Waste Analysis
  - Stakeholder Analysis
  - Root cause Analysis

- **Quantitative** Analysis
  - Flow Analysis
  - Queueing Analysis
  - Process Simulation

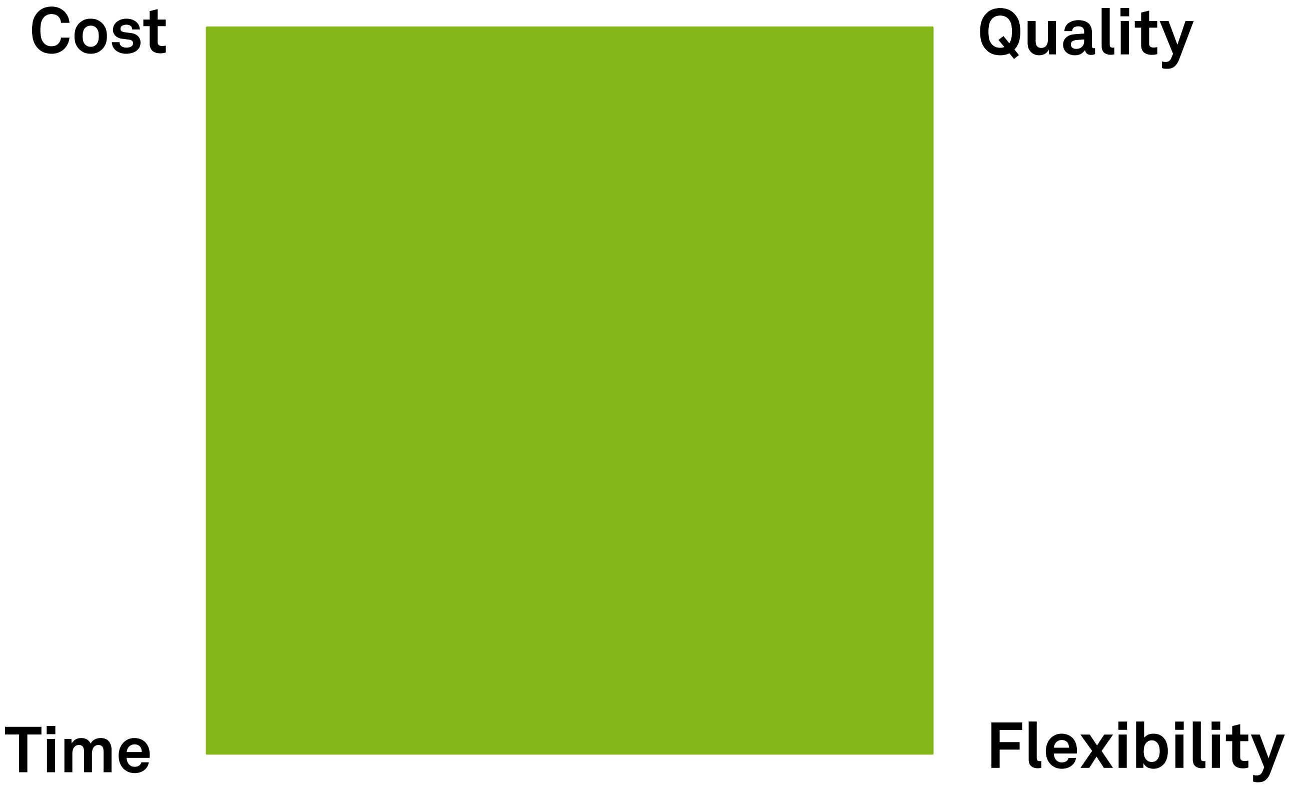
Dumas et al. (2018)

Process Performance Measures

Performance Dimensions

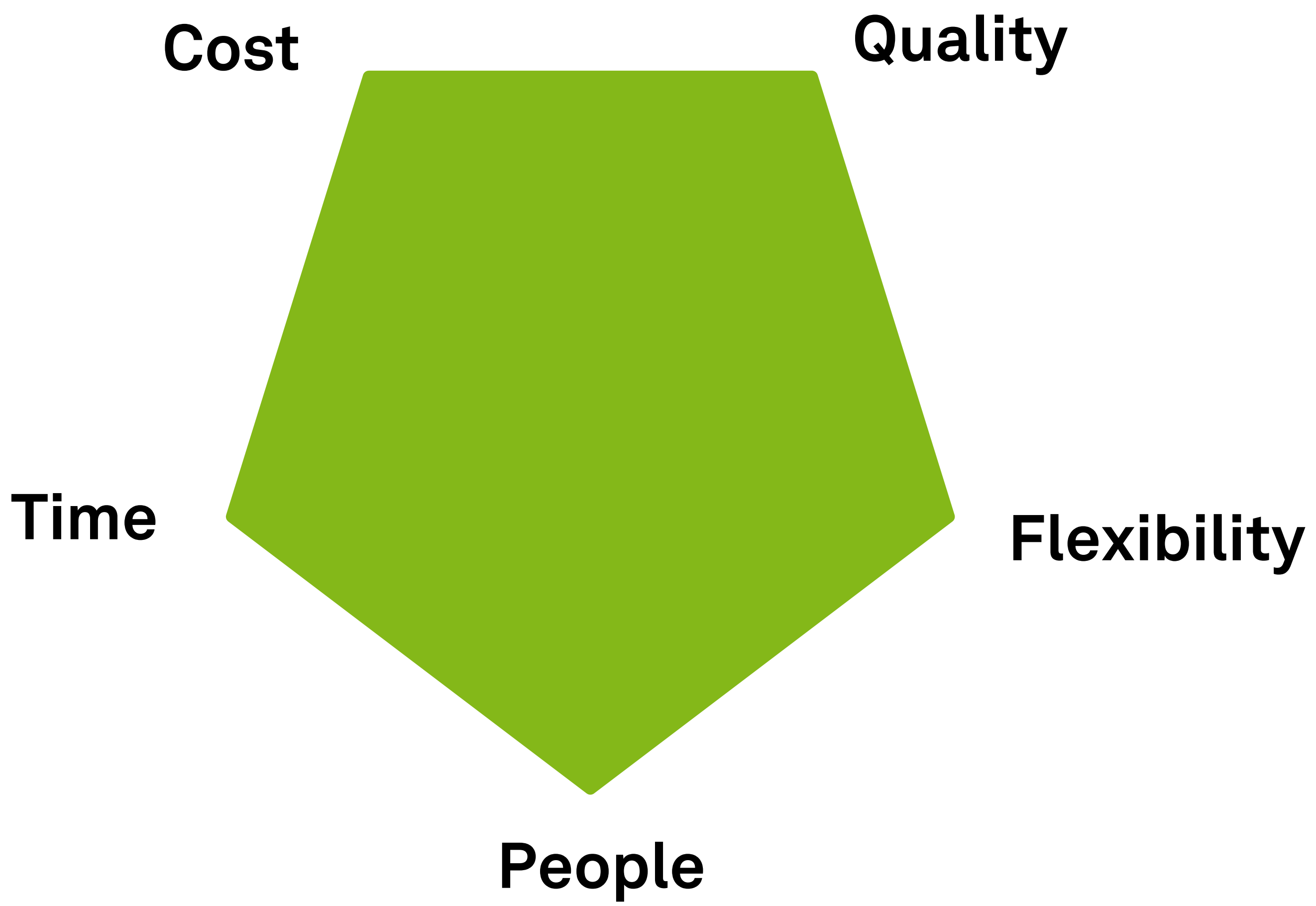


The Devil's Quadrangle: Process Performance Dimensions



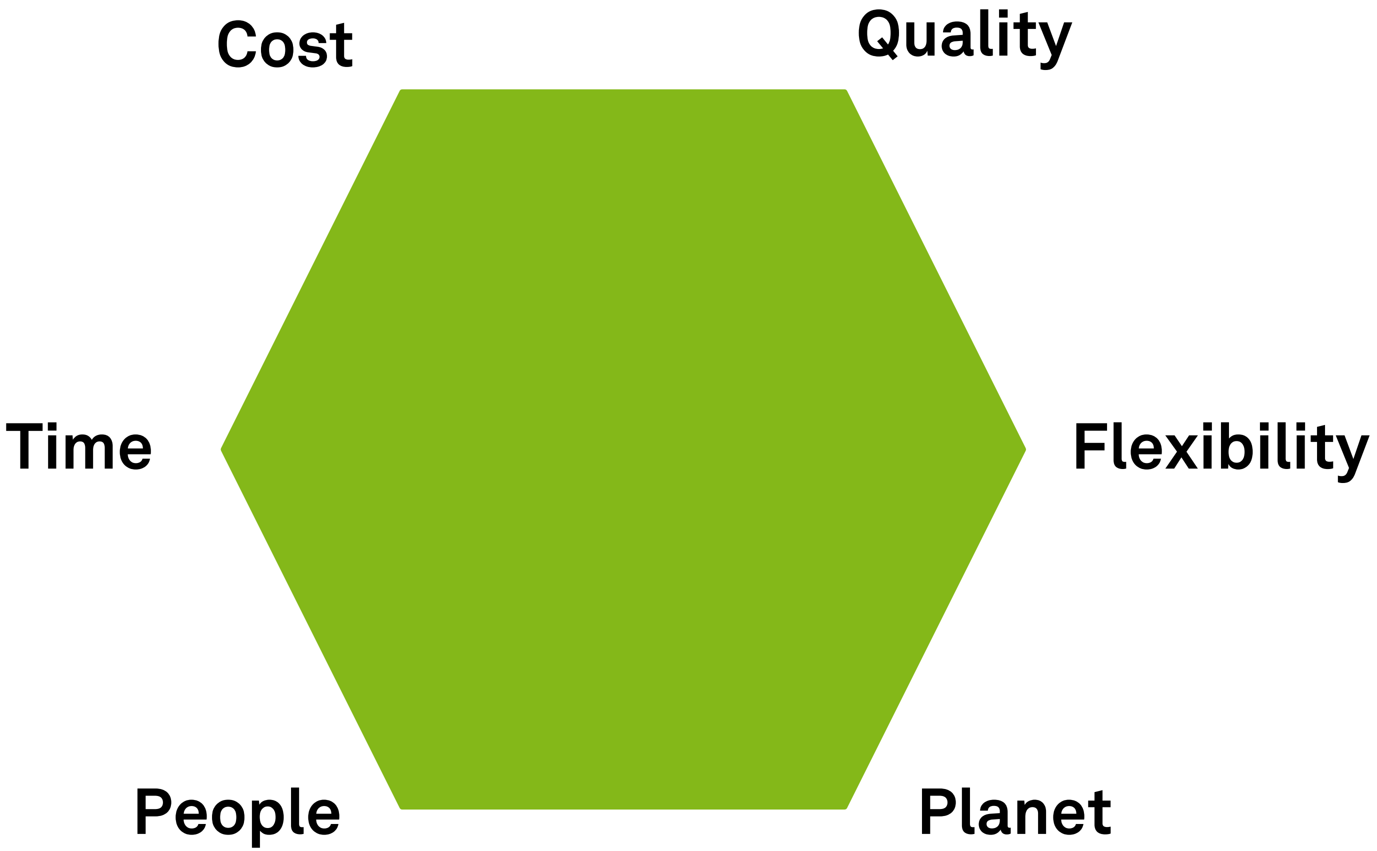
Dumas et al. (2018)

Goal Pentagon: More Process Performance Dimensions



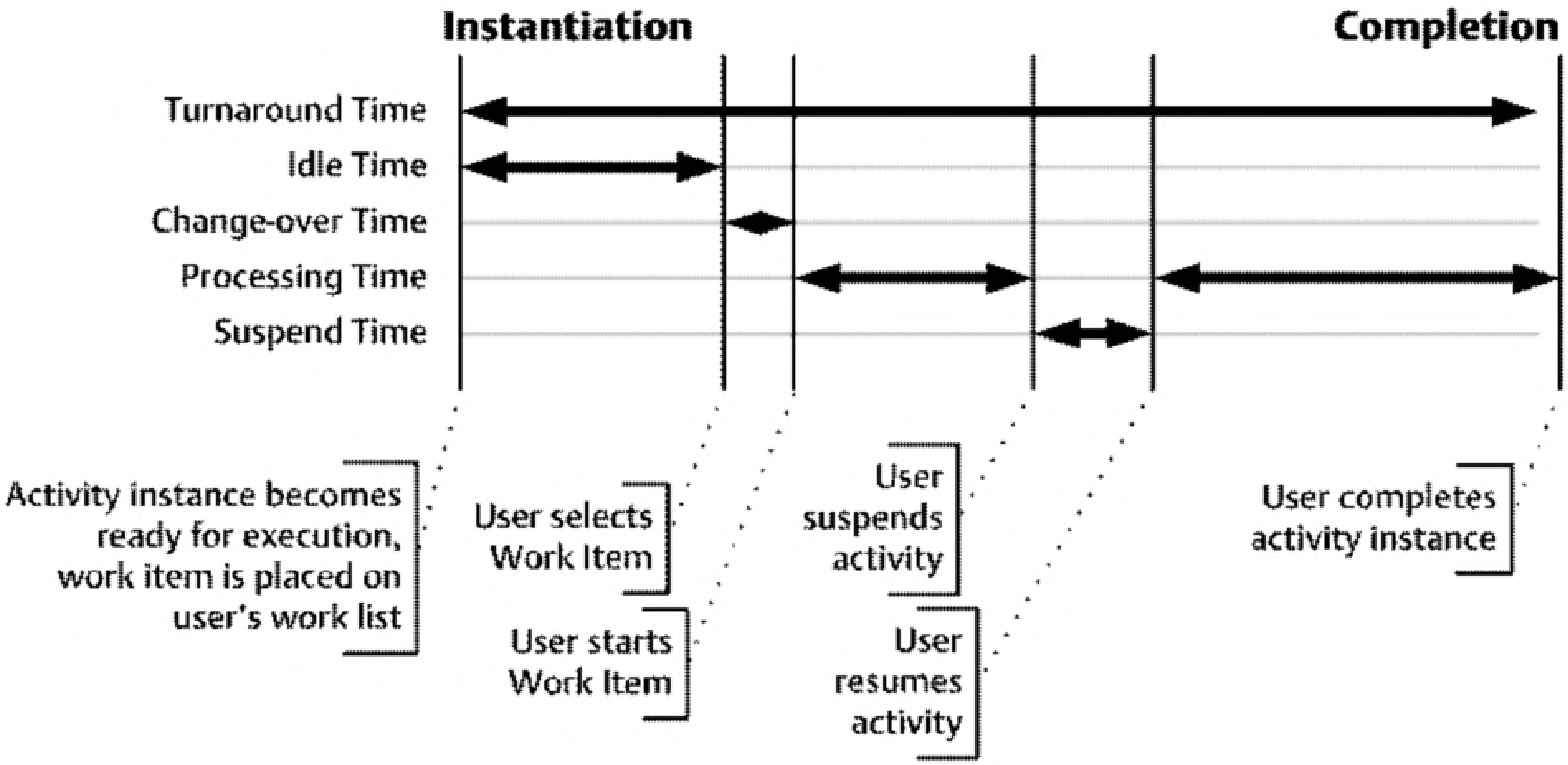
Francois et al. (2022)

Triple Bottom Line: More Performance Dimensions



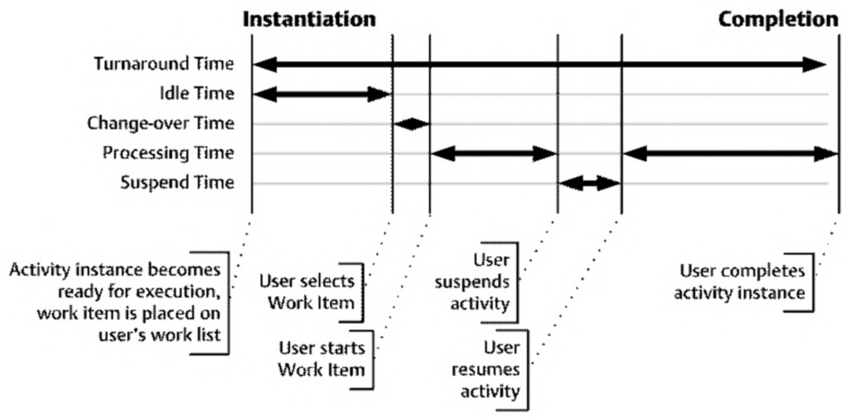
Dumas et al. (2018), Elkington (2018), François et al. (2022)

Time Data of a Process



zur Mühlen (2004)

Process Performance Measure: Time



- **Cycle time/ throughput time**
  – Time it takes to complete a process
- **Processing time**
  – Time a resource works on the process
- **Waiting time**
  – Time a resource does not work on the process
- **Time spent in non-value-adding tasks**
  – Time a resource does not work on value-adding activities in the process

– Typical scopes
  - average cycle time
  - maximal cycle time
  - meet service level agreement (SLA) time
  - find variation

Dumas et al. (2018)

Process Performance Measure: Cost

- Could also be turnover, revenue, or yield

- **Fixed**
  - Overhead cost that are hard to change
    - infrastructure and maintenance cost (utilization)

  → activity-based costing

- **Variable**
  - Operational cost related to productivity/ input & output
    - per execution
    - in particular labor cost and software services

Dumas et al. (2018)

Process Performance Measure: Quality

- **Internal quality**
  - As perceived by internal participants
    - level of control
    - level of variation/variants
    - context of work
    - errors

- **External quality**
  - As perceived by the client/ customer
  - Information the customer receives
    - Amount, relevance, quality, timeliness (could be time dimension as well)
  - e.g., churn rate or net promoter score

Dumas et al. (2018)

Process Performance Measure: Flexibility

- Ability to **react to changes**
- Flexibility of
  - **resources** (ability to execute many tasks/ new tasks)
  - **process** (ability to handle various cases and changing workloads)
  - **management** (ability to change rules/ allocation)
  - **organization** (ability to change the structure and responsiveness to demands of market or business partners

- **Run time** vs. **build time** flexibility

Dumas et al. (2018)

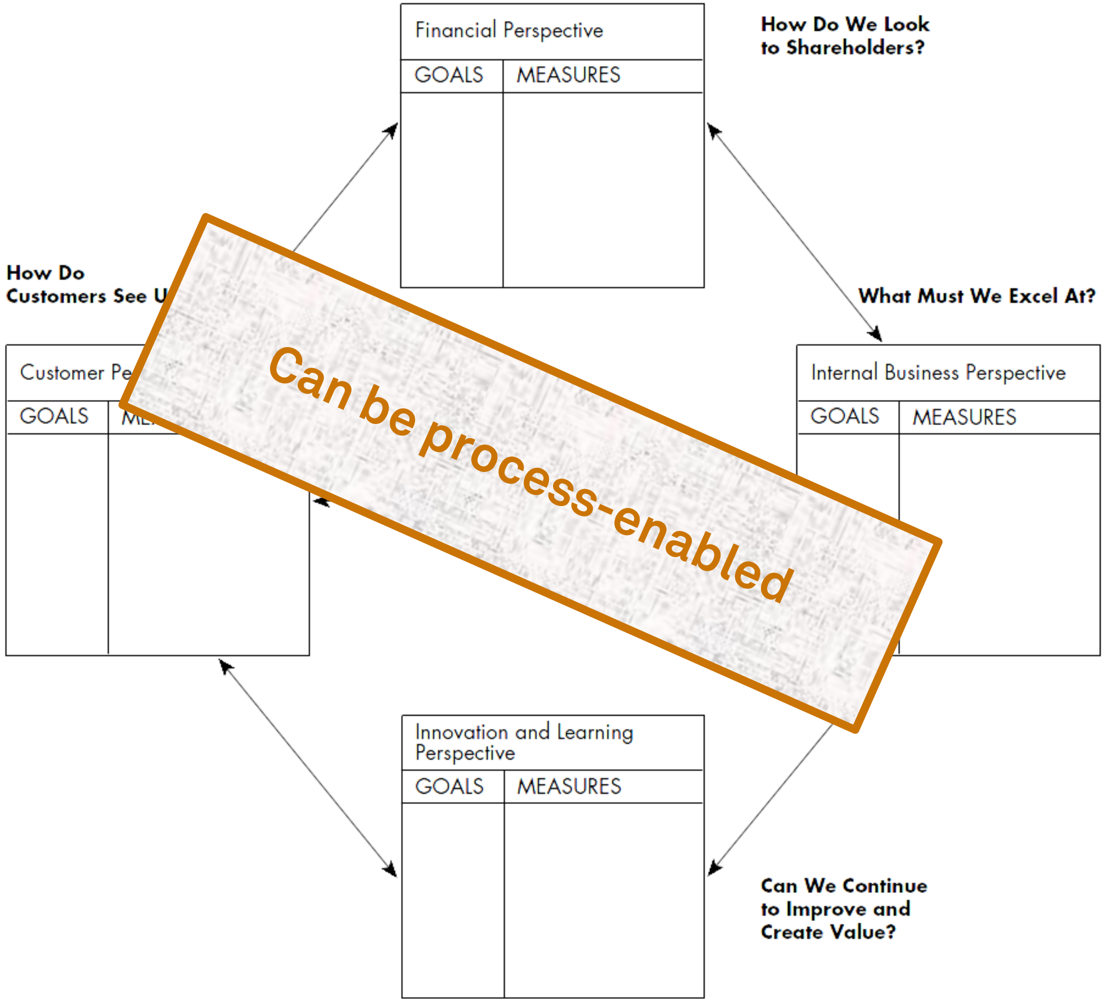
Definition of Process Performance Measures

1. Formulate **performance objectives** of the process at a high level, in the form of a **desirable state** that the process should ideally reach
   – "customers should be served in less than 30 minutes"
2. For each performance objective, identify the relevant **performance dimension(s)** and **aggregation function(s)**, and from there, define one or more **performance measures** for the objective in question
   – "percentage of customers served in less than 30 minutes"
3. Refine this performance measure using a **performance target** and attaching a **timeframe**
   – "99% of customers served in less than 30 minutes until 31/12/2023"

Dumas et al. (2018)

Balanced Scorecard

- Focuses on the **strategic agenda** of the organization
- Focused set of **measures** to monitor performance against **objectives**
- Mix of **financial** and **non-financial** data items
  - Originally: financial, customer, internal process, learning & growth
- Portfolio of initiatives to impact performance



Kaplan, Norton (1992)

Measure Derivation via Reference Models

- Reference model is used as a template to design the process performance measures

- Examples:
  - Process Classification Framework (PCF)
  - Information Technology Infrastructure Library (ITIL)
  - Supply Chain Operations Reference Model (SCOR)

Dumas et al. (2018)

Flow Analysis

Flow Analysis

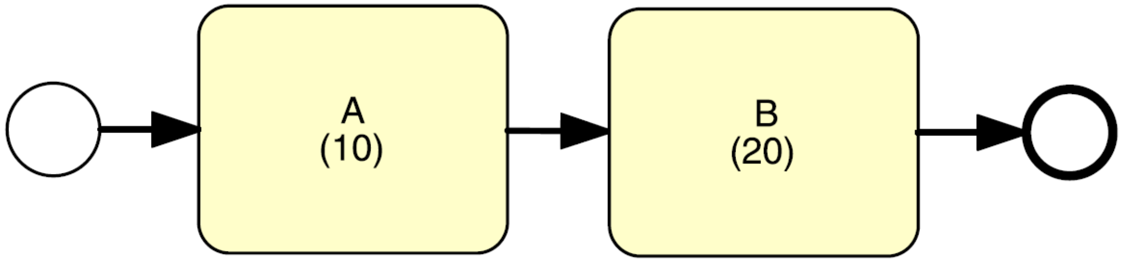
- Family of techniques to estimate **overall performance** based on some knowledge about the performance of its **tasks**

- **Cycle time analysis** calculates the average cycle time for a process or a process fragment

- Recap
  - Cycle time: Difference between a process start and end time

Dumas et al. (2018)

Performance of a Sequence



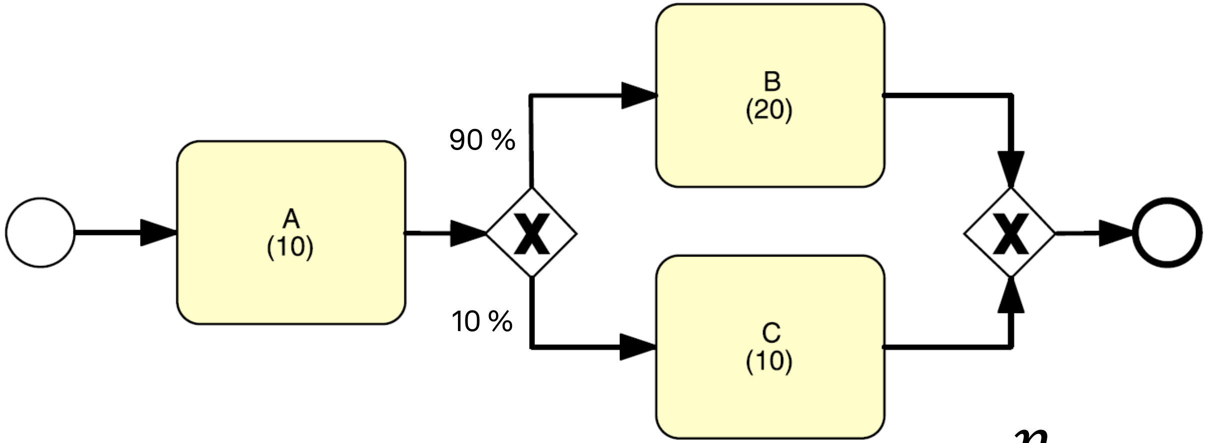
A + B = 10 + 20 = 30

$$CT = \sum_{i=1}^{n} T_i$$

$T_i = cycle\ time\ of\ a\ process\ fragment$
$CT = cycle\ time\ of\ a\ process$

Dumas et al. (2018)

Performance of a XOR Block



$A + p_1 * B + p_2 * C =$
$10 + 0.9 * 20 + 0.1 * 10 = 29$

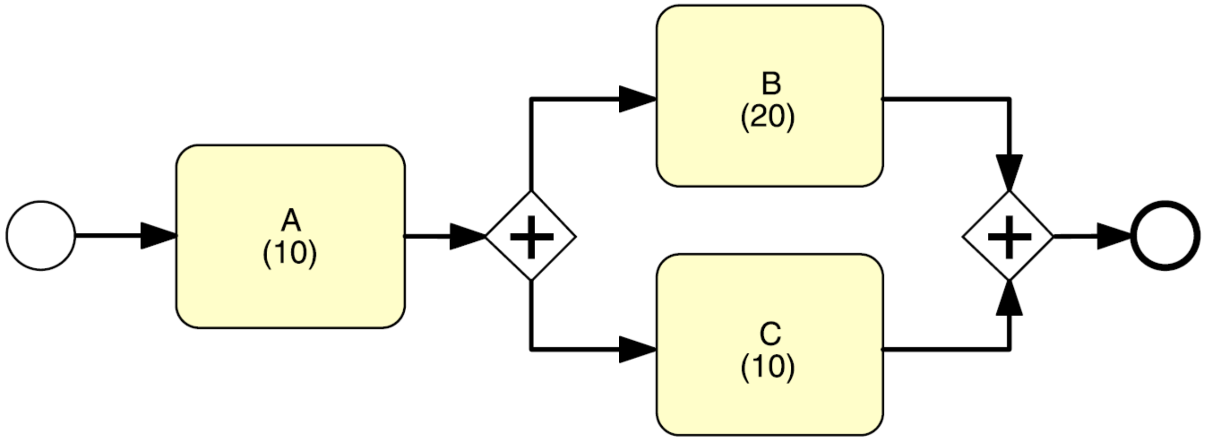
$$CT = \sum_{i=1}^{n} p_i \times T_i$$

$p_i = branching\ probability\ of\ a\ process\ fragment$

$T_i = cycle\ time\ of\ a\ process\ fragment$
$CT = cycle\ time\ of\ a\ process$

Dumas et al. (2018)

Performance of a AND Block



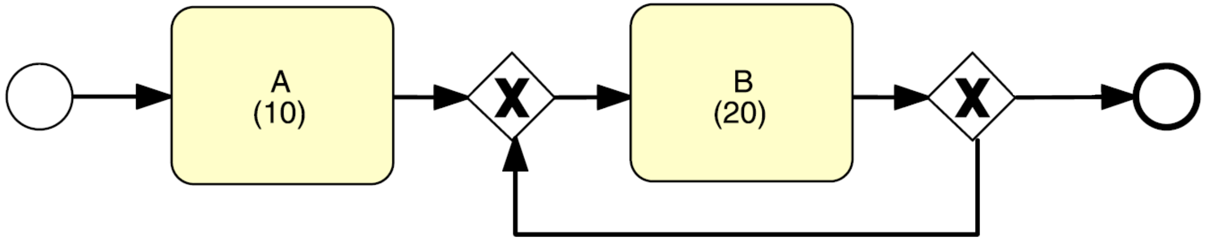
A + B = 10 + 20 = 30

$$CT = Max(T_1, T_2, \ldots, T_n)$$

$T_i = cycle\ time\ of\ a\ process\ fragment$
$CT = cycle\ time\ of\ a\ process$

Dumas et al. (2018)

Performance of a Loop Block (Rework)



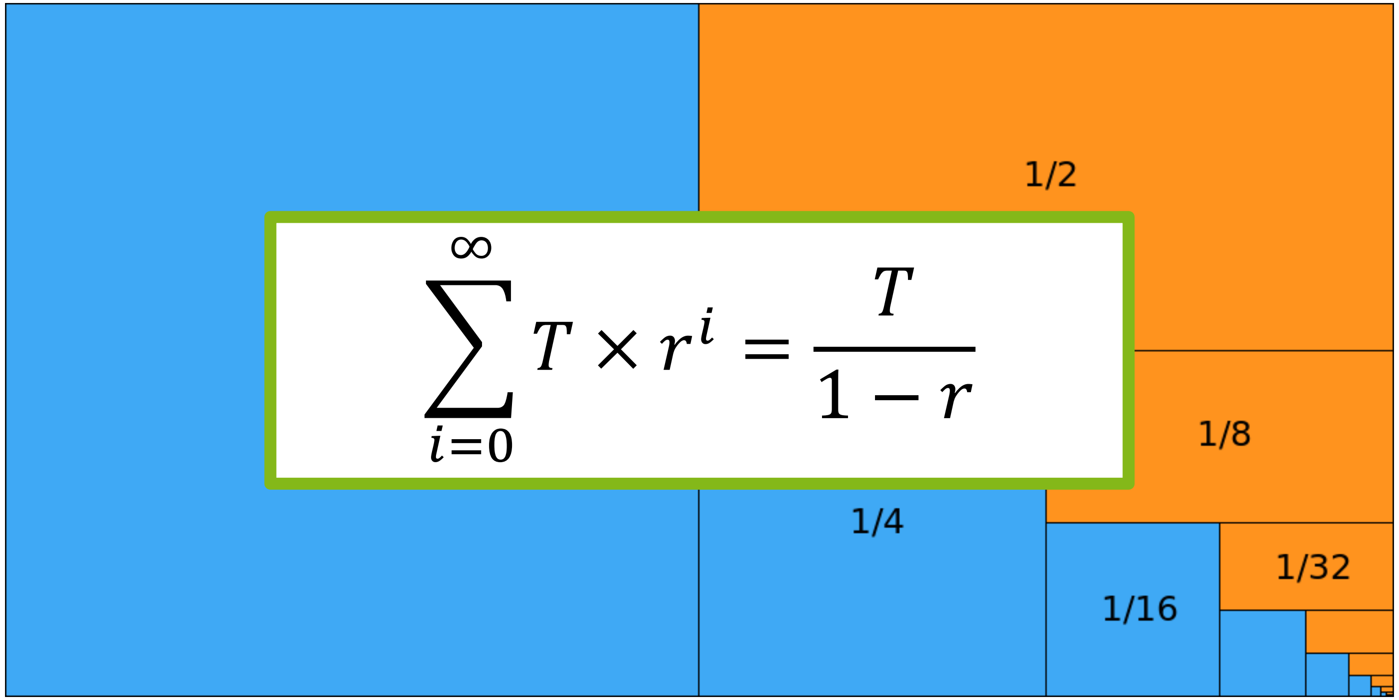
$A + B + B*r + B*r^2 + \ldots + B*r^N =$
$10 + 20 + 4 + 0{,}8 + \ldots = ???$

$$CT = \sum_{i=0}^{\infty} T \times r^i$$

$r = rework\ probability$
$T_i = cycle\ time\ of\ a\ process\ fragment$
$CT = cycle\ time\ of\ a\ process$

Dumas et al. (2018)

Geometric Series (50 % Example)

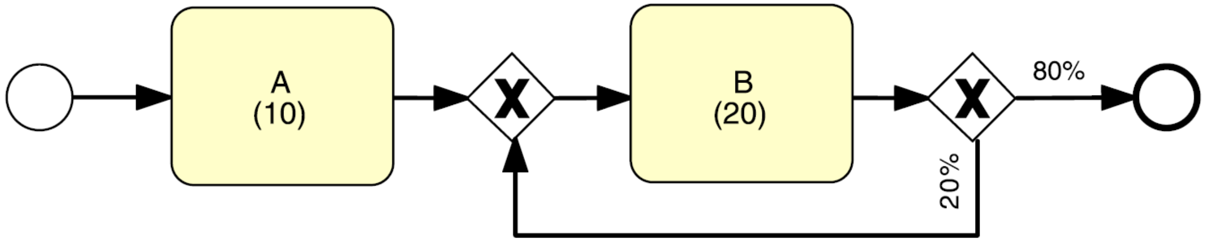


$$\sum_{i=0}^{\infty} T \times r^i = \frac{T}{1-r}$$

$r = rework\ probability$
$T_i = cycle\ time\ of\ a\ process\ fragment$
$CT = cycle\ time\ of\ a\ process$

Dumas et al. (2018)

Performance of a Loop Block (Rework)



A + B/(1-r) = 10 + 20/(1-0,2) = 35

$$CT = \frac{T}{1-r}$$

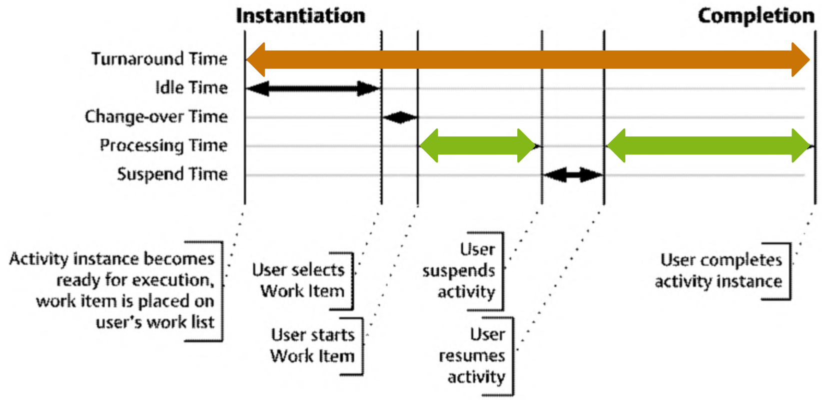
$r = rework\ probability$
$T_i = cycle\ time\ of\ a\ process\ fragment$
$CT = cycle\ time\ of\ a\ process$

Assumes uniform probabilities and durations for every rework loop

Dumas et al. (2018)

Cycle Time Efficiency

- Ratio of **theoretical processing time relative to overall processing time**
- Theoretical cycle time does not have waiting/ idle time



$$CTE = \frac{TCT}{CT}$$

$TCT = theoretical\ cycle\ time$

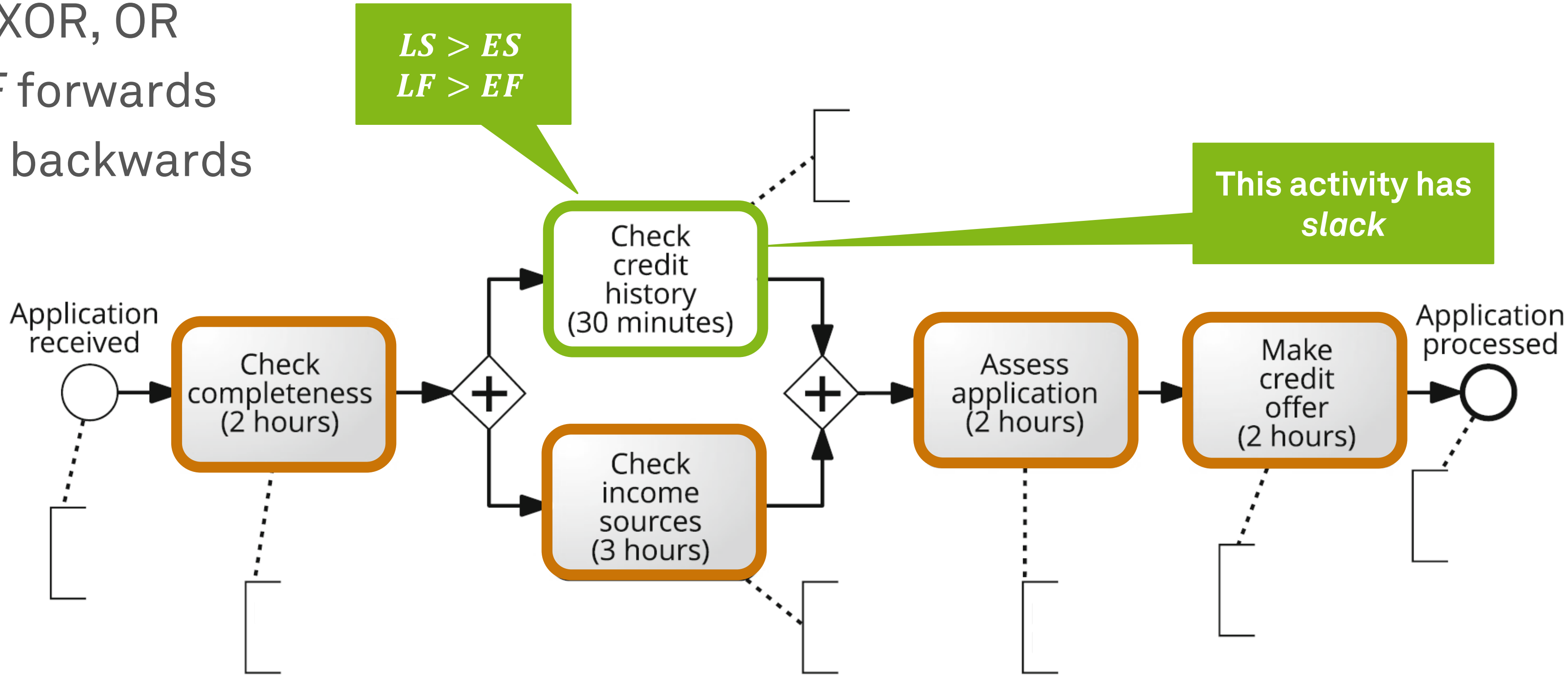
$CT = cycle\ time\ of\ a\ process$

$CTE = cycle\ time\ efficiency$

Dumas et al. (2018)

Critical Path Method

- Identify tasks that contribute to the theoretical cycle time
  - Remove or simplify XOR, OR
  - Calculate *ES* and *EF* forwards
  - Calculate *LS* and *LF* backwards



$ES = early\ start$
$EF = early\ finish$
$LS = late\ start$
$LF = late\ finish$

Dumas et al. (2018)

Little's Law for What-If Analysis

- Average **cycle time of a process**
- Average number of **new process instances per time unit** (arrival rate)
- Average number of **process instances active at a given time** (work-in-progress, started but not completed)
  - → If cycle time or arrival rate increases, WIP increases, and the process slows down
  - → If the arrival rate increases, the cycle time must decrease or the WIP increases as well

$$WIP = \lambda \times CT$$

Holds for any stable process

$CT = cycle\ time\ of\ a\ process$
$\lambda = arrival\ rate$
$WIP = work{-}in{-}process$

Arrival rate can be substituted by throughput (average completed instances per time unit)

Dumas et al. (2018)

Capacity and Bottlenecks

- **Theoretical capacity** is the maximum number of instances that can be completed per time unit given a set of resources

$$\mu_p = \frac{uc}{ul}$$

$p = resource\ pool$
$uc = unit\ capacity$
$ul = unit\ load$

- **Resource utilization** (occupation rate) of a pool is the share of working vs. idle workers

$$\rho_p = \frac{\lambda}{\mu_p}$$

$\lambda = arrival\ rate$

- The pool that reaches 100% resource utilization first is called a **bottleneck** and determines the overall theoretical capacity of the process

Dumas et al. (2018)

Summary Flow Analysis

- Flow Analysis is not limited to **time** analysis
- Can be used to estimate **error rates** or **capacity requirements**
- Can be used to estimate **cost**
  - Sequence, XOR, loop are calculated similar
  - AND is calculated like a sequence
- **Limitations**
  - Does not work on any process model (blocks required)
  - OR gateways are more complicated
  - Average cycle times must be known (interviews/ observations/ logs)
  - Load behavior (resource contention)

    → **When resource utilization > 90%, waiting time increases steeply**

Dumas et al. (2018)

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Queueing Analysis



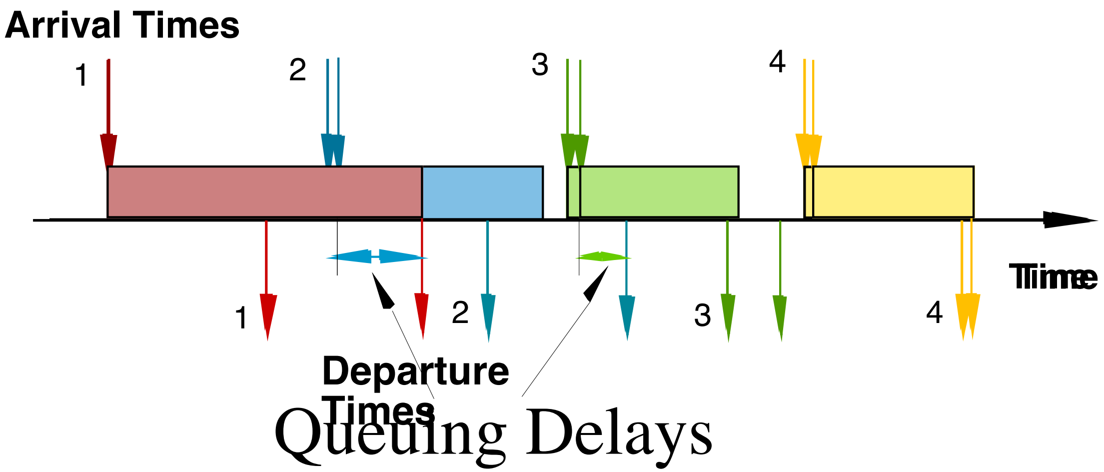

Reasons for Contention

- Not all traffic is deterministic
- Arrival rate is just an average
- Variable but **spaced apart** traffic
- **Burstiness** (variability in service times and/or inter-arrival times)
- **Job size** variation
- **High utilization** interference

Dumas et al. (2018)

Motivation Queueing Analysis

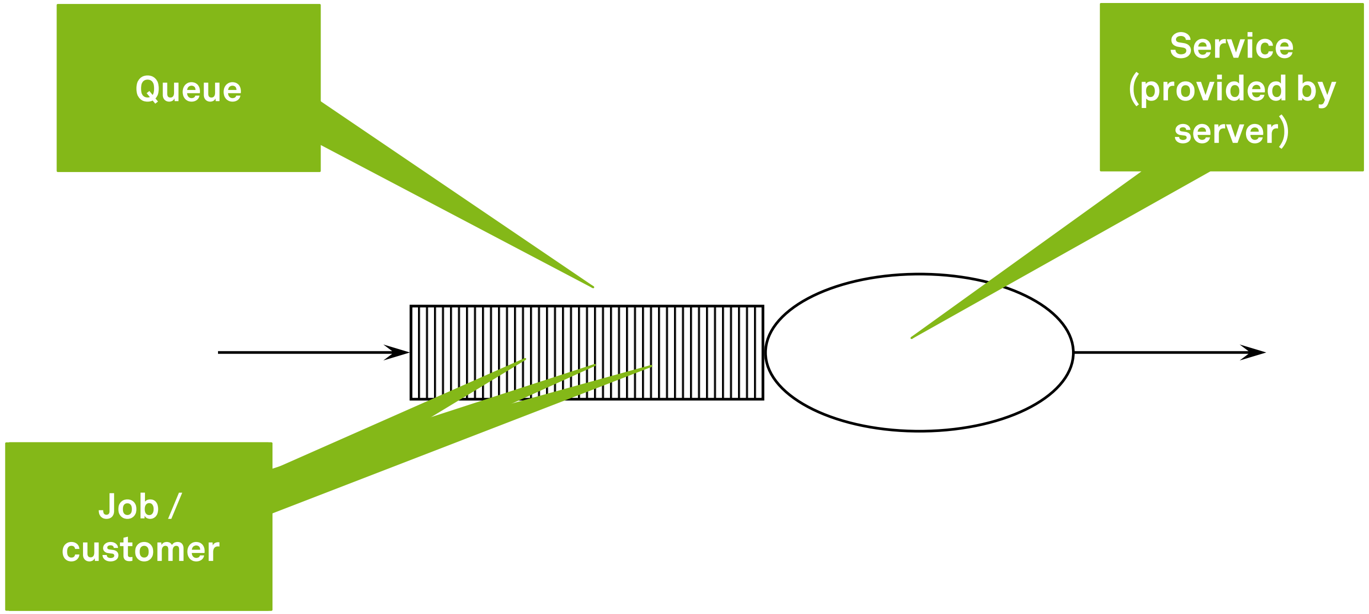
- **Capacity problems** are very common in industry and one of the main drivers of process redesign
  - Need to balance the cost of increased capacity against the gains of increased productivity and service
- Queuing and waiting time analysis is particularly important in **service systems**
  - Large costs of waiting and of lost sales due to waiting



Dumas et al. (2018)

Queues

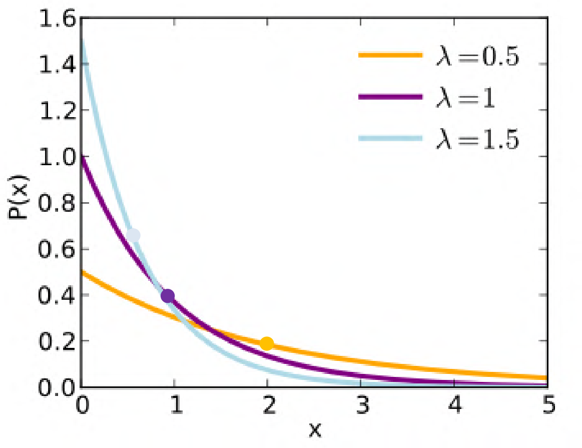
- Flow analysis **does not consider waiting times** due to resource contention
- **Queueing analysis** and **simulation** address these limitations and have a broader applicability

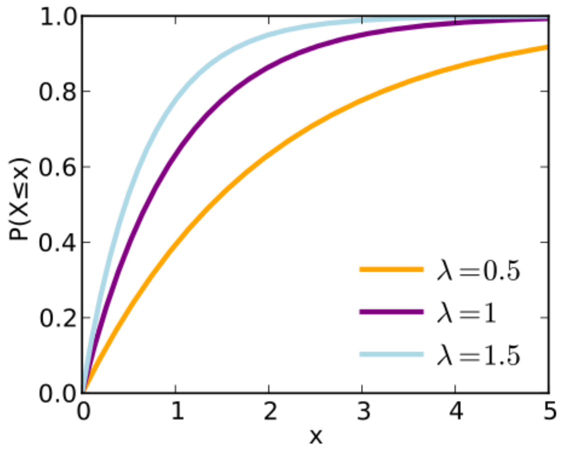


Dumas et al. (2018)

M/M/1 and M/M/c Queues

- Queue length in a system having 1 or *c* servers
- Arrivals and jobs are determined by a **Poisson process** (→ M)
  - **independent, identically distributed**, and (negative) **exponential**, uses **FIFO**



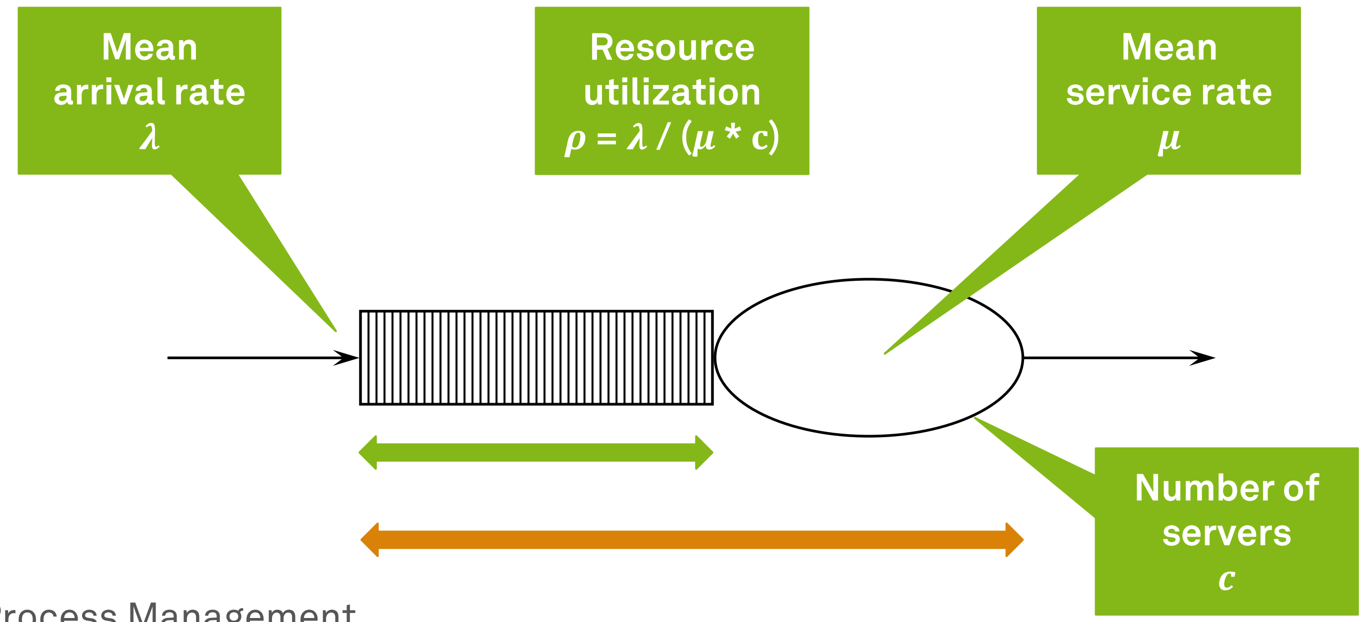


$$\lambda = (mean)\ arrival\ rate$$
$$\frac{1}{\lambda} = (mean)\ inter\text{-}arrival\ rate$$

Dumas et al. (2018)

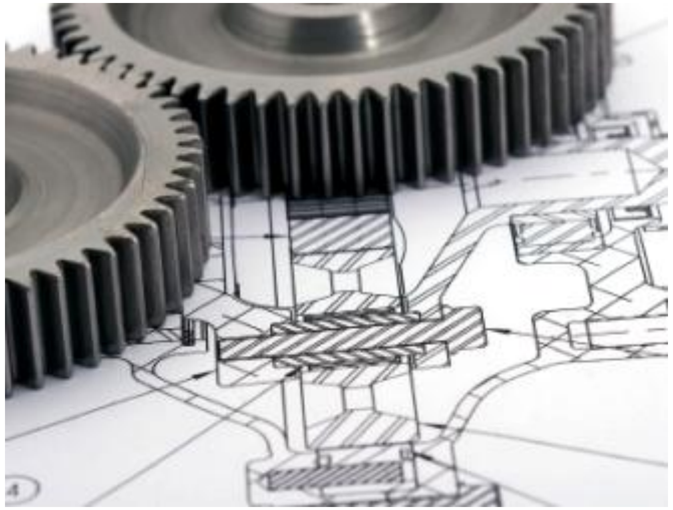
M/M/1 and M/M/c Systems

W_q = average time in queue (i.e. waiting time)
L_q = average number in queue (i.e. length of queue)
W = average time in system (i.e. cycle time)
L = average number in system (i.e. work in progress)



Dumas et al. (2018)

Example: Build-to-order



- *“Every 20 days, I receive 1 order.”*
- *“Every 10 days, I complete 1 order.”*
- *“What is my waiting time and my cycle time?”*
- *“What is my length of queue and my work-in-progress?”*

M/M/1 Systems

$\lambda = 0{,}05\ (per\ day)$

$\frac{1}{\lambda} = 20\ (every\ 20\ days)$

$\mu = 0{,}1\ (per\ day)$

$\frac{1}{\mu} = 10\ (every\ 10\ days)$

$\rho = 0{,}5\ (50\ \%)$

Length of queue

$$W_q = \frac{L_q}{\lambda}$$

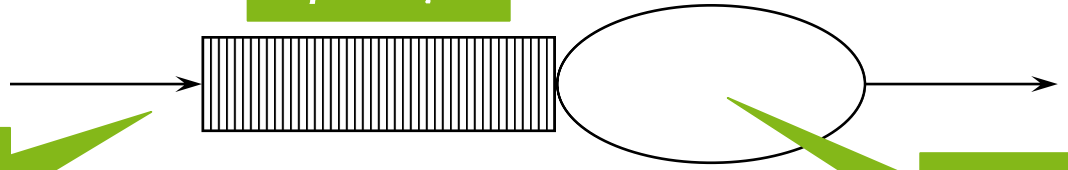
$$L_q = \frac{\rho^2}{(1-\rho)}$$

$$W = W_q + \frac{1}{\mu}$$

$$L = \lambda W$$

$0{,}5^2 / (1 - 0{,}5) = 0{,}5$ jobs

Occupation rate
$\rho = \lambda / \mu$



Mean arrival rate
λ

Mean service rate
μ

Dumas et al. (2018)

M/M/1 Systems

$$\lambda = 0{,}05 \ (per\ day)$$
$$\frac{1}{\lambda} = 20 \ (every\ 20\ days)$$
$$\mu = 0{,}1 \ (per\ day)$$
$$\frac{1}{\mu} = 10 \ (every\ 10\ days)$$

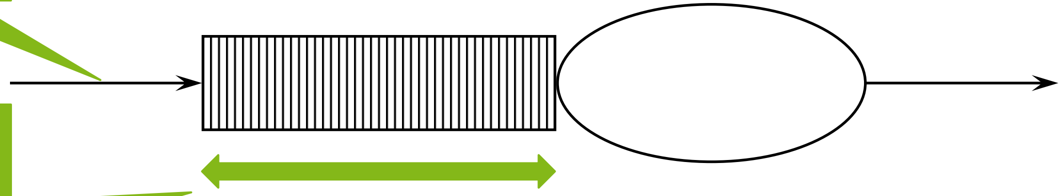
Waiting time

$$W_q = \frac{L_q}{\lambda}$$
$$L_q = \frac{\rho^2}{(1-\rho)}$$
$$W = W_q + \frac{1}{\mu}$$
$$L = \lambda W$$

0,5 / 0,05 = 10 days

$$\rho = 0{,}5 \ (50\ \%)$$
$$L_q = 0{,}5$$

Mean arrival rate λ



Length of queue L_q

Dumas et al. (2018)

M/M/1 Systems

$\lambda = 0{,}05 \ (per \ day)$

$\frac{1}{\lambda} = 20 \ (every \ 20 \ days)$

$\mu = 0{,}1 \ (per \ day)$

$\frac{1}{\mu} = 10 \ (every \ 10 \ days)$

cycle time

$$W_q = \frac{L_q}{\lambda}$$

$$L_q = \frac{\rho^2}{(1-\rho)}$$

$$W = W_q + \frac{1}{\mu}$$

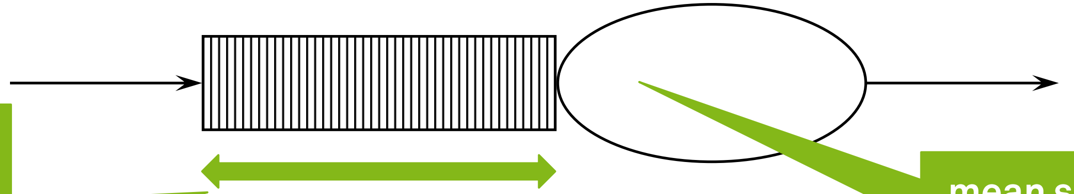
$$L = \lambda W$$

10 + 10 = 20 days

$\rho = 0{,}5 \ (50 \ \%)$

$L_q = 0{,}5$

$W_q = 10$



time in queue W_q

mean service rate μ

Dumas et al. (2018)

M/M/1 Systems

$\lambda = 0{,}05 \ (per\ day)$

$\frac{1}{\lambda} = 20 \ (every\ 20\ days)$

$\mu = 0{,}1 \ (per\ day)$

$\frac{1}{\mu} = 10 \ (every\ 10\ days)$

$\rho = 0{,}5 \ (50\ \%)$

$L_q = 0{,}5$

$W_q = 10$

$W = 20$

$$W_q = \frac{L_q}{\lambda}$$

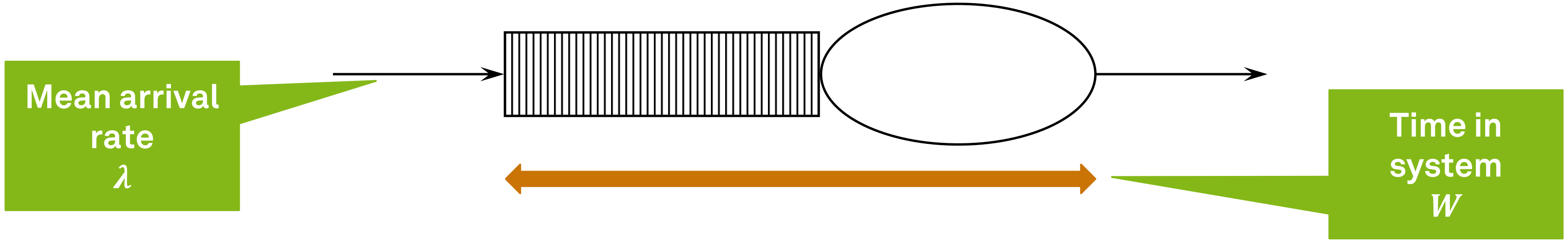
$$L_q = \frac{\rho^2}{(1-\rho)}$$

$$W = W_q + \frac{1}{\mu}$$

$$L = \lambda W$$

Work-in-progress

0,05 * 20 = 1 job



Dumas et al. (2018)

M/M/c Systems

$$W_q = \frac{L_q}{\lambda}$$

L_q is rather complex for M/M/c systems

$$W = W_q + \frac{1}{\mu}$$

$$L = \lambda W$$

$$L_q = \frac{(\lambda/\mu)^c \rho}{c!(1-\rho)^2 \left(\frac{(\lambda/\mu)^c}{c!(1-\rho)} + \sum_{n=0}^{c-1} \frac{(\lambda/\mu)^n}{n!} \right)}$$

Thank God, nobody does M/M/c without computers...

Dumas et al. (2018)

Summary Queueing Analysis

- Does not have to use negative exponential distribution

- **Limitations**
  - Generally not applicable when system includes parallel activities
    - One activity at a time
  - Requires case-by-case mathematical analysis
    - Queueing networks are available though
  - Assumes steady-state (valid only for long-term analysis)
  - Does not work for cost or quality measures

Dumas et al. (2018)

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Process Simulation

Process Simulation

- Uses **hypothetical instances** to create a **synthetic log**
- **Model** the process
- Enhance the process model with **simulation information**
  - Processing time (probability distribution for each task)
    - fixed / exponential (mean) / normal (mean + std dev)
  - Cost / value-add of tasks
  - Branching probabilities
  - Resource assignment
- **Run** the simulation
  - Define mean arrival rate (and distribution)
  - Define start date
  - Define end date / duration / number of instances

Dumas et al. (2018)

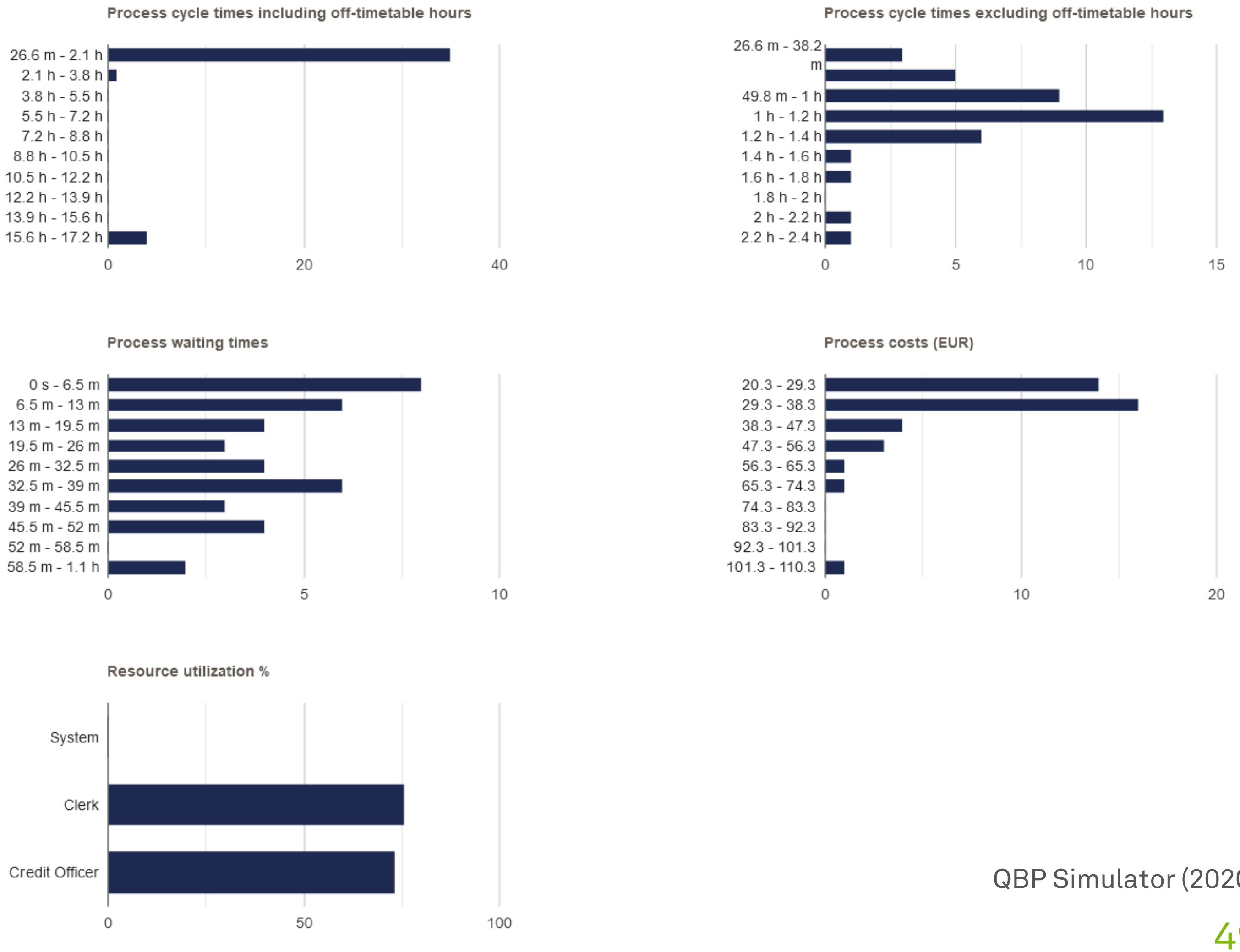
Process Simulation

- **Analyze** the simulation outputs
  - Process duration and cost stats and histograms
  - Waiting times (per activity)
  - Resource utilization (per resource)
- **Repeat** iteratively for alternative scenarios (caveat: stochasticity)
- **Cross-check** against expert advice
- **Limitations**
  - Results are based on models and **simplified** assumptions
  - Results depend on **accuracy** of input numbers, do sensitivity checks
  - **Human** resources are not robots, do credibility checks with stakeholders

Dumas et al. (2018)

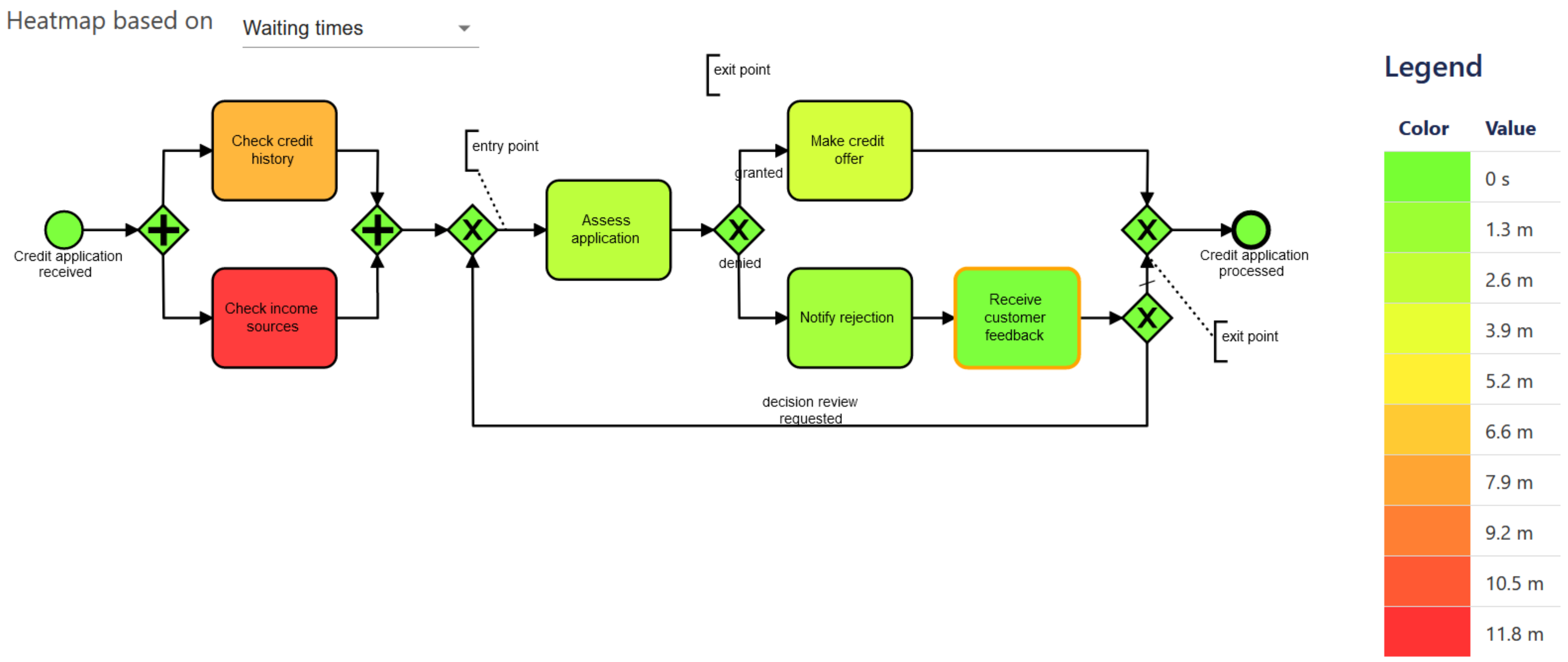
BIMP Example Output

- Other tools include
  - Appian
  - ARIS
  - IBM BPM
  - Oracle Business Process Analysis
  - Signavio Process Manager



QBP Simulator (2020)

BIMP Process Model Heatmap



QBP Simulator (2020)

Learning Goals of Today

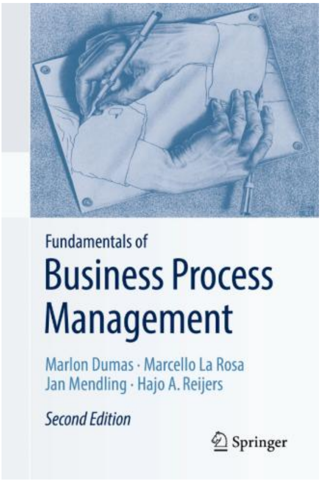
- Understand process performance measures
- Understand and be able to use flow analysis
- Understand and be able to use queueing analysis
- Understand process simulation

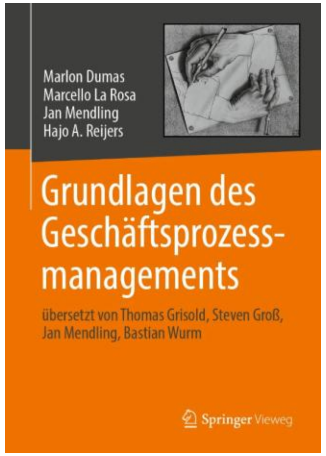
Recap Questions

- What are the four process performance dimensions of the Devil’s Quadrangle? What are exemplary measures for each dimension and how could the quadrangle be extended in the future?
- What do we measure with flow analysis? What does it abstract from?
- What is resource contention and how can we measure it?

Literature

- **Section 2:** Process Identification
- **Section 7:** Quantitative Process Analysis

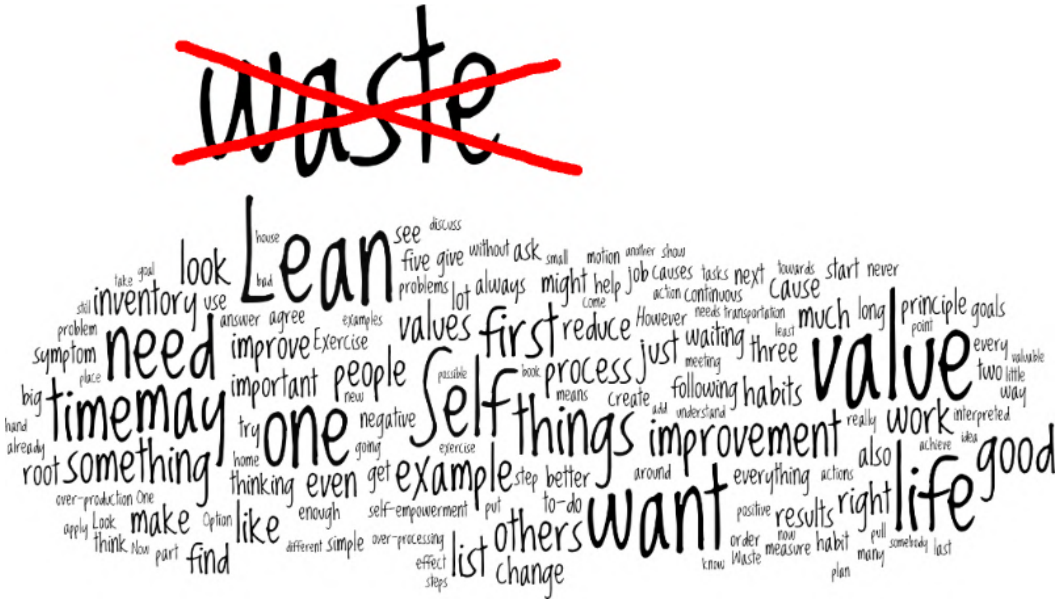




Roadmap

Introduction	10.04. Introduction	17.04. Process Identification			
Process Discovery & Modeling	24.04. Essentials of BPMN	01.05. Public Holiday	08.05. Advanced BPMN I	15.05. Advanced BPMN II	22.05. Process Discovery
Process Analysis & Redesign	29.05. Public Holiday	05.06. Qualitative Analysis	12.06. Quantitative Analysis	19.06. Public Holiday	26.06.2025 Process Redesign
Application of BPM	03.07. Process Technology	10.07. Process Monitoring and Mining	15.07. BPM as an Enterprise Capability		

Any Questions about Quantitative Process Analysis?



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Business Process Management

Process Redesign

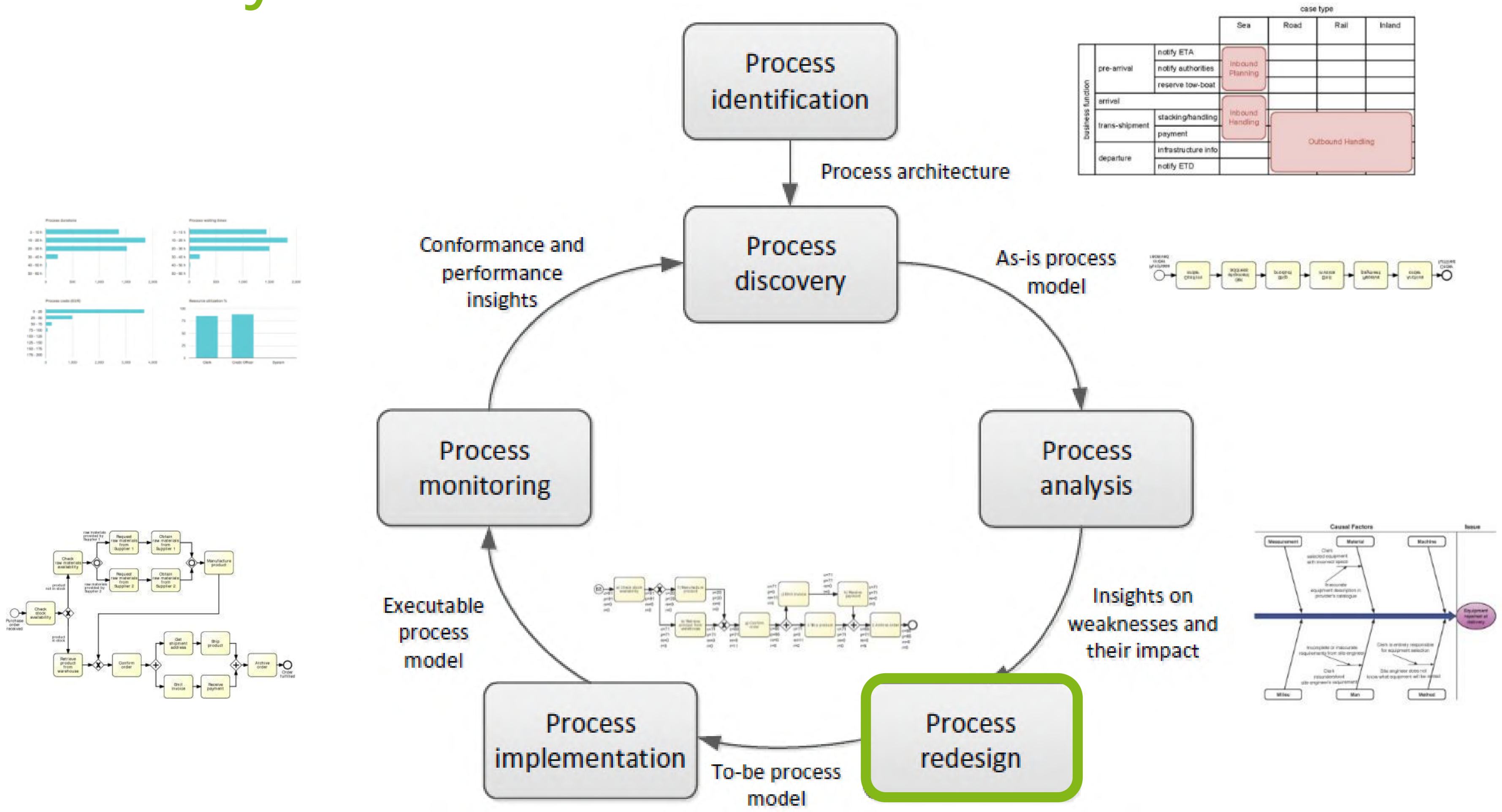
Roadmap

Introduction	10.04. Introduction	17.04. Process Identification			
Process Discovery & Modeling	24.04. Essentials of BPMN	01.05. Public Holiday	08.05. Advanced BPMN I	15.05. Advanced BPMN II	22.05. Process Discovery
Process Analysis & Redesign	29.05. Public Holiday	05.06. Qualitative Analysis	12.06. Quantitative Analysis	19.06. Public Holiday	26.06.2025 Process Redesign
Application of BPM	03.07. Process Technology	10.07. Process Monitoring and Mining	15.07. BPM as an Enterprise Capability		

Learning Goals of Today

- Understand the essence of process redesign
- Understand the process redesign orbit
- Know and be able to use redesign techniques

The BPM Lifecycle



Dumas et al. (2018)

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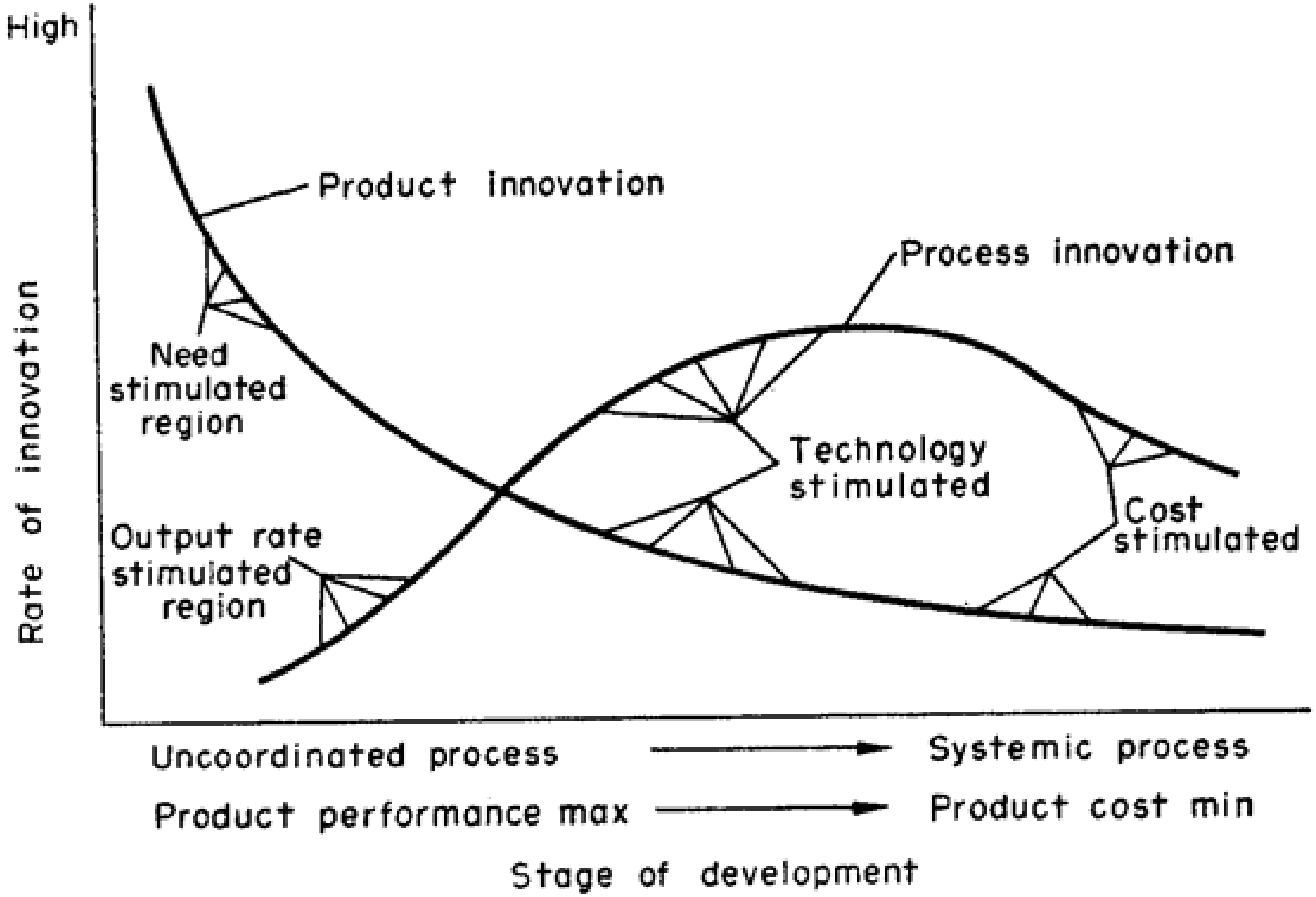
Process Redesign




Product and Process Innovation



Utterback & Abernathy (1975)

Motives of Process Redesign

- Process innovation
  - Complements product innovation
  - **Positive and proactive motive to process redesign**
- Organizational entropy
  - All processes evolve over time
  - Processes become more complex, and performance deteriorates
  - **Negative and reactive motive to process redesign**
- Process redesign does not only focus on existing but also on new processes

Dumas et al. (2018)

Aspects of Process Redesign

- Internal or external **customers**
- Business process **operation** view
  - Number and nature of activities
- Business process **behavior** view
  - Ordering and scheduling of activities
- **Organization** and the participants
  - Organization structure (groups, roles, etc.) and organization population (individuals)
- **Information** used and created
- **Technology** used
- **External environment**

Dumas et al. (2018)

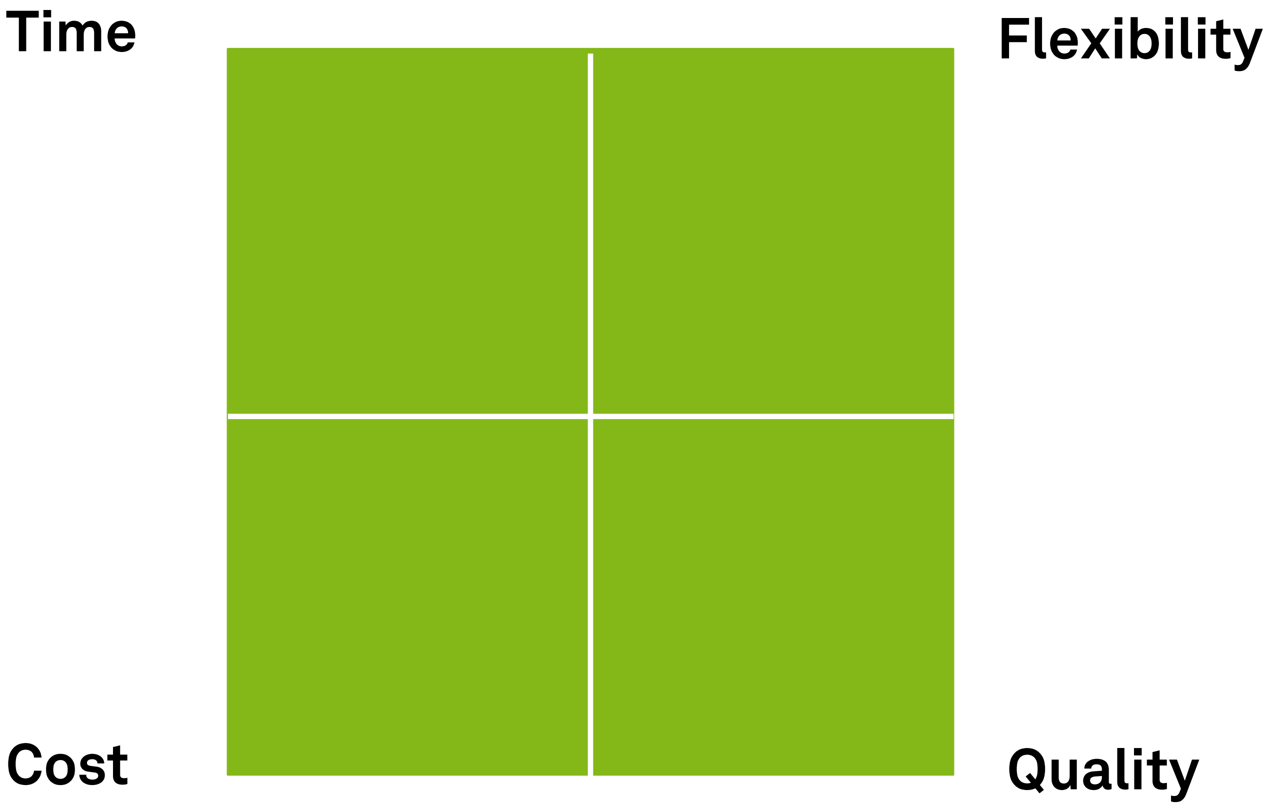
Problems with Redesign

- **Complex** structures
  - not transparent: badly controllable, fault intolerant
- System **proliferation**
  - overlap, difficult information exchange, uncomfortable, bad maintainability
- **Inflexible** structure
  - change = pulling a house of cards

- Process was perhaps **once designed**, but **organic evolution degraded it** for years…

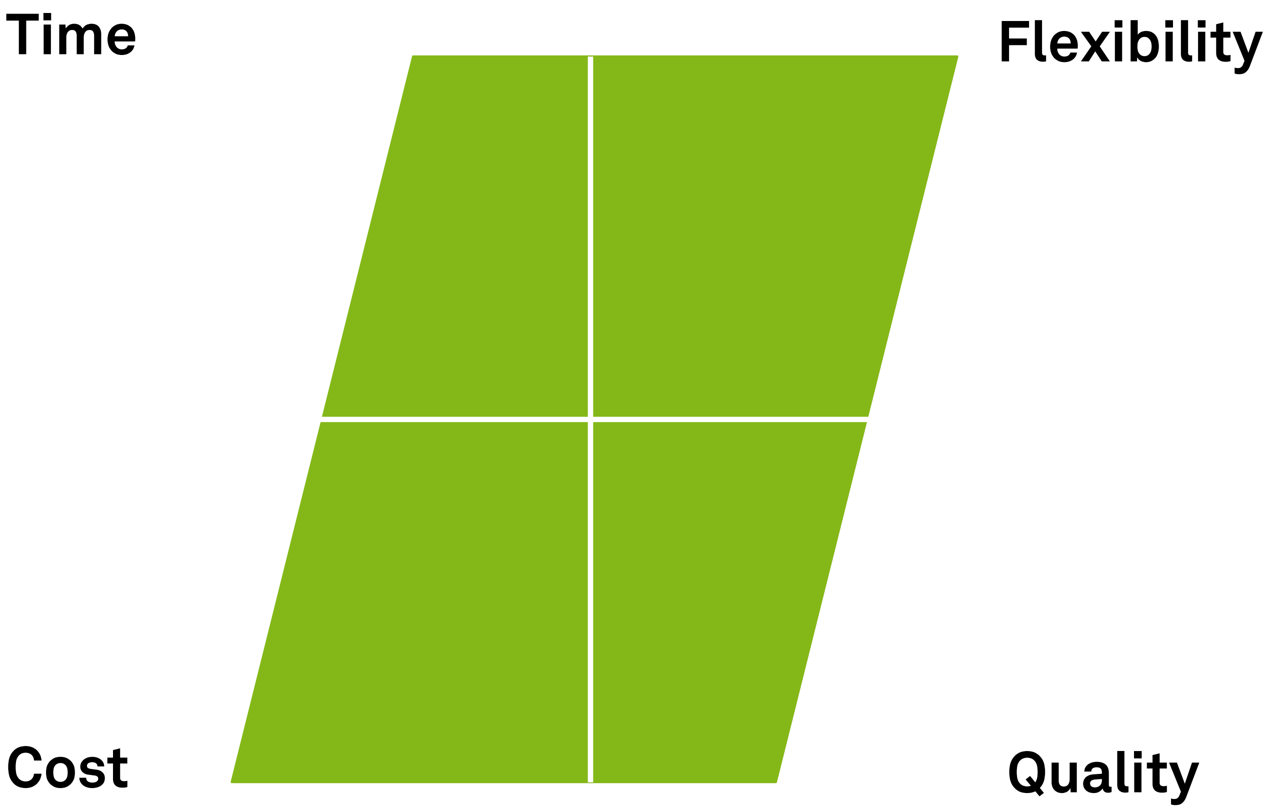
Dumas et al. (2018)

The Devil's Quadrangle



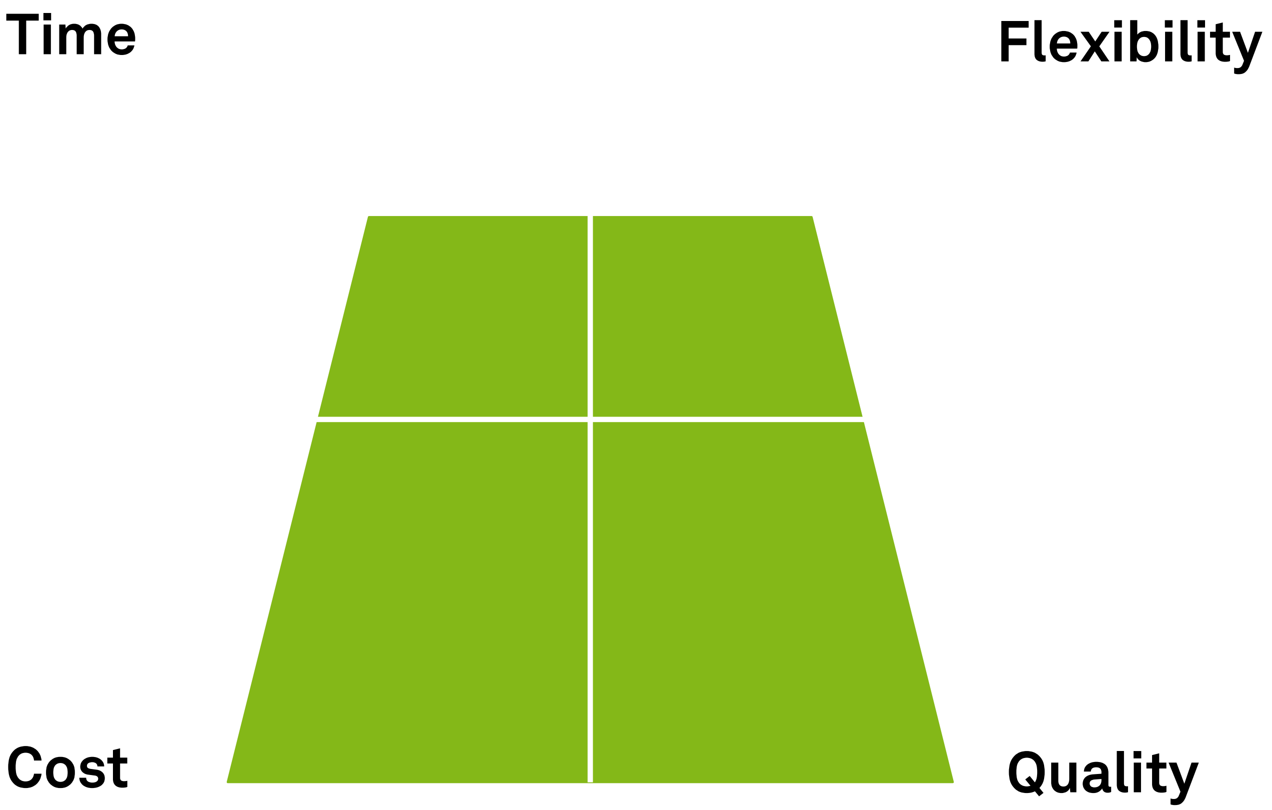
Dumas et al. (2018)

The Devil's Quadrangle – Example



Dumas et al. (2018)

The Devil's Quadrangle – Example



Dumas et al. (2018)

How to Redesign?

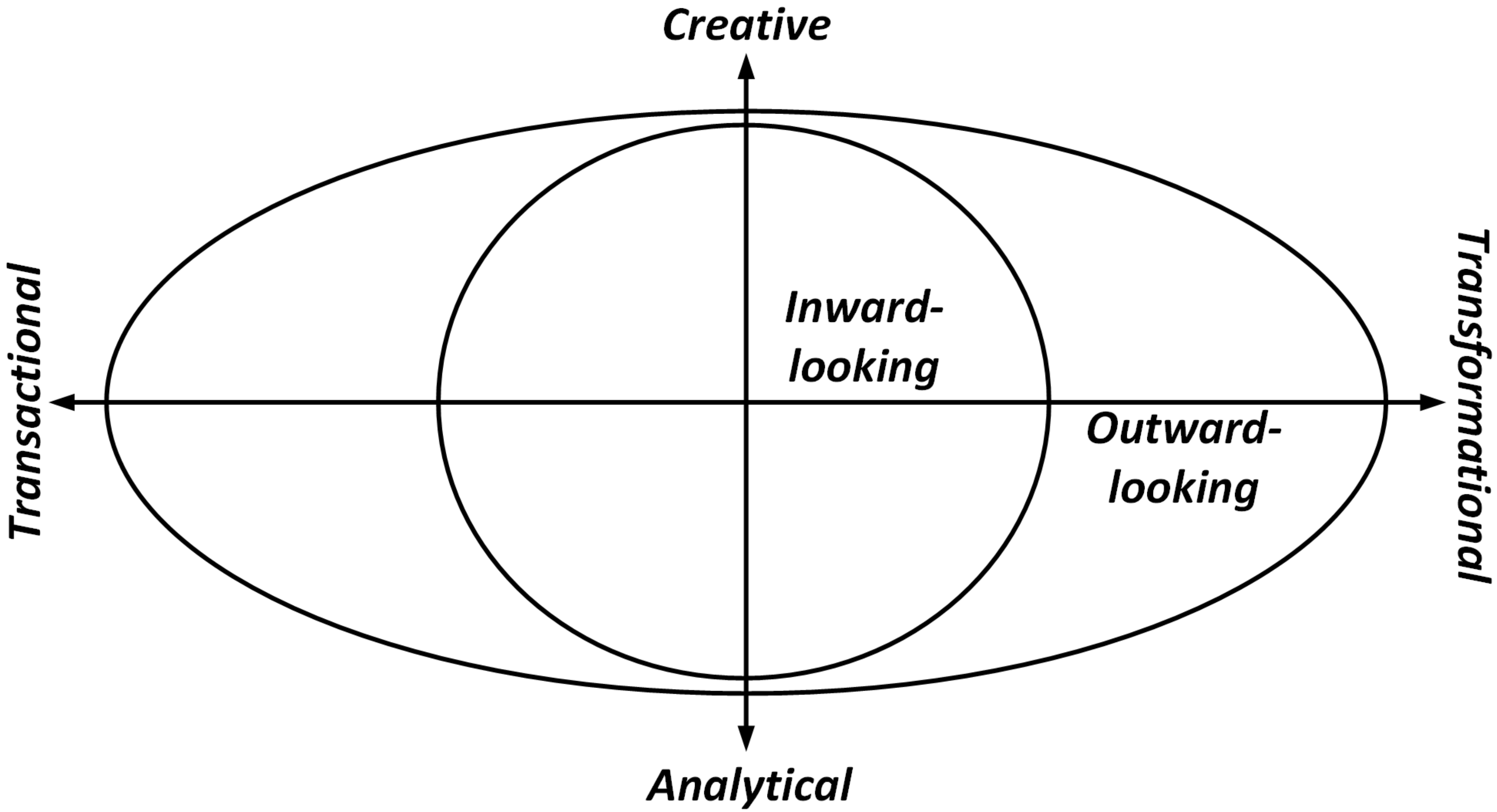
- Ambition and Situation?
  - **Revolutionary / Exploration**
    - Fundamental rethinking and radical redesign
  - **Evolutionary / Exploitation**
    - Incremental adaptation of process
  - Starting from
    - Clean slate
    - Existing process
    - Reference model/ blueprint

- How to redesign?
  - **Method, technique, tool**

- **There is no silver bullet!**

Dumas et al. (2018)

Process Redesign Orbit

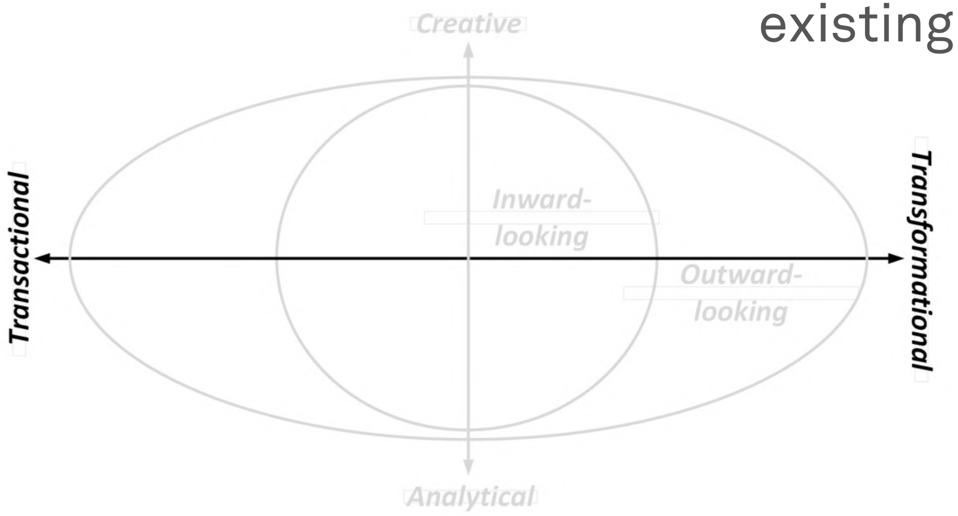


Dumas et al. (2013)

Process Redesign Orbit – Ambition

- **Transactional** Methods
  - Seek to identify problems and resolve them **incrementally**, one step at a time
  - Do not challenge the current process structure

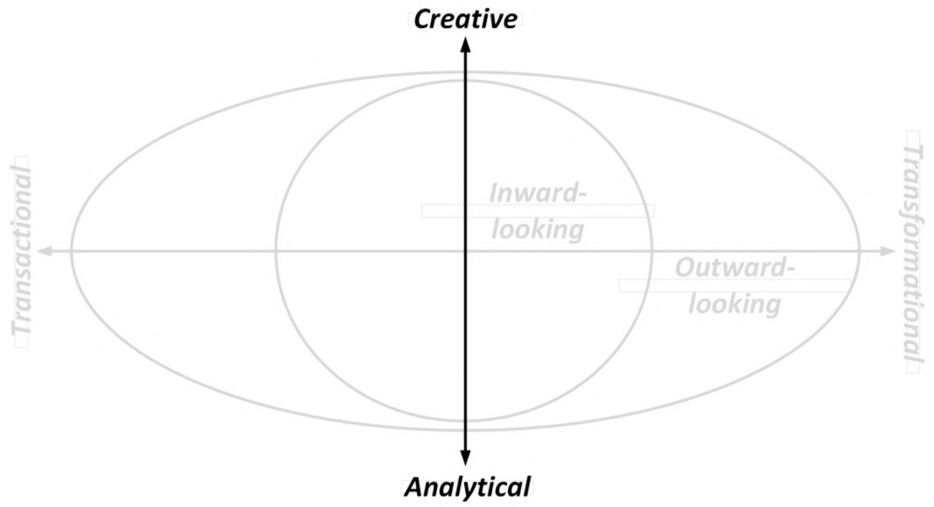
- **Transformational** Methods
  - Aim to achieve **breakthrough innovation**
  - Put into question the fundamental assumptions and principles of the existing process structure



Dumas et al. (2018)

Process Redesign Orbit – Nature

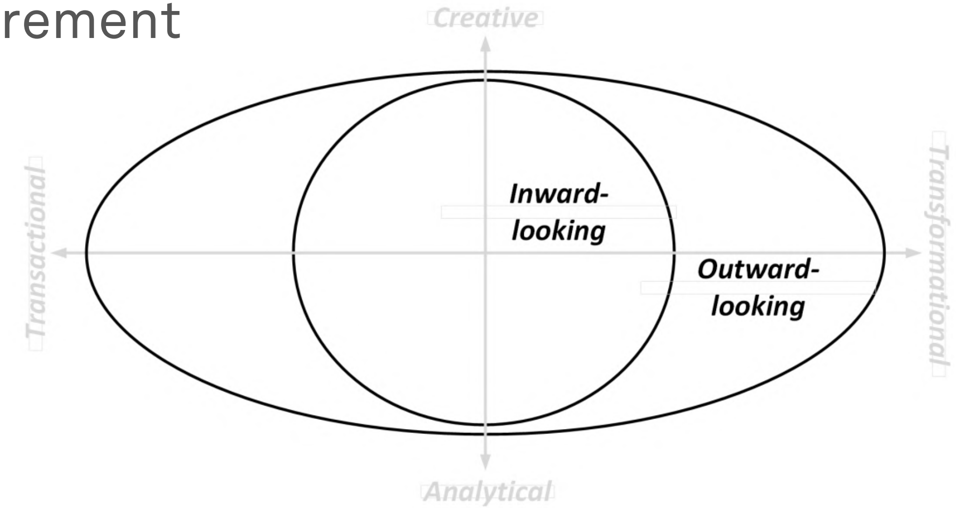
- **Analytical** Methods
  - Tend to have a strong mathematical and quantitative focus
  - Embrace **tools** and **technology**
- **Creative** Methods
  - Rely on (human) creativity and ingenuity
  - Embrace **group dynamics**



Dumas et al. (2018)

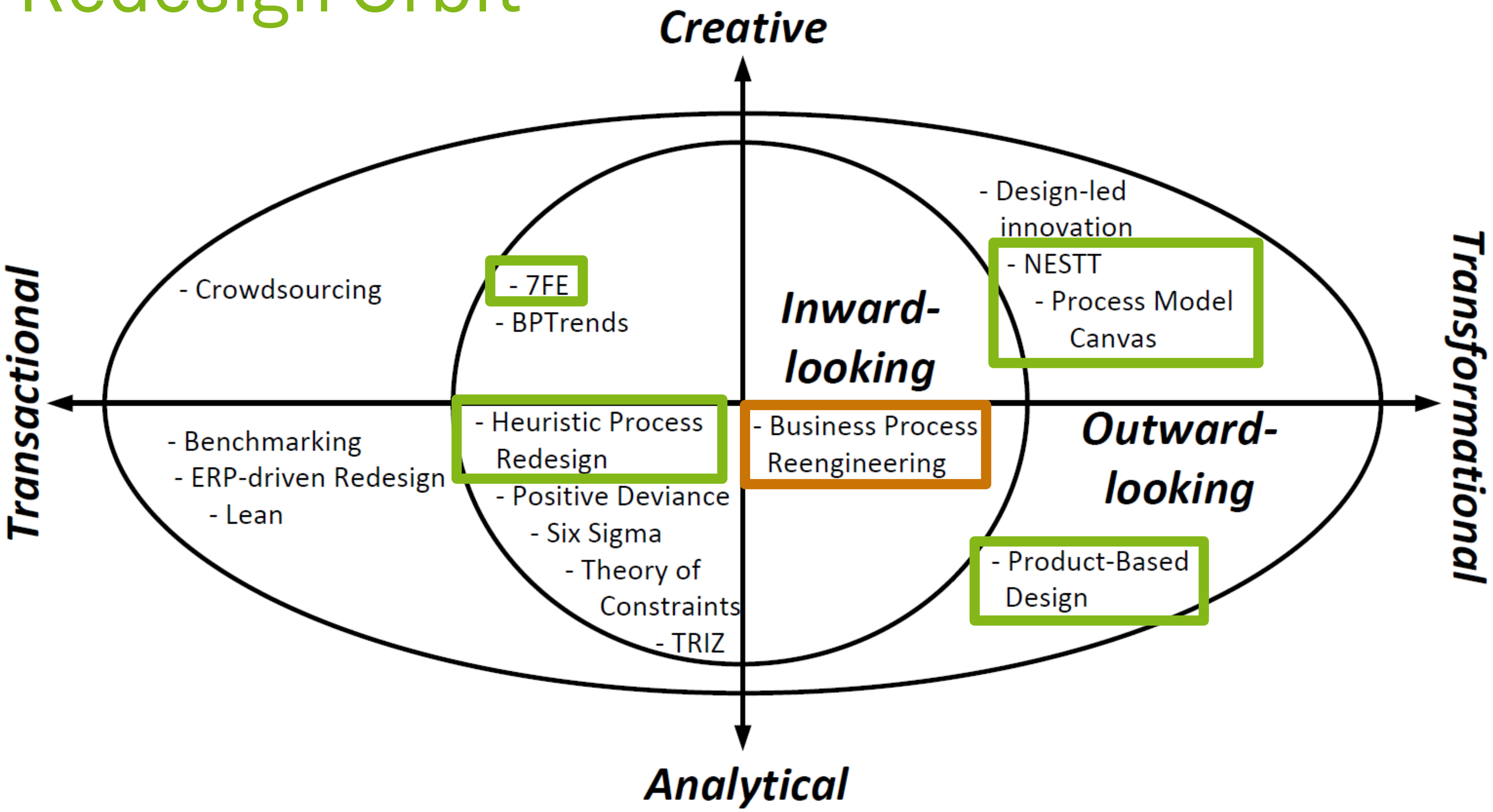
Process Redesign Orbit – Perspective

- **Inward-looking** Methods
  - Consider the process from the perspective of the **internal organization**
  - Draw from objectives and performance measurement
- **Outward-looking** Methods
  - Consider the process from an **outsider's perspective**
  - Are driven by external opportunities and developments



Dumas et al. (2018)

Process Redesign Orbit



Dumas et al. (2013)

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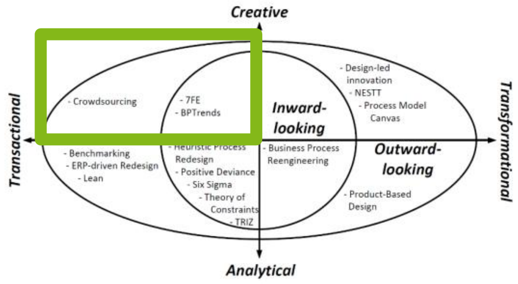
Creative Transactional Methods




Creative Transactional Methods



- Unleashing the creativity of people

- Inward-looking
  - **7FE**
  - BPTrends

- Emphasis on engaging professionals that already play a role within the business process

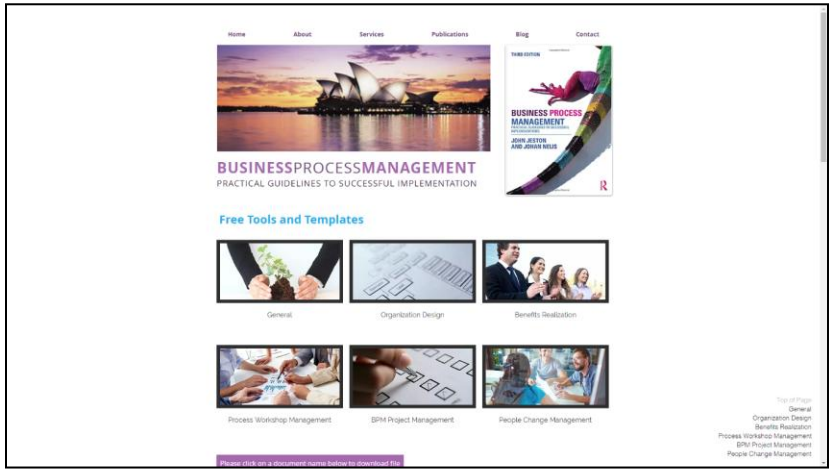
- Outward-looking
  - Crowdsourcing

- Tapping into the skills and knowledge of people outside their organizational borders

Dumas et al. (2018)

7FE

- Framework for BPM projects or even BPM programs
- Consists of several phases
  - **Innovate** phase deals with the technical challenge of process redesign
- Method suggests conducting **workshops**



Jeston & Nelis (2013), Dumas et al. (2018), managementbyprocess.com (2015)

Stages of the Innovate Phase

- Prepare
  - Collect inputs for workshops
    - strategy link, goals and measures, constraints, timeframe
- Generate
  - Generate ideas with facilitator, converge to consensus
    - Redesign from **customer perspective**
    - Identify the **customer journey**
    - Identify (issues at) customer **touch points**
    - Use **mystery shopper**
    - Guide through **problematic elements** and **generic solutions**
  - Determine scenarios
- Validate
  - Test effectiveness and feasibility (with simulation)

Jeston & Nelis (2013), Dumas et al. (2018)

7FE

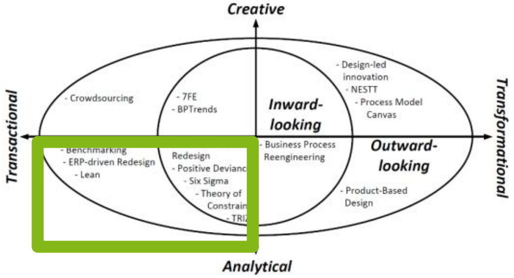
- Facilitators…
  - …need to ask lots of "what if" and "why this" questions
  - …should not accept what he or she is told (the first time)
  - …must look for the second 'right' answer
  - …will regularly change the question and come it at from a different direction
  - …shall challenge the rules of the process
  - …need to rely on intuition

- …need vast experience

Jeston & Nelis (2013), Dumas et al. (2018)

Analytical Transactional Methods

Analytical Transactional Methods



- Employing all kinds of tools, involving statistics
- strongly rationalized

- Inward-looking
  - Six Sigma
  - Theory of Constraints
  - TRIZ
  - Positive Deviance
  - **Heuristic Process Redesign**

- Focus on existing process in an organization as a starting point

- Outward-looking
  - Benchmarking
  - ERP-driven Redesign
  - Lean

- Take an external perspective on process redesign

Dumas et al. (2018)

Heuristic Process Redesign Procedure

- Uses a list of 29 redesign heuristics to determine improvement actions
- **Initiate**
  - Set up project
  - Create an understanding of the existing situation (**as-is**)
  - Set the **performance goals** for the redesign project
- **Design**
  - **Identify suitable heuristic** and **use heuristics** to determine potential improvement
  - **Generate scenarios** (to-be) as design alternatives
- **Evaluate**
  - Qualitative and/or quantitative **analysis of** redesign **scenarios**
- Execute in highly **iterative and overlapping** ways

Dumas et al. (2018)

Examples of Time-based Redesign Heuristics

Parallelism

- *"Put activities in parallel"*
- Activities are often ordered in a **strictly sequential** way
- Some activities may well be carried out in an **arbitrary order** or even **simultaneously**
- With less restrictive choice on the order, a business process can be carried out **faster**

Case-based work

- *"Remove batch-processing and periodic activities"*
- **Sources of delays**
  - batch that is only processed once all its items are available
  - periodic triggers
- Removing such constraints is in general a good way to **speed up** a process

Dumas et al. (2018)

Examples of Cost-based Redesign Heuristics

Activity elimination

- *“Eliminate unnecessary activities”*
- Organizational entropy entails that processes **build up unnecessary steps** over time
- Removing unnecessary activities is an effective way to **reduce the cost** of handling a case

Empower

- *“Give workers decision-making authority”*
- Often other roles people **authorize** the outcomes of activities
- Empowerment to take decisions autonomously may remove some work of middle managers and **reduce cost**

Dumas et al. (2018)

Examples of Quality-based Redesign Heuristics

Triage

- *“Split an activity into alternative versions”*
- Alternative versions of an activity can better deal with **variants of cases**
  - Deal with a sub-category of cases
  - Exploit characteristics of a resource
- By aligning work more specifically to a particular case, the **quality of work improves**

Case assignment

- *“Let participants perform as many steps as possible”*
- A resource becomes acquainted with the case
- Make this participant the **preferred resource** for any work in this case
- This knowledge can be leveraged to deliver a **high standard of work**

Dumas et al. (2018)

Examples of Flexibility-based Redesign Heuristics

Flexible assignment

- *"Keep generic participants free for as long as possible"*
- Assign cases to the most specialized resource first
- Maximizes the **likelihood to commit** the free, more general participant to another work item
- Process **stays flexible** with respect to assigning work

(Virtualized) Centralization

- *"Let geographically dispersed resources act as if they are centralized"*
- Exploit the benefits of a BPMS
- Geographic allocation becomes irrelevant
- Resources can be **committed more flexibly**

Dumas et al. (2018)

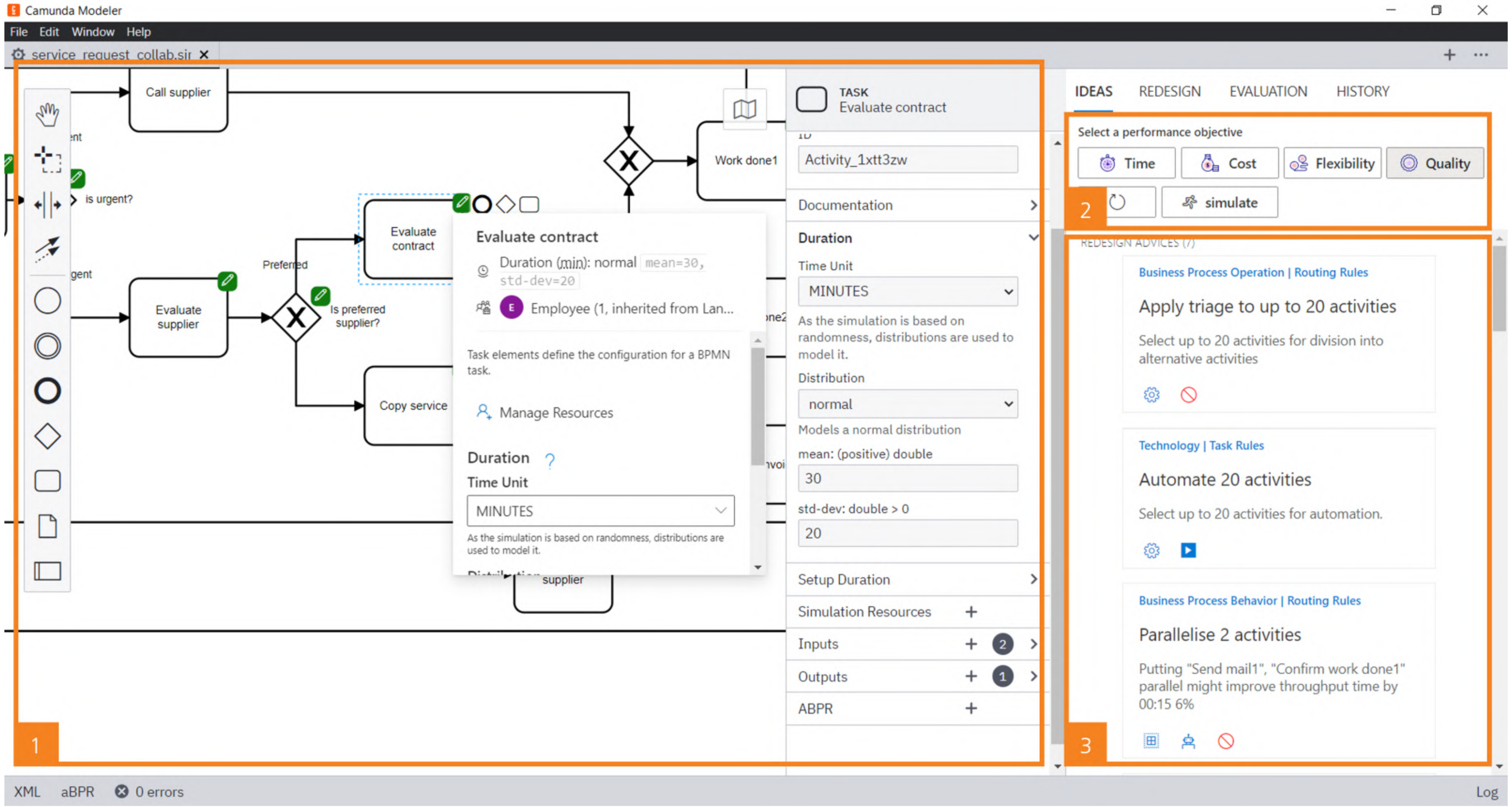
Examples of Redesign Heuristics

Dimension	Heuristic	Time	Cost	Quality	Flex.
Time	Parallelism	+			
	Case-based work	+			
Cost	Activity elimination	+	+		
	Empower	+	+		+
Quality	Triage			+	
	Case assignment	+		+	
Flexibility	Flexible assignment	+		+	+
	Centralization	+			+

Redesign Heuristics can also have negative effects (see Devil's Quadrangle)

Dumas et al. (2018)

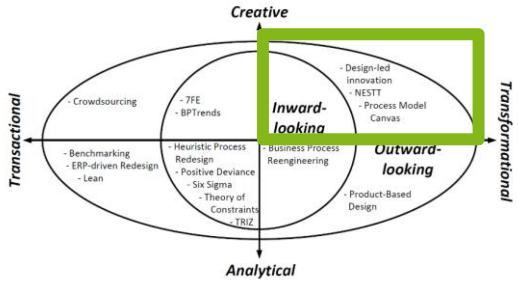
A Tool for Assisted Business Process Redesign



Fehrer et al. (2022)

Creative Transformational Methods

Creative Transformational Methods



- Unleashing the creativity of people

- Inward-looking
  - *(True transformations hardly emanate from reasoning from an internal perspective only)*

- Emphasis on engaging professionals that already play a role within the business process

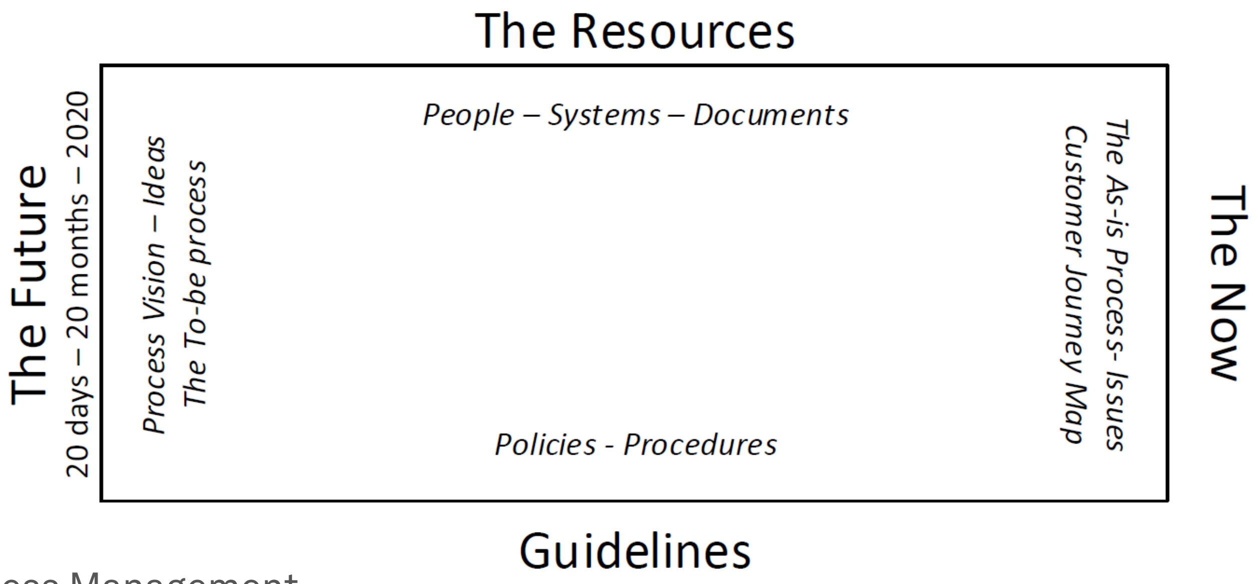
- Outward-looking
  - BPM Billboard
  - Design-led innovation
  - **NESTT**
  - **Process Model Canvas**

- Tapping into the skills and knowledge of people outside their organizational borders

Dumas et al. (2018)

NESTT

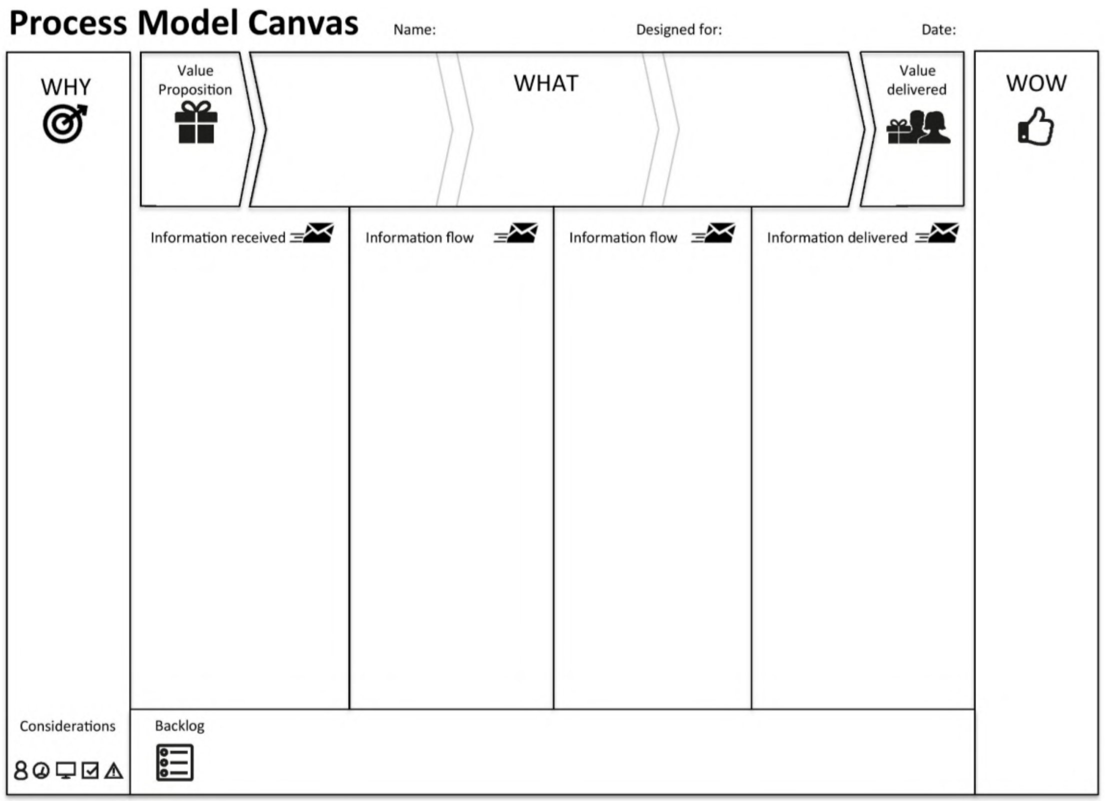
- **Stages**
  - Navigate, Expand, Strengthen, Tune/Take-off
- **Workshop** setting
  - 8 to 10 people use the walls and the floor of the room to visualize and address different viewpoints



Dumas et al. (2018), Rosemann (2018)

Process Model Canvas

- Allows firms to reason about the **value proposition** behind their processes
- Questions
  - Why?
  - What?
  - Wow!?

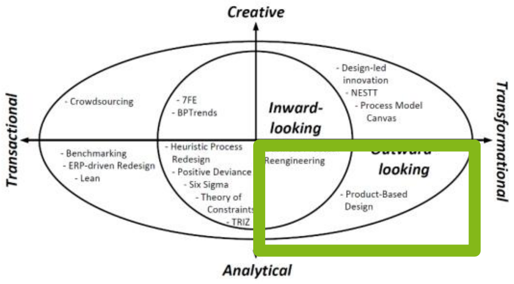


Dumas et al. (2018), processmodelcanvas.net (2022)

Analytical Transformational Methods




Analytical Transformational Methods



- Employing all kinds of tools, involving statistics
- strongly rationalized

- Inward-looking
  - **Business process reengineering**
- Focus on existing process in an organization as a starting point

- Outward-looking
  - **Product-based design**
- Take an external perspective on process redesign

Dumas et al. (2018)

Business Process Reengineering

Caveat: Has practically been superseded with more transactional approaches

- **Clean slate approach**
  - Successful organizations do not rely on piecemeal improvement
  - Break away from ingrained patterns, integrate cross-functionally
- **Information technology**
  - Successful organizations do not only automate, they innovate
- **Product (or service)** is central concept rather than process
- **Principles**
  - Make sure information is fresh from the source
  - Integrate information capture and processing with the real work
  - Let those that have interest in the output drive the change, not only participate
  - Put decision points at the place where work is performed

Dumas et al. (2018), Hammer (1990)

Product-based Design

- Idea to **ignore the existing process** and purely **consider the features of the product or service**
- Develop the **leanest, most performative** process possible

- Developed to design processes that produce informational products
- **Product data model** (bill-of-material) is the main vehicle to determine the best process structure to create and deliver that product

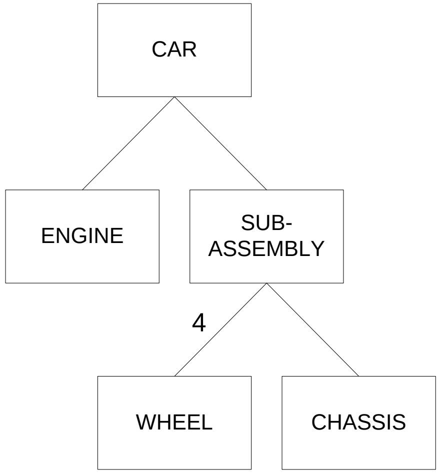
Dumas et al. (2018), Reijers et al. (2003)

Product-based Design Procedure

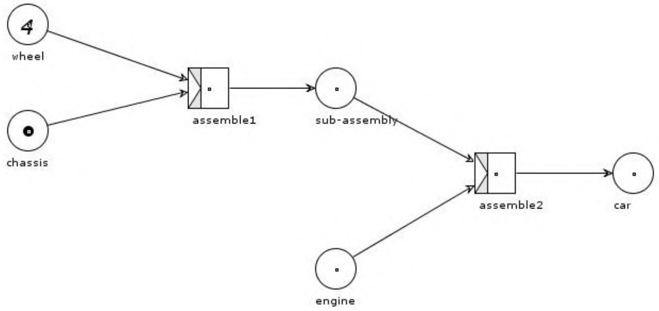
- **Scoping**
  - Select **subject** of redesign
  - Identify performance targets and limitations
- **Analysis**
  - Create and study product data model
- **Design**
  - Derivate (multiple) designs using elements and operations
- **Evaluation**
  - Verify, validate, and select
- Execute in an **iterative** ways

Dumas et al. (2018)

Manufacturing Process Model

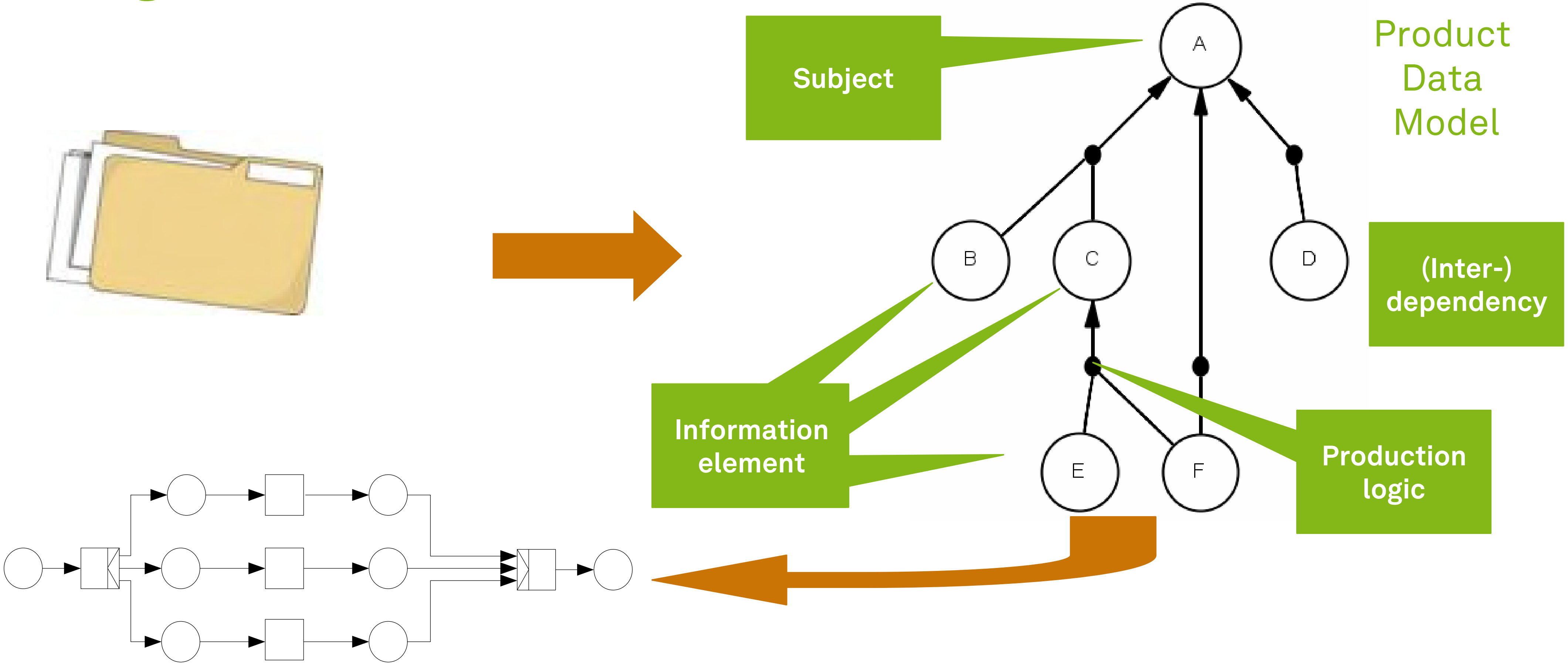


Bill-of-material



Dumas et al. (2018)

Exploiting the Product Data Model



Dumas et al. (2018)

Scoping – Information Elements

- Information elements should be **adequate in size**
  - Information element is too large if different parts of it are used in different production rules
  - Information element should not contain irrelevant information
- Information elements should **not be** necessarily **associated with** their **physical manifestation**
  - Avoid information elements like “computer screen output”
- Information elements may be **atomic** or **composite**
  - The type of a composite information element is a composed type

Dumas et al. (2013)

Scoping – Production Logic

- The full **specification is** a direct **validation** on the inputs of the involved production rule
  - Forgotten inputs or bad data types can be detected
- An **analysis** of the **production logic** is **relevant for** the estimation of **performance** characteristics
  - Labor and computer cost, speed, accuracy, etc.
- Production logic of **algorithmic nature** can be **used** as a **functional specification** for the BPM system that can execute this production rule
- If the production logic is **not totally algorithmic**, it is **likely** that a **human operator** must execute it in practice
  - The production logic can be used to develop task instructions

Dumas et al. (2013)

Analysis

- **Source analysis**
  - Identify the sources of all the leaf elements in the product data model
  - Relevant points of comparison for each leaf are
    - cost of obtaining it, delivery speed of the information, availability of the specific information and reliability of the provided information
- **Production analysis**
  - Estimate the involved cost, speed, quality, and flexibility
  - Select the right set and the right execution order of production rules given a set of performance targets
- **Fraction analysis**
  - Study the distribution of information element values
  - Model the rule with the widest applicability first (reduce workload)

Dumas et al. (2013)

Design

- **Correct**
  - Activities may only be incorporated in a process model in a way such that the **dependencies** between the information elements in the product data model **are respected**
- **Complete**
  - A process design must **cover all the information elements** that have been identified in a product data model
- Sound
  - Verification and validation
- Fast vs. efficient
  - Speed vs. resource utilization
- Secure

Dumas et al. (2013)

Learning Goals of Today

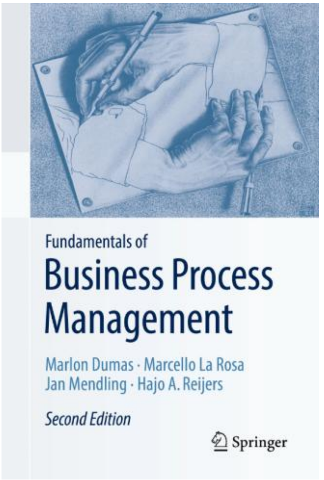
- Understand the essence of process redesign
- Understand the process redesign orbit
- Know and be able to use redesign techniques

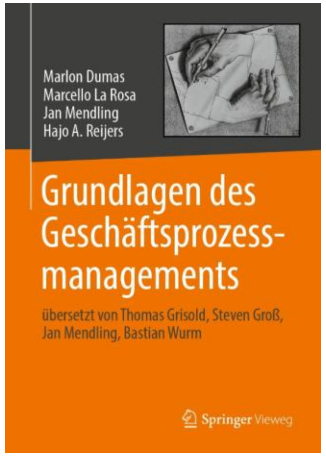
Recap Questions

- What is the process redesign orbit and which three dimensions does it distinguish?
- How can heuristic process redesign be classified in the process redesign orbit and how does it work?
- What is product-based design? How does it work?

Literature

- **Section 8**: Process Redesign





- www.managementbyprocess.com
- www.processmodelcanvas.net

Roadmap

Introduction	10.04. Introduction	17.04. Process Identification			
Process Discovery & Modeling	24.04. Essentials of BPMN	01.05. Public Holiday	08.05. Advanced BPMN I	15.05. Advanced BPMN II	22.05. Process Discovery
Process Analysis & Redesign	29.05. Public Holiday	05.06. Qualitative Analysis	12.06. Quantitative Analysis	19.06. Public Holiday	26.06.2025 Process Redesign
Application of BPM	03.07. Process Technology	10.07. Process Monitoring and Mining	15.07. BPM as an Enterprise Capability		

Any Questions about Process Redesign?

http://www.reestore.com/

ec.cs.tu-dortmund.de

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Department of Computer Science

Business Process Management

Process Technology

Roadmap

Introduction	10.04. Introduction	17.04. Process Identification			
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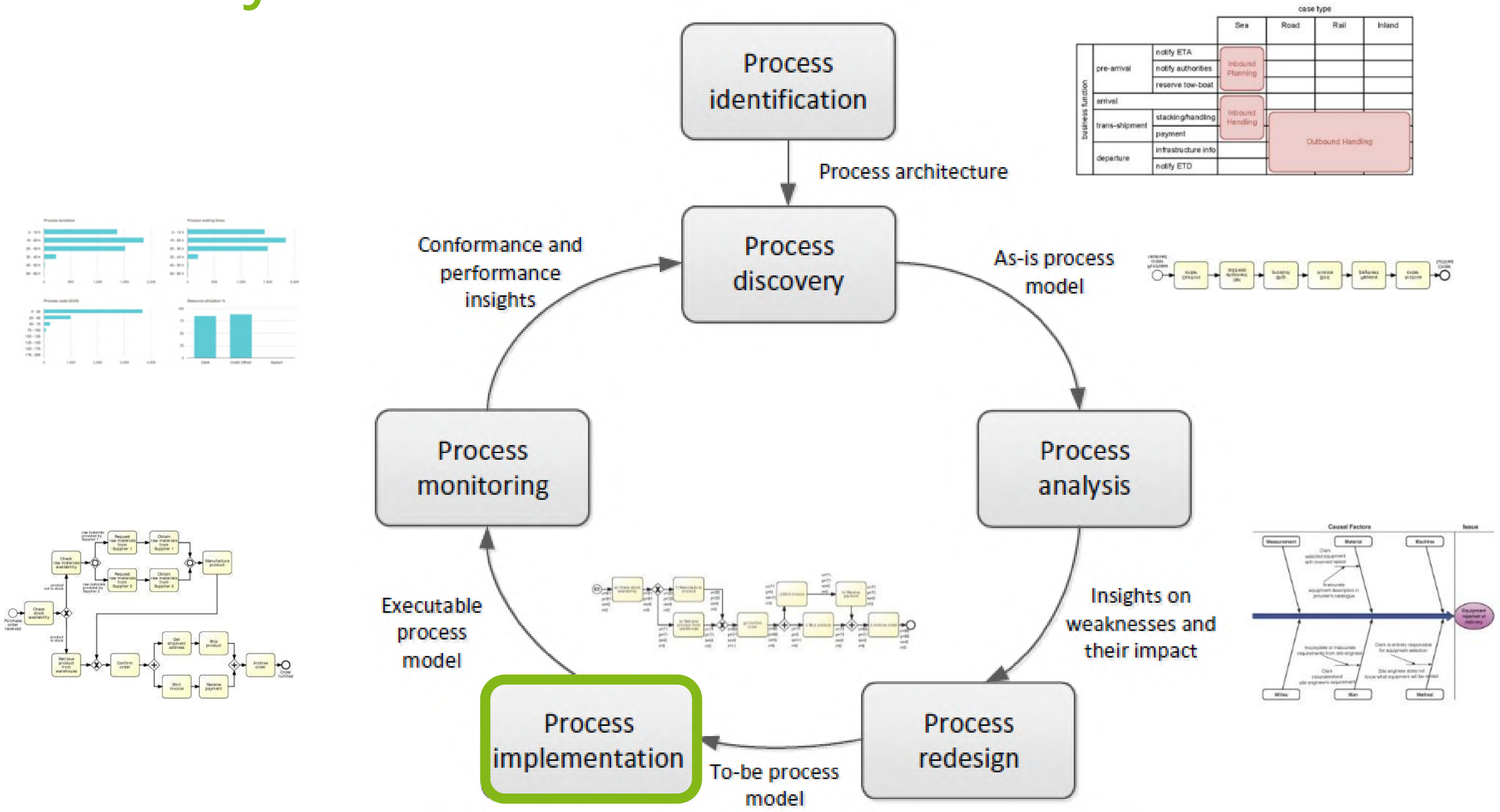
Learning Goals of Today

- Understand the evolution of process technology
- Understand workflow execution semantics
- Understand BPMS
- Understand RPA

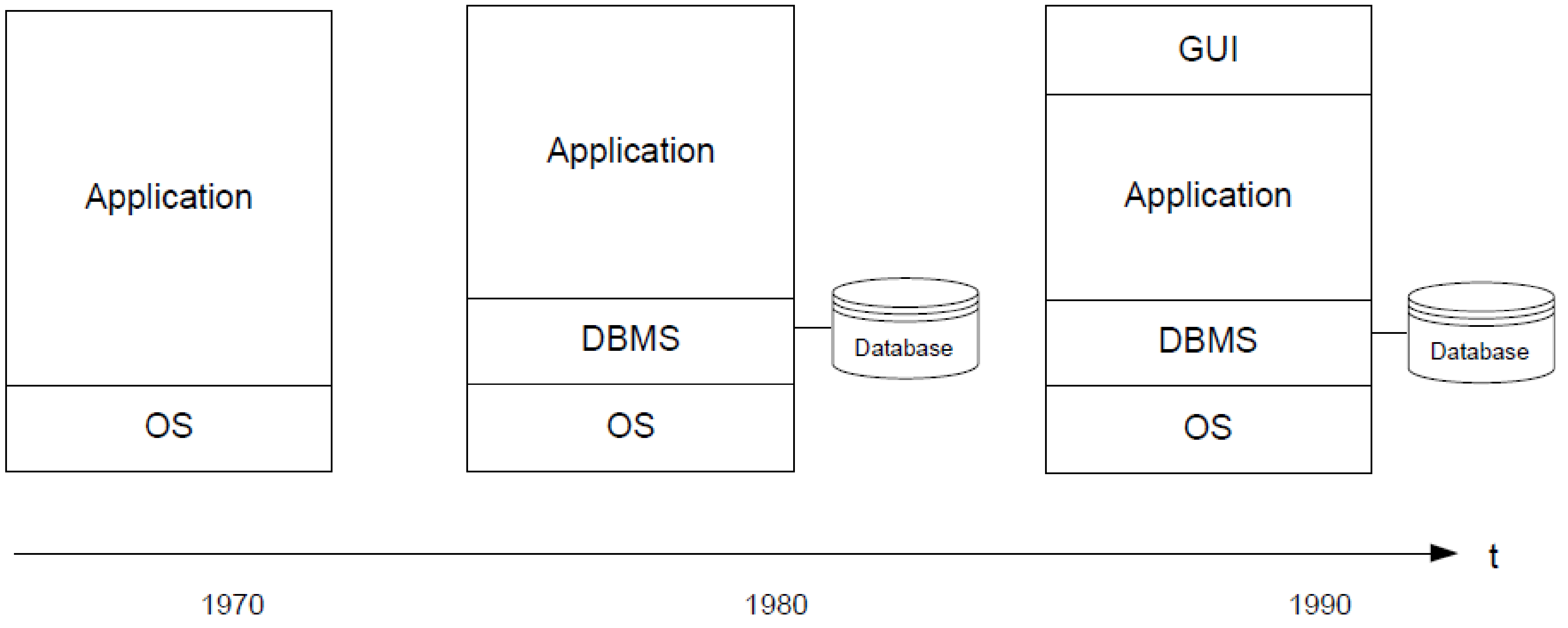
The BPM Lifecycle



Dumas et al. (2018)

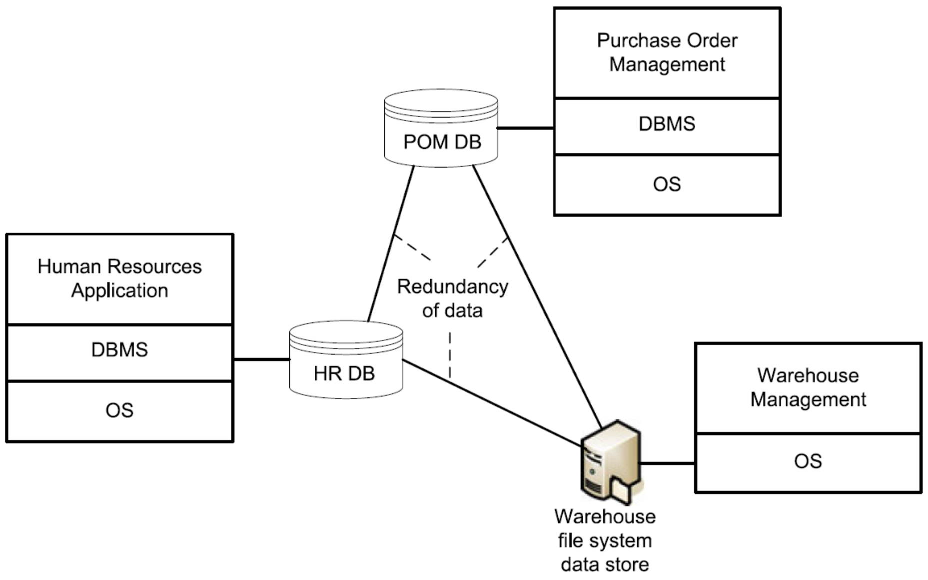
Evolution of Enterprise Software

Software Applications Before BPM



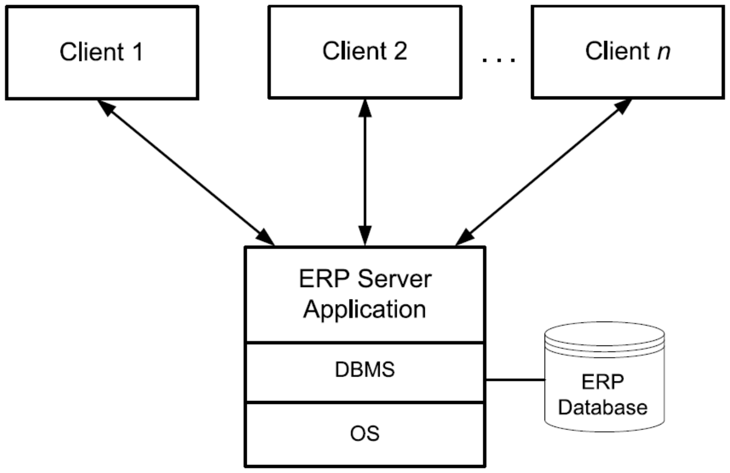
Weske (2019)

Enterprise Software Before BPM



Before SAP R/3

SAP R/3



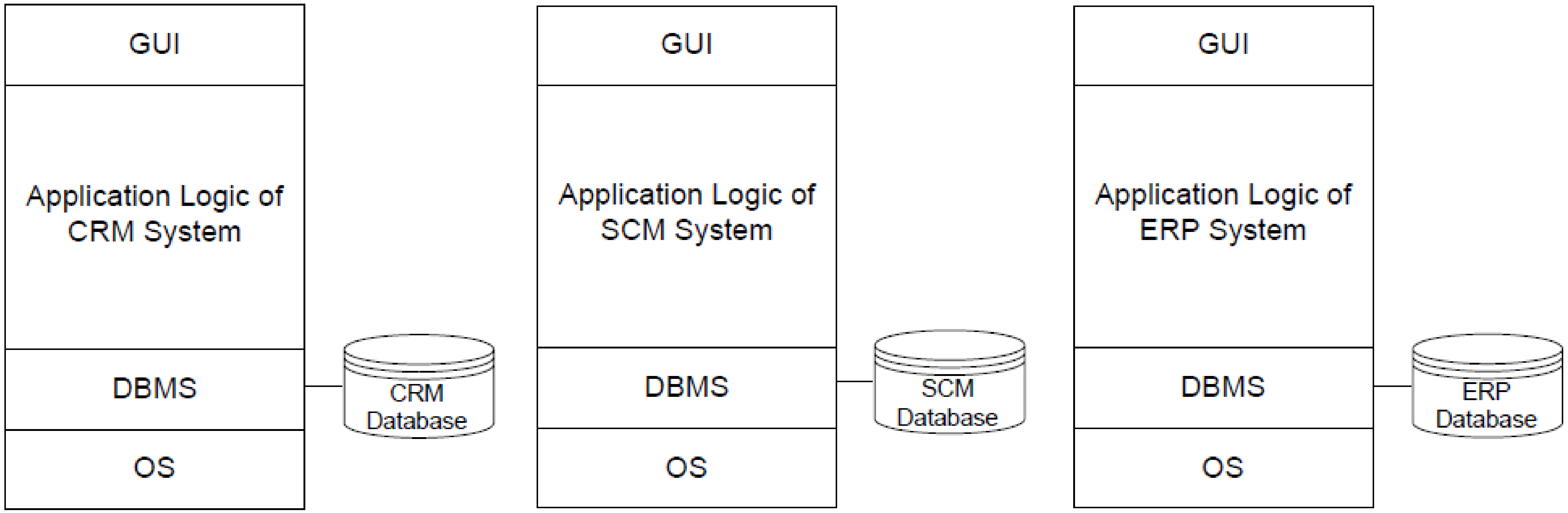
Weske (2019)

Domain-Specific Process-Aware Information Systems

- **Enterprise resource planning (ERP)** systems
  - Provide generic business functionality, which is required across various industries
    - accounting, controlling, human resource management, production, etc.
- **Customer relationship management (CRM)** systems
  - Support operational and analytical marketing and sales
    - sales and marketing activities related to pricing, distribution, and campaigning
- **Supply chain management (SCM)** systems
  - Focus on support of logistics operations
    - freight and transportation, warehousing, storage and inventory
- Product lifecycle management (PLM) systems
  - support the various processes of the lifecycle of a product

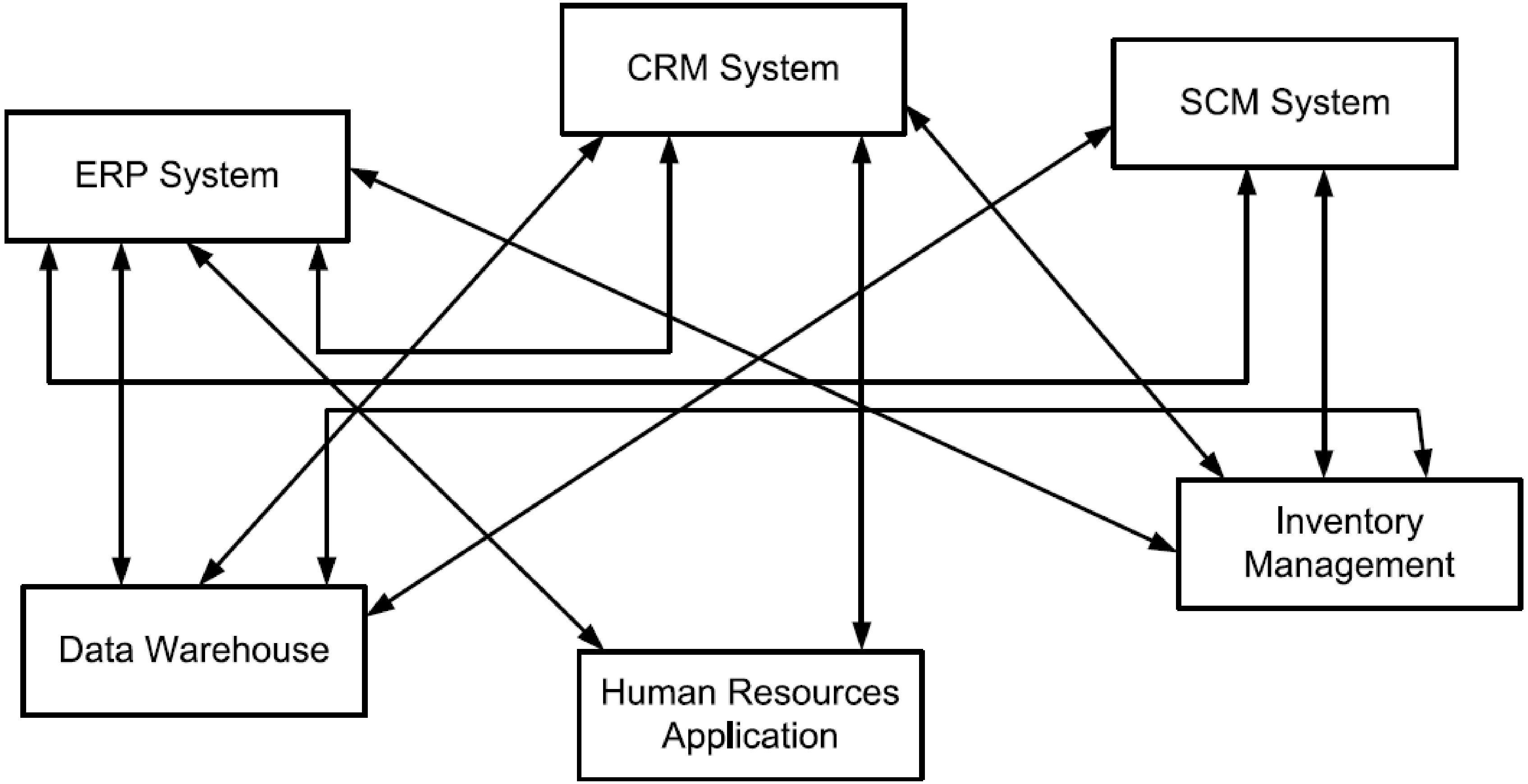
Dumas et al. (2018), Janiesch (2021)

Enterprise Software Before BPM



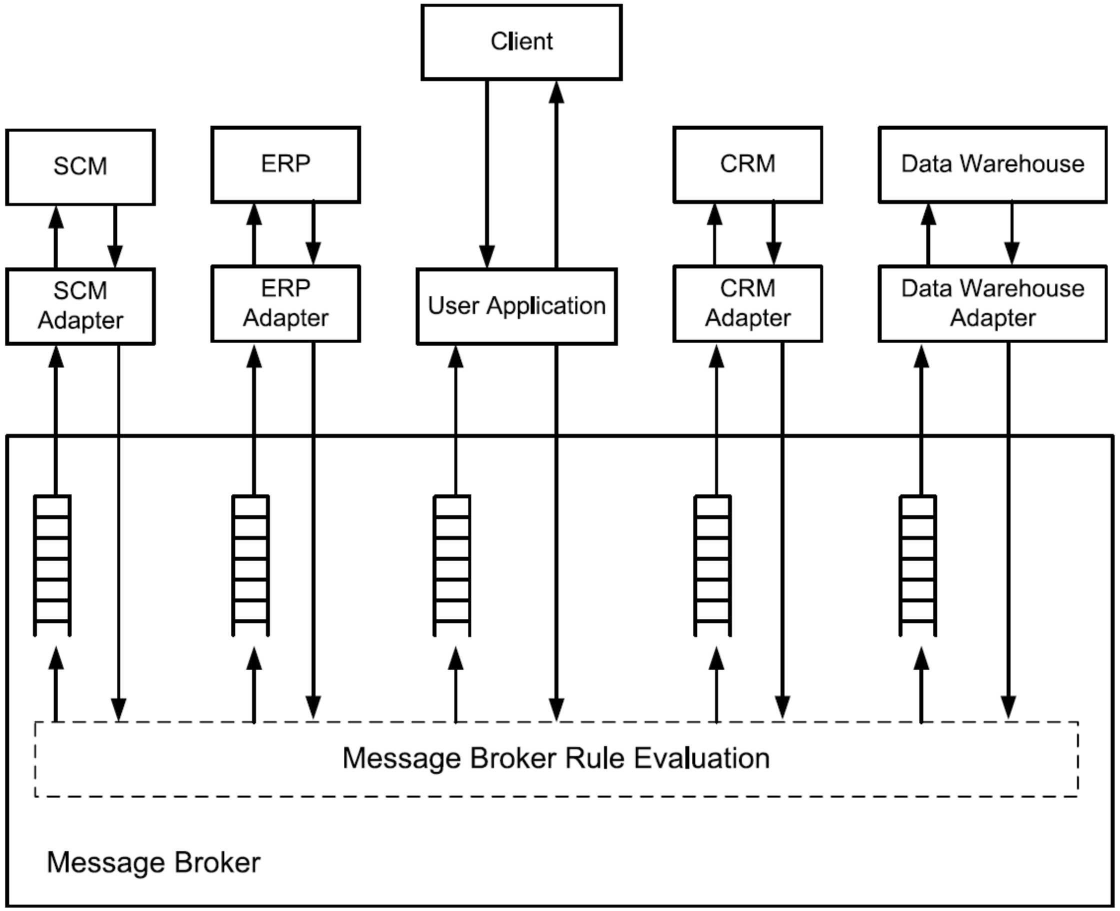
Weske (2019)

Enterprise Software Integration (1)



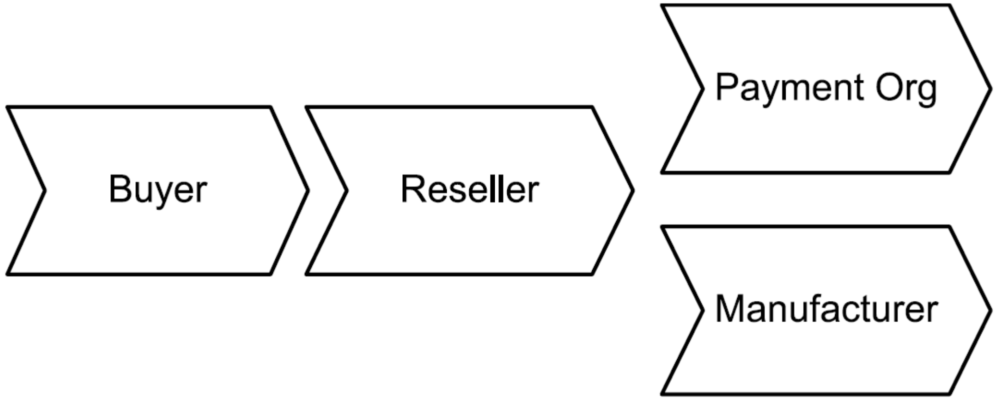
Weske (2019)

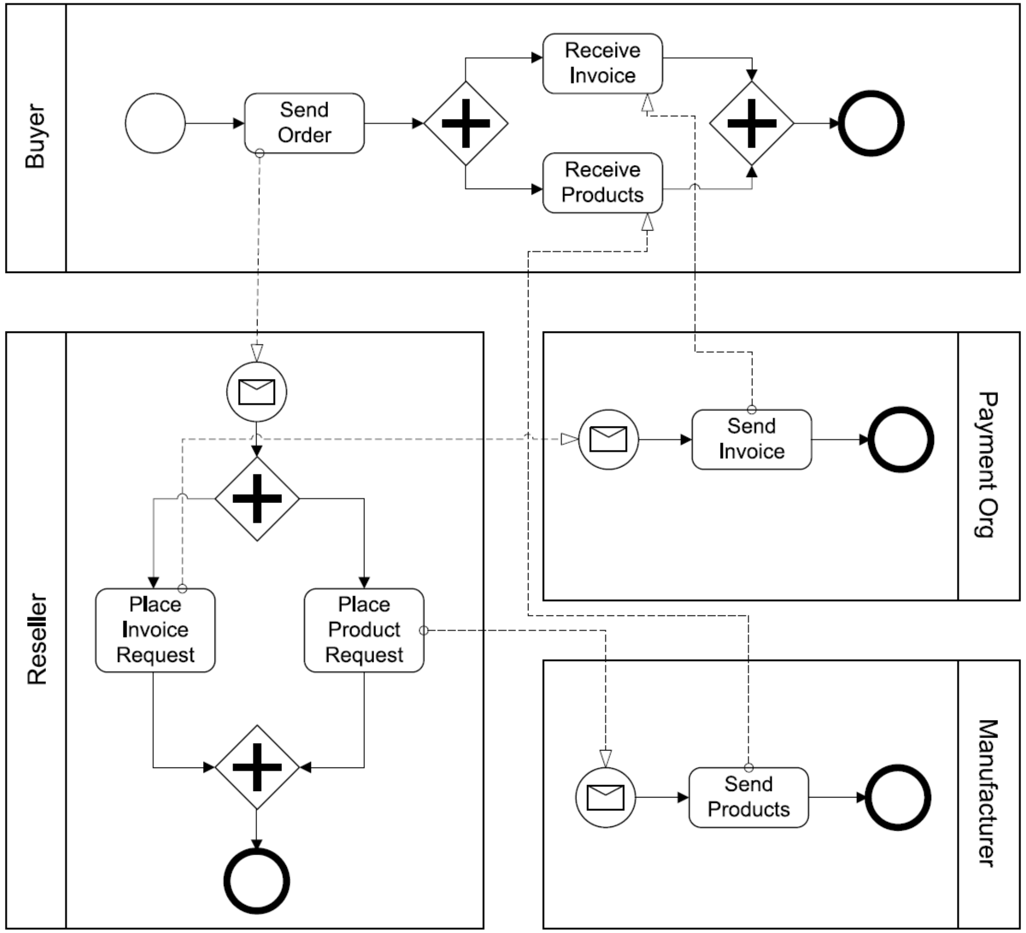
Enterprise Software Integration (2)



Weske (2019)

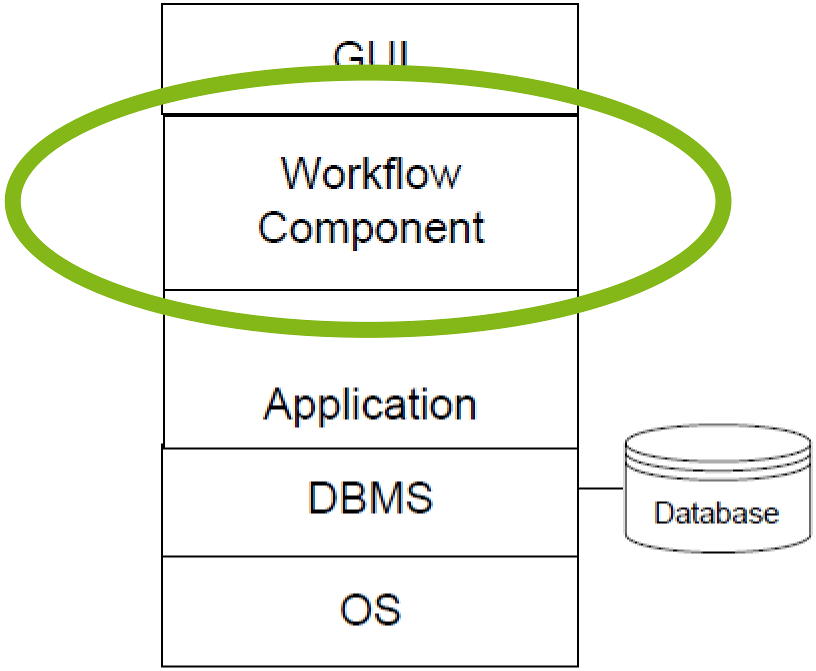
Value and Process Orientation





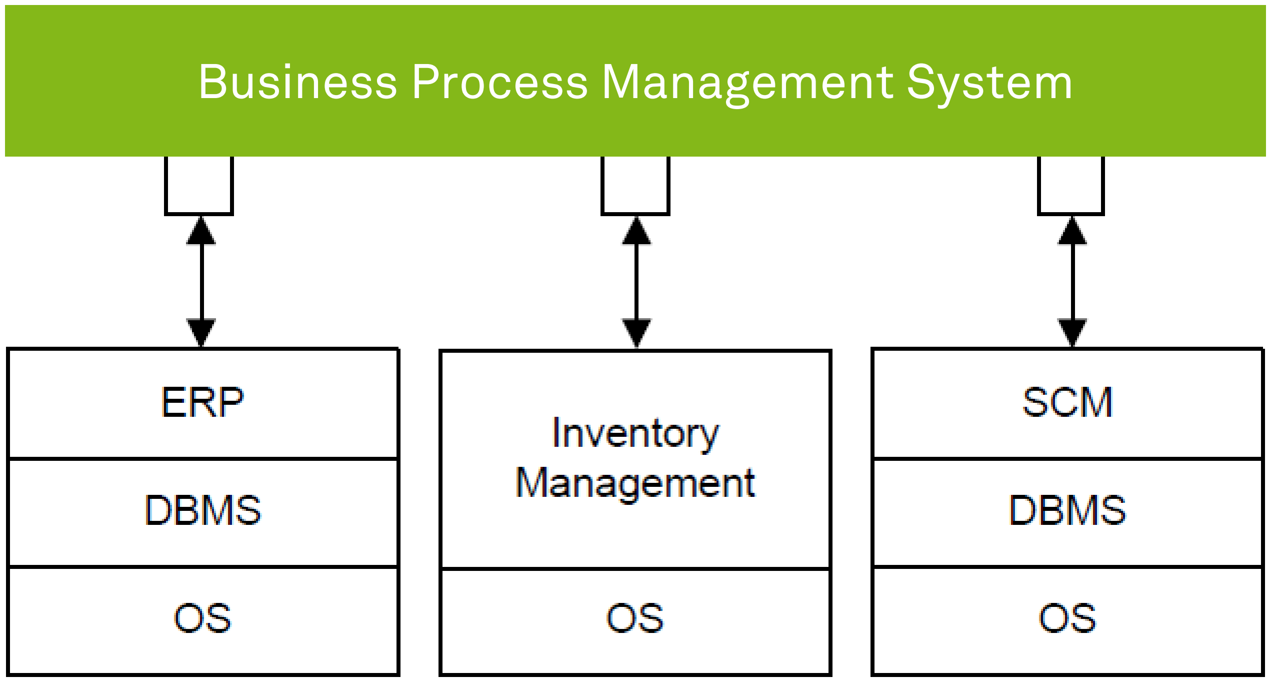
Weske (2019)

BPM/ Workflow: It's Another Layer!



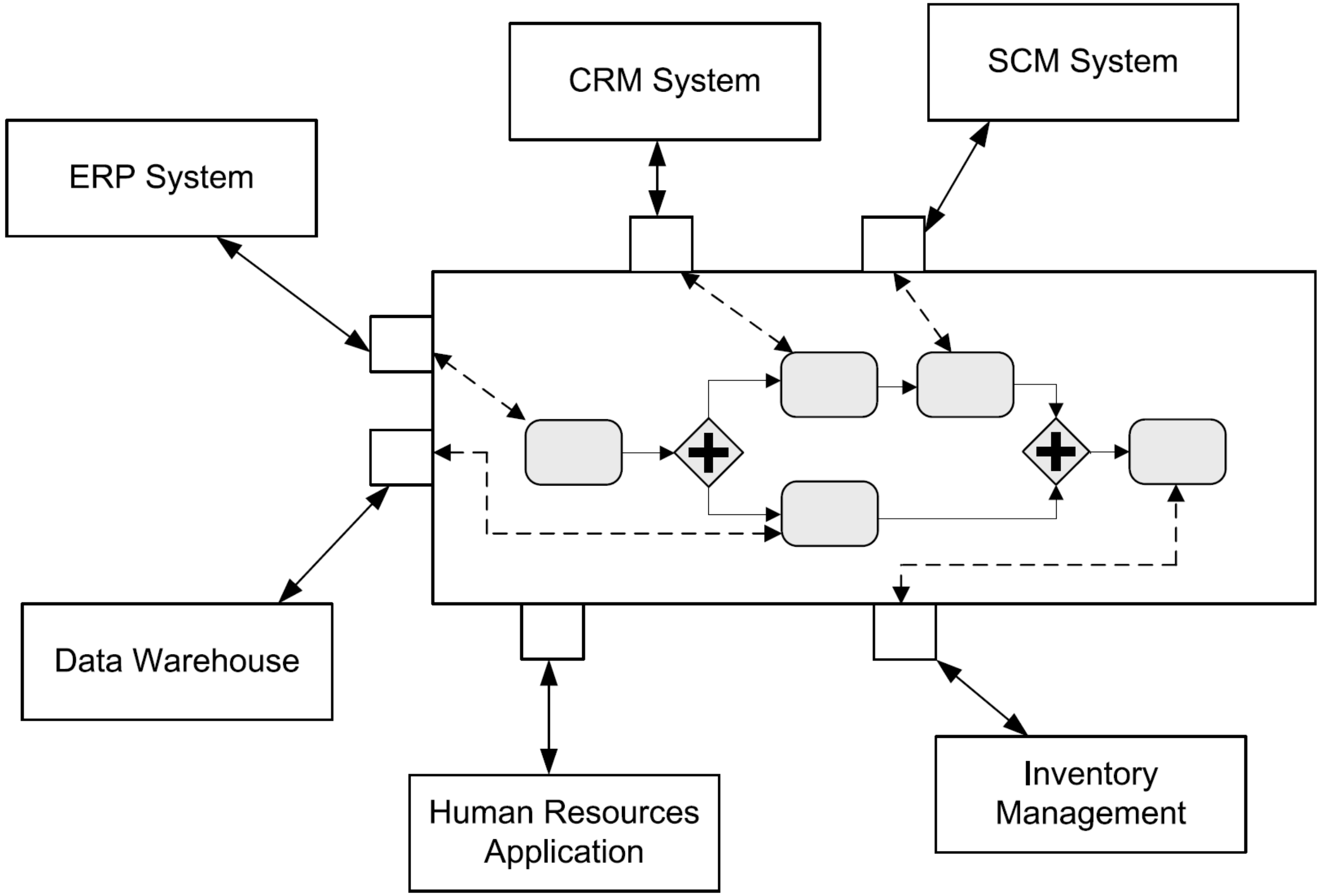
Weske (2019)

Process Orchestration with BPMS



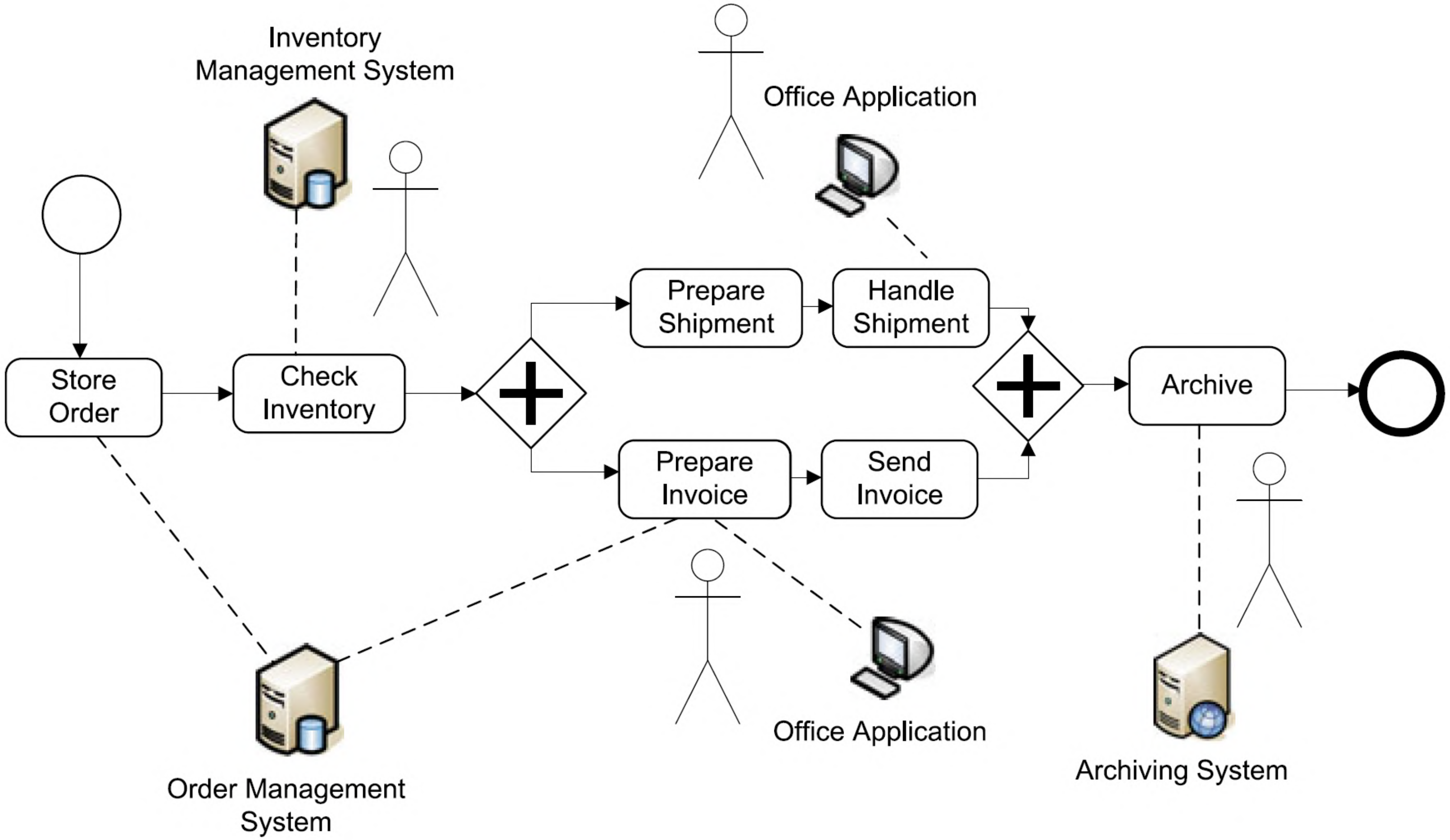
Weske (2019)

Process Orchestration with BPMS in Detail



Weske (2019)

Human Interaction



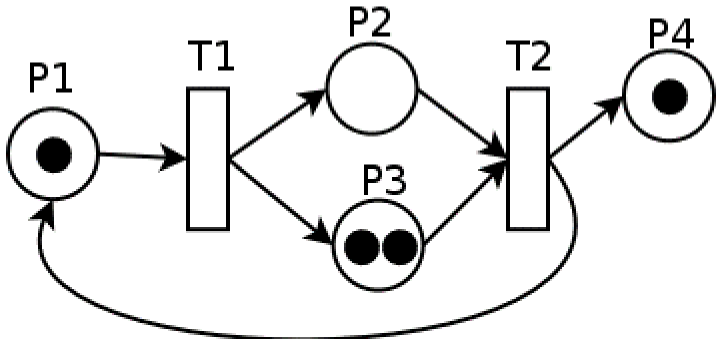
Weske (2019)

Execution Semantics




Execution Semantics

- Execution semantics are described by means of **token flows** (comparable to tokens in Petri-nets)
- Tokens flow along **sequence flow**
- Tokens are **produced and consumed** by activities, events and gateways



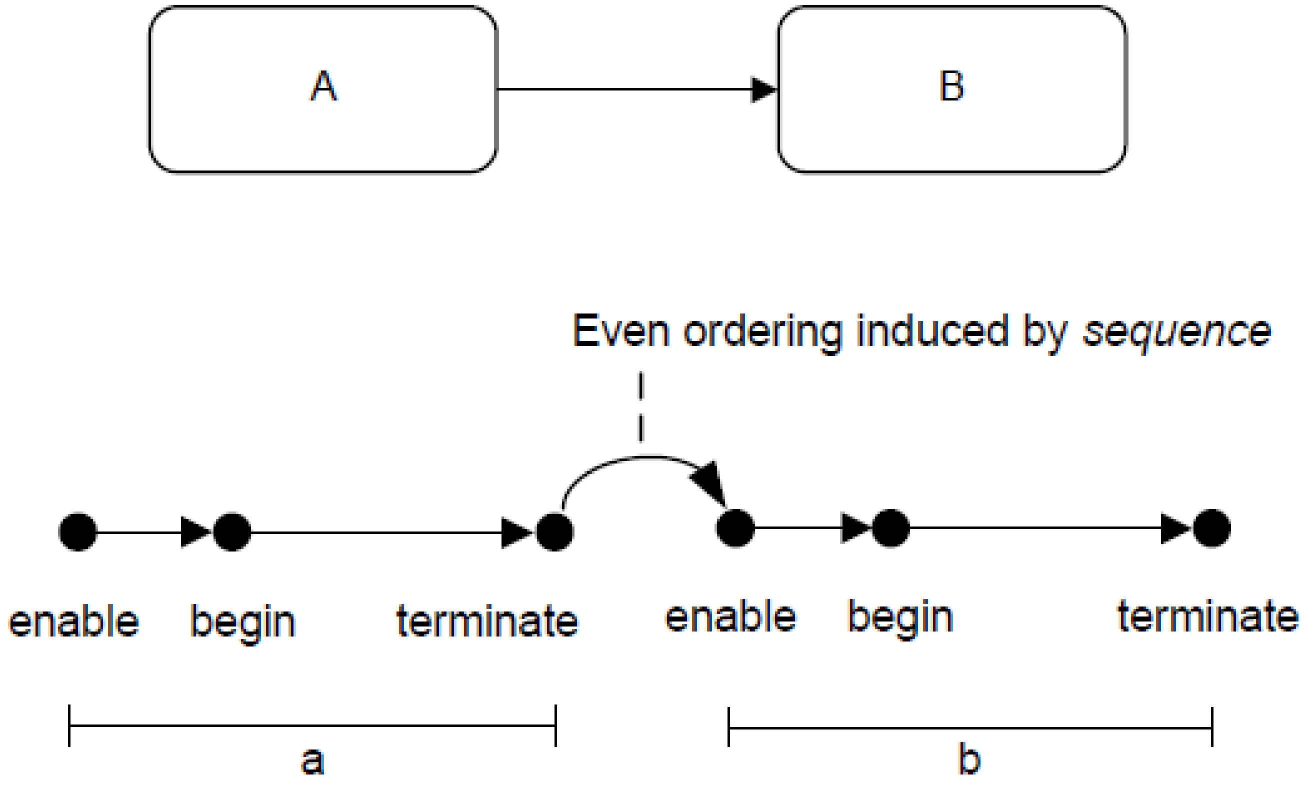
Weske (2019)

Process Lifecycle

- **Instantiation**
  - When one of its start events is triggered
  - Exception: If start event is part of an existing conversation that conversation is joined
- **Completed**
  - When all of the following hold
    - No remaining token
    - No active activity
    - If process was instantiated via parallel event-based gateway all events must have occurred

Object Management Group (2011)

State Changes of a Process Instance



Weske (2019)

Example for State Changes of a Process Instance

n new event

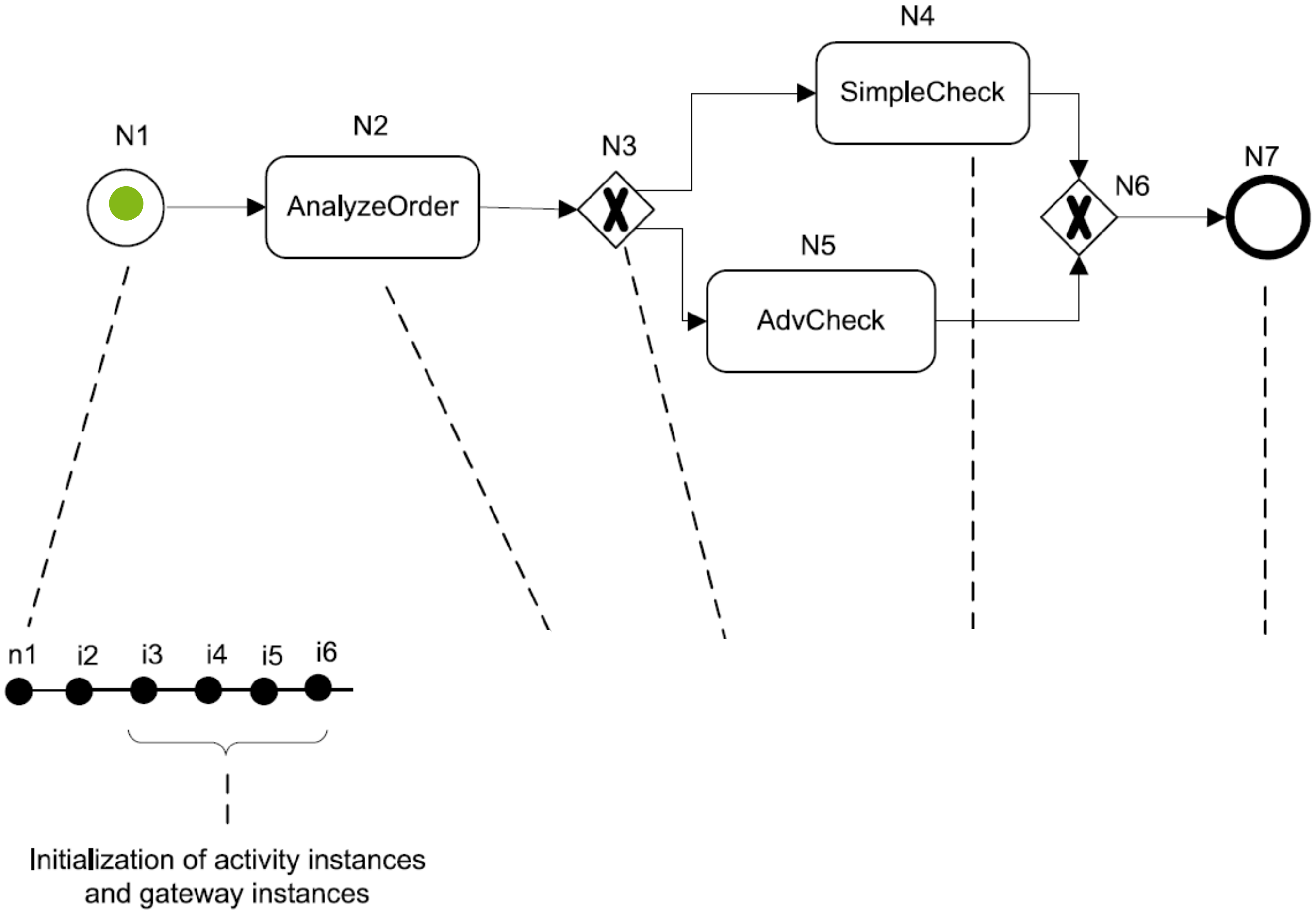
i initialize

e enable

b begin

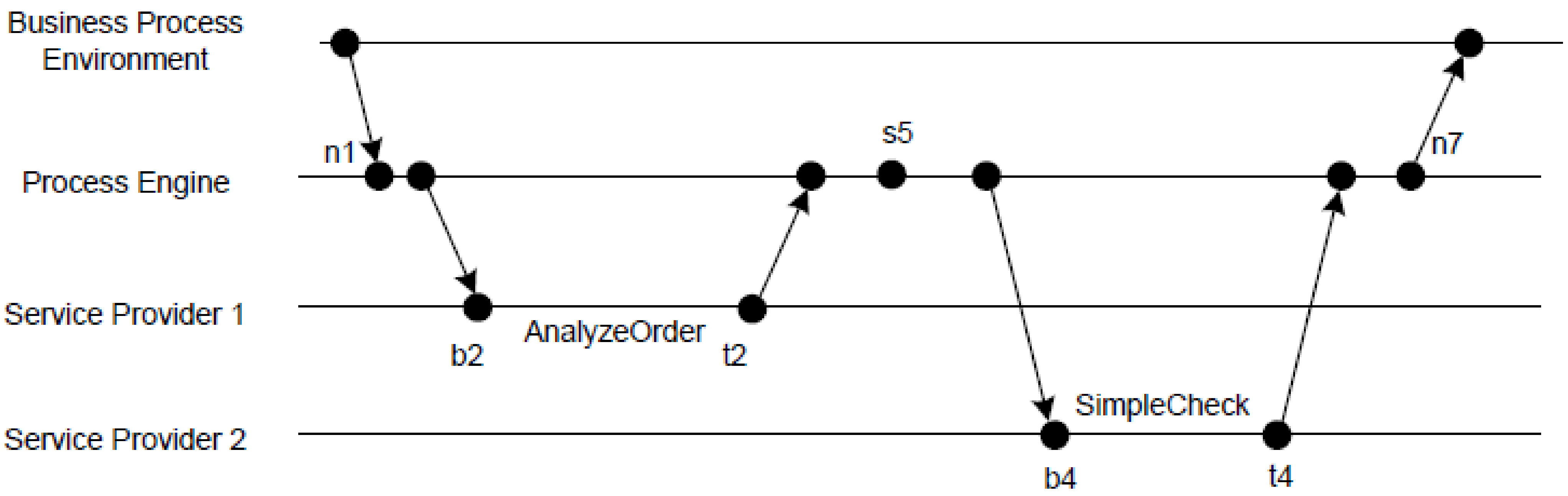
t terminate

s skipped



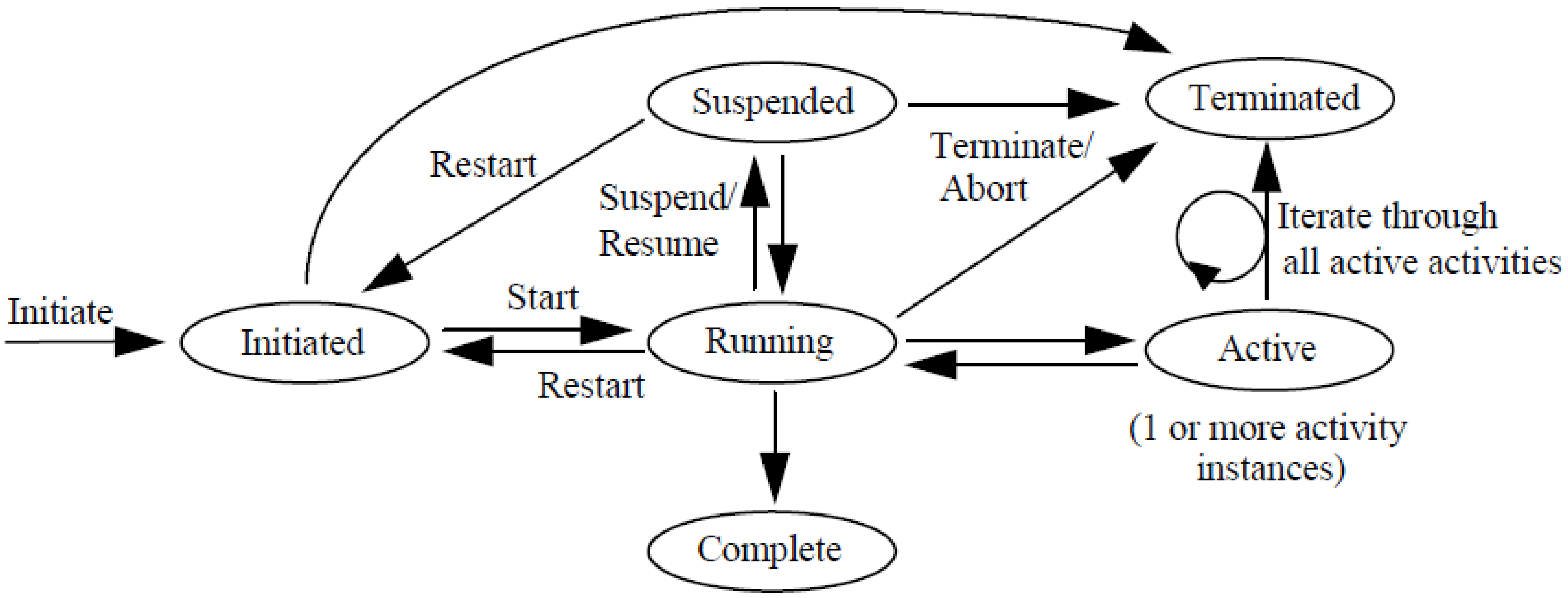
Weske (2019)

Changes to Business Process Execution Environment



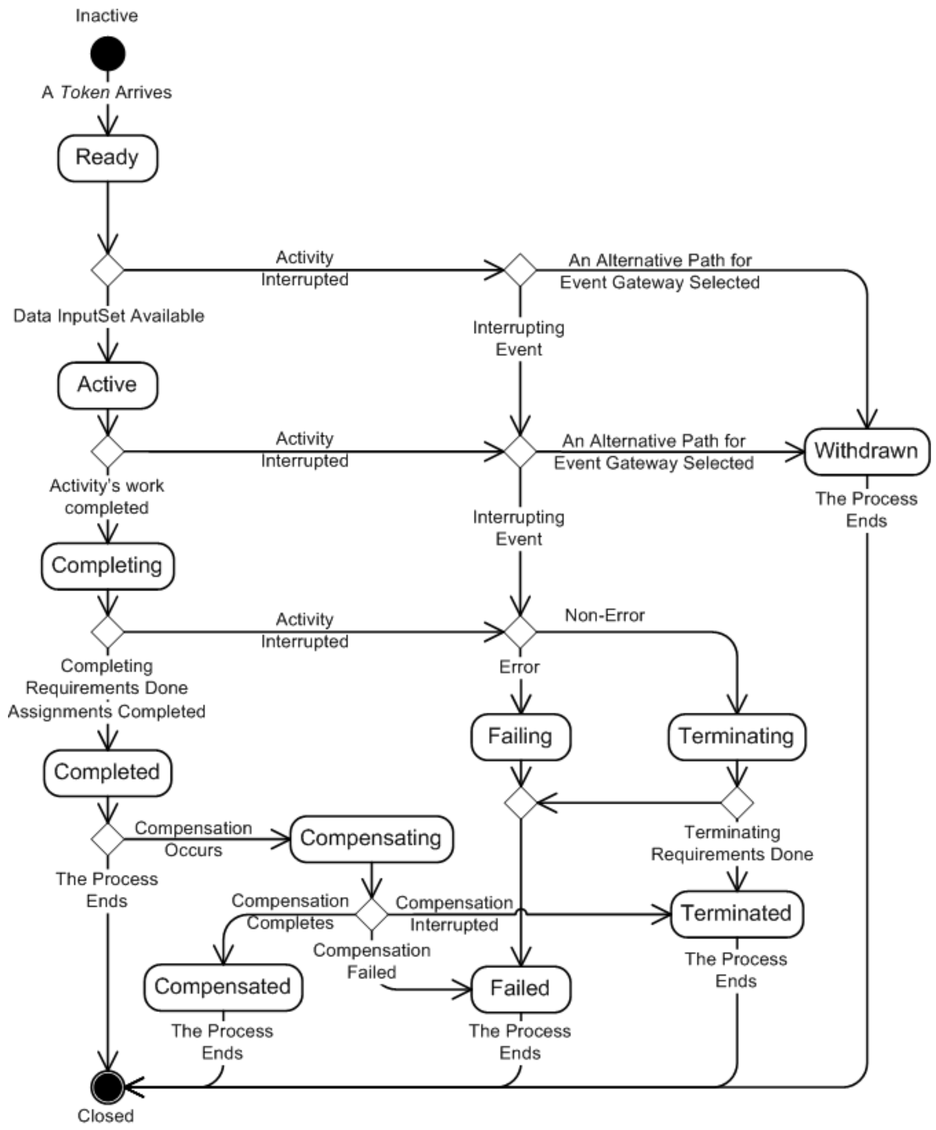
Weske (2019)

States of an Activity



Janiesch (2021), Workflow Management Coaliton (1995)

Lifecycle of an Activity in BPMN



Object Management Group (2011)

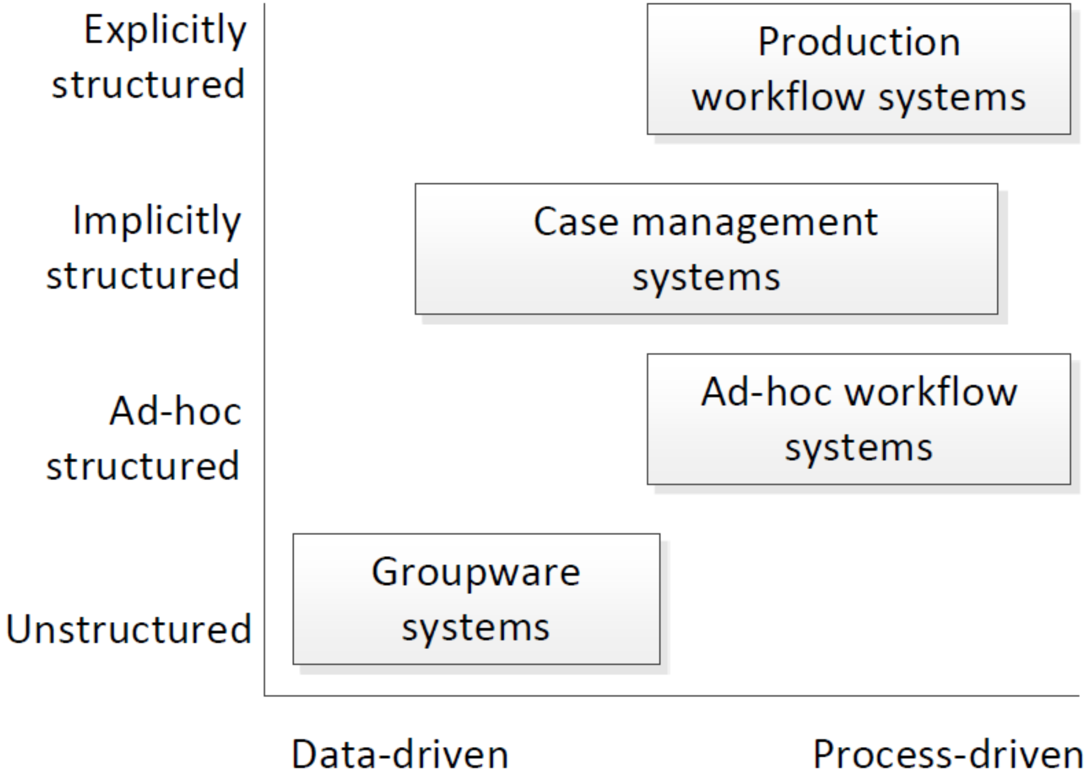
Business Process Management Systems

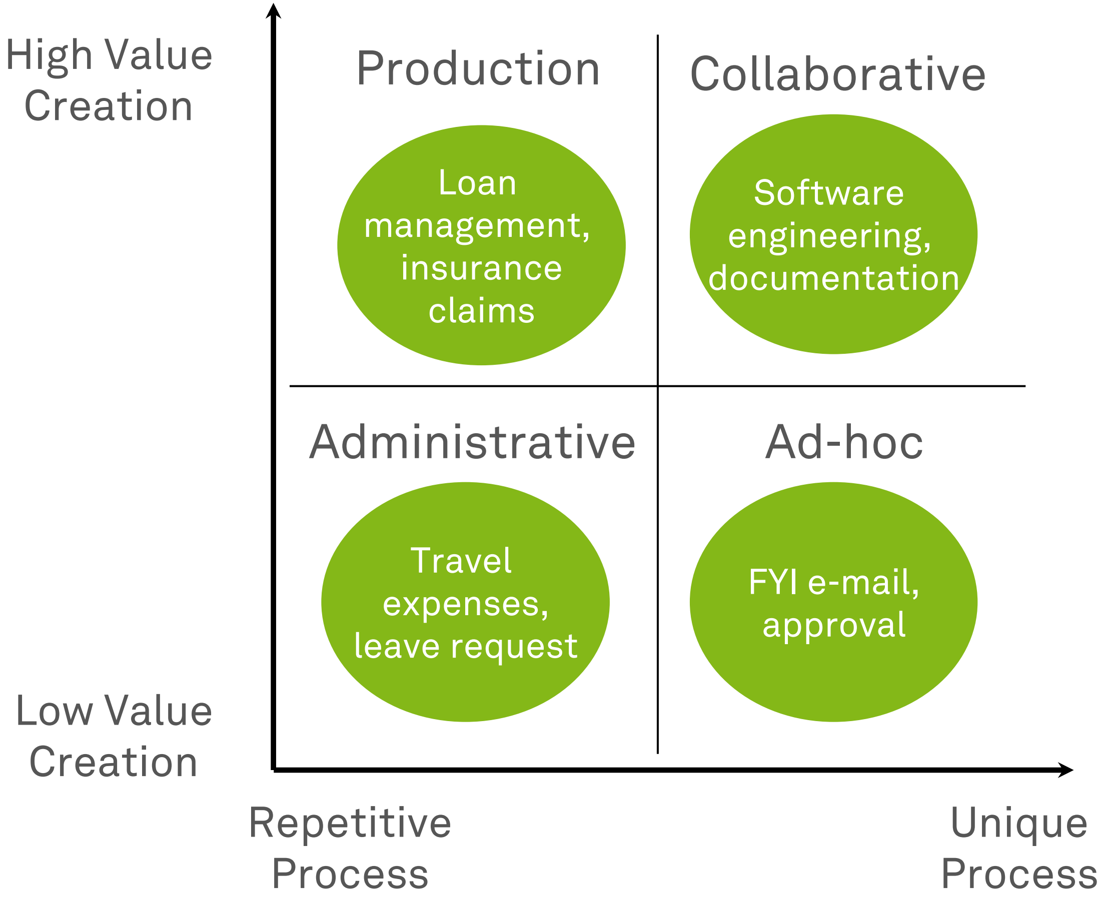
Types of Business Process Management Systems (1)

- **Groupware** systems
  - Do not directly support business processes in a strict sense but allows users to share documents and information
- **Ad hoc workflow systems** systems
  - On-the-fly process definitions that can be created and adapted even during execution
- **Production workflow** systems
  - Most prominent type, distinguished into administrative and transaction processing
  - Work is routed strictly based on process models; data typically handled by complementary DBMS
- (Adaptive) **case management** systems
  - Supports processes that are neither tightly nor completely specified

Dumas et al. (2018)

Types of Business Process Management Systems (2)

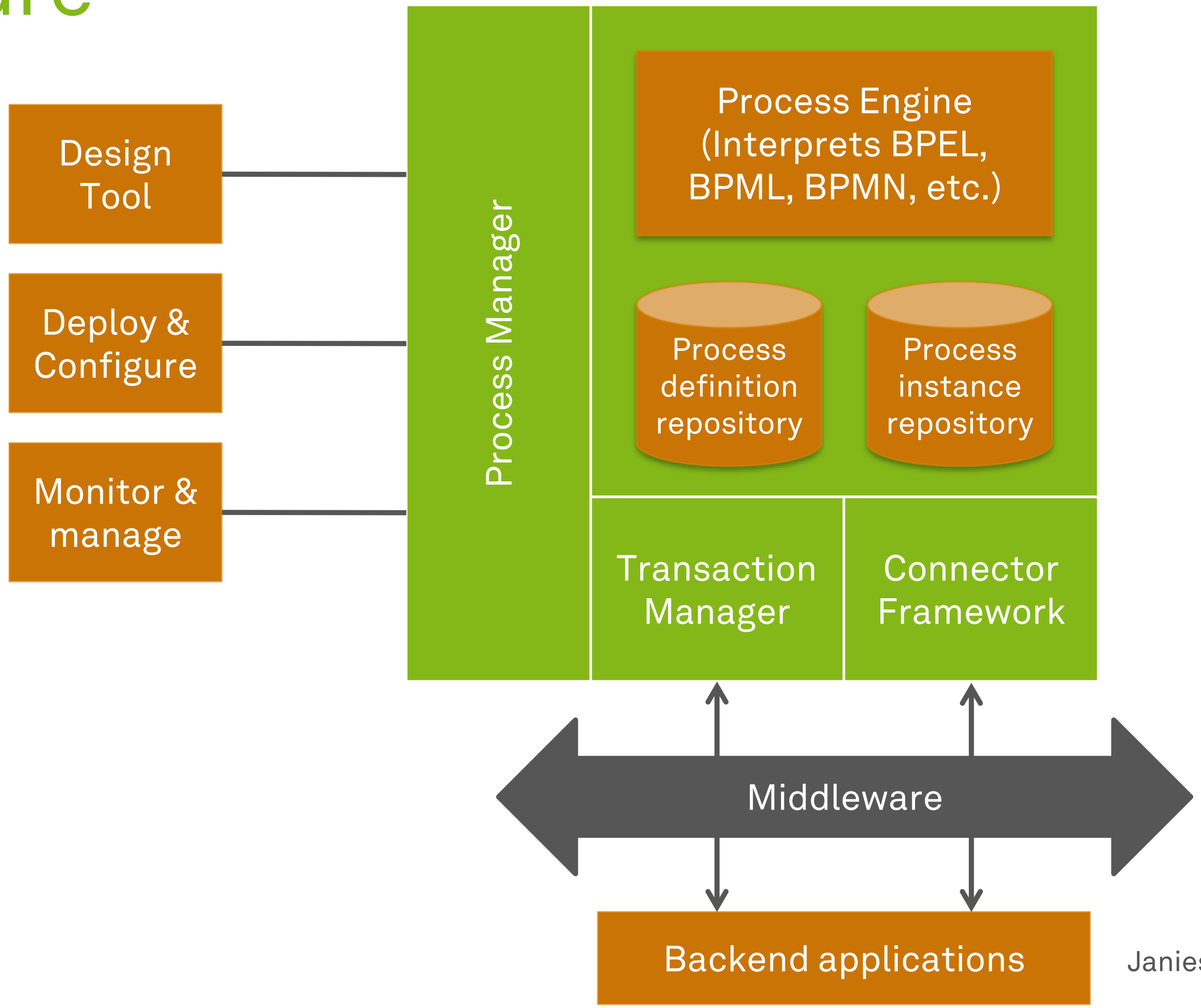




Dumas et al. (2018)

Generic BPMS Architecture

- Provides the **technical platform** for realizing BPM
  - **Process engine**, executing digital process models
  - **Build time** tools for process modeling, simulation and deployment
  - **Run time** facilities for process monitoring and management



Janiesch (2021)

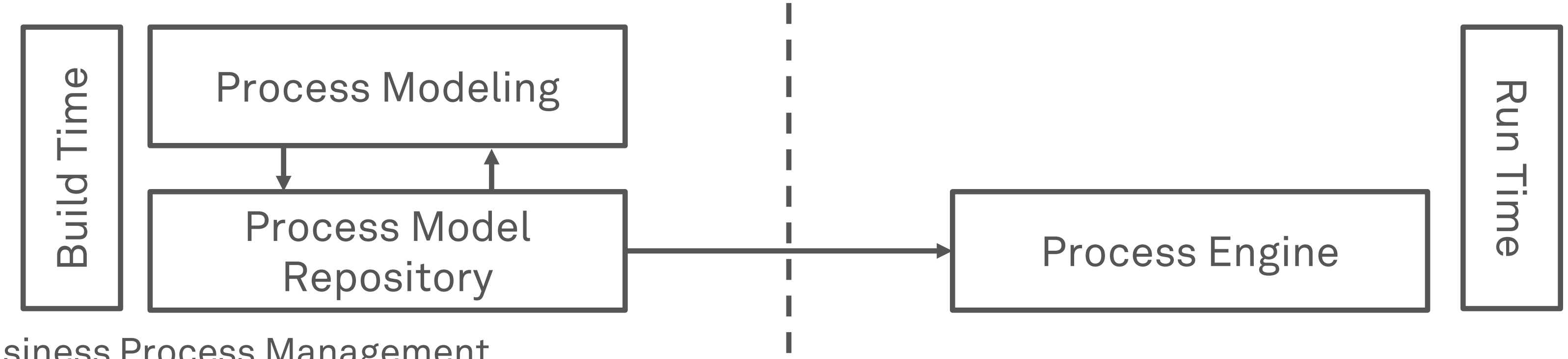
BPMS Architecture

- BPMS standard components
  - **Tooling**
    - Graphically model processes and control BPMS
  - **Repository**
    - Store models in a process definition language
  - **Process Engine**
    - Execute process in a process engine

- BPMS might support additional functionality like
  - Support of different modeling languages
  - Simulation and other analysis methods for process models
  - Support for application integration (EAI middleware features)
  - Monitoring and management of process instances

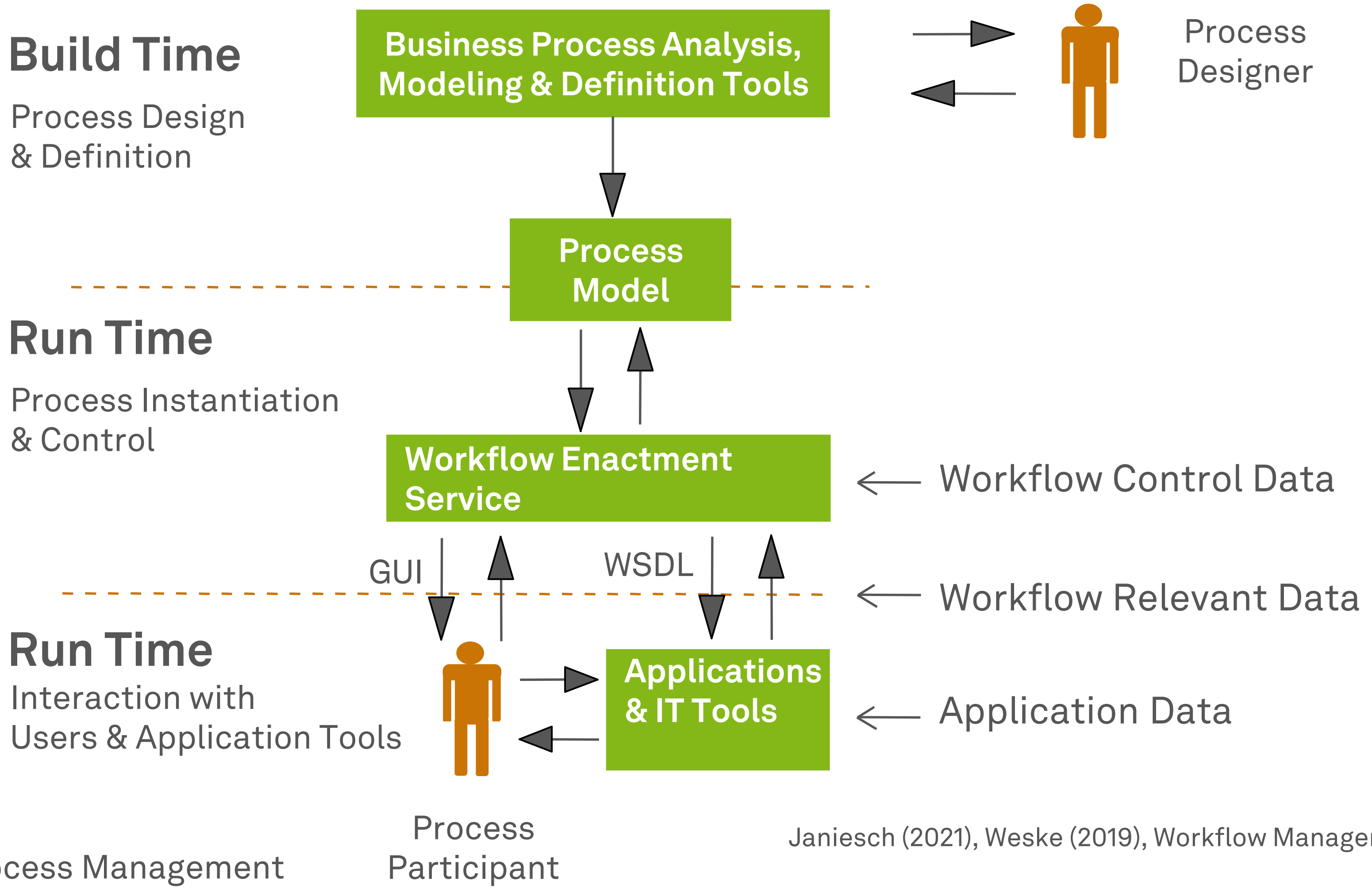
Build Time and Run Time

- A process model has to be **deployed** on the BPMS process model repository in order to be started
- A BPMS creates a **process instance** once the process model is **started**
- A process instance typically **lives** in the BPMS process engine's main **memory**
- The process engine **decides** which activities are executed and **communicates** with client applications
- Traditionally, process instances are **independent of the process model** once being executed



Janiesch (2021), Weske (2019)

BPMS Architecture and Workflow Data



Janiesch (2021), Weske (2019), Workflow Management Coalition (1995)

Advantages of a BPMS

- Workload **Reduction**
  - Straight-through processing
  - Less coordination
  - Less gathering of relevant information
- Flexible System **Integration**
  - Generic functionality of process layer
  - Easier to change process logic
  - Island automation
- Execution **Transparency**
  - Transparency of operational information
  - Transparency of historic information
- Rule **Enforcement**
  - Reducing freedom of executing process
  - Enforce separation of duties
  - Implement control tasks

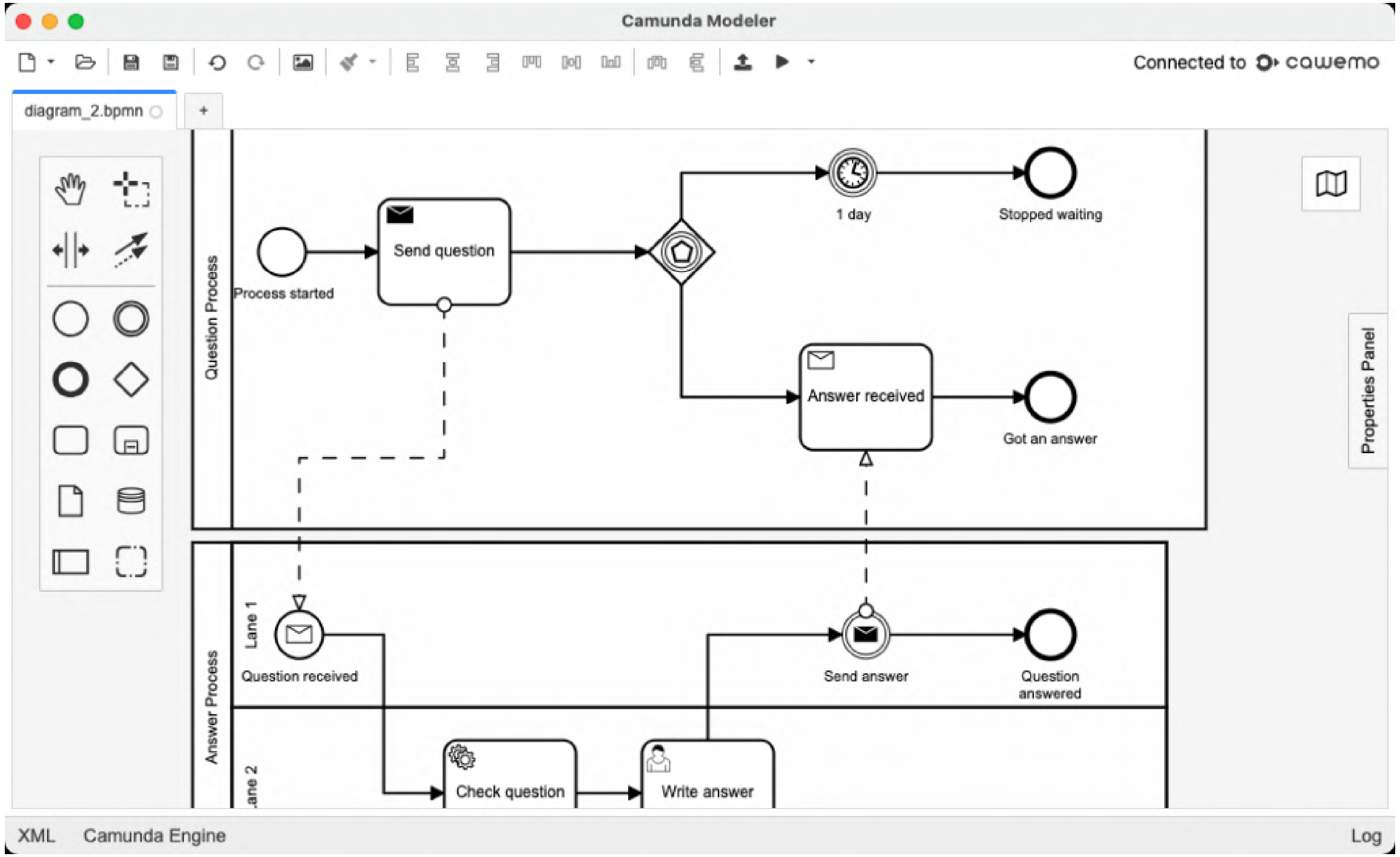
Dumas et al. (2018)

Challenges of BPMS Introduction

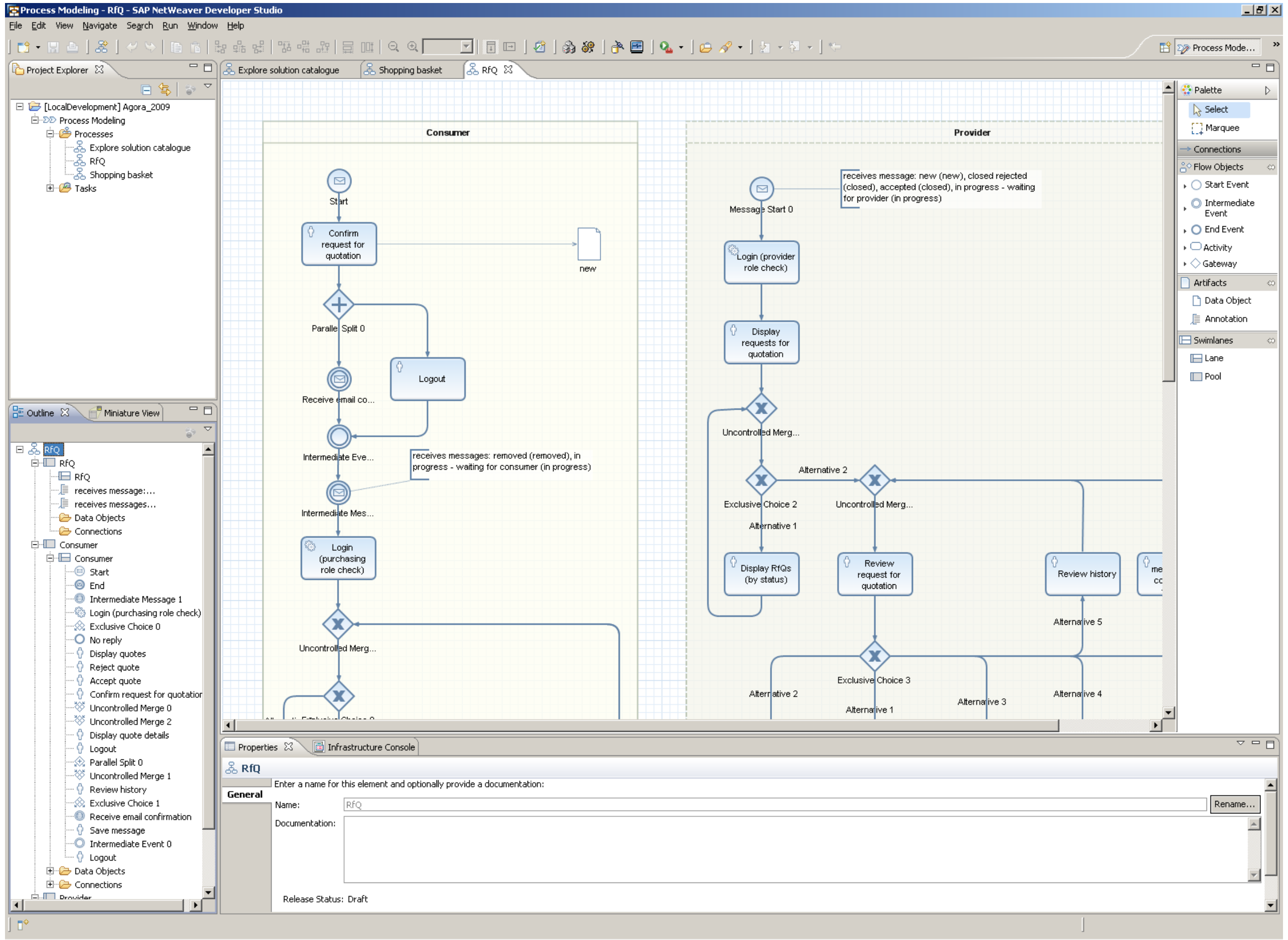
- Technical
  - Applications often not developed from a business process perspective
  - Screen scraping might be required to integrate input and output from **legacy software** (→RPA)
  - **Batch processing systems** do not work with a case concept
  - → Middleware, enterprise application integration (EAI), service-oriented architecture (SOA) and Web service solutions support **integration**
- Organizational
  - **Complexity** due to exceptions
  - Adjust to pace of organizational change
  - Rollout strategy: potential fears of process participants
  - Strong management commitment needed
  - → **Change Management** to mitigate transformation issue

Dumas et al. (2018)

BPMN Tools (1) – Camunda Platform 7



BPMN Tools (2) – SAP NetWeaver BPM



technische universität
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Chair of Enterprise Computing
Department of Computer Science

Robotic Process Automation




Hyperautomation...

- ...is a **business-centric** automation approach...
  - not driven by the IT department
- ...that uses **one or more** technologies...
  - not necessarily enterprise or heavyweight BPM software
- ...to **rapidly automate and augment** business processes...
  - it does not involve change requests, budget evaluations, etc.
- ...in a **scalable** way.
  - elasticity independent of the human factor

hyper-
/ˈhaɪ pər/

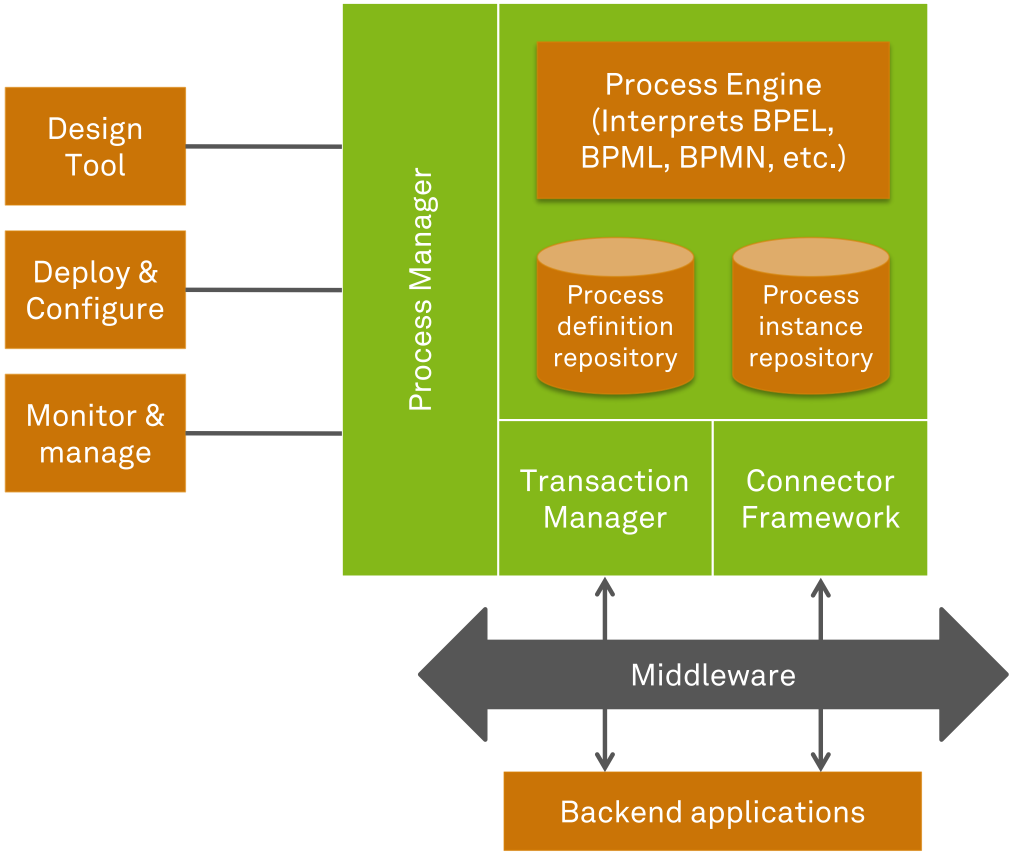
prefix
1. over; beyond; above
2. excess ; exaggeration (hyperbole)

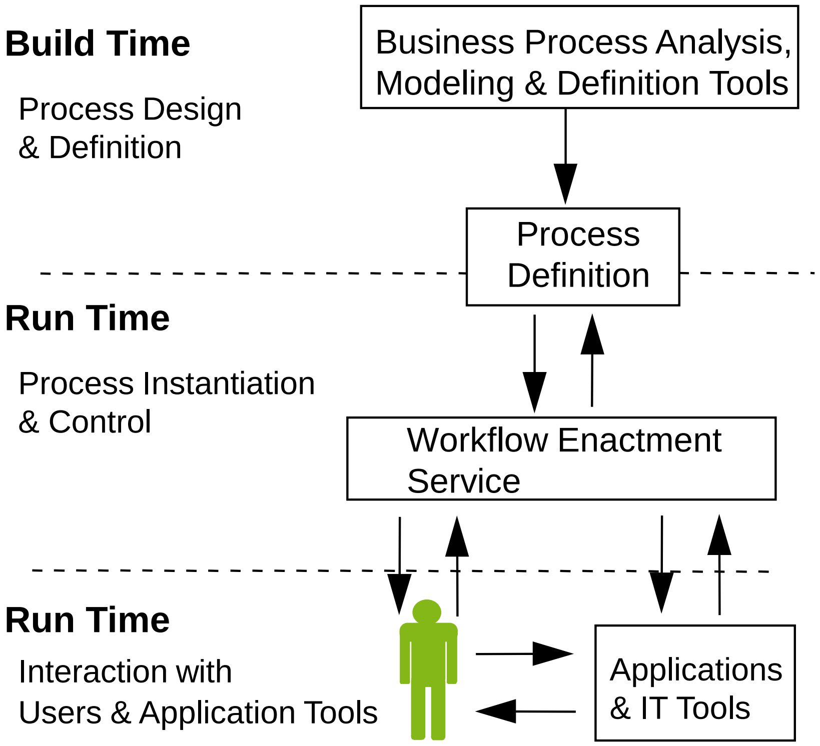
automation
/ˌɔ təˈmeɪ ʃən/

noun
1. the technique, method, or system of operating or controlling a process by highly automatic means, as by electronic devices, reducing human intervention to a minimum
2. ...

dictionary.com (2022), Reijers & Jonker (2020)

Digitalization with Traditional BPMS





Janiesch (2021), Workflow Management Coalition (1995)

Digitalization with Traditional BPMS

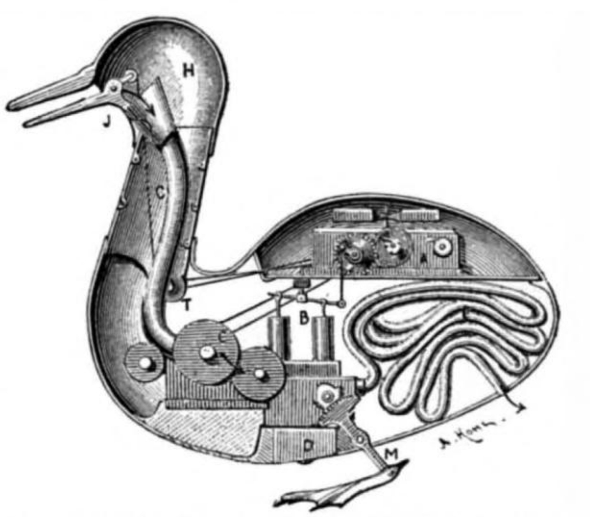
- **BPMS**
  - ...are typically implemented by the **IT department**
  - ...may use multiple technologies to implement activities, but only use **one technology** to rule them all (orchestration)
  - ...**take time** to implement
  - ...do **scale** ☺

Backend applications

Janiesch (2021), Workflow Management Coalition (1995)

Robotic Process Automation

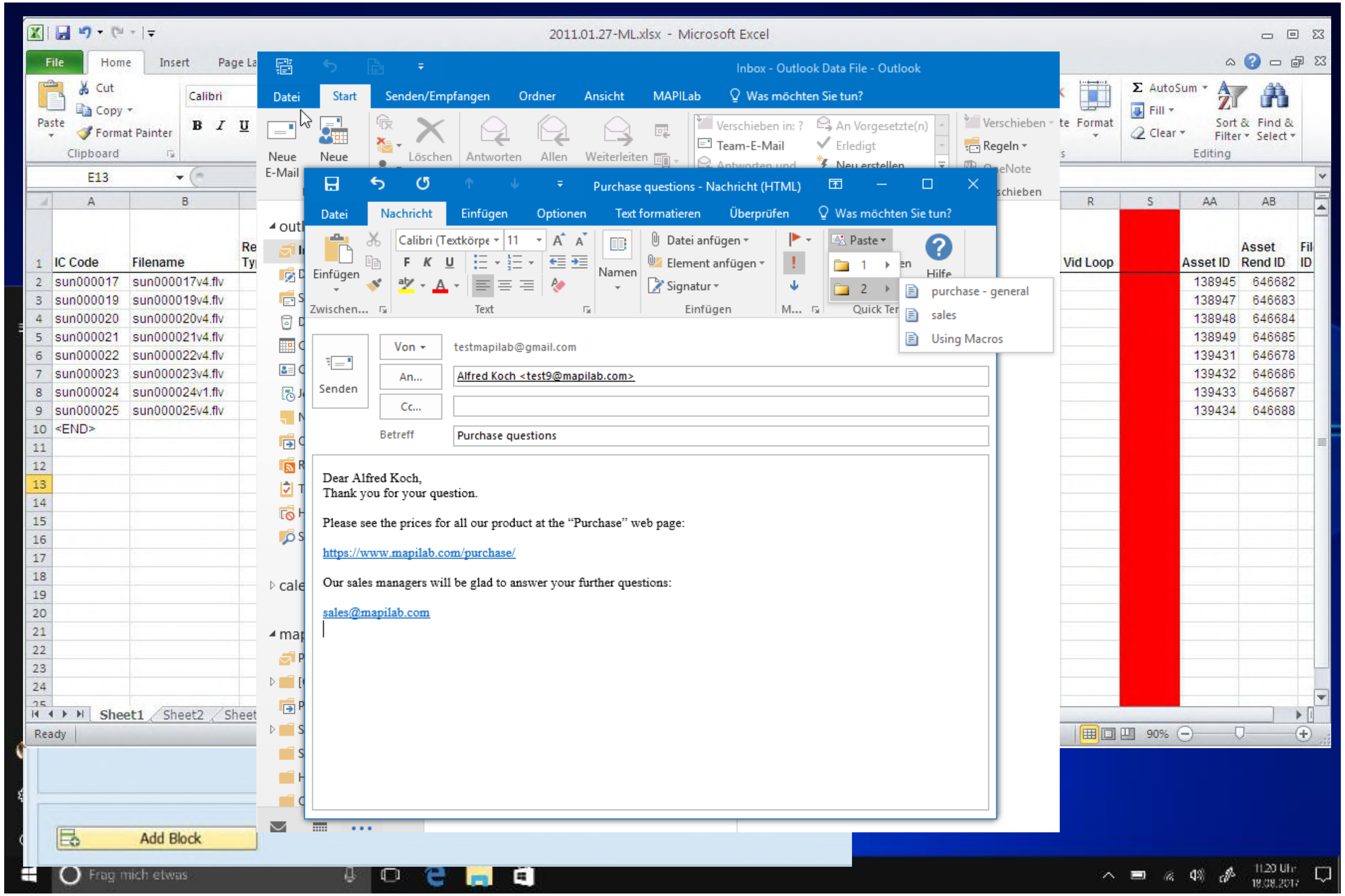
- General Idea
  - Develop an automation (=robot) that is **functionally equivalent** to a human (acting in a specific business process)
- Characteristics of (ro)bots
  - RPA bot is a **single instance** of an RPA system
  - RPA bots are **software** (robots)
  - RPA bots access only the **user interface**
  - RPA bots execute activities using **some rules**
  - RPA is **elastic**
- Result
  - Scalable automation without change in legacy systems



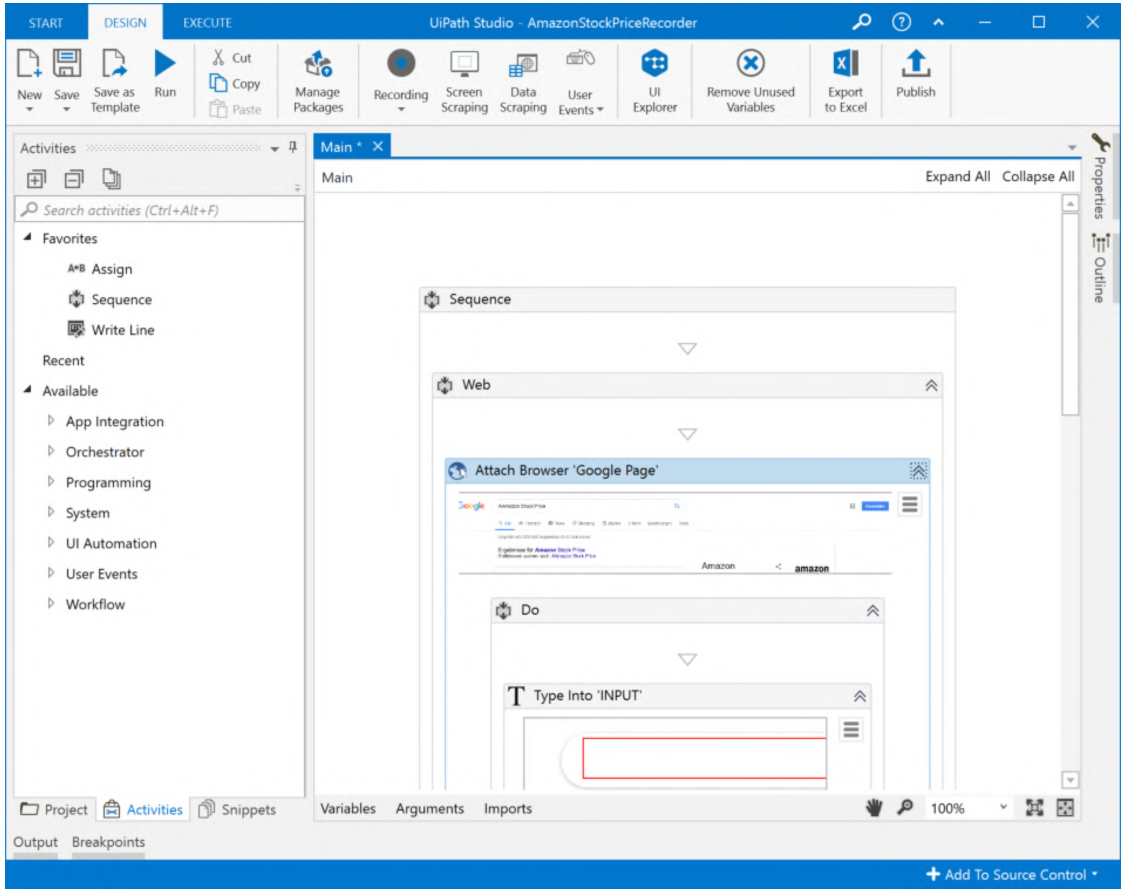
INTERIOR OF VAUCANSON'S AUTOMATIC DUCK.
A, clockwork; B, pump; C, mill for grinding grain; F, intestinal tube; J, bill; H, head; M, feet.

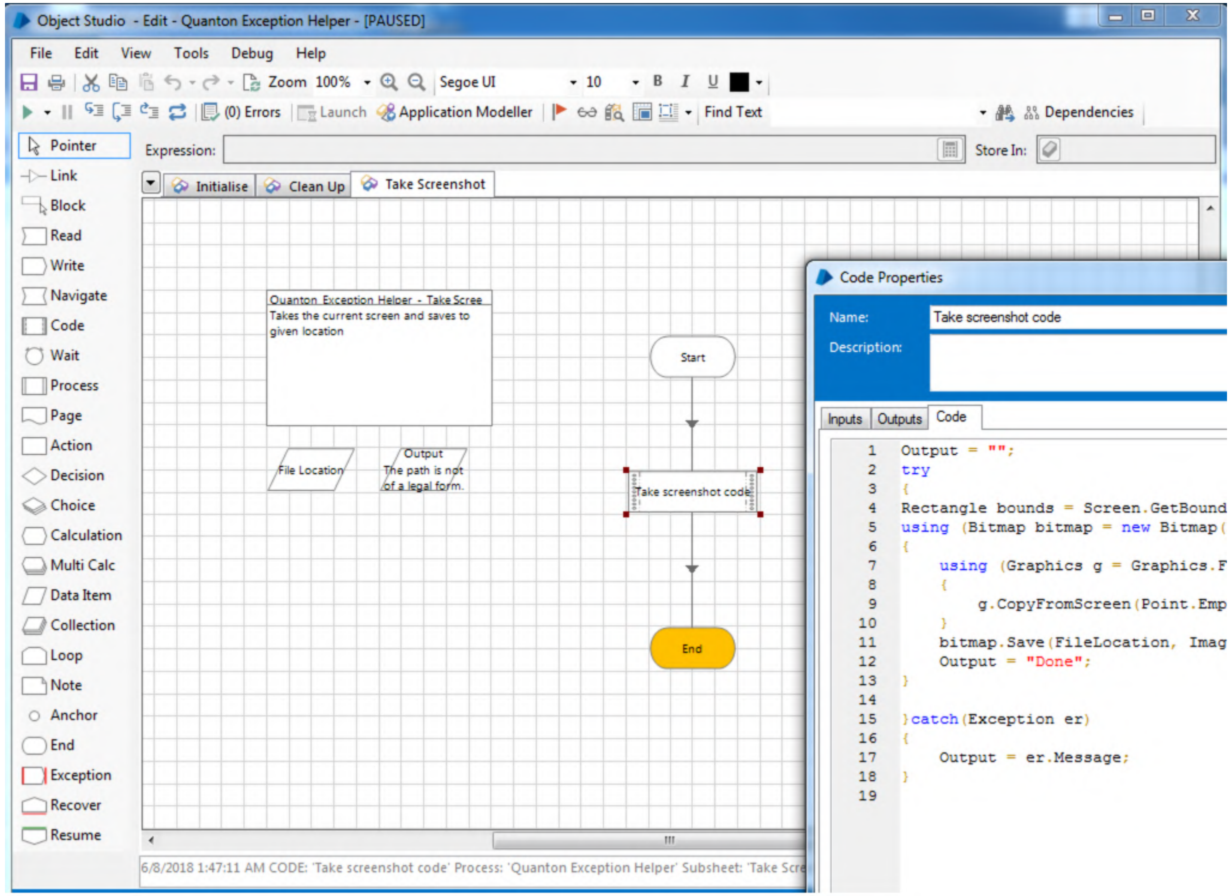
Czarnecki & Fettke (2021), Janiesch (2020), Scientific American (1899)

An RPA Workflow



RPA Software





UIPath (2022), Blue Prism (2022)

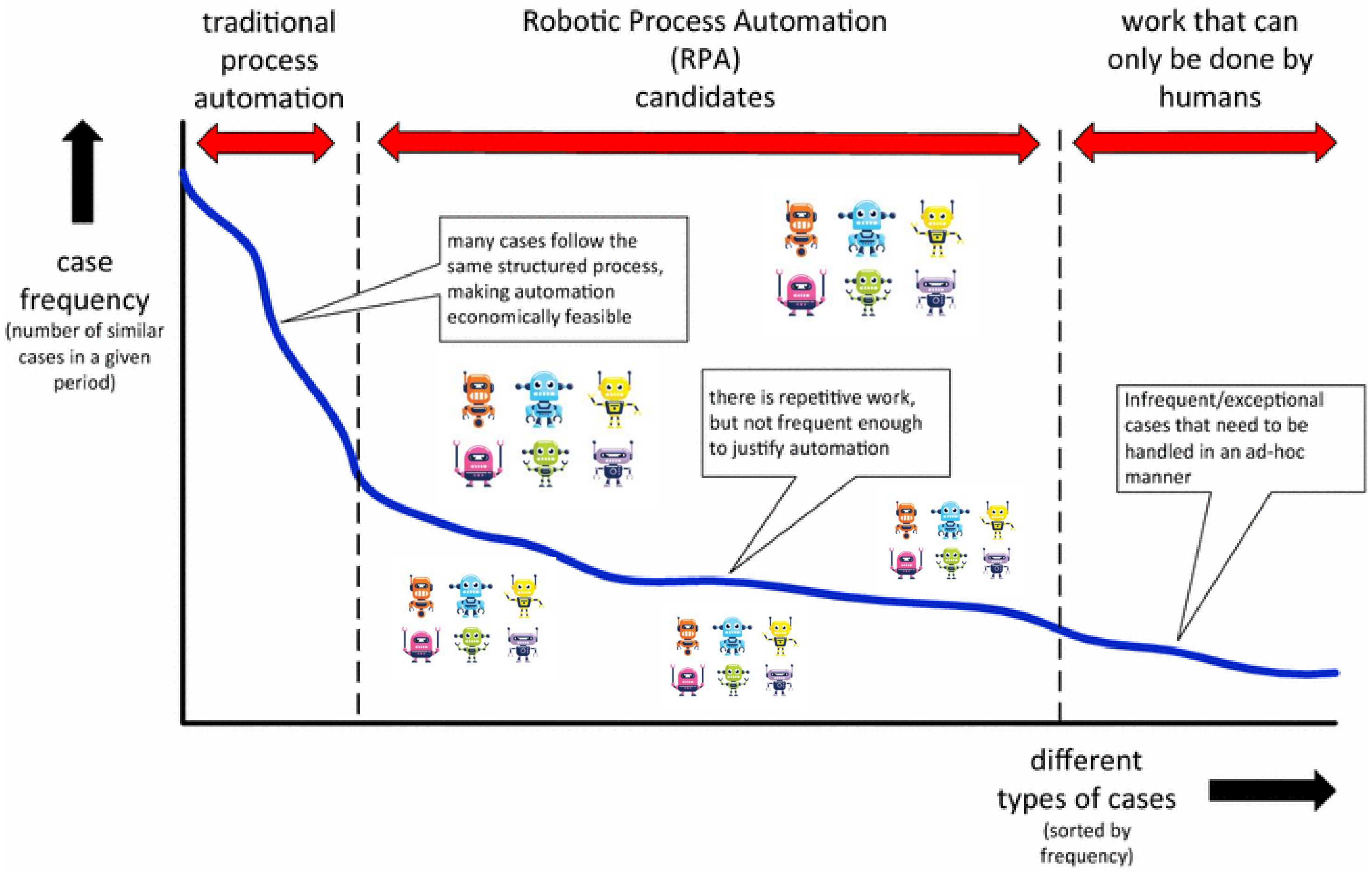
RPA vs. BPM

Digital yet manual tasks
Non disruptive
Rapid and agile
Business driven, DIY

End-to-end processes
Transformational
Long-term
IT-driven

Janiesch (2021), Reijers & Jonker (2020)

BPM and RPA Happily Ever After!



van der Aalst et al. (2018)

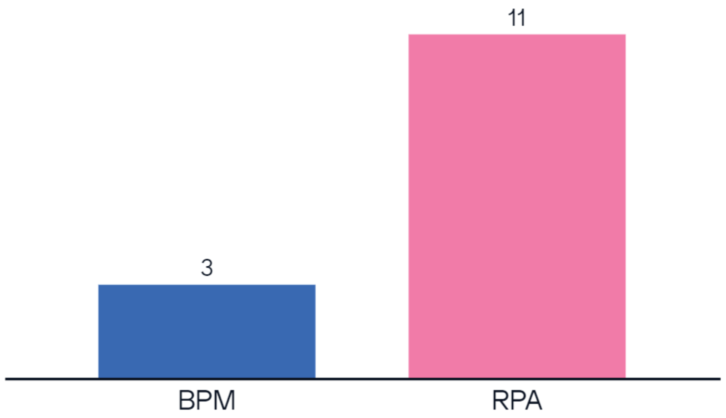
How would you decide? Municipal Health Services (MHS) in the Netherlands during COVID-19

- Covid-19 contact tracing researchers from the MHS complain about the use of system PZ
- People complain that it is illogical and cumbersome
- It has a tricky UI and UX, sometimes crashes and takes an hour to log in again
- The various MHS regions within the system cannot exchange information as they a technically separated, so in formation has to be passed on outside the system
- PZ is MHS's trusted program and cannot be replaced in an instant

- Would you rather have used **BPM** or **RPA** to alleviate the situation?

Reijers & Jonker (2020)

Exemplary Results



fast solution is needed

A fast and agile approach is needed to ensure functionality

Actually, I would have clicked both.

need a short term solution

The problems need to be solved within a short period of time

BPM is to complicated for non IT people to understand and implement in a short time

More appropriate

Reijers & Jonker (2020)

Learning Goals of Today

- Understand the evolution of process technology
- Understand workflow execution semantics
- Understand BPMS
- Understand RPA

Recap Questions

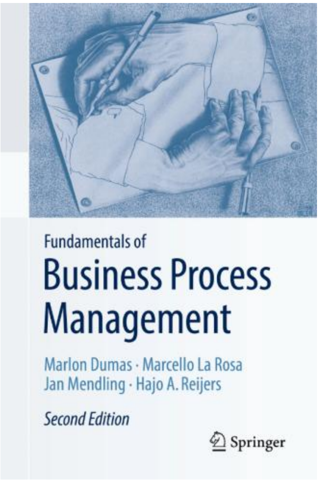
- What are execution semantics and how does a BPM system step through a process?
- What are the principal components of a generic BPM system and how are they arranged?
- What is the relation of robotic process automation to BPM systems?

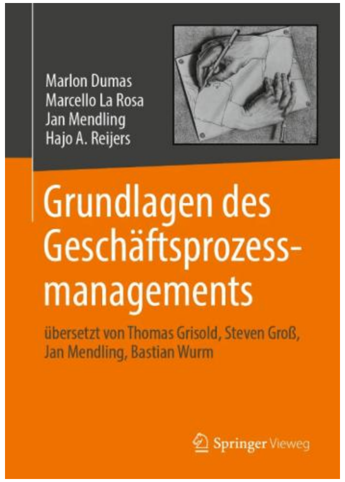
Literature

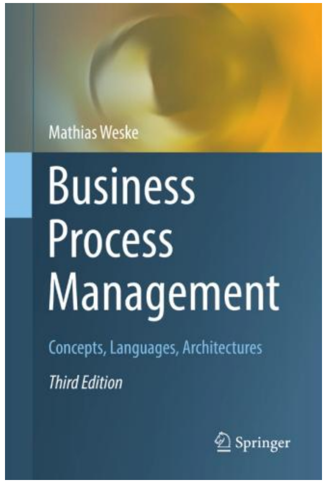
- **Section 9:** Process-Aware Information Systems

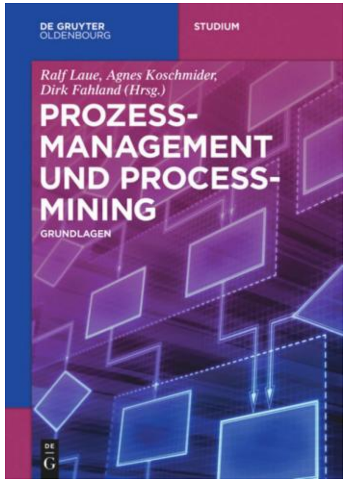
- **Section 2:** Evolution of Enterprise Systems Architectures
- **Section 3:** Business Process Modelling Foundation
- **Section 8:** Business Process Management Architectures

- **Section 9:** Geschäftsprozessmanagement-systeme und Robotic Process Automation









Roadmap

Introduction	10.04. Introduction	17.04. Process Identification			
Process Discovery & Modeling	24.04. Essentials of BPMN	01.05. Public Holiday	08.05. Advanced BPMN I	15.05. Advanced BPMN II	22.05. Process Discovery
Process Analysis & Redesign	29.05. Public Holiday	05.06. Qualitative Analysis	12.06. Quantitative Analysis	19.06. Public Holiday	26.06.2025 Process Redesign
Application of BPM	03.07. Process Technology	10.07. Process Monitoring and Mining	15.07. BPM as an Enterprise Capability		

Any Questions about Process Technology?

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Business Process Management

Process Monitoring and Process Mining



Roadmap

Introduction	10.04. Introduction	17.04. Process Identification			
Process Discovery & Modeling	24.04. Essentials of BPMN	01.05. Public Holiday	08.05. Advanced BPMN I	15.05. Advanced BPMN II	22.05. Process Discovery
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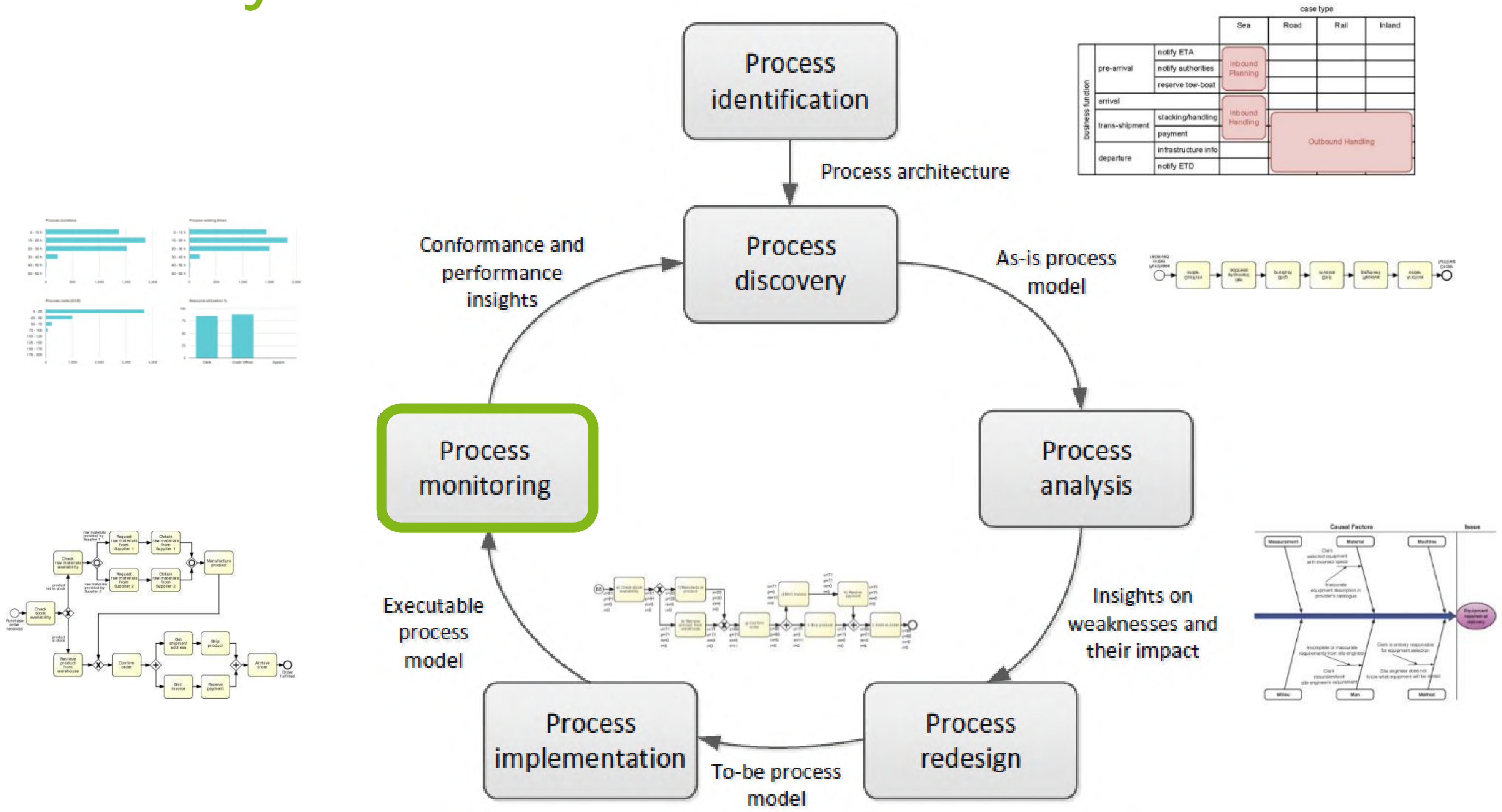
Learning Goals of Today

- Understand process monitoring with process dashboards
- Understand automated process discovery
- Understand process conformance checking
- Understand variants analysis and process performance mining

The BPM Lifecycle



Dumas et al. (2018)

Process Monitoring (1)

- Use the data generated by the execution of a business process
  - **Event record** captures state change in execution of **work items**
  - Collection of time-stamped event records is an **event log**
- Extract insights about the actual **model** and **performance** of the process
- Verify its **conformance** with respect to norms, policies, or regulations

- Analysis can be done
  - Post-mortem (**offline**) or on-the-fly (**online**)
- Techniques are
  - **Statistics**-based (dashboards) or **model**-based (process mining)

Dumas et al. (2018)

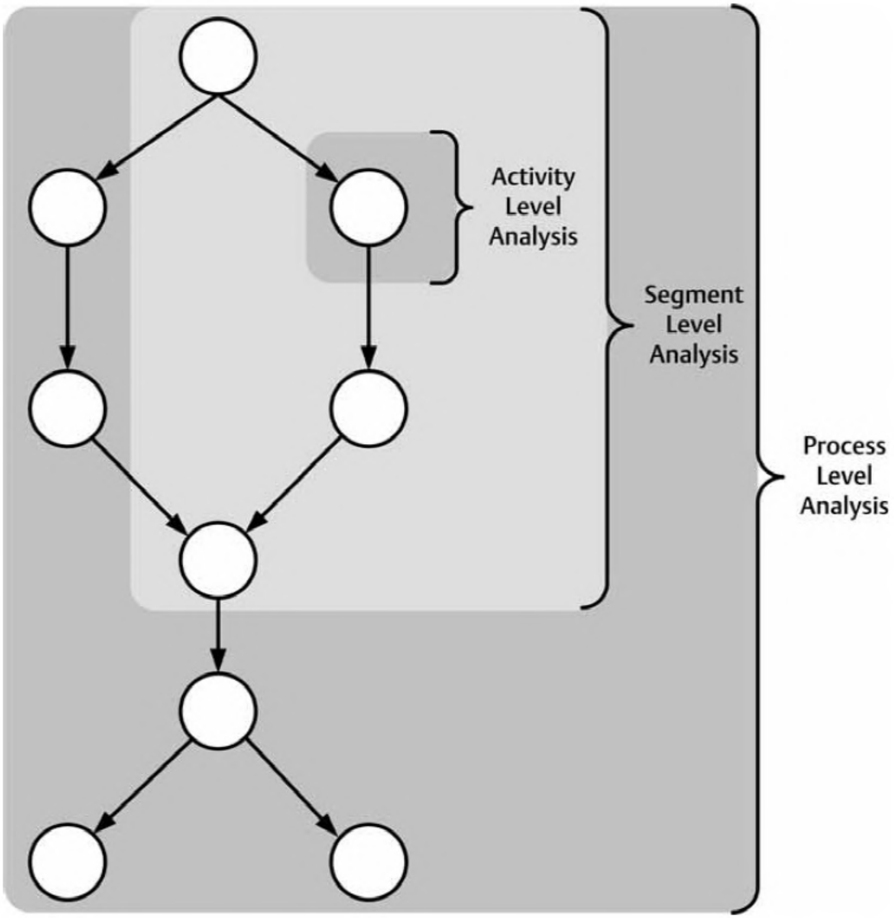
Process Monitoring (2)

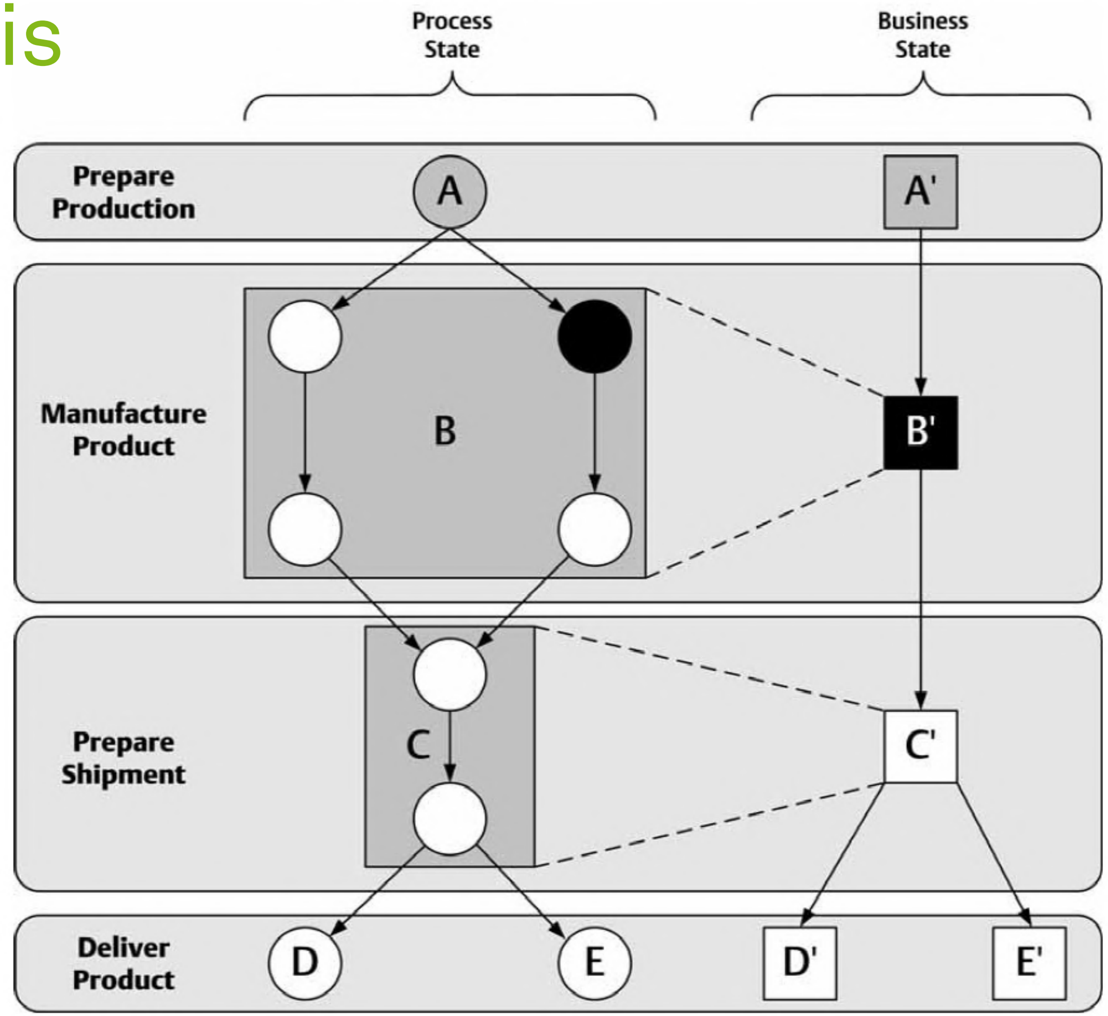
- **Offline Process Monitoring**
  - Covers **historic** process executions
  - Provides a picture of
    - the **performance** of the process
    - the **reasons** for
      - poor performance or
      - for undesirable performance variations
    - the **conformance** of the process with respect to certain rules or expected behaviors

- **Online Process Monitoring**
  - Covers **currently running** process instances
  - Produces **real-time** pictures of
    - the **performance** of ongoing cases
  - Generate **alarms** or trigger **counteractions**
    - when certain **performance objectives** or **compliance rules** are **not fulfilled**

Dumas et al. (2018)

Granularity of Process Analysis





zur Mühlen (2004)

Process Monitoring Dimensions

Attribute	Possible Values					
Focus	Technical Information	Business Information				Presentation
Presentation	Active	Passive				Presentation
Timeframe	Running Processes	Completed Processes				Presentation
Aggregation	Single Instance	Multiple Instances				Presentation
Data Scope	Process	Process + Business Objects	Enterprise			Information
Object	Event	Activity	Process	Resource	Business Object	Information
Process Scope	Activity	Segment	Process	Process Chain		Information

zur Mühlen (2004)

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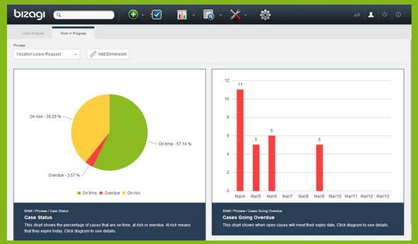
Statistics-based Process Monitoring with Dashboards

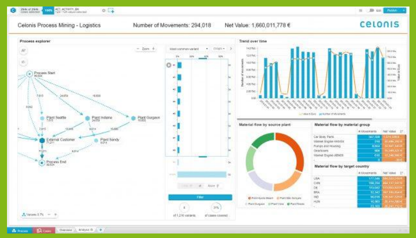
Types of Process Dashboards

- Process dashboards are often provided **off-the-shelf** by BPMS

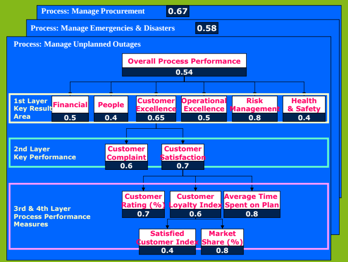
Operational dashboards (real-time)



Tactical dashboards (historical)

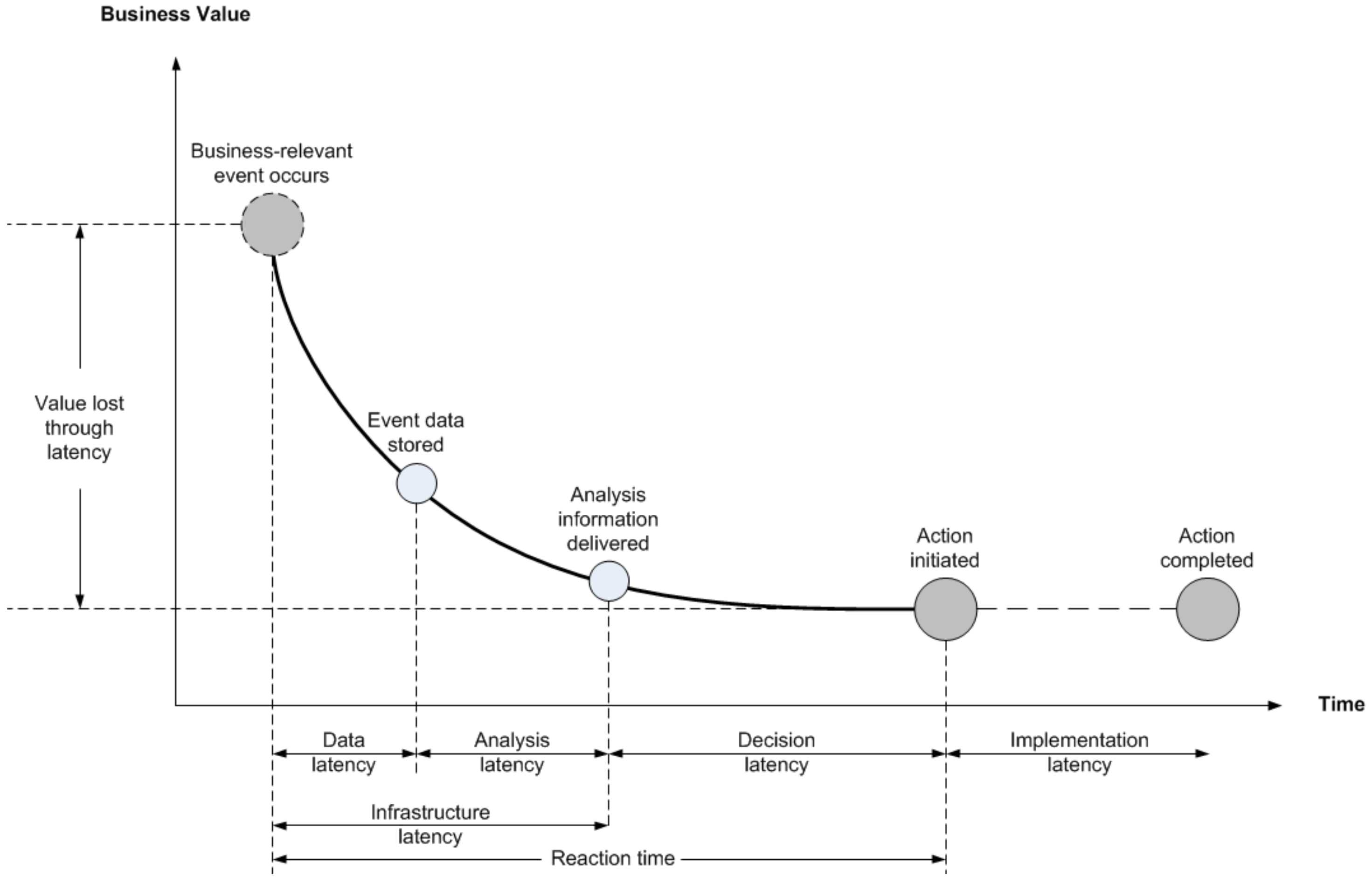


Strategic dashboards (historical)



Dumas et al. (2018)

Information Latency



zur Mühlen, Shapiro (2010)

Operational Dashboards

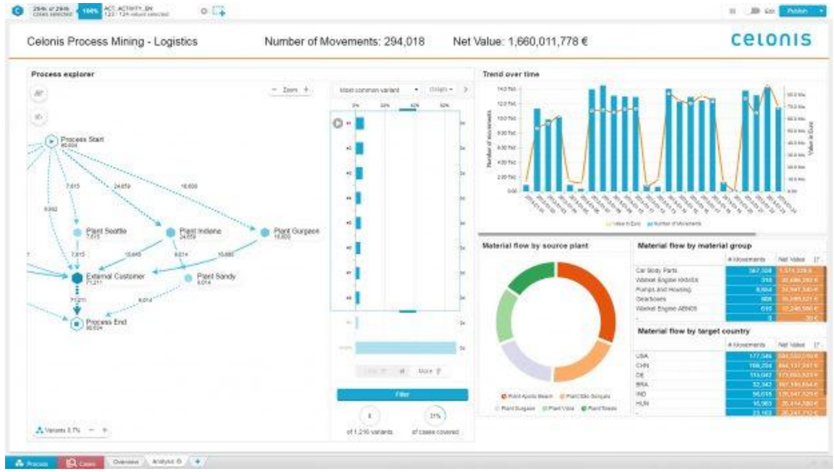
- Aimed at
  - Process **participants** and **operational managers**
  - **Ongoing** or recently completed cases
- Emphasis on **detect-and-respond**
  - E.g., work-in-progress, resource load, problematic cases (e.g., overdue/at-risk cases)
- Focus on
  - Level of resources
  - Tasks rather than cases

Dumas et al. (2018)

Tactical Dashboards

- Aimed at
  - Process **owners** and **functional managers**, **process analysts**
  - Process over a **relatively long period of time** (e.g., month, quarter, year)
- Emphasis on **analysis and management**
  - E.g., detecting bottlenecks
- Focus on
  - Process level (may be decomposable to task level)

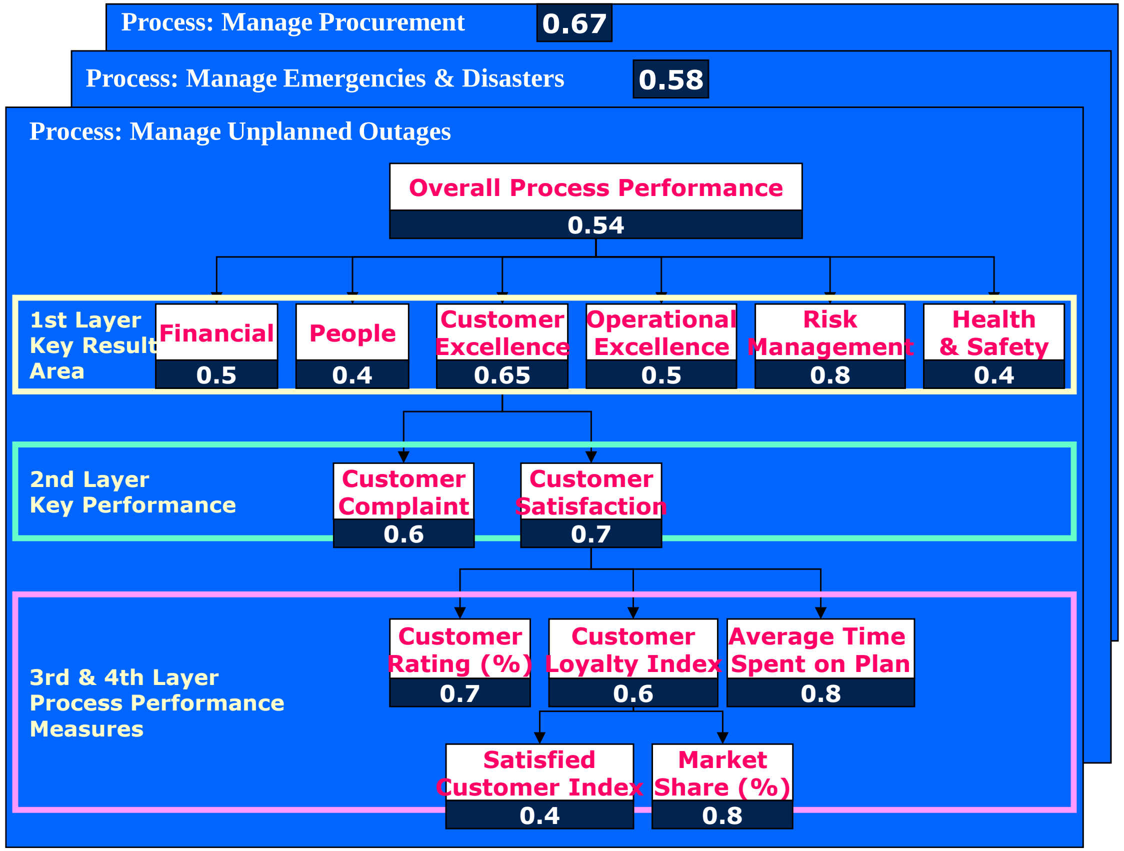
- **Longitudinal** view
  - Variation of a given performance measure over time
- **Cross-sectional** view
  - Distribution of a performance measure according to a given classification of cases



Dumas et al. (2018)

Strategic Dashboards

- Aimed at
  - **Executive managers**
- Emphasis on **high-level** performance and **link to strategic objectives**
  - Aggregating individual processes into entire **groups of processes**
  - Aggregating multiple performance measures into a **single measure**
- Focus on
  - Groups of processes
  - Performance dimensions



Dumas et al. (2018)

Monitoring Multiple BPMS with Business Activity Monitoring

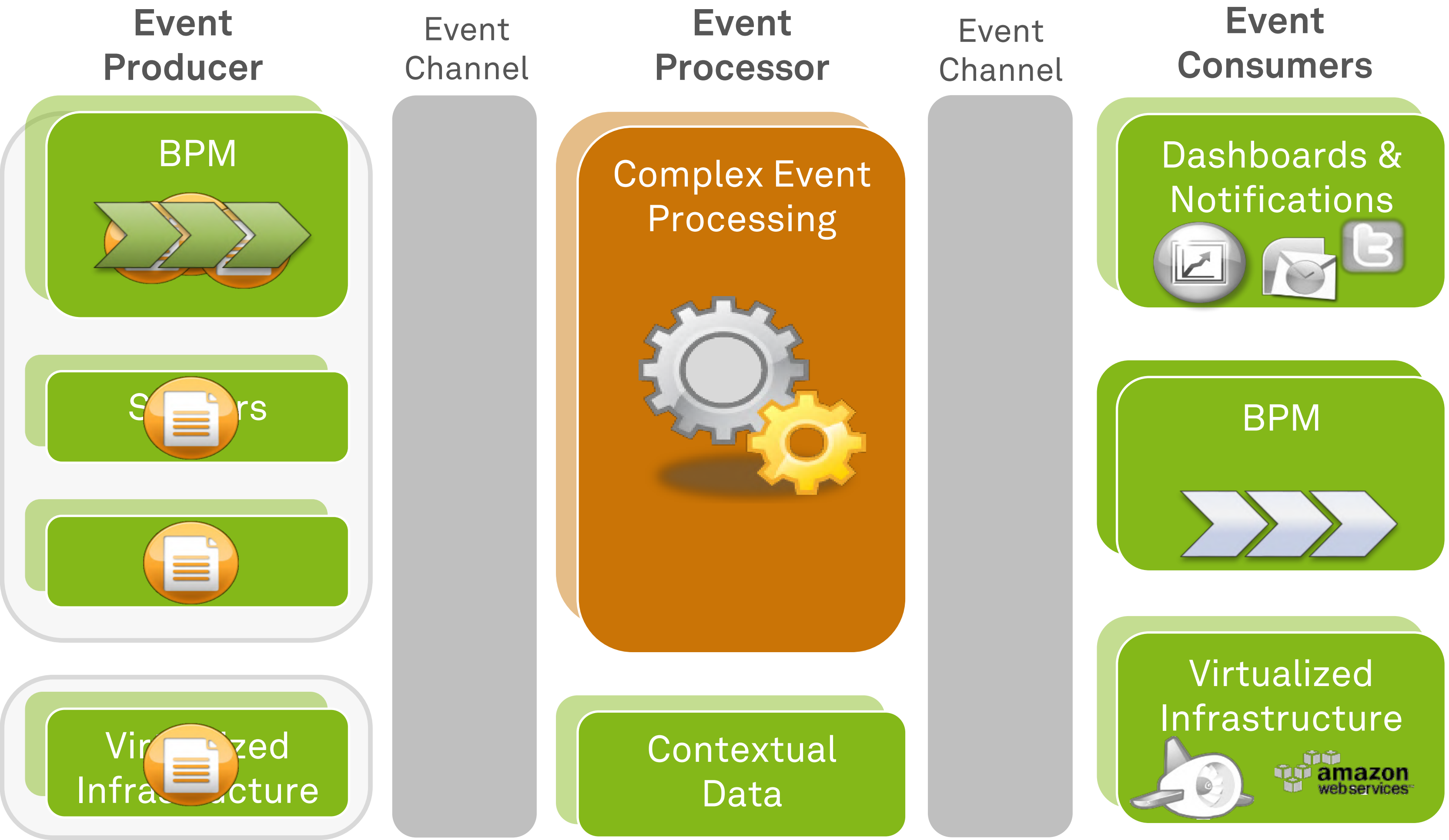
- **BAM** is **Gartner's** term defining
  - how we can provide **real-time access** to critical business performance indicators
  - to improve the **speed and effectiveness of business operations**

- Unlike traditional real-time monitoring, BAM draws its information from
  - **multiple application systems** and
  - other **internal** and **external** (inter-enterprise) sources
  - **enabling** a broader and richer view of business activities

- **There is no universal BAM software with a process focus on the market**

Gartner (2002)

Real-time Event-driven Architecture



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Model-based Process Monitoring with Process Mining

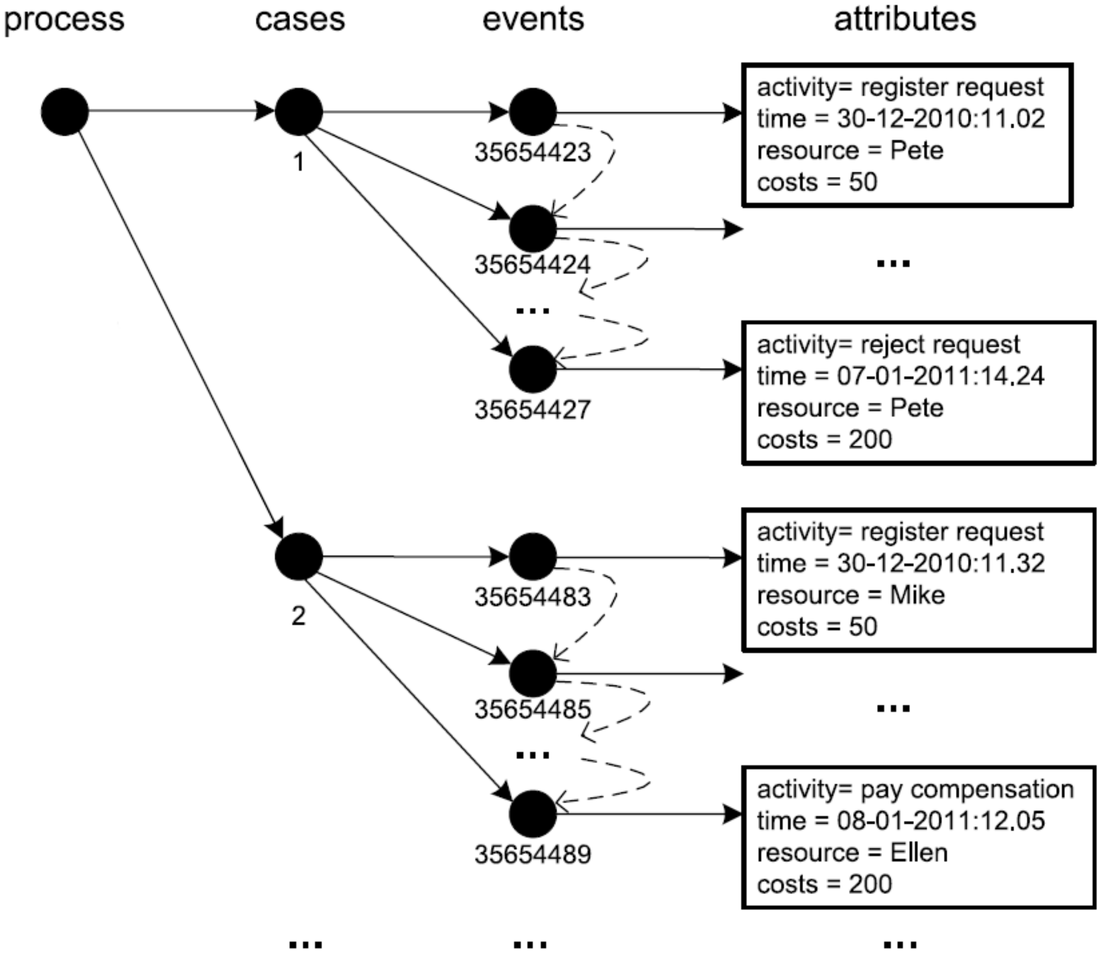
Process Mining

- “Process mining aims to **exploit event data** in a meaningful way”
- “Process mining techniques use event data to **discover** processes, check **compliance**, analyze **bottlenecks**, compare process **variants**, and suggest improvements”
- “Process mining adds the **process perspective** to machine learning and data mining”
- “Process mining brings together **traditional model-based process analysis** and **data-centric analysis techniques**”

- **Process mining can be performed exploratively or question-driven**

van der Aalst (2016), Dumas et al. (2018)

Structure of an Event Log



Event Record

Minimum attributes: Case ID, timestamp, event class

Event Log

Trace (ordered sequence)

Case ID	Event ID	Timestamp	Activity	Resource
1	Ch-4680555556-1	2012-07-30 11:14	Check stock availability	SYS1
1	Re-5972222222-1	2012-07-30 14:20	Retrieve pr	k
1	Co-6319444444-1	2012-07-30 15:10	Confirm ord	uck
1	Ge-6402777778-1	2012-07-30 15:22	Get shippin	2
1	Em-6555555556-1	2012-07-30 15:44	Emit invoic	2
1	Re-4180555556-1	2012-08-04 10:02	Receive pay	2
1	Sh-4659722222-1	2012-08-05 11:11	Ship product	Susi
1	Ar-3833333333-1	2012-08-06 09:12	Archive order	DMS
2	Ch-4055555556-2	2012-08-01 09:44	Check stock availability	SYS1
2	Ch-4208333333-2	2012-08-01 10:06	Check materials availability	SYS1
2	Re-4666666667-2	2012-08-01 11:12	Request raw materials	Ringo
2	Ob-3263888889-2	2012-08-03 07:50	Obtain raw materials	Olaf
2	Ma-6131944444-2	2012-08-04 14:43	Manufacture product	SYS1
2	Co-6187615741-2	2012-08-04 14:51	Confirm order	Conny
2	Em-6388888889-2	2012-08-04 15:20	Emit invoice	SYS2
2	Ge-6439814815-2	2012-08-04 15:27	Get shipping address	SYS2
2	Sh-7277777778-2	2012-08-04 17:28	Ship product	Sara
2	Re-3611111111-2	2012-08-07 08:40	Receive payment	SYS2
2	Ar-3680555556-2	2012-08-07 08:50	Archive order	DMS
3	Ch-4208333333-3	2012-08-02 10:06	Check stock availability	SYS1
3	Ch-4243055556-3	2012-08-02 10:11	Check materials availability	SYS1
3	Ma-6694444444-3	2012-08-02 16:04	Manufacture product	SYS1
3	Co-6751157407-3	2012-08-02 16:12	Confirm order	Chuck

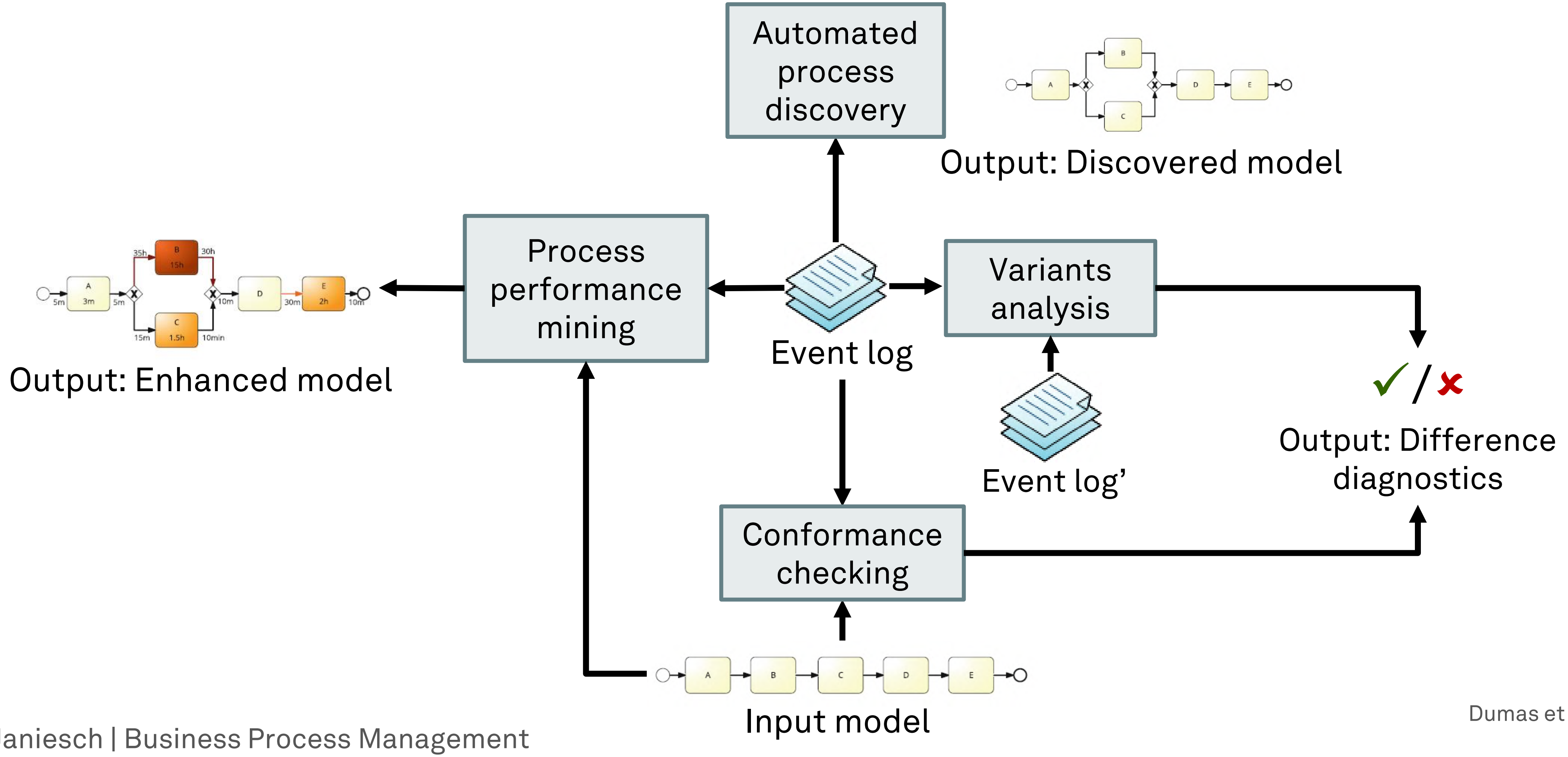
van der Aalst (2016), Dumas et al. (2018)

Process Mining Challenges

- **Correlation** challenge
  - Identify the case an event belongs to
- **Timestamps** challenge
  - Logging is often delayed until the system has idle time: sequential events with the same timestamp, logs from different BPMS/time zones
- **Longevity** challenge
  - Long running processes might be too slow for snapshot window
- **Granularity** challenge
  - Abstraction of model might be different from log
- **Scoping** challenge
  - Information system does not directly produce event logs, logs are synthesized

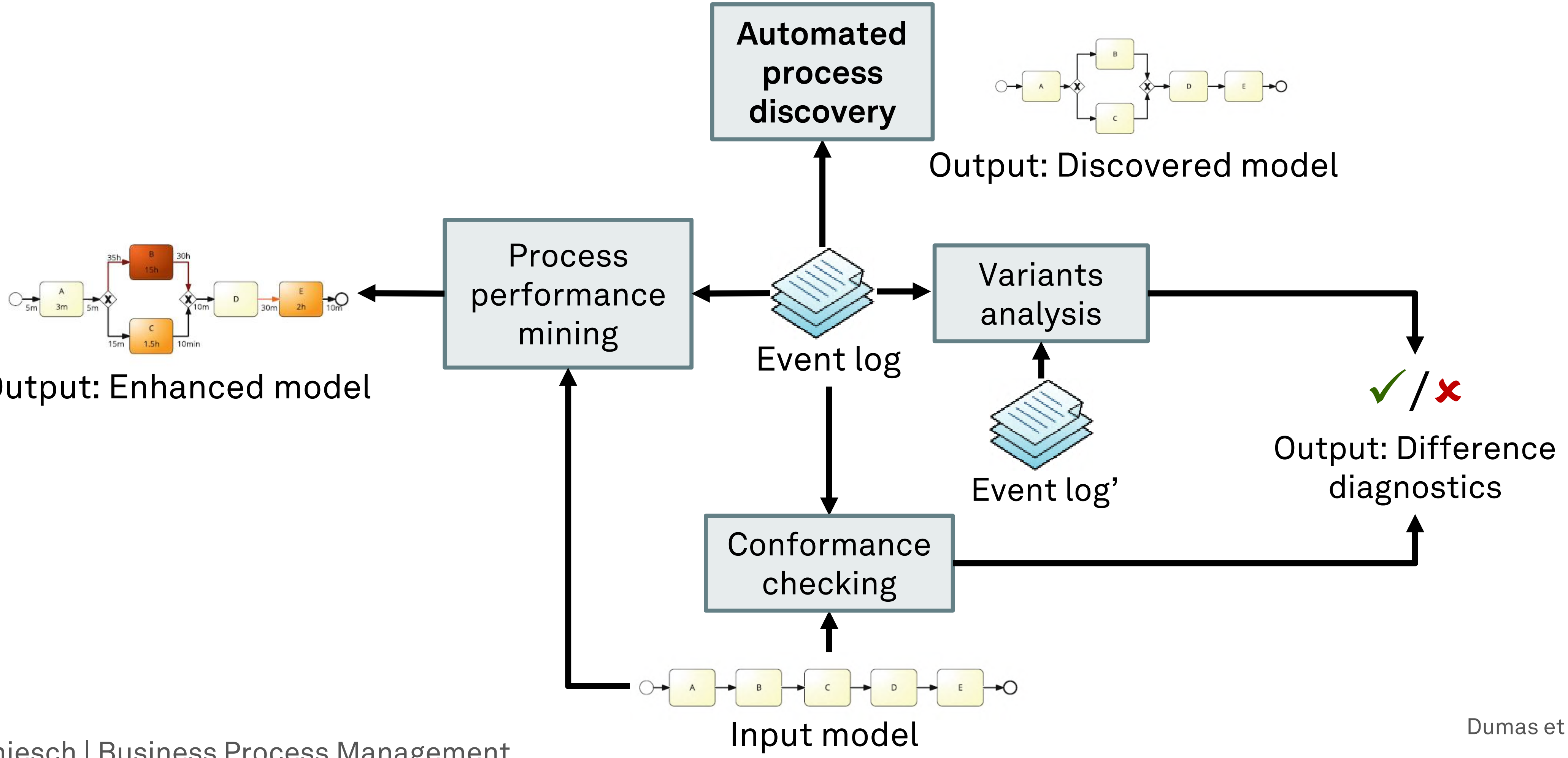
Dumas et al. (2013, 2018)

Process Mining Techniques



Dumas et al. (2018)

Process Mining Techniques



Dumas et al. (2018)

Automated Process Discovery

- Goal
  - Construct a process model that **captures the behavior** of an event log in a **representative way**
- Construction
  - **Automatically and generically** using an algorithm
  - Should make minimal assumptions about properties of the log and the resulting process model
- Result
  - Constructed process model should be able to **replay** the cases of the event log and
  - **Forbid behavior** not found in the logs

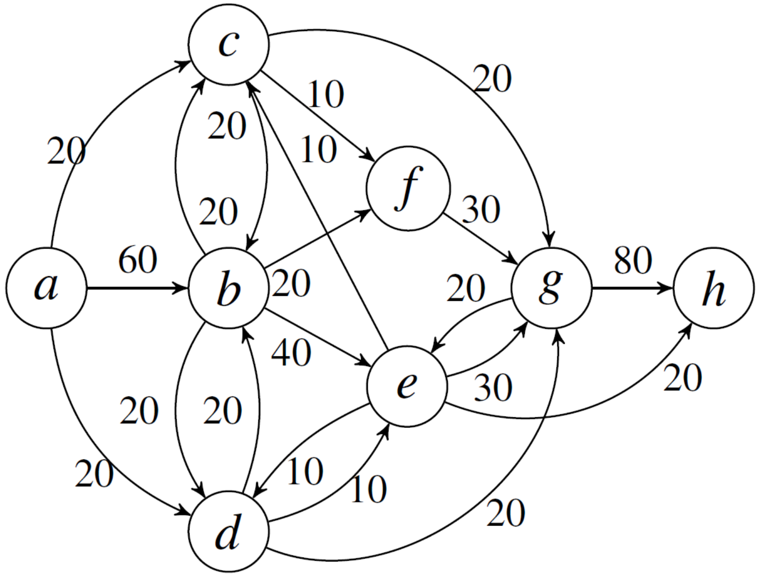
Dumas et al. (2018)

Dependency Graph

- aka **directly-follows graph** aka **process map**
- Each **activity** is represented by one node
- An **arc** from activity A to activity B means that B is directly followed by A in at least one trace in the log
- Arcs in a process map can be annotated with
  - Absolute/ relative frequency (percentage)
  - Temporal delay
- **Does not allow us to distinguish whether two tasks are in parallel, in a loop, or in some other relation**

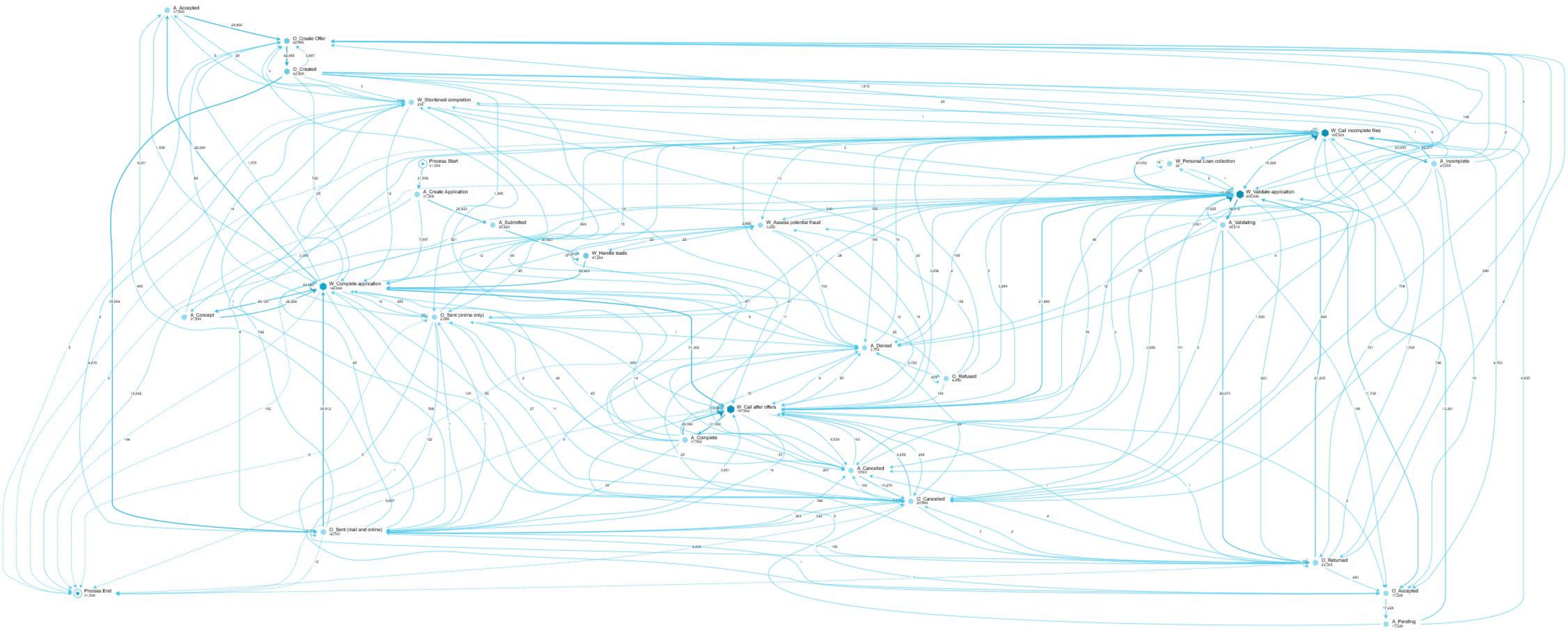
Event log:

10: a,b,c,g,e,h	10: a,b,e,d,g,h
10: a,b,c,f,g,h	10: a,c,b,e,g,h
10: a,b,d,g,e,h	10: a,c,b,f,g,h
10: a,b,d,e,g,h	10: a,d,b,e,g,h
10: a,b,e,c,g,h	10: a,d,b,f,g,h



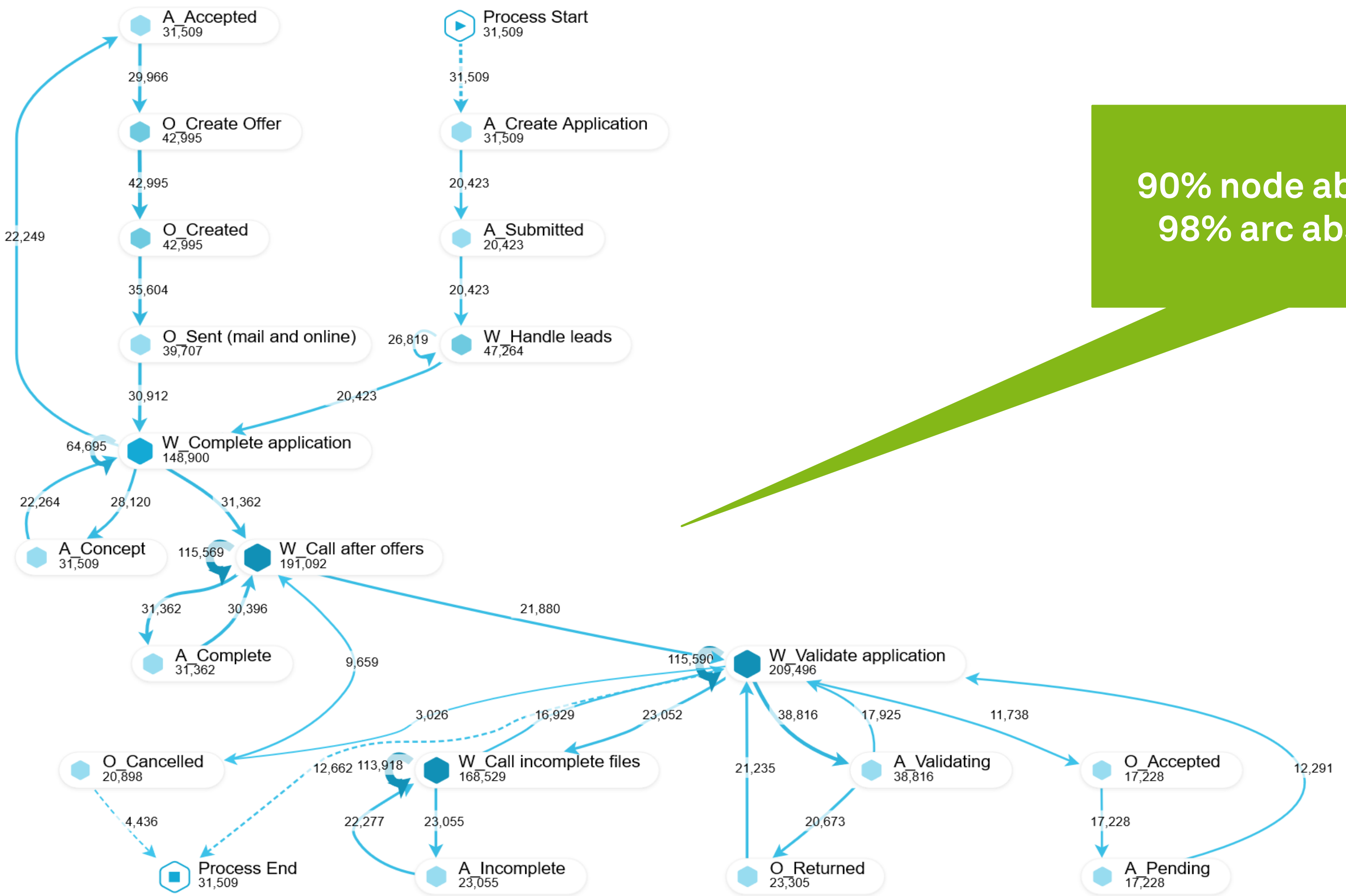
Dumas et al. (2018)

Real-life Process Map in Celonis



Dumas et al. (2018)

Real-life Process Map in Celonis (Filtered)



90% node abstraction, 98% arc abstraction

Dumas et al. (2018)

Abstraction and Filtering

- Abstraction
  - **Only show** the most frequent **nodes** or **arcs** of a dependency graph
- Event Log Filtering
  - Only show traces that **fulfil** (or do not fulfil) **a condition**
  - **Event filters**
    - Retain only **events** that fulfil a condition (e.g., certain event class)
  - **Event pair filters**
    - Retain only **events** that fulfil a condition **in respect to other events** (e.g., follows)
  - **Trace filters**
    - Retain only **traces** that fulfil a given condition (e.g., contain certain events or endpoints, in relation to performance measures)

Dumas et al. (2018)

Overview of Discovery Algorithms

- **Alpha miner** (α-miner)
  - Simple, limited, not robust
- **Heuristics miner**
  - Uses frequencies, detects self- & short-loops, handles incomplete and noisy logs
  - **Structured heuristic miner** further converts models to block-structured models
- **Inductive miner**
  - Divides dependency graphs in block structures using a different approach
- **Split miner**
  - Discovers sound but not always block-structured models
- **Genetic miner**
  - Iterative, non-deterministic procedure

van der Aalst (2016), Dumas et al. (2018)

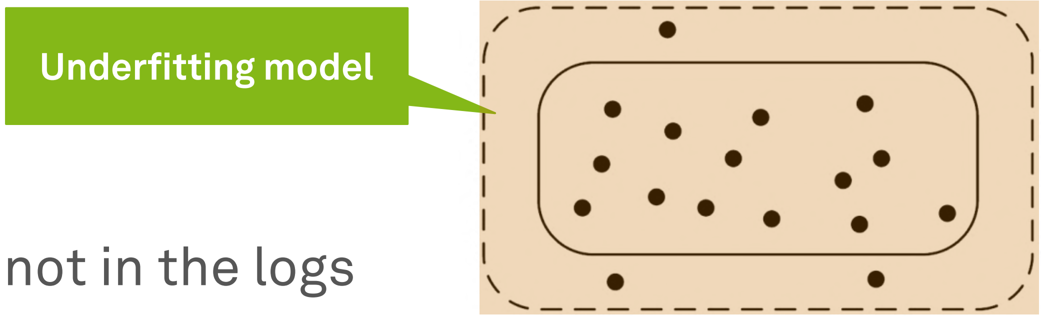
Characteristics of Advanced Process Discovery Algorithms

- **Representational Bias**
  - Inability to represent **concurrency**
  - Inability to deal with (arbitrary) **loops**
  - Inability to represent **silent actions**
  - Inability to represent **duplicate actions**
  - Inability to model (inclusive) **OR-splits/ joins**
  - Inability to represent **non-free-choice behavior**
  - Inability to represent **hierarchy**
- **Ability to Deal With Noise**
  - Deal with **errors** and **rare** behavior
- **Completeness Notion Assumed**
  - Too week → underfitting
  - Too strong (e.g., log contains all possible traces) → overfitting
- **Approach Used**

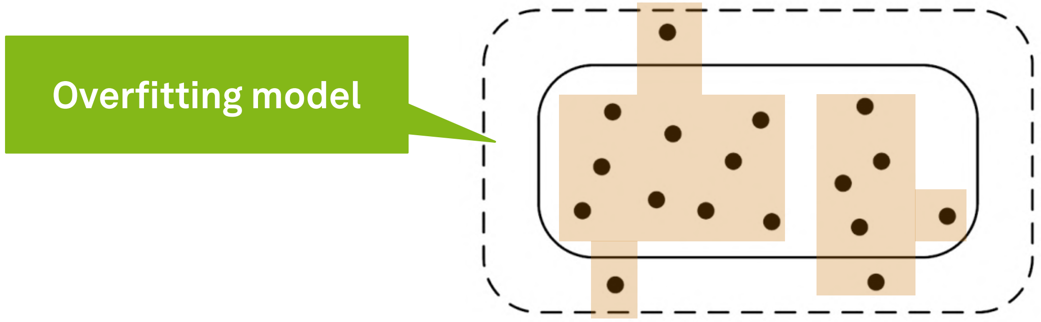
van der Aalst (2016)

Quality Measures for Automated Process Discovery (1)

- **Fitness**
  - Ability to replay behavior found in the logs
- **Precision**
  - Degree of behavior allowed by the model, but not in the logs
  - Do not **underfit**



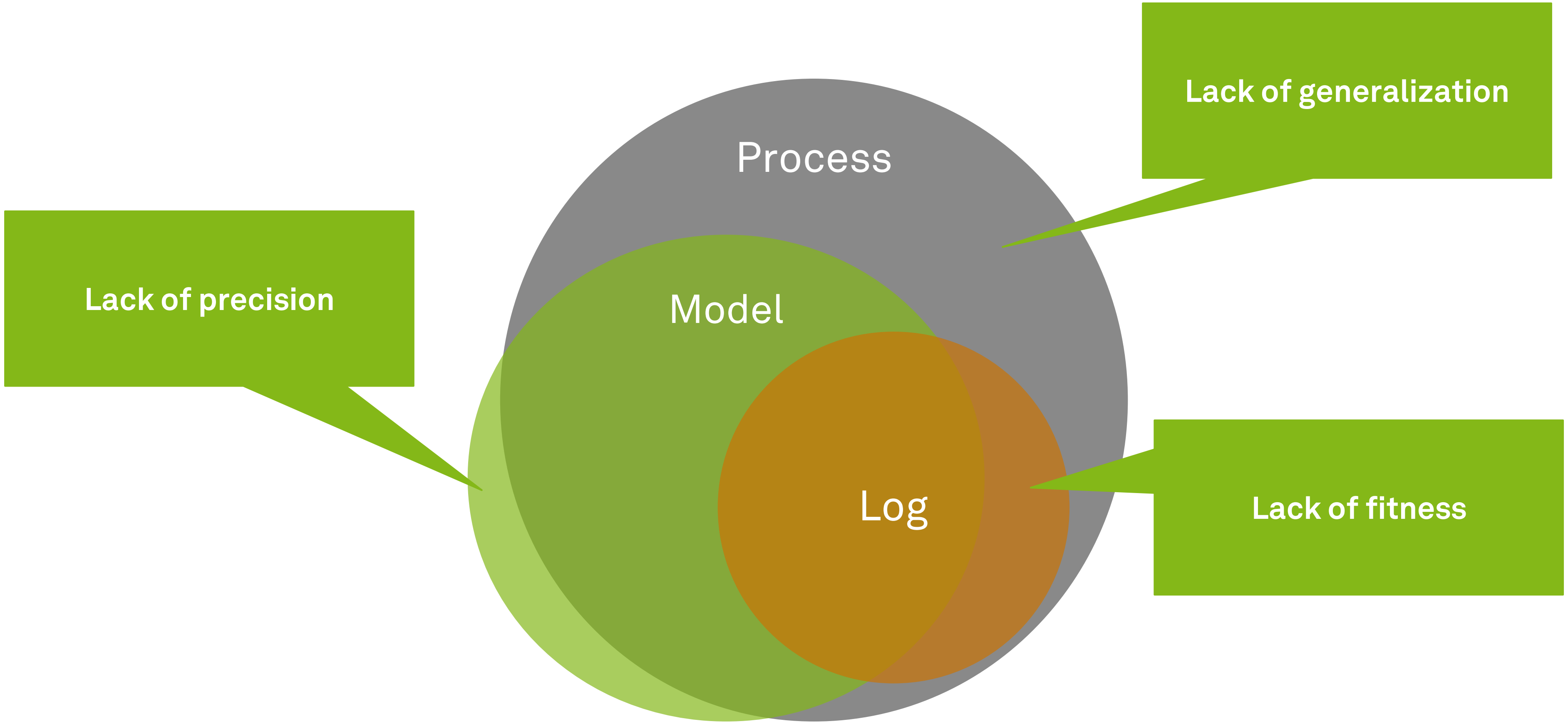
- **Generalization**
  - Work with incomplete behavior
  - Do not **overfit**



- **Simplicity/ Complexity (Occam's razor)**
  - Difficulty of understanding the model (e.g., number of nodes, arcs, etc.)
  - **The simplest model to explain the log's behavior, is the best model**

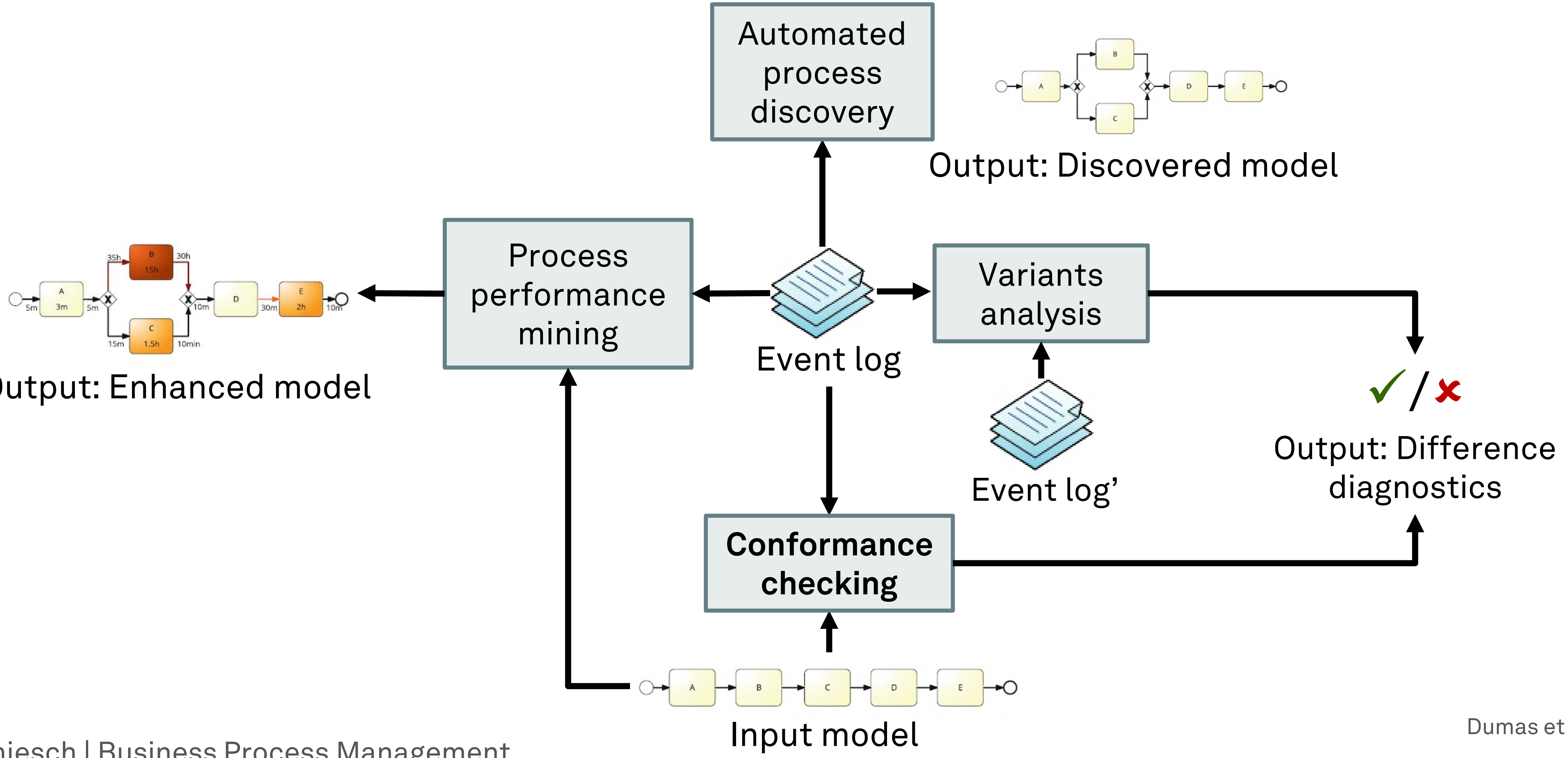
van der Aalst, (2016), Dumas et al. (2013, 2018)

Quality Measures for Automated Process Discovery (2)



Dumas et al. (2018)

Process Mining Techniques



Dumas et al. (2018)

Conformance Checking

- Goal
  - Determine whether the execution of a process **follows predefined rules or constraints**, or if it **violates** them
- Analysis
  - With families of conformance checking **techniques**
    - Replay, trace alignment, and behavioral alignment
  - Explicit **constraints** or normative process **model**
    - Perspectives: control flow, data, and resources
- Result
  - **Violations** that relate to one of the three process perspectives in isolation or in combination

Dumas et al. (2018)

Conformance Checking of Control Flow

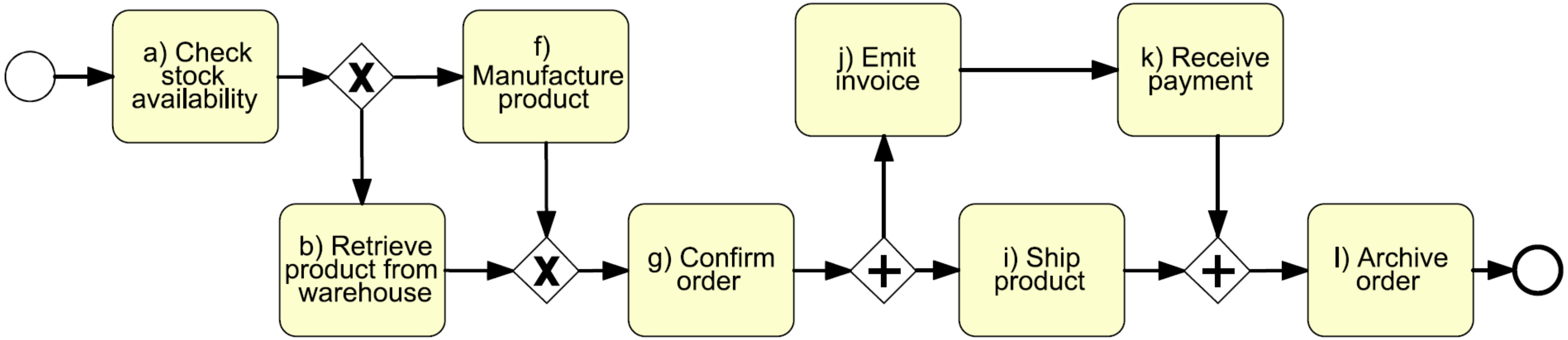
- **Explicit Constraints**
  - **Mandatoriness constraint**
    - Tasks are required from a control perspective
      - e.g., “approval task”
  - **Exclusiveness constraint**
    - Tasks that relate to a decision
    - Two or more tasks cannot co-occur in the same case
      - e.g., “buy” vs. “rent”
  - **Ordering constraint**
    - Two tasks must always occur in a specific order
      - E.g., “approval” → “order”

- **Normative process model**
  - **Unfitting** log behavior
    - Behavior observed in the log that is not allowed by the model
  - **Additional** model behavior
    - Behavior allowed in the model but never observed in the log

Dumas et al. (2018)

Normative Process Model Replay

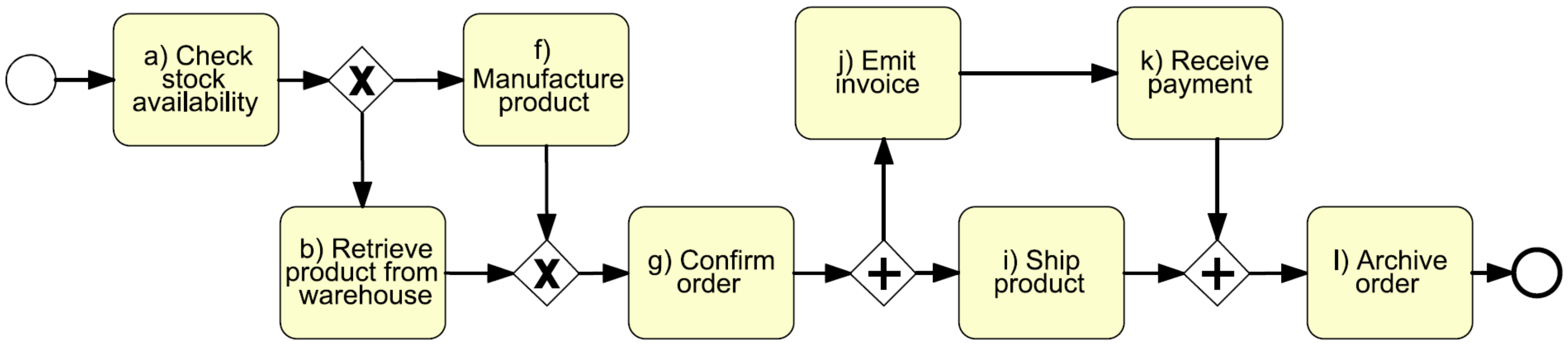
- For each trace in the workflow log
  - **Replay with tokens**
  - **Record** at each step whether an activity was **allowed** to be executed according to the rules of the model
- Is (a, b, g, j, i, k, l) valid?



- Is (a, b, i, j, k, l) valid, too? If not, what does it violate?

Dumas et al. (2018)

Replay of Tokens in a Normative Process Model



consumed c
produced p
missing m
remaining r

- How many misplaced tokens in (a, b, i, j, k, l)?
  - Number of tokens that are **correctly consumed *c***
  - Number of tokens that are **correctly produced *p***
  - Number of tokens that are **missing for executing *m*** the next activity in the log
  - Number of tokens **remaining unconsumed *r*** after executing the final activity in the log

Dumas et al. (2018)

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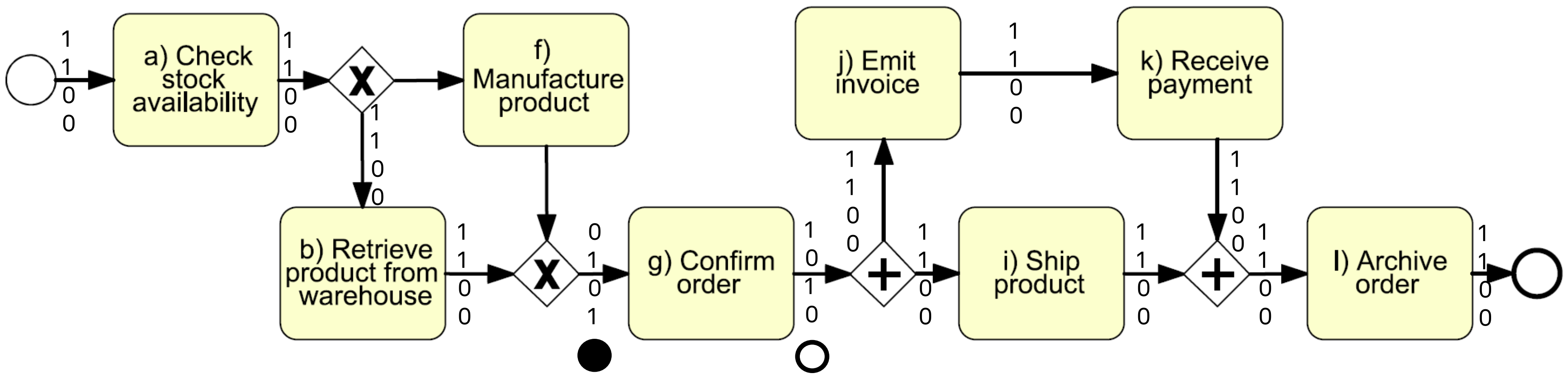

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Dumas et al. (2018)

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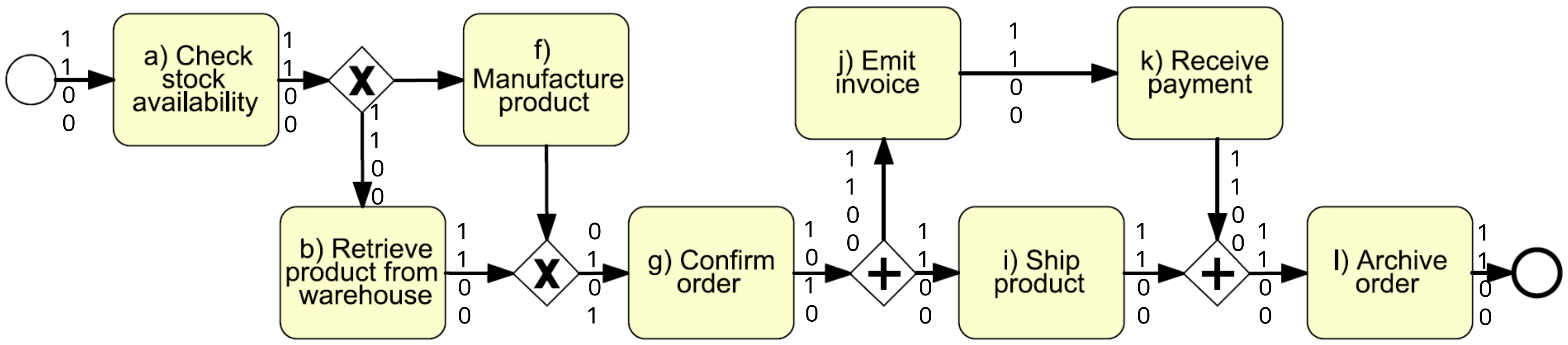


consumed c
produced p
missing m
remaining r

$$fitness = 0.9166$$

$$fitness = \frac{1}{2}\left(1 - \frac{m}{c}\right) + \frac{1}{2}\left(1 - \frac{r}{p}\right)$$

- How many misplaced tokens in (a, b, i, j, k, l)?
  - Number of tokens that are **correctly consumed *c* *(12)***
  - Number of tokens that are **correctly produced *p* *(12)***
  - Number of tokens that are **missing for executing *m*** the next activity in the log ***(1)***
  - Number of tokens **remaining unconsumed *r*** after executing the final activity in the log ***(1)***

Dumas et al. (2018)

Trace Alignment and Behavioral Alignment

- With **replay**, the recovery from an error is done locally
- Replay does **not identify all errors at once** that explain the unfitting trace

- **Trace Alignment**
  - Identifies the **closest corresponding trace** that can be parsed by the model
  - Shows the points of divergence
  - **Does not explicitly handle concurrent tasks nor cyclic behavior**
- **Behavioral Alignment**
  - Aligns the entire event log against the process model at once
  - Identifies differences that cannot be observed at individual traces level

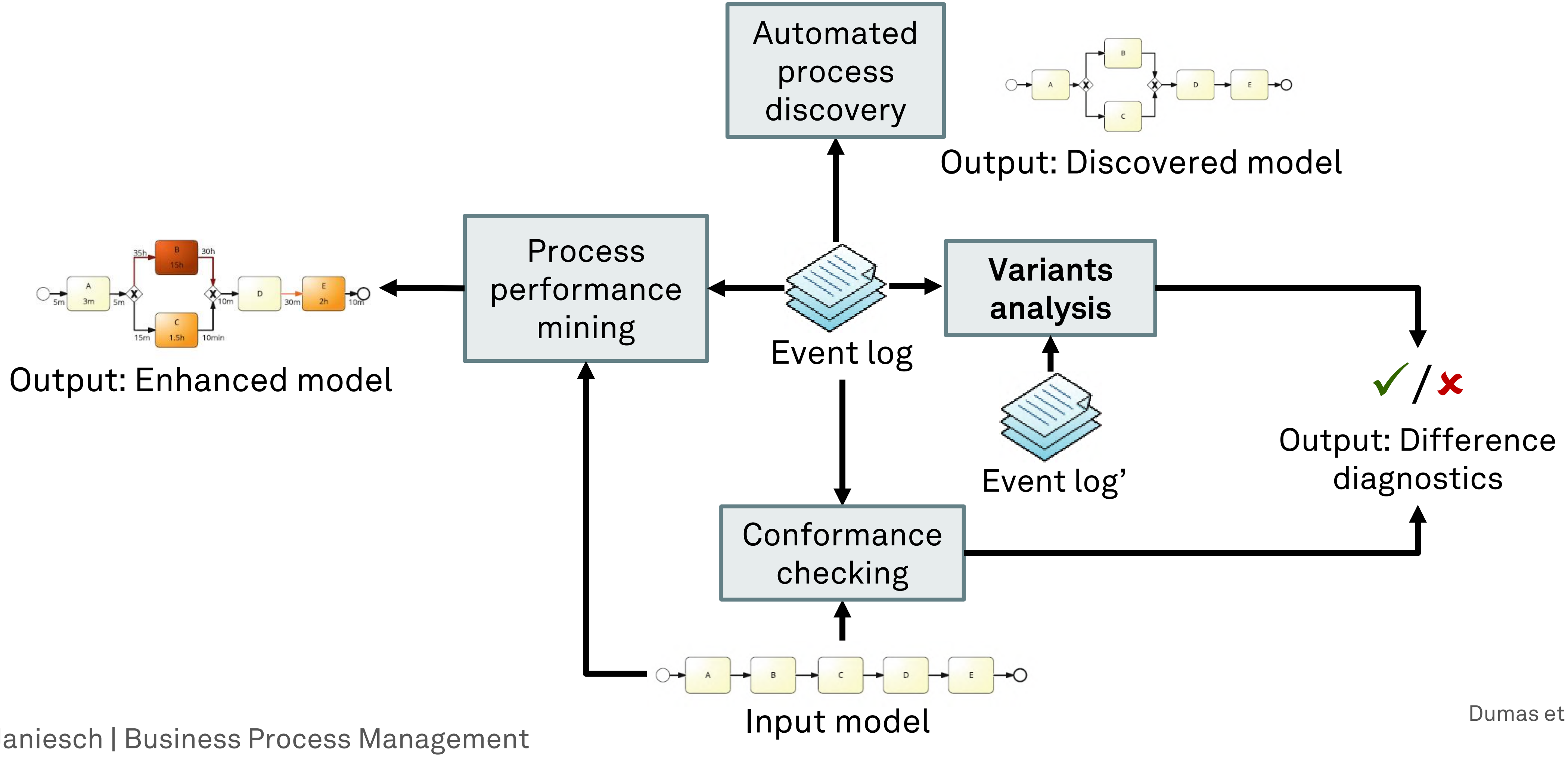
Dumas et al. (2018)

Conformance of Data and Resources

- Participants usually require **permissions** to execute certain activities
- Permissions are bundled for specific **roles**
- Violations of permissions can be checked by searching for each **activity conducted** by a participant whether or not an **appropriate role or permission** existed
- Specific control rules which require two different persons to approve a business transaction are called **separation of duties** constraints
  - These rules do not necessarily involve supervisors

Dumas et al. (2018)

Process Mining Techniques



Dumas et al. (2018)

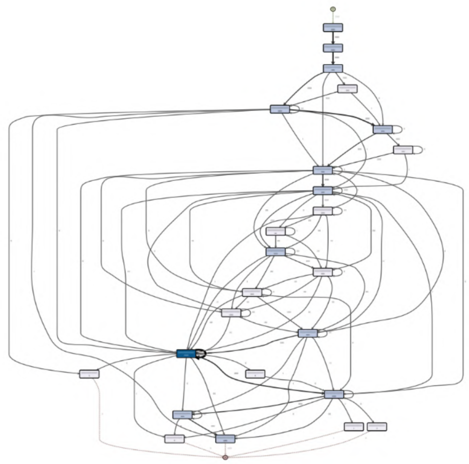
Variants Analysis

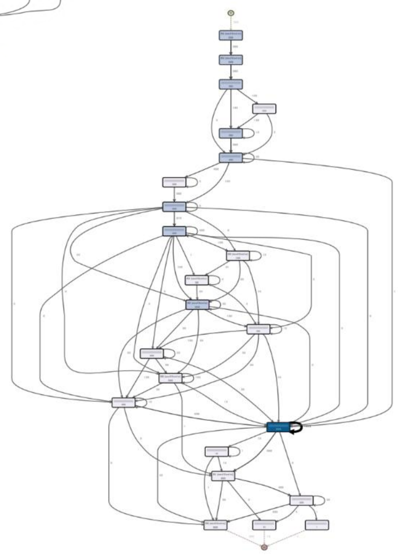
- Goal
  - Determine the **root cause for (performance) variation** between cases
- Analysis
  - Differences across **subsets of cases** in a process
    - Split based on time, resources, business data/attributes
    - Deviance mining: Comparison of normal performance with undesirable performance
- Result
  - **Characteristics** that are commonly found in the distinct cases

Dumas et al. (2018)

Variants Analysis

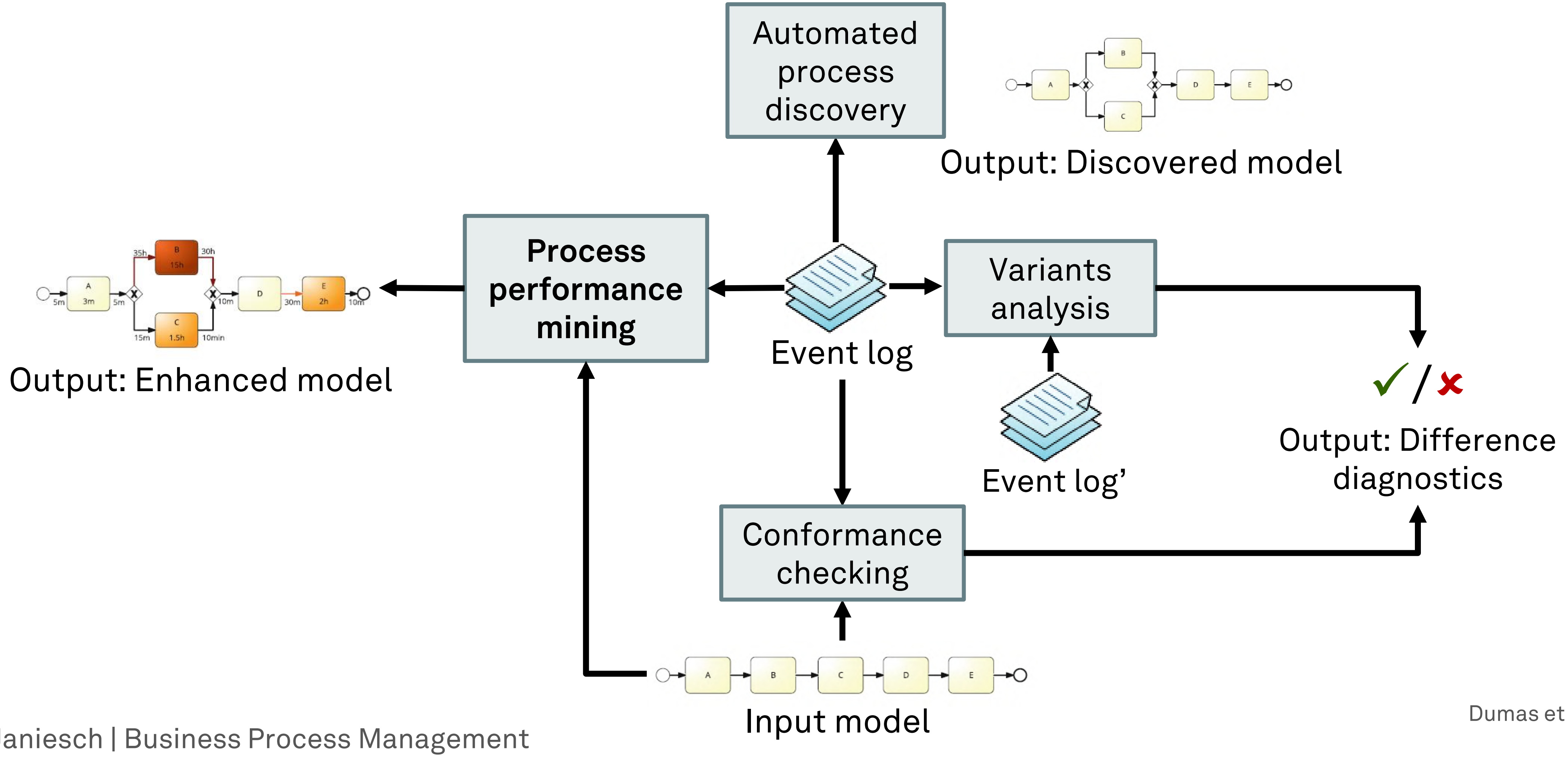
- **Manual Comparison**
  - Only suitable for small or filtered event logs
- **Log Delta Analysis**
  - Compares two event logs and produces a **set of patterns or statements** that are common in one event log and uncommon in the other event log
- **Association Rules**
  - Generally used to transform an event log into patterns
  - **Discriminative sequence mining** compares two event logs and identifies **association rules**, which are common in one event log and uncommon in the event other log





van Beest et al. (2015), Dumas et al. (2018)

Process Mining Techniques



Dumas et al. (2018)

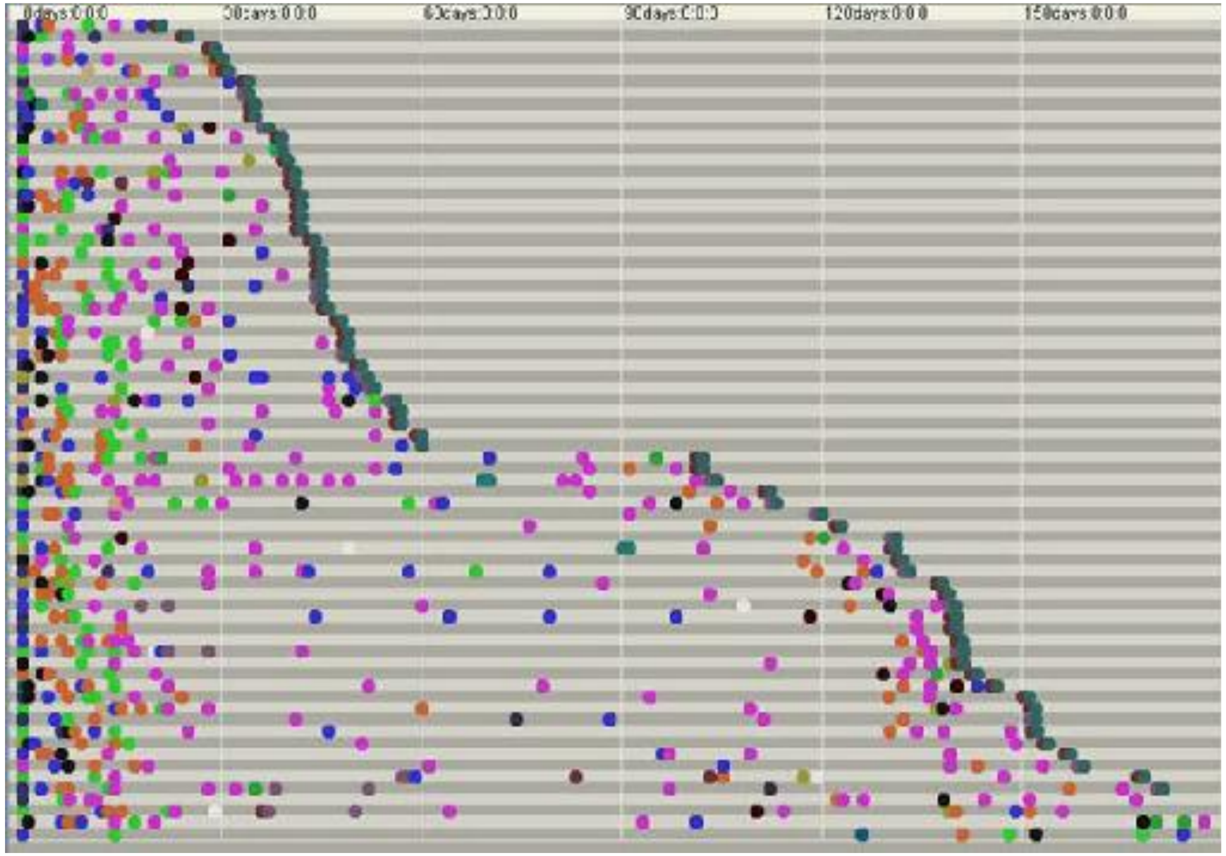
Process Performance Mining

- Goal
  - **Quantify and visualize** the performance of a process automatically
- Analysis
  - Can be organized around the devil’s quadrangle of time, cost, quality, and flexibility
- Result
  - Quantified measures that evaluate a process

Dumas et al. (2018)

Process Performance Mining: Time

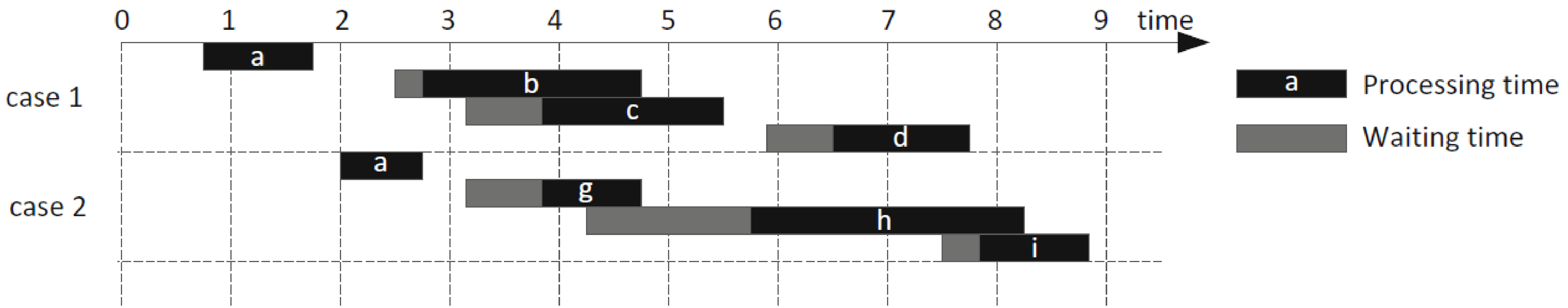
- **Dotted chart**
  - One line per trace, each line contains points, one point per event
  - Each event type is mapped to a color
  - Position of the point denotes its occurrence time in (relative) scale
  - Birds-eye view of the timing of different events (e.g., activity end times)
  - **Does not allow one to see the *processing times* of activities**



Dumas et al. (2018)

Process Performance Mining: Time

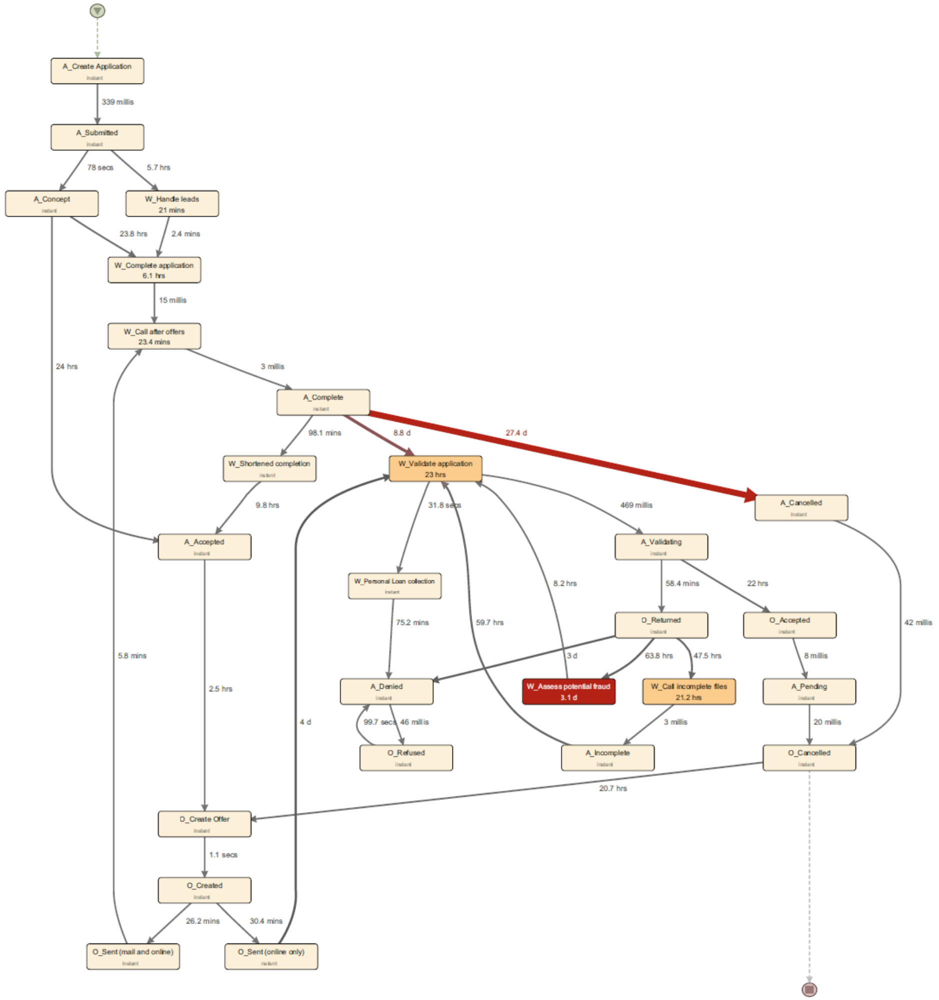
- **Timeline chart**
  - One line per trace, each line contains segments capturing the start and end of tasks
  - Captures process time (unlike dotted charts)
  - Good to show “representative” traces
  - **Not scalable for large event logs**



Dumas et al. (2018)

Process Performance Mining: Time

- **Performance-enhanced dependency graphs**
  - Nodes are color-coded with respect to a performance measure
  - Typically, edge color denotes waiting time, node color denotes processing time
  - Typically, nodes represent activities, but they may represent resources and then arcs denote hands-offs between resources



Dumas et al. (2018)

Process Performance Mining: Cost

- **Direct cost** of labor
  - Salaries, only 10% today in automated production
- **Overhead** cost of labor
  - Education & training, installation & maintenance, marketing & support

- **Activity-based Costing** allocates indirect cost to an activity's direct cost
  - All business activities must be broken down into their discrete components
  - Different cost (product, service, resource) can be determined based on **cost drivers**
  - Costs are assigned to adequate **activity centers** to determine total activity cost
  - Cost of activities are related to the products or processes to be measured

Cooper & Kaplan (1988), Dumas et al. (2018)

Process Performance Mining: Quality

- Quality is often **not directly visible** from event logs

- Repetitions are a good indication for a lack of quality
- They typically occur when a task has not been completed successfully

Cycle time of a rework block

$$CT = \frac{T}{1-r}$$

$$r = 1 - \frac{T}{CT}$$

Rework probability

- Alternative: track repetitions based on the assignment of tasks to resources (e.g., via ticketing system)

Dumas et al. (2018)

Process Performance Mining: Flexibility

- Compare the desired level of flexibility with the actual flexibility
- A **lack of flexibility** hints at a **lack of resilience**
- A **surplus of flexibility** hints at a **lack of standardization**
- Suggested Measures
  - **Distinct executions**
    - Traces *DE* with unique sequences in event log *L*
    - Issues with **concurrency**

$$DE = |L|$$

  - **Degree of optionality**
    - Tasks T_{opt} in a sequence of tasks *T* are optional

$$OPT = \frac{T_{opt}}{T}$$

Dumas et al. (2018)

Learning Goals of Today

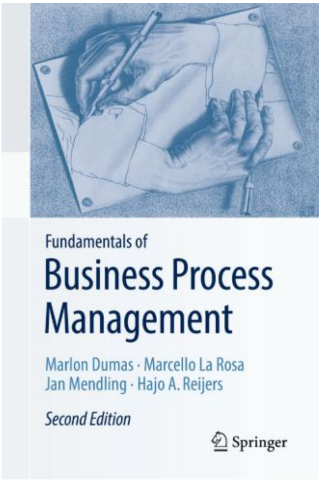
- Understand process monitoring with process dashboards
- Understand process discovery
- Understand process conformance checking
- Understand variants analysis and process performance mining

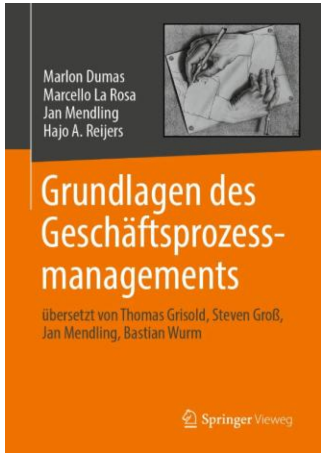
Recap Questions

- What is the difference between offline and online process monitoring? Which kinds of dashboards are best suited for which mode?
- How are the four distinct techniques of process mining related and how can they work together to optimize processes?

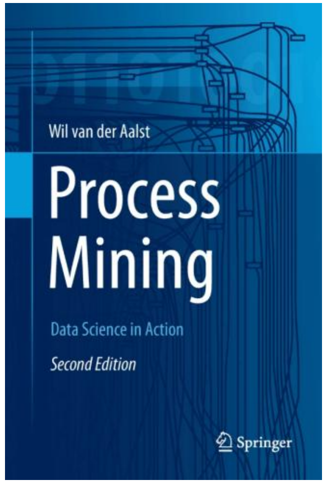
Literature

- **Section 11:** Process Monitoring





- **Section 2:** Process Mining: The Missing Link
- **Section 6:** Process Discovery: An Introduction
- **Section 8:** Conformance Checking



Roadmap

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Any Questions about Process Monitoring and Process Mining?

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Department of Computer Science

Business Process Management

BPM as an Enterprise Capability



Roadmap

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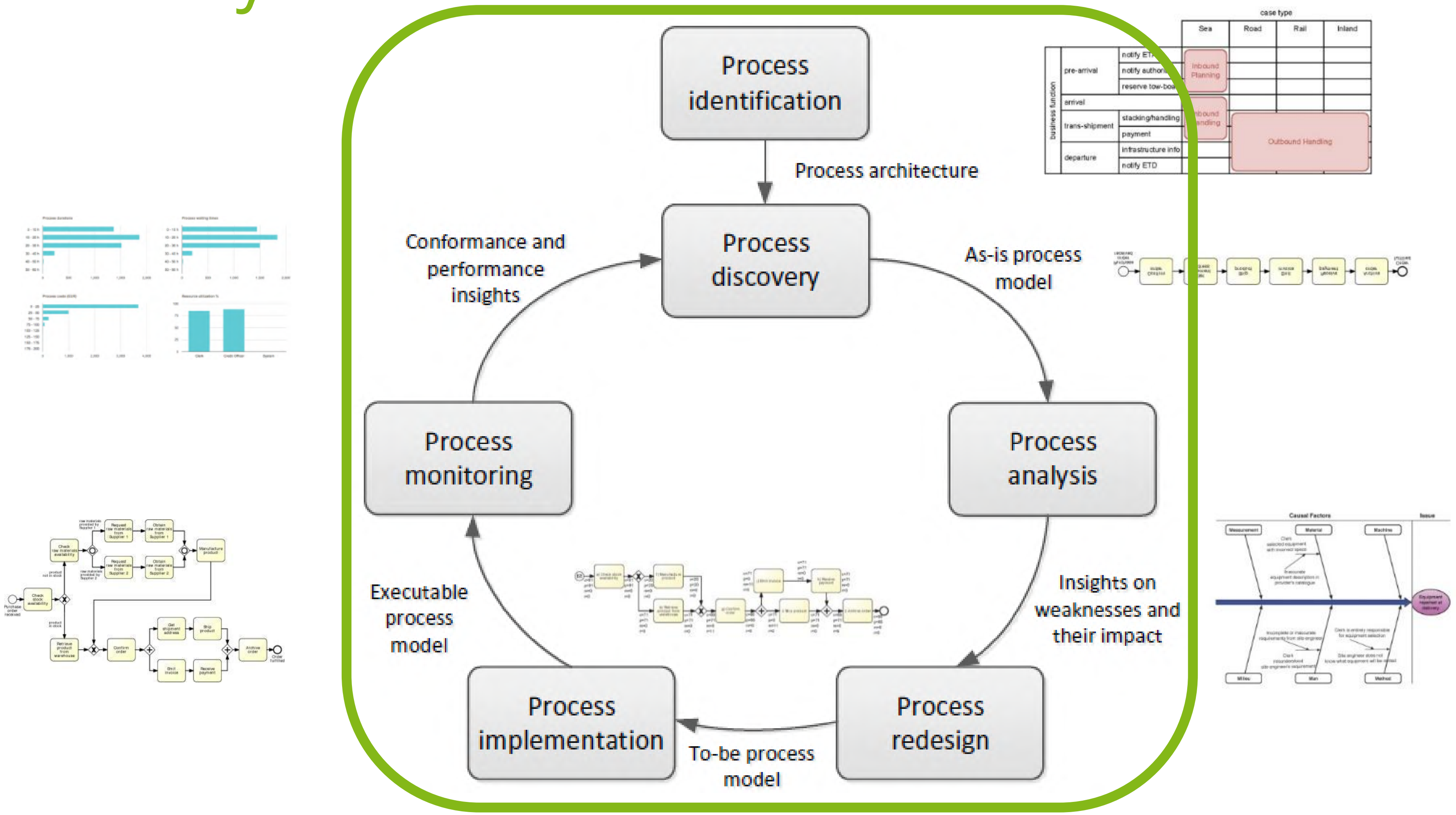
Learning Goals of Today

- Understand that BPM success is multifaceted
- Understand the factors of the Six Core Elements of BPM
- Understand (BPM) maturity models and assessment

The BPM Lifecycle



Dumas et al. (2018)

BPM Success

- We discussed
  - a range of methods and related techniques
  - software tools and systems
  - measurement of BPM projects

- **BPM projects still fail!**

- **What does it take to successfully manage a BPM program?**

Dumas et al. (2018)

Common Reasons for Failure of BPM Programs

- Sole focus on BPM methods and tools, **not on business goals**
  - BPM is a means to an end, not an end in itself
- Belief that BPM is the **single source of truth**
  - BPM needs to be aligned with other management concepts
- BPM projects that are managed as **isolated silos**
  - BPM is a holistic management concept that looks at activities, people, and technology
  - Different BPM projects should not be managed in isolation
- Overall **inability to change**
  - Transferring a great to-be model into an operational business process requires not only software but strategic alignment, change management, and consideration of company culture

Dumas et al. (2018)

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Maturity Models and the Six Core Elements of BPM

BPM Requires a Robust Frame of Operation

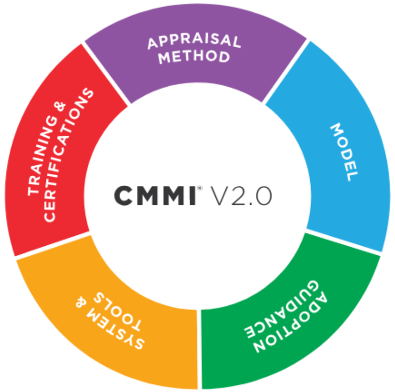
- **Project and program management**
  - Manages scope and flavors in different areas of an organization by safeguarding
    - required competencies
    - suitable BPM approaches
    - BPM tailoring
- **Complexity management**
  - BPM is holistic and comprehensive by nature and needs decomposition
- **Standards management**
- **Strategy management**

Rosemann & vom Brocke (2015)

Generic Maturity Models

- **Set of structured levels** describing how well the behaviors, practices and processes of an organization can **reliably and sustainably produce required outcomes** (e.g., quality, effectiveness)
- Can be a benchmark for comparison and an aid to understanding as well as a means to continuous improvement
- Maturity models typically contain
  - **Maturity levels**
  - Key processes
  - Goals or targets
- There is a wide variety of maturity models, most build on the **Capability Maturity Model Integration (CMMI)** staged maturity levels

CMMI Levels of Capability and Performance



Level	*Continuous Representation Capability Levels*	*Staged Representation Maturity Levels*
Level 0	Incomplete	
Level 1	Performed	Initial
Level 2	Managed	Managed
Level 3	Defined	Defined
Level 4		Quantitatively Managed
Level 5		Optimizing

Information Systems Audit and Control Association (ISACA) (2022)

BPM Maturity Model

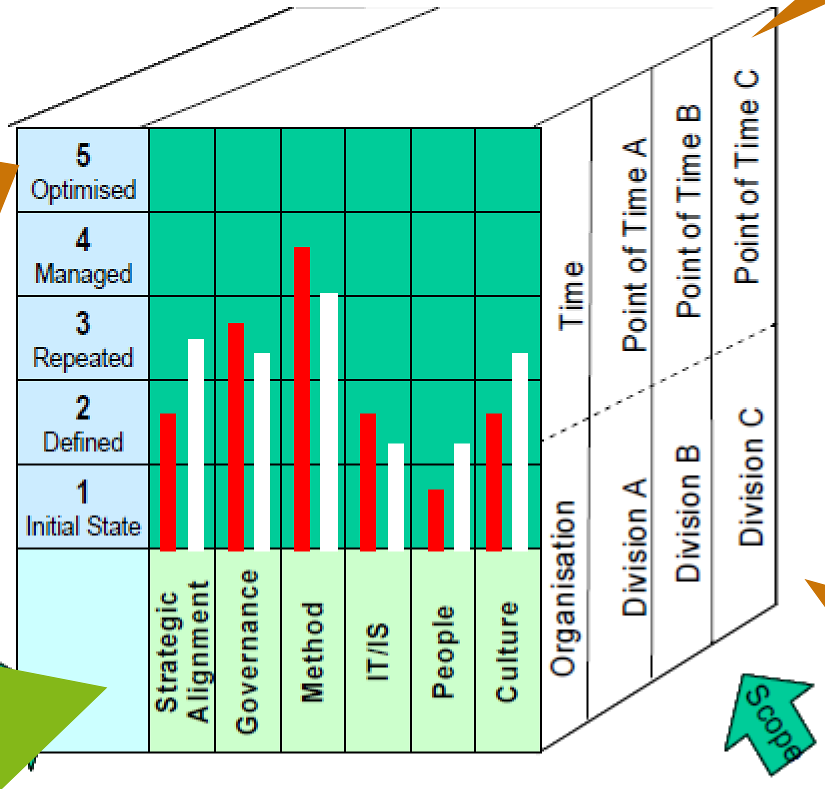
- **BPM Maturity Model**
  - Addresses the requirements and complexities identified within Business Process Management
- **Focus**
  - Corporate business processes
- **Scope**
  - Organizational entity and time
- **Multi-dimensional model**

Rosemann & de Bruin (2005)

BPM Maturity Model

Maturity Stage
- Pre-defined maturity stage ranging from 1 (low) to 5 (high)

Factor
- Specific, measurable and independent element, which reflects a funda-mental and distinct charac-teristic of BPM



Key
Coverage
Proficiency

Scope – Time
- The point in time at which the model is applied

Coverage
- Extent to which BPM practices extend through the organizational entity

Proficiency
- Perceived goodness of BPM practices in the organizational entity

Scope – Organizational Entity
- Organizational entity which defines the unit of analysis and to which the model is being applied, e.g., a division, a business unit, a subsidiary

Rosemann & de Bruin (2005)

Six Core Elements of BPM – Overview

Strategic Alignment	Governance	Methods	Information Technology	People	Culture	Factors
Process Improvement Planning	Process Management Decision Making	Process Design & Modelling	Process Design & Modelling	Process Skills & Expertise	Responsiveness to Process Change	Capability Areas
Strategy & Process Capability Linkage	Process Roles and Responsibilities	Process Implementation & Execution	Process Implementation & Execution	Process Management Knowledge	Process Values & Beliefs	
Enterprise Process Architecture	Process Metrics & Performance Linkage	Process Monitoring & Control	Process Monitoring & Control	Process Education	Process Attitudes & Behaviors	
Process Measures	Process Related Standards	Process Improvement & Innovation	Process Improvement & Innovation	Process Collaboration	Leadership Attention to Process	
Process Customers & Stakeholders	Process Management Compliance	Process Program & Project Management	Process Program & Project Management	Process Management Leaders	Process Management Social Networks	

The models has six factors representing a **critical success factor** each

Each factor has five capability areas as a further level of detail

Rosemann & vom Brocke (2015)

Six Core Elements of BPM – Factors (1)

- **Strategic Alignment**
  - BPM contributes to the organization's **superordinate, strategic goals**
  - Related capabilities include
    - the **assessment** of processes and process management initiatives according to their **fit** with the overall corporate **strategy**
- **Governance**
  - BPM must be implemented in the **organizational structure**
  - Related capabilities include
    - the assignment of BPM-related tasks to **stakeholders** and applying specific **principles and rules** to define the required **responsibilities and controls** along the entire business-process lifecycle

Rosemann & vom Brocke (2015), Dumas et al. (2018), vom Brocke & Mendling (2018)

Six Core Elements of BPM – Factors (2)

- **Methods**
  - BPM must be supported by methods for **process design, analysis, implementation, execution, and monitoring**
  - Related capabilities include
    - **selecting** the appropriate BPM methods, tools, and techniques and **adapting** and combining them according to the organization's **requirements**
- **Information Technology**
  - BPM must use technology, particularly **process-aware information systems (PAIS)**, as the basis for process design and implementation
  - Related capabilities include
    - the ability to select, implement, and use relevant **PAIS** that cover, e.g., workflow management or process mining solutions

Rosemann & vom Brocke (2015), Dumas et al. (2018), vom Brocke & Mendling (2018)

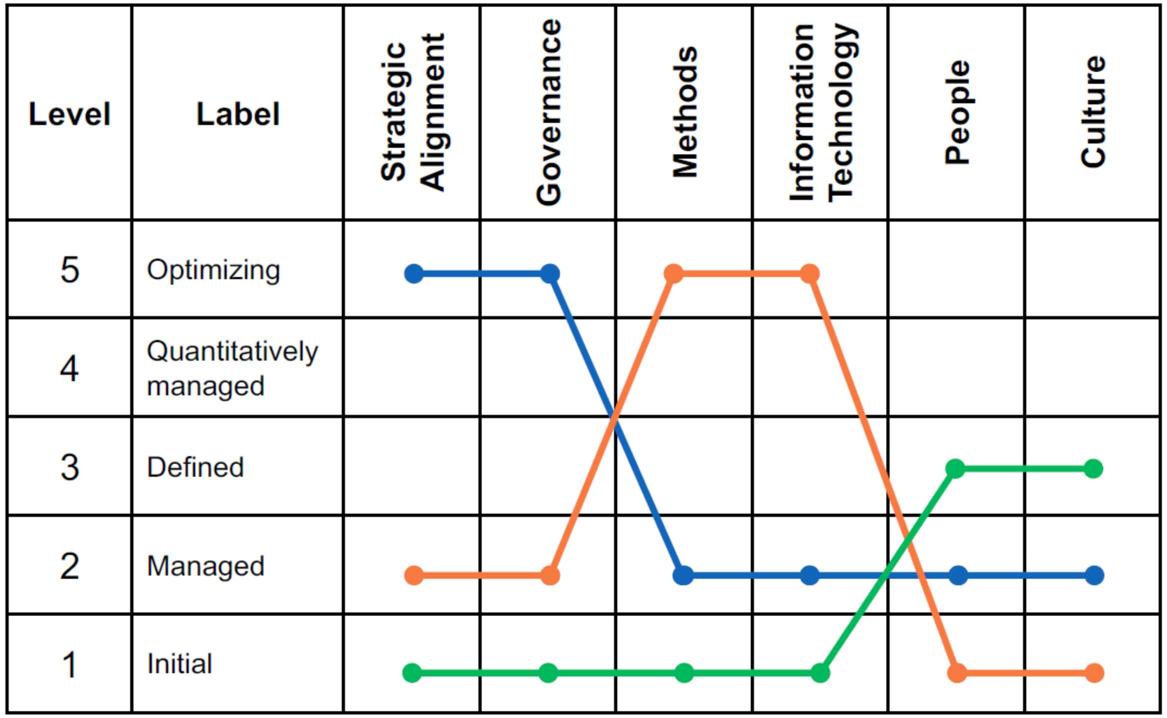
Six Core Elements of BPM – Factors (3)

- **People**
  - BPM must consider **employees' qualifications** in the discipline of BPM and their **expertise** with relevant business processes
  - Related capabilities include
    - assessing the **human-resources impact** of BPM initiatives and programs that facilitate the development of **process-related skills** throughout the organization
- **Culture**
  - BPM must be met with a **common value system** that supports process improvement and innovation
  - Related capabilities include
    - the ability to **assess** the organizational **culture's values** and the ability to derive measures to **develop** these values accordingly

Rosemann & vom Brocke (2015), Dumas et al. (2018), vom Brocke & Mendling (2018)

Patterns of BPM Maturity

- **Blue**
  - Typical for top-down-driven BPM organizations with executive support, sponsored by CFO/CEO
- **Orange**
  - Typical for IT-driven organization, sponsored by CIO
- **Green**
  - Typical for organizations affected by rules-based governance and heavy unionization



Dumas et al. (2018)

Case: BPM Maturity Assessment in Insurance

Critical Success Factor	Implementation Stages			
	Foundation	*Capability*	*Business Architecture*	*The Last Mile*
Strategic Alignment	Level 1	Level 1	Level 2 *Drive synergies*	Level 3-4 *Group-wide approach*
Governance	Level 1	Level 1	Level 2 *IT-wide approach*	Level 3-4 *Group-wide approach*
Methods	Level 2 *Workshop and observation methods*	Level 3 Integrate into System Dev Lifecycle	Level 4 *Integrate into Enterprise Architecture*	Level 5 *Integrate into Service Architecture*
Information Technology	Level 2 *ARIS for modeling*	Level 3 *ARIS for simulation and analytics*	Level 4 *Link to Enterprise Architect*	Level 5 *Link with Enterprise Service Bus and BPMS*
People	Level 1 *Establish BPM Team*	Level 2 *Extend into App Dev Team*	Level 3 *Integrate into Ent Arch Team*	Level 4 *Whole organization*
Culture	Level 1 *Basic process awareness*	Level 2 *Integrate into IT culture*	Level 3 *Become part of Enterprise planning*	Level 4 *Integrate into org culture*

Application focus → *Service focus* →

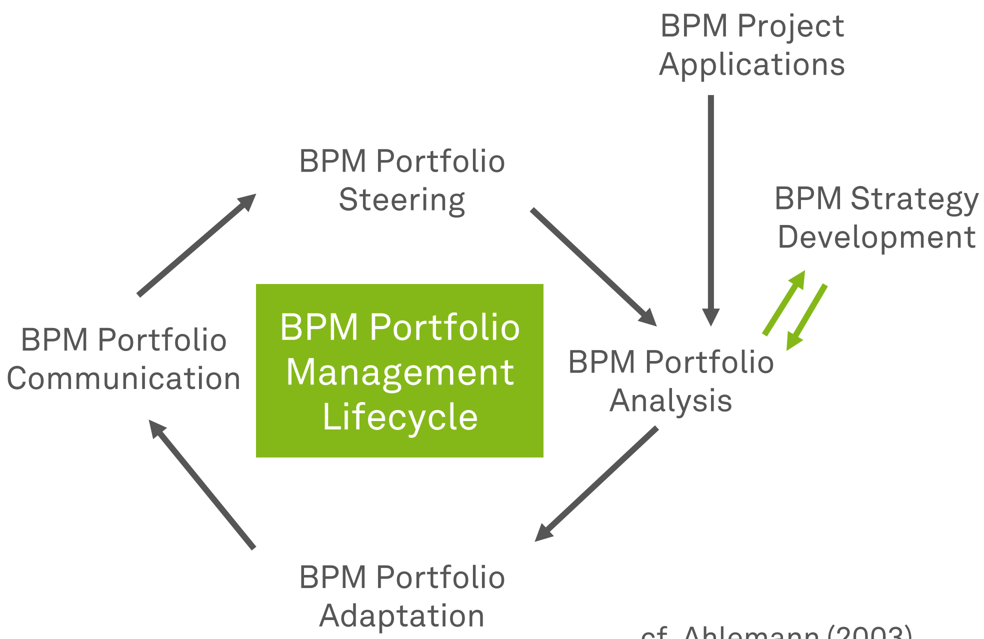
- Assessment is framed in the context of a **BPM roadmap**, which describes desired levels of maturity over time
- The company in this example follows the **orange** pattern
  – Driver for BPM is methods and IT, rather than strategy

Dumas et al. (2018)

Portfolio Management in BPM Programs

- Goal is the **planning and control of all BPM projects and programs**
- Totality of all stand-alone projects of a strategic business unit grouped into programs

- BPM portfolio management
  - … pursues the **strategic corporate objectives**
  - … **evaluates and prioritizes** the BPM projects
  - … **ensures** the efficient use of resources
  - … **plans and controls** the projects in a portfolio

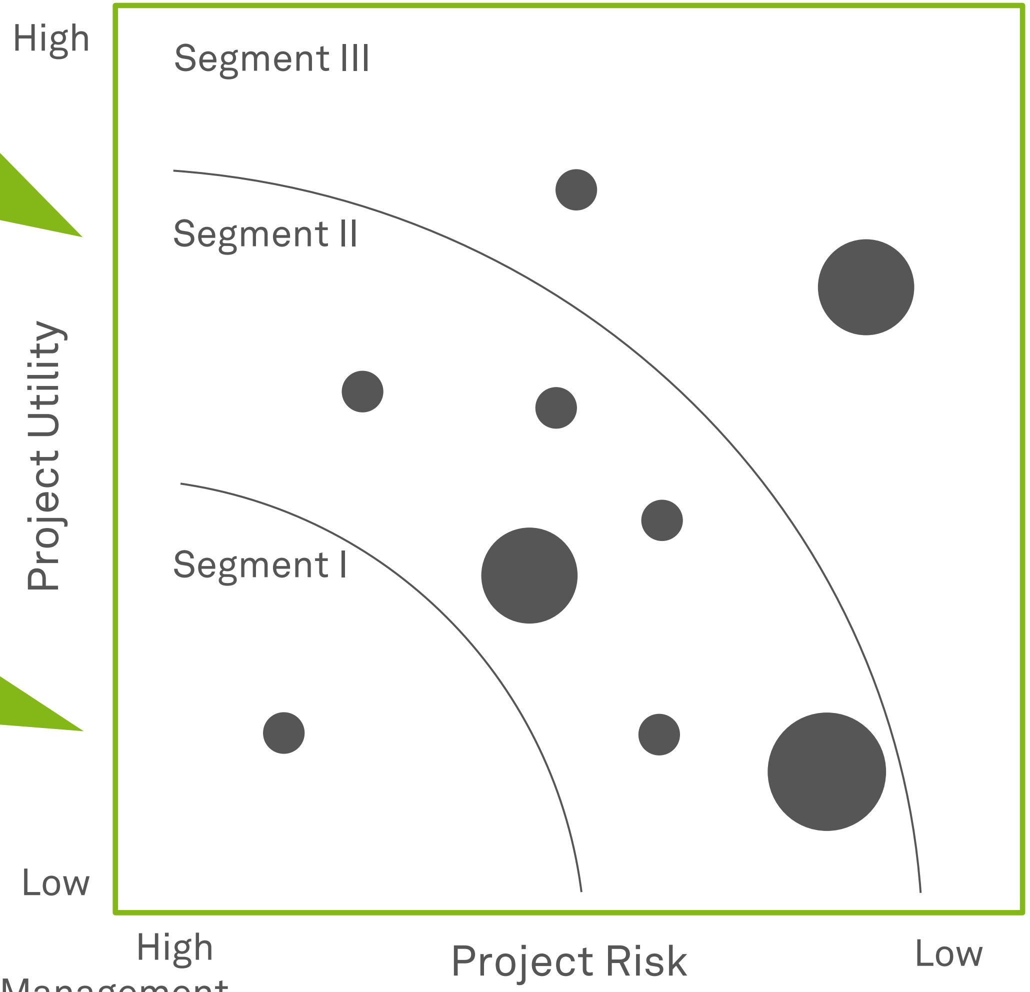


cf. Ahlemann (2003)

BPM Project Portfolio Matrix

Segment II encompasses those projects with mid-utility or mid-risk, which require prioritization as they typically fight for resources to be approved

Segment I indicates high risk, low utility projects, which may never see the light of day



Segment III contains high utility and/or low risk projects, which will be approved first

Circle size can be used as a 3rd dimension (e.g., alignment with corporate strategy)

cf. Schmidt (2010)

Ten Principles of Good BPM

1. Principle of Context Awareness
   – BPM is not a cookbook
2. Principle of Continuity
   – BPM is not a one-off project
3. Principle of Enablement
   – BPM is not be limited to firefighting
4. Principle of Holism
   – BPM does not have an isolated focus
5. Principle of Institutionalization
   – BPM is not an ad hoc responsibility
6. Principle of Involvement
   – BPM does not neglect employee participation
7. Principle of Joint Understanding
   – BPM is not only the language of experts
8. Principle of Purpose
   – BPM is not an end in itself
9. Principle of Simplicity
   – BPM should not be over-engineered
10. Principle of Technology Appropriation
    – Technology management is not an afterthought

Dumas et al. (2018)

Learning Goals of Today

- Understand that BPM success is multifaceted
- Understand the factors of the Six Core Elements of BPM
- Understand (BPM) maturity models and assessment

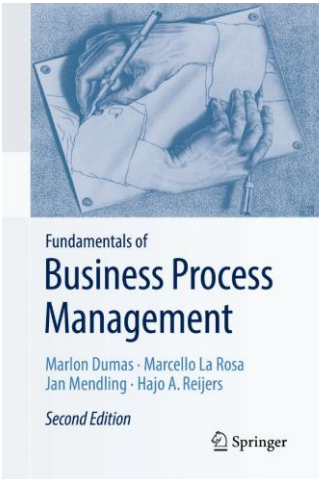
Recap Questions

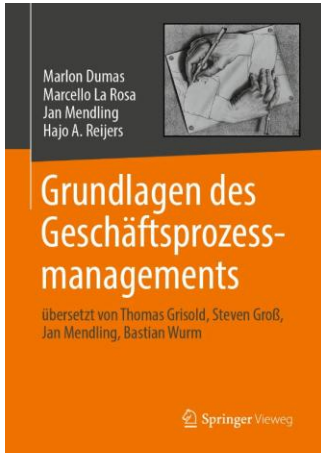
- What is the Six Core Factors of BPM framework for and what are the six factors about?
- Why are maturity models with respect to BPM necessary and how do they commonly look like?
- What is BPM project portfolio management and howe can it be organized?

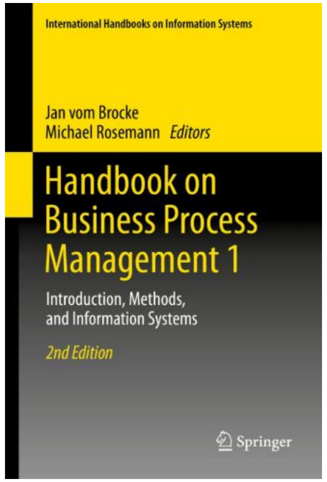
Literature

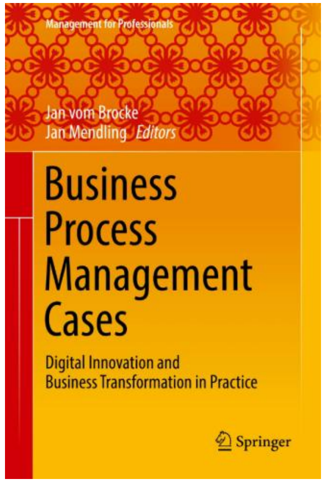
- **Section 12:** BPM as an Enterprise Capability

- **Section 5:** The Six Core Elements of Business Process Management









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